

Lucretius

ON THE NATURE OF THINGS

Translated by Ian Johnston
Vancouver Island University
Nanaimo, British Columbia

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TRANSLATOR'S NOTE

This translation is based primarily on the Latin text of H. A. J. Munro, Fourth Revised Edition (London 1900). However, I have not followed all of Munro's editorial decisions, especially where the removal and rearrangement of lines are concerned, and often I have made use of the suggestions of other editors about particular words, the arrangement of lines, and missing lines.

For the convenience of the reader who wishes to consult the Latin text, I have included the line numbers of the Latin text of William Ellery Leonard, because that is the most readily accessible version on the internet (at Perseus), even though there are some discrepancies between the line numbers in his text and in Munro's. In the text of this translation, the numbers in square brackets refer to the line numbers in Leonard's Latin text; those without brackets refer to the English text. In the count, successive partial lines count as a single line.

I have supplied footnotes for two reasons: first, to inform the reader of a few details of my editorial decisions about the Latin text and, second, to provide a general commentary of some help to the reader encountering Lucretius for the first time.

BACKGROUND NOTE

Titus Lucretius Carus (c. 99 BC to c. 55 BC) was a Roman poet and author of *De Rerum Natura* [*On the Nature of Things*], which he appears to have completed but failed to revise and fully prepare for the reader. We assume from the words of the poem that Lucretius was a friend of Memmius, a prominent Roman political figure, to whom the work is addressed. Other than that, we know virtually nothing about him, other than a scurrilous story circulated four hundred years after his death that he was driven mad by a love potion, created his poem in lucid intervals, and then killed himself.

On the Nature of Things is a long celebration of the philosophy of Epicurus, a vision of life which claims that all natural phenomena are to be understood in terms of material particles, that gods play no role in natural events or human affairs and have nothing to do with creating or sustaining the world, that the immortality of soul is a myth fabricated by traditional religions for their own absurd and cruel purposes, and that the highest goal of life is the avoidance of unnecessary pain and the pursuit of appropriate pleasures, especially through contemplation. The poem is thus a long, impassioned plea for what we would now call classical humanism.

Most of *On the Nature of Things* is taken up with a wide-ranging materialist explanation of natural phenomena based on atomic theory, so that we can understand how the world works without reference to divine planning or intervention and can accept how we human beings, like all other things, including minds and souls, are made up of material stuff which combined when we were born and which will dissolve back into particles when we die (as will the earth and our cosmos eventually). The notions of the immortality of the soul and of an afterlife of rewards and punishments are therefore specious.

It is important to recognize, however, that the greatness of the poem does not stem primarily from its contributions to our scientific knowledge or from any complex

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philosophical arguments. It is a magnificent poem because it conveys to us the excitement and passion of the speaker's feelings for these materialistic ideas and the urgency with which he pursues his ethical mission of persuading his readers to live better lives. It is the most famous, long-lasting, and influential endorsement of Epicurean philosophy in our culture.

For Lucretius, the world is in constant motion, driven by the mechanical forces of production and dissolution, and intensely vital. At the heart of it lies the random movement of basic particles, so that there is nothing deterministic about why things occur the way they do. Nature has its regular phenomena, of course, but its essence is the unpredictable motions of elementary particles. These can make our existence precarious and short-lived, but nature is also intensely beautiful, awe-inspiring, and worthy of contemplation. We should have the courage to accept this condition and reorient our lives so that we are not misled by false ambitions, unnecessary fears, and superstitions.

The poem has for centuries been extremely popular and influential. It played an important role in the development of Latin poetry before Virgil and, after it was rediscovered, became an essential text in those centuries when a knowledge of Latin literature was an essential part of an educated person's agenda. The list of those who have expressed their admiration for and debt to Lucretius reads like a Who's Who of Western culture.

BOOK ONE

[Invocation to Venus; plea for peace; dedication to Memmius; tribute to Epicurus; tyranny of religion; example of Iphigeneia; importance of resisting religion with reason; tribute to Ennius; Lucretius defines his task, acknowledges difficulty of using Latin; first principle: nothing is made of nothing; second principle: nothing is reduced to nothing; existence of invisible particles; presence of empty space (void); explanation of movement; sense experience as criterion of truth; no third form of nature; properties and accidents; time does not exist; primary elements are permanent; basic particles make hard and soft objects; primary particles cannot be broken up; criticism of Heraclitus; tribute to and criticism of Empedocles; criticism of Anaxagoras; analogy of elements to letters in words; infinity of matter and space; no common pull to the centre.]

Mother of Aeneas' sons, joy of men and gods, nourishing Venus, who beneath the stars that glide across the sky, crams full of life ship-bearing seas and fruitful lands—through you are conceived all families of living things which rise up to gaze upon the splendour of the sun, and when you approach, goddess, winds and sky clouds scurry off; for your sake, artful earth puts forth sweet flowers; for you, smooth seas smile, calm sky pours glittering light, and once day's face reveals the spring, winds blow freely from the west, bringing fertility, and air-born birds whose heart your power strikes give first signs of you, goddess, and your approach.¹ Then herds of wild beasts leap in carefree fields, swim through raging rivers—so seized with joy and eagerness, all follow you, no matter where you lead—from there through seas and mountains, roaring streams, leafy homes of birds, and fields now turning green, as you inspire all hearts with tempting love and, through desire, bring out new generations, each in accordance with its kind. And because you, by yourself, guide natural things and lacking your support nothing rises in the godlike regions of the light, and nothing rich and worthy of our love comes into being, I yearn for you to be my partner as I write, attempting verses on the nature of things, for my Memmius, whom you, goddess, have willed at all times to be excellent, a splendid man in everything he does.² So for him, divine lady, give these words all the more everlasting grace. Bring in a universal lull meanwhile that calms all brutal works of war on sea and land, since you alone can succor mortal men	10 20 30	[10] [20] [30]	INVOCATION TO VENUS
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¹Aeneas is the legendary founder of the Roman people, and Aeneas's sons are the Romans. The goddess of love, Venus, is his mother.

²Gaius Memmius was a leading politician in Rome (tribune in 66 BC), and evidently a friend of Lucretius (other than the poem we have no reliable details of their relationship). When his political career collapsed, he retired to Athens and Mytilene. He died around 49 BC.

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with tranquil peace, for Mars, the lord of war,
who controls the savage acts of battle,
will often hurl himself onto your breasts, 40
conquered by the eternal wound of love,
and there, with his smooth neck leaning back,
he gazes up, goddess, his mouth open,
and feeds his eyes, greedy with love, on you;
as he reclines, his breath hangs on your lips.
While he is there, goddess, from above allow
your sacred body to flow around him.
O splendid lady, let pleasing words pour
from your lips, seeking sweet peace for Romans, [40]
since at a time of crisis in our land, 50
we cannot do this work with peace of mind,
nor in these events can the noble son
of Memmius neglect the common good.¹

For the whole nature of gods, in itself, NATURE OF THE GODS
must for all time enjoy the utmost peace—
far removed and long cut off from us
and our affairs, and free from any pain,
free from dangers, strong in its own power,
and needing nothing from us, such nature
will not give in to those good things we do 60
nor will it be moved by our resentment.²
As for the rest, you must direct yourself,
with unbiased ears and a judicious mind
quite free from care, to proper reasoning, [50]
so that you do not scorn and throw away INSTRUCTIONS TO MEMMIUS
my gifts to you, laid out with true good will,
before you grasp them. For I will begin
to set down for you the highest matters
of heaven and gods, and I will disclose
the first principles of matter, the ones 70
nature uses to produce, increase, sustain
all things, and into which she changes them
once more, when they disintegrate. These things,
in explanatory accounts of them,
we are accustomed to call “materials”
and “the generating bodies of things”—
to name them “seeds of things,” using the term [60]
“primordial elements,” since they come first,

¹Lucretius appears to have written these lines at a time of growing political crisis in Rome, during the consulship of Caesar and his political alliance with Pompey (c. 60 BC). He had already lived through the civil war between Sulla and Marius (in 82 BC).

²The passage “For the whole nature of the gods . . . resentment” (54 to 61 in the English) reappears in Book 2 (line 646 in the Latin). Many editors and translators omit these lines from this opening part of the poem. It seems likely, too, that after line 54 (line 43 in the Latin) a few lines have been lost, in which a transition is made to Memmius.

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and from these things all objects are derived.¹

When to all eyes men's life lay foully crushed
throughout the land beneath the heavy burden
of religion, who, from heavenly regions
would show her head, menacing mortal men
with her hideous face, a Greek man was the first
who dared raise his mortal eyes against her,
the first one to oppose her, undeterred
by stories of the gods, by lightning strikes
or menacing rumbles from the heavens.
Instead, with even greater eagerness
he roused his spirit's keen intelligence,
to answer his desire to be the first
to break the narrow bolts of nature's doors.²
And so the living power of his mind
won out, and he moved forward, far beyond
the flaming bulwarks of the world, and then,
in his mind and spirit, made his way through
the boundless immensity of all things.³
From there, triumphant, he brings back to us
what can come into being and what cannot,
and finally the processes by which
the power of each thing has boundary stones,
a deep-set limit. And so religion,
in its turn cast down, is thrown underfoot.
This victory makes us heaven's equals.

80

EPICURUS

90 [70]

100

But I fear in these matters you perhaps
may think you move into first principles
of a wicked way of thinking, starting
down an impious road—whereas, in fact
that same religion has too often spawned
profane and criminals acts, like that time
at Aulis, when leaders chosen by the Greeks,
preeminent men, horribly defiled
the virgin Trivia's altar with the blood
of Iphianassa.⁴ Once the bands of wool

[80]

SACRIFICE OF
IPHIANASSA AT AULIS

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¹Lucretius for some reason wishes to avoid the Greek word *atom* and its Latin equivalent, *atomus*. It may be that, given his desire to show how his Latin, in spite of its limitations, is capable of explaining “obscure” Greek ideas, he does not wish to use a Greek word very familiar to many of his readers. Whatever his reasons, I have not used the word *atom* in the text of this translation for the reason given above and also because the English word *atom* immediately conveys to the modern reader a great deal more information than the Greek word did to Lucretius or to his readers (e.g., electrons, protons, nucleus, and so on).

²The “Greek man” is Epicurus (341-270 BC), a Greek philosopher, founder of the school of philosophical thought called Epicureanism. None of his work remains, except for some fragments.

³Lucretius commonly uses the term world (*mundus*) to refer to the part of the universe more or less visible from earth. It does not mean earth, which is part of this world, or the entire universe, which contains many worlds. As Lucretius makes clear later in the poem, this world is a sphere enclosed in fiery aether. Hence, as Bailey observes, the expression about the bulwarks of the world is to be taken literally.

⁴Homer gives Agamemnon's eldest child the name Iphianassa. However, the girl is usually called Iphigenia. Smith suggests that Lucretius uses the Homeric name in order to give his poem more epic weight.

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were wrapped around the young girl's hair and hung
down both cheeks equally, and once she saw
her father standing right beside the altars
looking gloomy and, there at his side, priests [90]
hiding the knife, with people gazing on,
weeping at the sight of her, she sank down, 120
kneeling on the ground, struck dumb with terror.
The hapless girl had been the very first
to award the king the name of father,
but at such a time that was no help to her.
For men's hands lifted her and bore her on,
trembling, to the altars—and not so that,
with a solemn ritual completed,
a loud bridal hymn could now escort her,
but so she, quite pure in her defilement,
even at the time of her own wedding, 130
might fall a wretched victim to a blow
from her father's hand in that sacrifice,
to ensure a happy, successful trip [100]
was granted to the fleet. That shows how much
religion could turn mankind to evil.

And even for you the time will come when, RELIGIOUS SUPERSTITIONS
overpowered by prophets' horror stories,
you seek to move away from us. No doubt,
they can now make up many dreams for you
which could disturb a life of principle 140
and with fear upset all your good fortune—
and rightly so. For if men could perceive
there is a set limit to their misfortunes,
they would, with some reason, have strength enough
to resist religion and prophets' threats.
But now, since we must fear that, when we die,
we will be punished for eternity,
there is no means, no possibility, [110]
of fighting back. For people do not know
the nature of the soul—whether it is born 150
with them, or, by contrast, is inserted
at their birth, whether it perishes with us,
dissolved in death, or whether it visits
the shades of Orcus, his enormous pools,
or whether, as our Ennius said in song,
it sets itself, by divine influence,
in other animals. He first brought back
from lovely Helicon a wreath of leaves

Agamemnon, the leader of Greek expedition to Troy had offended the goddess Artemis, who then sent contrary winds to prevent the fleet assembled at Aulis from sailing. The prophet Calchas told Agamemnon he would have to sacrifice his daughter in order to get favourable winds. In some versions of the story Agamemnon lured Iphigenia to Aulis by telling her she was going to be married to Achilles. Agamemnon sacrificed his daughter, and the fleet sailed to Troy. Trivia is another name for the Greek goddess Artemis or her Roman equivalent, Diana.

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that never fades—its fame is spoken of
by families of men in Italy.¹ 160

And yet after this, Ennius explains,
setting it down in deathless poetry,
there are truly regions in Acheron [120]
where our souls and bodies do not remain,
but only certain phantoms, strangely pale.
From there, he says, in front of him arose
the ghost of always flourishing Homer,
which started to shed salty tears and then
to describe in words the nature of things.

And so we must with proper reasoning 170
look into celestial matters—explain
the reasons for the wandering of the sun
and of the moon, the force which brings about
everything that happens on the earth.

And, in particular, we must employ [130]
keen reasoning, as well, to look into
what makes up the soul, the nature of mind,
and what it is that comes into our minds
and terrifies us when we are awake
and suffering some disease or in deep sleep, 180
so that we seem to see and hear right there,
before our eyes, those who have met their deaths,
whose bones the earth now holds in its embrace.

I am not unaware how difficult
it is to clarify in Latin verse
obscure matters discovered by the Greeks,
above all since we must deal with many things
employing new words, because our language
is impoverished and the subject new.
But your own excellence and the pleasure 190 [140]
I look forward to from your sweet friendship
are prompting me to finish any work,
no matter how demanding, urging me
to stay awake throughout the peaceful night,
seeking words and verse where I can at last
hold up a clear light for your mind, and you
can see into the hidden core of things.

And so this terror, this darkness of mind,
must be dispelled, not by rays from the sun
or bright shafts of daylight, but by reason
and the face of nature. And we will start
to weave her first principle as follows:
nothing is ever brought forth by the gods 200 [150]

FIRST PRINCIPLE: NOTHING CAN
COME FROM NOTHING

¹Quintus Ennius (239-169 BC) was a Latin poet and playwright, none of whose works survives except in fragments. He was considered the first great Latin poet. Orcus is the Roman god of the underworld.

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from nothing.¹ That is, of course, how, through fear,
all mortal men are held in check—they view
many things done on earth and in the sky,
effects whose causes they cannot see at all,
and so they assume that such things happen
because of gods. Hence, once we understand
that nothing can be produced from nothing, 210
then we shall more accurately follow
what we are looking for, how everything
can be created and all work can be done
without any assistance from the gods.

For if things were made from nothing, each type [160]
could be produced from any other thing,
with no seed required. To start with, humans
could spring up from the sea, races of fish
arise from land, and birds burst from the sky.
Domestic beasts, other cattle, all kinds 220
of savage creatures of uncertain birth
would live in farm land and the wilderness.
The same fruits would not be produced from trees
with no alterations—no, they would change,
and any tree could carry any fruit.
In fact, were there no procreant bodies
for each one, how could anything possess
a fixed and constant mother? But now, because
each object is produced from certain seeds,
it grows out of them and comes to regions 230
of the light from places in which its stuff, [170]
the primary elements of each, belongs.
For this reason, it is impossible
for all things to be produced from all things,
since there are in specific substances
powers which make those substances distinct.

And why do we see roses coming out
in spring, grain when it gets hot, and grape vines
ripening under autumn's influence,
if not because, when certain seeds of things 240
have fused together at their proper time,
whatever is created then appears,
while the season favours it, and the earth,
full of life, safely brings out tender things
to regions of the light? But if these things [180]
were made from nothing, then they would spring up
suddenly at random, at strange moments
of the year, because then there would not be
any primal matter which could be checked

¹This is the most important basic principle of Epicurean materialism: everything is composed of matter and must be made by the actions of matter, without divine miracles which produce a physical object out of nothing at all.

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from a productive union at a time 250
that was unfavourable. And what is more,
if they could increase in size from nothing,
there would be no need of time for growing
once seeds had joined together. For young men
might suddenly be produced from infants,
and groves of trees might come up from the ground,
arising unexpectedly. These things,
quite obviously, just do not happen—
all things mature gradually [at set times],
as is appropriate, [since they all grow] 260
from certain seeds, and as they get bigger,
they maintain their kind, so you can understand [190]
that every individual thing is fed
and grows from its own particular stuff.¹
And what is more, without seasonal rains
during the year, the earth could not produce
her delightful fruits. Then, too, without food
animal nature could not reproduce
the species and maintain its life. From this,
you can all the more easily believe 270
that many things have many elements
in common—just as we see with letters,
which are the same in many words—rather
than thinking any substance could exist
without its primary matter.²

And further,
why could nature not have created men
so big that they could make their way on foot [200]
across the sea, with their hands tear down
great mountains, and in life expectancy
outlast many human generations, 280
unless the reason is that certain stuff
has been designed to make specific things,
and that determines what can be produced?
Therefore, we must acknowledge that nothing
can be produced from nothing, since with things
there is a need for seeds, from which each one
is made and can be brought into the air,
into the gentle winds. And finally,
since we perceive that cultivated lands
are preferable to those left on their own 290

¹I follow Munro's suggested emendation of the text in lines 188-189 of the Latin. The additional words are in square brackets.

²Lucretius here introduces one of his favourite analogies, comparing the letters of the alphabet used in the formation of words with the primary particles used in the formation of substances. The analogy is all the more pertinent in Latin because the word *elementum* (plural *elementa*) refers to both letters and particles.

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and, when worked by hand, yield better produce,
we clearly see that there are in the earth [210]
primordial elements of things, which we,
by turning over fertile ground with ploughs
and taming the land's soil, stir into birth.
If there were no seeds, you might well observe
that things become much better on their own
without our work.

To this we can also add
that nature dissolves all things back again
into their own elements and does not
turn matter into nothing.¹ If anything
were destined to die, including the parts
of which it is composed, then all matter
would be quickly snatched away before our eyes
and vanish. For no force would be needed
which could bring about the dissolution
of its parts and sever their connection. [220]
As it is now, since everything consists
of ageless seeds, nature does not let us
witness the death of anything, until
force intervenes to cut it into pieces 310
with some blow or to penetrate inside,
through the empty spaces, and dissolve it.
And if time totally destroys those things
it takes away by aging, consuming
all their matter, how does Venus send back
into the light of life those families
of creatures, each according to its kind?
When they are restored, how does artful earth
offer them food, nourish, and strengthen them, 320
meeting each one's needs? How do its own springs
and distant rivers flowing far and wide [230]
keep the sea supplied? How does the aether
feed the stars?² The infinity of time
and days gone by should have destroyed all things
made up of mortal elements. But if
those elements which make up and renew
the total sum of things have been around
though all the ages of those years long past,
then we can be assured they do possess 330
an immortal nature. And thus, no things
can be converted into nothing.

Indeed, unless some everlasting stuff
kept substances more or less connected PERMANENCE OF FIRST ELEMENTS

¹The second basic principle of Epicurean materialism is stated here: no substance can be reduced to nothing.

²Aether (or ether) is the material stuff which fills space, surrounding and containing all planets and stars. Since the stars are burning fires, they must be fed.

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in a mutual matrix, one common force [240]
and cause could generally destroy all things,
for then, in fact, a touch would be enough
to kill, as is obvious, if there were
no substance in a body which endured,
if it were linked-up seeds which any force 340
was bound to break apart. But as it is,
since different networks of first elements
unite together and since their substance
endures forever, things continue on,
their bodies unimpaired, until the time
an opposing force with sufficient strength,
a power which can undo their structure,
encounters them. Thus, there is no substance
which is reduced to nothing—but all things,
once dissolved, go back to material stuff. 350

Lastly, the rains vanish, when the aether, [250]
our father, has poured them into the lap
of earth, our mother. But then glistening crops
spring up, the branches on the trees turn green,
and trees themselves grow bigger and become
weighed down with fruit. Moreover, from this rain
our race is fed, as well as those of beasts.
Thus, we see happy cities filled with youth
and leafy woods full of young birds singing
on every side, and fat, weary cattle 360
set their bodies down in joyful pastures,
and dazzling white liquid milk flows out
from swollen udders; thus, new offspring play
on unsure limbs, frolic on tender grass, [260]
with fresh milk stirring their young hearts. And so,
what seems to disappear does not all leave—
nature renews one thing from another
and does not allow objects to be born
without the help of something else that died.

Come, I have been teaching you that matter 370
cannot be created out of nothing
and, in the same way, once it is produced,
cannot be reduced to nothing, and yet,
in case you should perhaps still start to doubt
my words, because our eyes cannot perceive
the elementary particles of things,
learn more about those bodies you yourself
must grant exist in what cannot be seen. [270]
First of all, the power of wind, once roused,
lashes harbours, annihilates huge ships, 380
scatters clouds. Sometimes in swift, whirling storms,
it drives across the plains, covering them
with giant trees, and assaults mountain tops

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with blasts that splinter wood—that's how fiercely
the wind howls out in passionate anger,
screaming and threatening with a frantic howl.
And therefore we can have no doubt that winds,
although invisible, are bodies, too.
They sweep sea and land as well as sky clouds,
jolt and ravage them with sudden whirlwinds. 390
They rush on ahead and spread destruction, [280]
just as water, whose nature is delicate,
suddenly carried in a flooding stream,
gorged with massive run-off from heavy rains
down towering mountains, races on, hurling
broken branches of the trees together,
whole trees, as well—strong bridges cannot stand
against the sudden power of the flood
as it charges on. In that way, swollen
with so much rain, the river then attacks, 400
with its massed, violent force, foundations
of the bridge—with a mighty roar it spreads
devastation, rolling immense boulders
underneath its waves, obliterating
whatever blocks its flow. And that, therefore,
must be how blasts of wind are carried, too. [290]
When, like powerful rivers, they swoop down
any place they wish, they drive things forward
and pummel them with repeated onslaughts.
Sometimes they seize things in a twisting swirl 410
and carry objects instantly away
in a spiraling vortex. That is why,
to make the point again, winds are bodies,
although unseen, for in the way they act
and in what they do, we find they rival
great streams, which clearly are material stuff.

Then, too, we sense the different smells of things,
yet never glimpse them coming to our nostrils.
Our eyes do not perceive a fiery heat, [300]
nor can they see the cold. As for voices, 420
we are not used to viewing them. But still,
all must consist of corporeal stuff,
since they can strike our senses, for unless
there is bodily substance, no object
can touch or itself be touched. Moreover,
clothes hung up on a beach with breaking waves
get wet, but these same garments, once spread out
dry off in sunlight, yet no one has seen
how water moisture makes its way to them
or how, by contrast, influenced by heat, 430
it escapes again. The moisture, therefore,
is broken up in tiny particles
our eyes cannot through any means make out. [310]

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There's more: with many yearly solar orbits,
a ring worn on the finger, through long use,
wears out underneath, and dripping water
falling from the eaves hollows out a stone;
and on a ploughshare, the blade's curving edge,
though composed of iron, when used in farm land,
thanks to some concealed effect, gets smaller. 440

We know people's feet wear down paving stones,
and bronze statues beside the gates reveal
that their right hands are being eroded
by people touching them so frequently,
when they salute them and then walk on by.
So we see these things are getting smaller,
as they are rubbed, but the jealous nature
of our vision prevents our noticing
at any moment matter moving off. [320]

Finally, whatever material stuff 450
time and nature little by little add
to things, forcing them gradually to grow,
the sharpness of our straining eyes can see
none of it, nor, once more, what wastes away
through old age and decay. Nor can you see
what rocks hanging by the sea and eaten
by corrosive salt lose in each moment.
Hence, nature works with unseen particles.

However, nature does not hold all things
in corporeal matter densely packed 460
on every side. For in material stuff [330]
there is a void—in many instances
a useful point to know; it will not let you
roam around in doubt, always seeking out
the total sum of things and losing faith
in what I say. So, then, there is a void—
intangible, empty, vacant. If not,
if this were not there, there would be no way
that anything could move, because substance
has this property—it stands in the way, 470
it obstructs—this would be present in it
all the time, acting against everything.
Therefore, nothing at all could move forward,
since nothing else would first make room for it
by giving way. But now, on sea and land, [340]
and in the celestial sky, we notice
before our eyes many things being shifted
in various ways by various means, and these,
if there were no void, would not so much lack
restless motion, of which they were deprived, 480
as have no means at all of being born,

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since matter, everywhere a compact mass,
would have remained inert.

Besides, although
things may be thought as solid as you please,
nonetheless, from what follows you may see
they have matter made up of elements
spaced far apart. For in rocks and caverns
liquid moisture flows, and every object
weeps many drops; food gets distributed
through the whole body in all living things; 490 [350]
orchards grow and, in due time, deliver
their fruit, because nourishment is sent up
from the lowest roots to the entire plant
through all the trunks and branches; voices move
through walls, fly through closed rooms in houses;
stiff frost penetrates right into our bones.
If there were no empty spaces through which
these substances could pass, there is no way
you would notice things like that occurring.

And then, why do we see some things weigh more 500
than other things, when there is no difference
in their size? For if in a ball of wool [360]
there is just as much matter as in lead,
they should weigh the same, since material stuff
has the property of pushing all things down,
but, by contrast, the nature of a void
continues on without weighing anything.
And so, the object which is just as large
and yet seems lighter clearly demonstrates
that it contains in it more empty space; 510
whereas, the heavier object indicates
that it has more material stuff inside
and far less void. Thus, there can be no doubt
the thing which we, with our keen argument,
are seeking out, what we describe as void,
exists, mixed in with solid matter.

I am forced, in dealing with these issues, [370]
to counter in advance what some men teach,
so that they cannot lead you from the truth.
They assert that when a fish moves forward 520
water gives way, opens liquid channels,
because fish leave behind an empty space,
into which water, as it moves aside,
can flow—in this way, other substances
can also move among themselves, change spots,
although all matter is completely packed.
This concept clearly has been taken up
through faulty reasoning. For where, I ask,

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could the fish move forward, if the water
did not give them room? In what direction
could the water shift aside, if the fish
could not swim forward? So we must, therefore,
either deny that any substance moves,
or else assert that material substance
has empty space mixed with it—from that fact
each thing's motion gets its initial start.
Lastly, if two wide bodies placed together
quickly separate, then quite obviously
the air must occupy all empty space
which is created there between the two. 540
And yet, however fast the flow of air,
as it blows in from all around, it still
would not be able to fill all the space
at once—air must fill one location first [390]
and then take over every place in turn.
Now, once these bodies have shifted apart,
if someone perhaps thinks that this occurs
because the air has made itself compact,
he is in error, for then a vacuum
is produced which did not exist before, 550
and, in the same way, what was previously
empty space is filled. In such a process
air cannot become more dense, and even if
that were possible, it would not, I think,
be able, without empty space, to draw
into itself and keep its parts united.¹

For this reason, though you may hesitate
and call many things in doubt, nonetheless,
you must grant there is a void in matter.
Besides, I could remind you of the truth 560 [400]
of what I have described by scraping up
many arguments. But for a keen mind
these small tracks will be enough—using them,
you yourself can recognize the others.
Just as dogs, with their noses, often find
the lair of some wild beast which roams the hills,
once they have found the right tracks on the path,
although the den was hidden in the leaves,
you yourself will be able, on your own,
in these matters to understand one thing 570
after another, make your way inside
each obscure hiding place, and then from there
draw out the truth. But, Memmius, if you
are slack or shrink a little from these things, [410]

¹The point in this rather awkward example seems to be that the idea of air being compressed or made less dense requires one to believe in empty space between the basic particles of air. If there is no empty space and air is all compact particles, how could it be compressed? And if it could be, that would create a vacuum somewhere where there was no void before.

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I can make you the following simple pledge:
from the riches in my heart, my sweet tongue
will pour out cups drawn from such great sources,
that I fear a slow old age will steal up
across our limbs, unfastening those bands
of life in us, before the full supply
of arguments on any single subject
in these verses has poured into your ears.

But now, to get back to weaving in words
what I have started: all things in nature
thus in themselves are made up of two things,
material substances and empty space,
in which these substances are placed and move
in various directions. Matter exists—
sense perception shared by all tells us that.
If faith in sense is not first firmly set,
if it does not prevail, there is nothing
to which we can appeal in what we claim,
by any form of mental reasoning,
about the truth of things we cannot see.¹

Then, once again, if space, which we call void,
and place did not exist, then materials
could not be situated anywhere
or move at all in different directions,
a point we have considered just above.
Moreover, there is nothing you can claim
is separate from all matter and distinct
from empty space—some third form of nature,
as it were, which someone might discover.
Whatever will exist, must, in itself,
be something. No matter how large or small
its size may be, so long as it exists,
if it can make contact, however slight
and delicate, it will increase the sum
of substantial things and be included
in the total. If it cannot be touched,
cannot, in any of its parts, prevent
matter in motion from passing through it,
quite clearly it will be that empty space
we call a void. So, too, whatever stuff
inherently exists, will have to act
or else to suffer when other matter
acts upon it, or else it will be there
so matter can exist and act in it.
But nothing can act and be acted on
unless it has corporeal substance,

¹Central to this argument for Epicurean materialism is a faith in sense perception as the criterion of truth. Only by contact with material things (i.e., sense perception) do we learn what is true and test our theories about what we cannot sense. Lucretius returns to this basic principle a number of times.

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and nothing can offer room for motion
unless it is empty, vacant space. Thus,
apart from void and matter, there can be,
in the whole sum of things, no third nature
left by itself, which at any time might fall
under our senses or which anyone
could ascertain with mental reasoning.

For you will find whatever things we name
are either properties, that is, attached
to these two things, or else you will perceive
they are their accidents, mere chance results.
A property is something which cannot
ever be separated or cut off
without destroying something by its loss—
like a stone's weight, water's fluidity,
and a fire's heat.¹ But on the other hand,
with slavery, poverty, wealth, freedom,
warfare, harmony, and other things which,
whether present or absent, do not change
the nature of a thing, our custom is
to call them, as is fitting, accidents.

PROPERTIES AND ACCIDENTS

[450]

630

640

Then, too, time in itself does not exist.
From things themselves our senses comprehend
what has been accomplished in the past,
what is present now, then what will follow
afterwards. We must concede that no one
has a sense of time in and of itself,
apart from things in motion or at rest.
What's more, when people claim "the ravishment
of Tyndareus' daughter" or "the rout
of Trojan races in the war" are real,
we must take care they do not compel us
to say perhaps that in and of themselves
these things exist, when time, which cannot now
be summoned back, has carried away
men of that generation, those for whom
events like these were merely accidents.²
One could say that whatever things are done
are accidents—in one case of the Trojans,
in another of the place itself. Furthermore,
if there was no material stuff in things
and no place or space in which all actions
happen, then Helen's beauty would never
have lit the fire of love which then blazed through

TIME

[460]

HISTORICAL NARRATIVES

650

[470]

660

¹I follow Munro in omitting line 454 in the Latin.

²Tyndareus' daughter is a reference to Helen of Troy, who was carried off from her home in Sparta to Troy by Paris, a prince of Troy. The point of these historical examples is to stress that the only reality is physical matter and void. What happens to material things (as in historical events) is simply an "accident." Matter and space are primary because without them no "accidental" event would have occurred.

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the Phrygian chest of Paris, igniting
the glorious struggles of that savage war,
nor would the wooden horse have secretly
delivered in the night those sons of Greece
born from its belly, who then set on fire
the citadel of Troy. Thus, you can see
that each event has no being—does not,
in any fundamental way, exist
the way that corporeal matter does,
nor can we describe it as existing
in the same way as empty space—instead
you can with justice label all events
accidents of the body and the place
where each of them occurs.

670

[480]

Bodies, therefore,
are, in part, primary elements of things,
in part, those elements in combination.
There is no force which can eradicate
the primary elements—their solid stuff
will finally endure, although it seems
hard to think that one can find in matter
any object with a solid body.
For lightning from the heavens penetrates
walls of houses—noises and voices, too.
Iron thrust into fire glows white hot, and stones,
when subjected to fierce heat, crack apart.
When heated, gold loses hardness and melts,
and icy bronze, once overwhelmed with fire,
turns liquid. Heat and penetrating cold
flow through silver when, as is our custom,
we lift up our cups and our hands feel both,
as water drops pour out from up above.¹

PERMANENCE AND SOLIDITY
OF PRIMARY ELEMENTS
680

[490]

690

So, given all that, we see that nothing
in matter is firm. But since true reason
and the nature of matter require it,
just listen, while in a few lines we show
that things with solid, eternal bodies
do exist—we shall prove that they are seeds,
primary elements of matter, from which,
in the grand total of created things,
all objects now are made. To begin with,
since we have shown that nature has two parts,
consisting of two very different things,
matter and space, in which all things occur,
each of them must be purely what it is,
in and of itself. Where there is empty space—

700 [500]

¹Watson notes that Lucretius is referring here to the common habit of holding up a silver goblet with some wine in it, so that hot or cold water could be poured into it (hot in winter, cold in summer).

Book One

what we call a void—there is no matter, 710
and similarly, where there is matter
there is no way there can be empty space.
Thus, the primary elements are solid [510]
and without a void. Then, too, since there is
in created things a void, there must be
solid space around it. There is nothing
which, by proper reasoning, can be shown
to hide an empty space, contain a void
inside itself, unless you will concede
what holds it consists of something solid. 720
But nothing can contain a void in things
except material stuff in combination.
Thus, matter, which consists of solid bodies,
can be eternal, although all the rest
may be dissolved. Besides, if what we call
empty space did not exist, the universe [520]
would then be solid. But unless there were
certain bodies filling whatever space
they occupy, then all existing things
would consist of empty space, a vacuum. 730
Thus, there is no doubt that material stuff
is distinct from void. Both things alternate.
Space is not completely full of matter,
and yet not wholly empty. Hence, there are
certain elements which can fill their space
and mark off what is full from what is void.
These elements cannot be broken up
by an external blow or be dissolved
by piercing their inside, nor can they yield
to any other method one might try, 740 [530]
a point I showed you somewhat earlier.
For it does seem that without empty space
nothing could be smashed apart or broken
or cut in two and split, or let in moisture
or seeping cold, or penetrating fire—
actions by which all objects are destroyed.
The more each thing contains a void inside,
the more it falters under these attacks
deep within it. So if first elements
are, as I have shown, solid, without void, 750
then they must be eternal.

BASIC PARTICLES ARE ETERNAL

Furthermore,
if material stuff had not been eternal, [540]
all things would have been utterly reduced
to nothing long ago—and things we see
would have been reborn from nothing. But since,
as I have previously explained, nothing
can be produced from nothing and, further,
what has been produced cannot be reduced

Book One

to nothing, then first elements must be
made of everlasting stuff, into which, 760
when its time is over, every object
can be dissolved, so matter is produced
for the renewal of things. Thus, elements
are entirely solid—since otherwise
there is no way they could have been preserved
through ages of infinite time till now, [550]
in order to restore things once again.
Besides, if nature had set no limits
to things being destroyed, particles of stuff
by now would have been constantly reduced, 770
worn out by time gone by, so that nothing
made from them at any specific time
could complete its entire span of life.
We see that anything can be dissolved
more quickly than it can be assembled
once again, and therefore all those objects
which the long, endless succession of days
in times past had to this point smashed apart,
by demolishing and dissolving them,
could never, in the time that yet remains, 780 [560]
be restored. But now determined limits
have been clearly set to the destructions,
since we see all things are recreated
and, at the same time, a fixed period
assigned to things according to their kind,
in which they can attain their bloom of life.

To this we add that, though materials 790
consist of elements completely solid,
yet one can still explain how everything
which is soft could be created from them—
for example, air, water, earth, and fire—
the processes by which these are produced
and the force by which each one carries on,
because, briefly put, there is empty space
intermixed in things. But if, by contrast, [570]
the primary elements of things were soft,
no reason could be given for the way
strong flint and iron could be created,
for their whole nature would entirely lack
basic principles for its foundation. 800
Thus, elements are strong, simple, solid.
When they form more compact concentrations
then all things can contract and demonstrate
their strength and power.¹

¹If the primordial particles were soft, there would be no way of accounting for hard objects, because the basic stuff of matter would contradict this idea. The notion of hard basic particles and empty space, by contrast, allows one to explain the different qualities of “hard” and “soft.”

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And furthermore,
if there was no limit for breaking elements,
certain particles of material stuff
would still have had to last through endless time
without being attacked by any danger. [580]
But since they would exist as fragile stuff,
endowed that way by nature, then it seems 810
inconsistent that they could have lasted
an infinite time through all the ages
assaulted by countless blows. Moreover,
since limits have now been given for growth
of things, each in accordance with its kind,
and for the ways they keep a grasp on life,
since it has been determined and sanctioned
by laws of nature what each thing can do
and what it cannot, and since none of that
has changed and everything remains the same— 820
so much so that in their young different birds
display particular body markings
of their species and maintain the pattern— [590]
we can be sure as well that things must have
a body of unchanging matter. For if
the primordial elements of things could,
in any way, be overpowered and changed,
then we would also have no certainty
about what could or could not come to be
and, in addition, about the principle 830
by which each thing has its power defined,
its fixed boundary stones, and species could not,
time after time, bring back their parents' nature,
manner of life, food, and movements, each one
following its own kind.

In addition, BASIC ELEMENTS CANNOT BE
since there are always extreme particles, BROKEN DOWN
[which in objects are the tiniest things
we see, there should, in the same manner, be
a smallest point] in those things which our sense [600]
cannot perceive—and that point, quite clearly, 840
has no parts and consists of the smallest
element in nature; it has never been
isolated on its own and cannot be
in future, since it is itself a part,
the single primary part, of something else.¹

¹There is general agreement that some lines are missing before line 600 in the Latin. Following other translators and commentators I have used the two-line restoration by Munro, placed between square brackets. At this point Lucretius is establishing that there must be ultimately irreducible particles making up the smallest parts of corporeal matter. These small particles cannot exist by themselves and are bound indissolubly together, so that the smallest part of corporeal stuff, made up of combinations of particles, cannot be divided (just as an atom is made up of different parts; it is a compound of particles but cannot be broken down into those particles). In this analogy the logic is a bit odd: he claims that because visible

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Then other parts like it and still others
in a series fill, in a compact mass,
the substance of that corporeal stuff.
Because they cannot exist on their own,
these parts must adhere to certain places 850
where there is no way they can be detached.
Thus, primary basic stuff is purely solid—
a close-packed mass of smallest elements, [610]
not combined in an aggregate of parts,
but rather with a unitary force
which is eternal. Nature does not let
any part be separated from them
or diminished, reserving them as seeds
for objects. And furthermore, if there were
no smallest body, the minutest stuff 860
would be made up of infinite pieces,
since, as you see, the half of any part
will always have its own half, and nothing
will bring the process to an end. And thus,
between the total sum and the smallest things
what difference will there be? Nothing at all
will distinguish them, for though the universe, [620]
the total sum of things, is infinite,
the smallest particles there are will still
equally consist of infinite parts.¹ 870
But since true reasoning rejects this claim
and asserts the mind cannot believe it,
you must concede, admitting there are things,
the very smallest natural elements,
which have no pieces, and since these exist,
you must also grant that they are solid
and last forever. And then if nature,
creative mother of things, were accustomed
to forcing all things to be broken down
into smallest particles, she could not 880
restore things now from those same particles, [630]
because things not endowed with any parts
do not possess the properties required
for generative stuff—different bondings,
weights, collisions, combinations, motions,
through which all actions happen.²

objects have a minimum size, beyond which we cannot see them, then it is reasonable to conclude that invisible elementary particles must have a minimum size. He uses the same analogy a few pages later.

¹The logic here, though not uncommon, is erroneous, claiming, as it does, that anything which is made up of an infinite number of parts is equal to any other thing similarly composed.

²The argument here is that the infinite division of matter would eventually produce particles which lacked the range of properties essential to those physical actions which create the objects of this world (for example, an atom, which is an indivisible unity of smaller particles, if divided up into those particles, could not function as an atom has to do if compounds are to be created and things produced from those compounds).

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That is why
HERACLITUS
those who claim that the substance of matter
is fire and that the grand sum of all things
consists of fire alone seem to have strayed
far from valid reasoning. Of these men,
890
Heraclitus is the chief, the first one
to head the charge, a man celebrated
for obscure speech, but more with simpletons
than with serious Greeks seeking out the truth.¹ [640]
For foolish people would rather admire
and adore everything they see concealed
in cryptic sayings, and consider true
the ones with power to contact our ears
agreeably, painted with pretty sounds.
How, I ask, can substances be different
900
if they are made from fire, pure and unmixed?
There would be no point in making hot fire
more dense or rarefied, if parts of it
had the same nature all the fire still has.
In condensed parts heat might be more intense, [650]
and less where parts were scattered and dispersed.
But you can conceive nothing more than this
which could be created from such causes,
much less could the huge diversity of things
exist from fire compressed and rarefied. 910
There's more: if they admit there is a void
mixed into things, fires will then be able
to be condensed or be left rarefied,
but because they see many things in that
which contradict their doctrine, they avoid
admitting that pure empty space exists
in matter—afraid of complications,
they lose the true path and do not perceive,
by contrast, that without a void in matter [660]
all things become compressed, a single mass 920
produced from all things, and this mass could not
send out quickly from itself a single thing,
the way warming fire throws off light and heat.
And so you see that fire does not consist
of compressed parts.²

But if perhaps they think

¹Heraclitus (c. 535-475 BC) was an important and influential Ionian philosopher from Ephesus in Asia Minor, who taught (among other things) that fire is the single primordial element and that the world is continuously changing. Only fragments of his work remain.

²The “they” mentioned here are the followers of Heraclitus, those who believe that fire is the single, basic, primary stuff. The major objection is one commonly made against those materialists who tried to identify a single basic substance as the primary matter out of which all things are made (water, fire, air, and so on): What causes can one think up which could create the diversity of the world from this one substance? And the objection to the absence of a void in matter is that then, as demonstrated earlier, no particles could move, since all space would be occupied.

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that fire, combining in some other way,
can be extinguished and change its matter,
then clearly, if they do not at some point
check their faith in this, all heat will, of course,
utterly decline to nothing—all things 930
which are produced will be made from nothing.
For when something is changed and moves beyond [670]
its limits, that is the immediate death
of what it was before. Thus, in their belief,
as you see, something must be left unchanged,
so that matter does not wholly revert
to nothing and the full supply of things
does not come to bloom reborn from nothing.

Now therefore, since there are undoubtedly
particles whose nature always is the same, 940
and, when they come and go or modify
their arrangement, things then change their nature
and corporeal stuff converts itself,
you may understand that these particles,
these elements of matter, are not fire.
For there would be no point if some of them [680]
detached themselves and left, or if others
were added on, or if the arrangement
of some of them were changed—if all of them
still were to retain qualities of fire, 950
what they created would, in every case,
be nothing but fire. As I judge these things,
the truth is this: there are certain bodies
whose combinations, movements, arrangement,
positions, and shapes produce fire—and when
their arrangement changes, they change nature;
they are not like fire or anything else
which can send particles to our senses
and by contact affect our sense of touch.¹

Moreover, to say that all things are fire 960 [690]
and in the total quantity of things
no substance is real but fire, a statement
Heraclitus makes, seems totally absurd.
On the basis of his sense experience,
he goes against his senses, subverting
those things on which all concepts we believe
depend and through which he himself has come
to recognize what he calls fire. He thinks
his senses truly know that fire exists,
but does not think they know all other things 970
which are no less clear. This appears to me

¹This summary statement indicates the main point about the basic particles. They are not like any particular substance in nature, but their different combinations produce the various things we see (like fire). This idea enables one to explain how the same basic stuff can create such an enormous variety of objects.

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empty and inane. For what will we then
appeal to? What could be more sure to us
than our senses as a way of noting [700]
what is true and what is false? And besides,
why would anyone sooner get rid of
everything and then want fire to remain
the only substance, rather than claim fire
does not exist, but other stuff remains?
Both assertions seem equally absurd. 980

Thus, those who have thought the material
of stuff is fire and the whole sum of things
can be made of fire and those who have held
that air is the first principle through which
things are produced, those, too, who have maintained
that water on its own can fashion things
from itself or that earth makes all matter
and changes into natural substances [710]
of all things seem to have strayed very far,
a long way from the truth.¹ Add those as well 990
who compound the primordial stuff of things,
linking air and fire, and earth and water,
and those who think that all things can arise
from these four elements—from fire and earth
and air and water. Among them, first comes
Empedocles of Agrigentum, born
within the coasts of that three-sided island
around which the Ionian Sea, flowing
in huge twisting coves, shoots up salty foam
from its green surf, and the ocean, rushing 1000
in a narrow strait, with its waves divides
the island rim from shores of Italy. [720]
Huge Charybdis is here, and here the growls
of Etna threaten, as, in her anger,
she once more gathers up her flames, so that
her power may yet again vomit fires
bursting from her gullet and hurl once more
her luminous flames up to the heavens.²
This great region, although it seems worthy
of admiration by the human race 1010
for many reasons and, so people say,
is somewhere one must visit, its produce
richly fertile, and strongly defended
by the power of its people, this place,

¹Anaximenes of Miletus (c. 585-c. 525 BC) taught that the primary material of stuff was air; Thales of Miletus (c. 624-c. 546 BC), considered in many quarters the founder of philosophical and of scientific thinking, taught that it was water.

²Empedocles (c. 490-430 BC), a Greek philosopher who lived in Sicily, proposed the well-known theory of the four elements (earth, air, water, fire). Fragments of his work survive. The Ionian Sea in ancient times was often thought of as extending past south Italy to Sicily. Charybdis is a whirlpool in the strait between Italy and Sicily, Etna an active volcano on the island of Sicily.

Book One

nonetheless, does not seem to have contained
anything more excellent within it
than this man, anything more sanctified, [730]
wonderful, and loved. In fact, even now
the verses from his godlike heart set down
and expound his celebrated findings 1020
in such a way he hardly seems created
from the human race. But he and those men
we talked about above, far inferior
to him and lower by several degrees
in eminence, although they did find out,
in an excellent and inspired manner,
many things and furnished explanations,
as if from temples deep within their hearts,
with more sanctity, far more true reason,
than the Pythia speaking from the tripod 1030
of Phoebus and his laurel—nonetheless,
in dealing with first elements of things [740]
these men fell into error—being great men,
their heavy fall here was significant.¹

Firstly, because they allow for movement
but take empty space away from matter, ERRORS OF OTHER
and they leave soft and thin material stuff— MATERIALISTS
air, sunlight, fire, earth, animals, and plants—
but still do not mix any vacancies
into their matter, and finally they set 1040
no end at all to splitting elements,
no limit to their being broken up,
nor does matter, in any way, possess
some particles of minimum extent,
although we do see an ultimate point
in every object, which to our senses [750]
appears the smallest thing we can perceive,
so that you can infer from this that things
we cannot see have their ultimate points,
the smallest particles which make them up. 1050
In addition to this, since they assert
the first material stuff is soft, things which
we see being born are made entirely
of perishable substance, so the sum
of all matter must revert to nothing,
the full supply of objects must arise
and grow up from nothing. How far these claims
are from the truth you will know already.
What's more, in many ways their elements
are incompatible and venomous 1060
to one another. And thus, when they meet

¹The Pythia was the priestess of Apollo (also called Phoebus Apollo or Phoebus) at Delphi who issued prophecies in answers to questions. She sat on a tripod. The laurel was sacred to Apollo.

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they will either perish or run apart, [760]
like those moments when, once a storm begins,
we notice how the lightning, rain, and winds
scurry off in various directions.

Moreover, if everything is produced
from four elements and if all matter
dissolves again into these elements,
how can they be called the primordial stuff
of things, rather than reversing the idea— 1070
making things the primordial material
of these four elements? For they are made
from one another and change appearance
and their total nature with each other
all the time. If you happen to believe [770]
that the elements of fire and of earth
and airy breezes and drops of moisture
come together so that, in combining,
their natures do not change, then you will see
that nothing could be created from them, 1080
no living thing, no inanimate body,
like a tree. In fact, all things in this heap
of various materials piled up
will display their natures—air will appear
mixed together with earth, heat with moisture.
But primary elements producing things
must use a secret, hidden influence,
in case some factor may predominate [780]
which could resist and check created things
so they cannot exist with their true nature.¹ 1090

Moreover, these men, in fact, take their start
from heaven and its fires and then make fire
first change itself to windy air; from air
water is produced, then earth from water,
and all things revert back again from earth—
first moisture, then air, then heat. And these things
do not stop changing into one another,
passing from sky to earth, and then from earth
to aetherial stars. But there is no way
primordial stuff should do this. For something 1100
unchanging must remain, so as to stop [790]
all matter from being totally reduced
to nothing. For when something is transformed
and goes beyond the limits set for it

¹The point here is that the fundamental elements of things should have no individual characteristics which dominate in the production of things. The “nature” of something created emerges from the combination and arrangement of fundamental particles which make it up but which themselves have no overt characteristics (their influence is “secret” and “hidden”). The “four element” theory of Empedocles requires that the physical characteristics of air, water, earth, and fire enter into the objects which they form by combination.

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that brings instant death to what it was before.
And thus, since these four basic elements
we talked about above go through changes,
they must consist of other particles
which cannot be transformed in any way,
in order to prevent, as you can see, 1110
all things being utterly reduced to nothing.

Why not conclude instead that there exist
certain bodies endowed with such a nature
that, if they should happen to create fire,
the same elements, with a few removed
or added, their structure and motion changed,
could make breezy air, and in this manner
all matter be transformed to other things?
“But plain facts,” you say, “clearly indicate
all things grow and are nourished from the earth 1120
up into the air, and if the season
is not kind to them, bringing rain showers
at favourable times, so orchards sway
under moisture from the storm, if the sun,
for his part, does not favour them and bring
his heat, no crops or trees or living things
could grow.” That is quite true. And we also,
lacking help from soft moisture and dry food,
would lose our bodies—all life then would drain
from bones and sinews. There can be no doubt 1130
that certain substances help and feed us,
as certain other foods feed other things.
Since many common primary elements
of many things are evidently mixed
in several ways in many substances,
therefore various things provide nourishment
for other different things. But frequently
what really matters is what elements
combine with and how they are organized,
what motions they both impart and absorb 1140
amongst themselves, for the same elements
make up sky, sea, lands, rivers, and the sun,
the same elements form crops, trees, animals—
but moving and combined with different ones
in different ways.

And why not? Everywhere
in these very verses of mine you see
many words have many shared elements,
though you must admit that words and verses
differ in what they mean and how they sound.
That’s how much basic elements can do, 1150
if one merely changes their arrangement.
But the primordial elements of things

Book One

can make more combinations and, from that,
create the whole variety of things.

Now let us also scrutinize that work
by Anaxagoras, the one Greeks call
the *homoeomeria*—what we lack
in our native speech does not allow us
to proclaim that word in our own language,
but it is easy to describe in words
the matter it contains.¹ For first of all,
that *homoeomeria* of things,
as he calls it, works like this: bones are made
from minuscule, extremely tiny bones
and, in the same manner, flesh is produced
from tiny, minute particles of flesh,
blood by many drops of blood collecting.
Gold, he thinks, can be made of bits of gold,
earth form a compact mass from little earths,
fire from fires, water comes from water,
and similarly with all other things—
that's what he imagines and understands.
But he does not concede there is a void
anywhere in matter or a limit
to cutting matter up. And that is why,
with these two principles, he seems to me
to be as much in error as those men
we talked about above. Now add to this
that he conceives primordial elements
as too weak, if, indeed, they are primordial
when they exist with a given nature
similar to things themselves and, like them,
suffer and perish, and nothing saves them
from destruction. For what in them survives
violent pressure so they escape death
in the very jaws of doom? Which of them—
fire or water or air? Or blood or bone?
In my view, not one—for essentially,
all stuff will be just as perishable
as all those things we clearly see dying,
defeated by some force, before our eyes.

[830]
ANAXAGORAS AND
HOMOEOMERIA

1160

[840]

1170

1180

[850]

1190

But no matter can revert to nothing
or rise up out of nothing—I appeal
to what I have proved before. Furthermore,
since food feeds us and makes our bodies grow,
we can know that veins, blood, bones, [and sinew

[860]

¹Anaxagoras (c. 500 BC-428 BC), a Greek philosopher from Asia Minor, maintained that the central concept in nature was *nous* (mind) and that all things existed as infinitely small particles of themselves. *Homoeomeria* means composed of similar parts. Bailey notes that Lucretius has not fully understood Anaxagoras and that his criticism in this passage is crude and “beside the mark.”

Book One

are made of particles unlike themselves.]¹
Or if they say all food is a mixture
of materials and contains small bits
of sinews, bones, and veins, and particles 1200
of blood, it will then follow, one believes,
all nourishment, solid and liquid, too,
is made up of various materials,
a compound mix of bones, nerves, veins, and blood.
Besides, if all those bodies which grow up
from earth exist in earth, it must be the case
that earth consists of all the different things
springing up from earth. Apply this thinking, [870]
and you may use this language once again:
if fire, smoke, and ash lie concealed in wood, 1210
then wood must be made of up of substances
unlike itself. Further, all those bodies
which earth feeds, it makes grow [from materials
different in kind from those which come from earth.
So, too, those substances which wood sends out
are fed] by matter of a different sort
than those which come from wood.²

Here there remains
a slender chance to avoid the issue,
which Anaxagoras appropriates
for his own purposes, so he may claim 1220
that all things are secretly intermixed
with everything, but what people notice
is the one mixed in the most, the one placed
at the front and more readily perceived.
This, however, is very far removed [880]
from truthful reasoning. For in that case,
we would also expect that grain, when crushed
by force of threatening stone, would often show
some sign of blood or of those substances
nourished in our bodies. In the same way, 1230
we should also expect that, when we rub grass
between two stones, blood should often drip out,
water should frequently give off sweet drops

¹A line is missing after line 860 in the Latin. As many commentators and translators have done I insert (in square brackets) a translation of the Latin suggested by Lambinus.

²There is a missing line or two in the Latin after line 873. I have followed the suggestion of Munro, who inserts two lines (indicated by the square brackets). Lucretius is exploring a problem arising from Anaxagoras's ideas. As Munro explains, if crops and trees grow out of earth, then the earth does not consist of little particles of earth (as the theory demands), but of miniature trees, crops, and so on. If flames and ash come from wood, then wood does not consist of miniature particles of wood, but of tiny bits of flame and ash (i.e., things unlike or different from wood). The case is the same with food. If food supplies all the things needed for the different parts of the body, then food does not consist of tiny particles of food, but of minute bits of bone, sinew, blood, and so on. Or else the things which are produced from earth and wood (like crops and fire) must come from things unlike themselves. The parent material (earth, food, wood) cannot be made up both of small particles of itself and of small particles of all the things which that material produces or feeds or turns into.

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mixed with the rich taste of milk from udders
of wool-bearing sheep. And undoubtedly,
when we crumble clumps of earth, we should see
types of grass, grain, and leaves—very small ones—
hidden scattered in the soil. Finally, [890]
in pieces of wood which we break apart
we should see ash and smoke and fire hidden 1240
in tiny particles. Since obvious facts
show this does not occur, we may be sure
there is in substances no such mixture
of matter, but there must be common seeds
of many substances concealed in things,
combined in many ways.

“But,” you may say,
“often in high mountains it does happen
that with tall trees the very tops of them,
if they are close by, are rubbed together,
an action forced on them by strong south winds, 1250
and then, like a flower, a fire blossoms, [900]
and the trees burst out in flames.” That is true.
But fire is not contained inside the wood—
instead there are numerous seeds of heat,
and when rubbing brings these seeds together,
they make fire in the trees. However,
if ready-made flames were concealed in wood,
fires could not be hidden for very long—
they would consume the forest everywhere,
burn up the trees to ashes. Now, therefore, 1260
do you not see what we just said above,
that frequently the essential issue
is what these same primordial particles
are combined with and in what position
and what motions they impart and receive [910]
among themselves, that the same elements
interchanging things a little, produce
fire and wood? In the same way, words themselves
consist of elements a little changed
among themselves when we use different terms 1270
to denote *firs* and *fire*. And finally,
if you now think that all things you observe
in objects you perceive cannot be made
unless you assume primary elements
endowed with a nature like those objects,
then those primary elements of matter,
by this line of reasoning, will perish,
as you see. What will happen is like this:
convulsed with cackling laughter they will shake
and wet their face and cheeks with salty tears.¹ 1280 [920]

¹The logic of this mockery perhaps rests on the idea that (as Kelsey suggests) since matter contains all things in miniature, it also contains human beings, who will find these ideas so ridiculous that they will

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Come now, listen more clearly, and then learn
what still remains. I am not unaware
how obscure the issues are, but great hope
of praise with her sharp thrysus has smitten
my heart and with that has infused my breast
with sweet love of the Muses—inspired by that,
my mind alive, I am now wandering
through trackless regions of the Pierides,
where no man's foot has ever gone before.¹
It gives me joy to approach those fountains 1290
no one has tasted and to drink from them.
I love to pick fresh flowers and collect
a splendid garland for my head in places
where the Muses have not yet crowned the brows [930]
of any man. Firstly, because I teach
important things and seek to free the mind
from constricting fetters of religion.
And then because the verses I compose
about dark matters are so luminous,
investing all things with poetic grace. 1300
And that, too, does not seem unreasonable.
For just as healers, when they try to give
young children foul-tasting wormwood, first spread
sweet golden liquid honey round the cup,
so at this age the unsuspecting child, [940]
with honey on his lips, may be deceived
and in the meantime swallow down the drink
of bitter gall—he may have been misled,
but he is not hurt—with this deception
he may be restored instead, grow stronger. 1310
In the same way now, since this reasoning
seems generally too bitter for those men
who have not tried it and the common crowd
shrinks back in fear, I wanted to explain
my argument to you in these verses,
sweet-spoken Pierian song, as if I were
sprinkling it with poetry's sweet honey,
if, with such a method, I could perhaps
get your attention on my verse, until
you perceive the entire nature of things— 1320
how it is shaped and what its structure is. [950]

But since I have revealed that particles,
the most solid bits of matter, always
move to and fro and never-ending time

THE POET'S PURPOSE

THE SUM OF MATERIAL
STUFF IS INFINITE

laugh themselves to death. Some have suggested the jump in thought is so abrupt that there might be some lines missing.

¹The thrysus is a plant stalk used during ecstatic rites of the god Bacchus; here it refers to poetic inspiration. The Pierides is another name for the Muses, derived from the place near Mount Olympus where they were alleged to have been born.

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does not destroy them, come now, let us see
whether or not the total sum of them
has any limit. Let us survey as well
that empty region we have discovered,
or the place and space where all things happen,
and learn whether, in its entirety, 1330
it is wholly limited or stretches
to infinite, immeasurable depths.

All that exists, then, has no boundaries
in any direction, for if it did, THE MATERIAL UNIVERSE
it would have to have something outside it. IS LIMITLESS
We see there can be no end to something, [960]
unless there exists something beyond it
which sets that limit, so one may observe
where our natural senses cannot follow
any further. Now, since we must admit 1340
that nothing exists outside the total,
it has no boundary—it is without end,
without limit. And it does not matter
where in it you stand—whatever station
someone occupies, he leaves the total
just as infinite in all directions.

Further, if we suppose all existing space
is now finite and if a man ran through
to its ultimate limit and then hurled [970]
a flying spear, would that spear thrown full strength 1350
fly out very far in the direction
it was sent, or do you think that something
could stop and block it? For you must concede
and grant one of these two alternatives.
Either one of these cuts off your escape,
forcing you to agree the universe
lies open without limit. For whether
there is some object which obstructs the spear
and prevents it going out where it was sent
and reaching its goal, or whether that spear 1360
is carried forward, its flight did not start
from any limit.¹ I will continue [980]
in this manner: wherever you may place
the furthest edge, I will raise a question:
What then happens to the spear? As it stands,
there cannot be an end point anywhere,
and room to fly will always lengthen out
the escape route of the spear. Finally,
before our eyes, we perceive that objects
set fix boundaries for objects: mountains 1370

¹If the spear is blocked, then something beyond space is limiting its flight, and if the spear continues on, then obviously it is moving beyond the limits of space.

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are limited by air, air by mountains,
land limits sea, and, on the other hand,
sea limits all the land. But still, there is [1000]
nothing outside the universe which might
set boundaries in place.¹

And furthermore,
if all the space of the whole universe
were enclosed on all sides with set limits
and were finite, by now supplies of matter,
given their solid weight, would have flowed down
from all sides together to the bottom, 1380
and so underneath the vault of heaven
nothing could take place, and there would not be
a heaven at all or light from the sun,
because all material, by sinking down [990]
for countless years, would by this time lie there
in a common heap. But now, as you can see,
no rest is given to first particles
of matter, for there is no foundation,
no bottom, to which they could, as it were,
flow down and find a resting place. All things 1390
move everywhere, always in constant motion—
material stuff is stirred up and supplied
from down below out of infinite space.
This, therefore, is the nature of deep space
and its extent—bright lightning in its course
could not pass through it—though sliding forward
for unending tracts of time, its motion,
as it proceeded, would not diminish
the remaining distance it still had to go.
That shows how much immense space lies open 1400
on all sides for things, free from all limits
everywhere in all directions.

Besides,
nature herself makes sure the universe
cannot set limits to itself—she compels
matter to be enclosed within a void,
and void, in turn, to be bound by matter. [1010]
With this reciprocal relationship
she therefore makes the total infinite,
or else one of the two, if the other
did not limit it, in its unmixed form 1410
would then extend out beyond all measure.
[But I have shown above that space spreads out
without limit; thus, matter, too, must be
infinite, for if the void were endless,

¹I have followed Munro in transposing lines 998 to 1001 in the Latin to a position a few lines earlier. The line numbers in square brackets are therefore in an odd sequence.

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and the total sum of matter finite,]¹
neither sea nor earth nor sky's bright spaces,
nor mortal races, nor sacred bodies
of the gods could endure for very long,
not even for the short space of an hour,
since, with their combining masses forced apart, 1420
supplies of matter would be carried off
and scattered through huge areas of space,
or what is, in fact, more likely, matter
would never have united and therefore
would never have produced a single thing,
since, in its dispersed condition, it would [1020]
be incapable of forming compounds.
For clearly the first particles of things
did not all place themselves in due order
by their own planning or intelligence, 1430
nor did they through some agreement assign
the motions each of them should have. Instead,
since there are many of them and they change
in many ways through all the universe,
they are pushed, energized by collisions,
for a limitless length of time, and then,
having gone through every kind of motion
and combination, they at length fall into
those arrangements which make up and create
this totality of things, which also, 1440
once suitably set in patterned motion,
[1030]
has been preserved through many lengthy years.²
It makes rivers with large flows of water
refresh voracious seas, and earth, once warmed
by sun's heat, restore what it produces,
races of living creatures grow and thrive,
and, in the aether, gliding fires live on.
There would be no way they could act like this,
unless supplies of matter kept arising
from infinite space, stuff which they then use
to restore, over time, what has been lost. 1450
For just as the nature of all living things
loses bodily substance and decays,
as soon as it lacks food, so everything
would have to waste away, as soon as matter,
diverted for any reason from its path,

¹Many editors suggest there is a gap here of one or two lines. I follow the Latin suggested by Munro. The translation of these lines is in square brackets.

²Here Lucretius is firmly rejecting any form of inner vital cause in matter or of divine purposefulness in nature. The basic material stuff of things is formed by chance collisions and movements of primary particles over infinite time. Munro notes that Lucretius's phrase *magnum annos* (here translated as lengthy years) is probably a reference to the so-called Great Year, an enormous expanse of time equivalent to many thousands of solar years.

Book One

failed to provide abundant fresh supplies.¹

[1040]

Nor can external impacts from all sides
hold together the complete totality
of all materials which have united.
True, they can often strike and hold in place
some section, until other particles
arrive which can make up the total sum,
but still, sometimes particles are compelled
to bounce off and in that very moment
give the primary elements space and time
to escape, so they can be carried off,
free from being linked up in combinations.
Thus, to repeat myself, many particles
must spring up. And yet to be capable
of keeping the number of those impacts
at a sufficient level, there must be
infinite amounts of matter on all sides.²

1460

1470

[1050]

In these things, Memmius, stay far away
from having faith in what some people say—
that all matter presses to the centre
of the universe and for this reason
the substance of the world remains in place
without all those collisions from outside,
and that the bottom and the top cannot
in any way be forced to separate,
since all matter sinks towards the centre—
if you believe that anything can stand
upon itself—and that all heavy things
on the lower part of earth press upward
and remain there, placed upside down on earth,
just as we now see images of things
in water. Similarly, they believe
that animals walk around with their heads
hanging downward and cannot fall off earth
into a lower region of the sky,
any more than our bodies can fly up,
of their own accord, to some location
in the heavens. When they observe the sun,
we perceive night stars, and they share with us,
each in turn, time determined by the sky,

MATTER DOES NOT PRESS TO
THE CENTRE

1480

[1060]

1490

¹If space were infinite and the supply of matter finite, then matter would spread throughout infinite space and never combine.

²The supply of elementary particles must be infinite; otherwise these particles could not form lasting aggregates and compounds. They would be detached from combinations (by the impact of other particles striking objects) and spread themselves through infinite space, without being replaced in numbers sufficient to keep the combinations of matter intact.

Book One

and pass nights in length equal to our days.¹
But vain [error has made these dreams for fools,
which they embrace with faulty reasoning.
There can be no centre where all extends
an infinite distance.

1500 [1070]

And if, in fact,
a mid-point did exist, nothing at all
could rest there for that reason, any more
than it could be, for some other reason,
driven far away.]² For all place and space
which we call void must let heavy bodies
pass, without distinction, to wherever
their motion carries them, through the mid-point
or through some places not in the centre.
And there is not any spot where bodies,
once they have arrived there, can lose the force
of weight and stand motionless in the void,
and what is void must not provide support
for anything, but let material through,
as the nature of empty space demands.
Thus, matter cannot, through this reasoning,
be held in combinations, overcome
by some desire to move towards the centre.
Besides, they do not believe all bodies
press towards the centre, but only those
of earth and water, [both what comes to earth
as rain] and what the body of earth holds,
that is, water from the sea and great floods
from mountains. But at the same time they claim,
by contrast, that soft breezes of the air
and fire's heat diffuse out from the centre,
and that is why all the aether flickers
with constellations all round, and sun's flame
throughout the deep blue heavens gets its food,
because all heat flying from the centre
collects there. Nor could, they say, top branches
on the trees produce any leaves at all
[if nature did not gradually send food
to each of them from earth through stems and boughs.
The reasons they set down are incorrect—
and besides, they contradict each other.

THE UNIVERSE HAS NO CENTRE

1510

[1080]

1520

[1090]

1530

Since I have shown that space is infinite,

¹As Copley notes, Lucretius seems here to confuse gravity, the force which pulls particles to a common centre within a celestial system (a concept which he rejects) with the idea that the universe, being infinite, cannot have a centre.

²The lines in square brackets are the translation for three lines missing parts in the manuscripts. I have followed the Latin suggested by Kelsey and Munro.

Book One

and, with space infinite, matter must be, too,]¹ [1100]
so that world's walls do not, like wings of flame,
suddenly disperse, scattering themselves 1540
through the enormous void, and other parts
do not, in a similar way, follow them,
and the innermost regions of the sky
do not fall down and, underneath our feet,
earth does not at once withdraw and all things
disappear, with substances being dissolved
in piled-up ruins of sky and matter,
parts scattering through the cavernous void,
so in an instant nothing remained of them
but blind elements and abandoned space. 1550 [1110]
For wherever you first assume a lack
of primary particles, that place will be
the door of death for things, since through that place
the whole mass of material elements
will rush out and dissipate.

In this way,
if you understand these matters, led on
without much trouble, [you will be able
to recognize the rest all by yourself],
for one fact will clarify another,
and dark night will not blind you to the road 1560
or stop you seeing nature's final ends,
and things will light a lamp for others things.

¹A number of lines are missing here. I have adapted the English reconstruction suggested by Munro in order to maintain the sense of the passage.

BOOK TWO

[Importance of philosophy; properties of particles; motion caused by weight and impact; weight of particles; collisions, rebounds, combinations; density of matter formed by combinations; wandering particles; no divine providence; examples of matter moving in sunlight; no particles move upward on their own; swerve of particles in their descent; weight does not affect speed in empty space; swerve linked to free will; continuity of motion in particles; importance of the shape of particles; different shapes of particles linked to different sensations; shapes of particles are not infinite in number; compound matter has particles of different shapes; earth as mother of all things; reference to Cybele; not all combinations of all particles take place; nature of gods; particles lack colour, heat, cold, taste, smell, but create objects with these characteristics; sensible objects are produced from insensible particles; necessary existence of other worlds; natural life cycle of all things, including the earth; decline of the earth.]

How pleasant it is, when windstorms lash
the mighty seas, to gaze out from the land
upon another man in great distress—
not because you feel delightful pleasure
when anyone is forced to suffer pain,
but because it brings you joy to witness
misfortunes you yourself do not live through.

IMPORTANCE OF PHILOSOPHY

It is also sweet to watch great armies,
opposing forces in a war, drawn up
in the field, when you are in no danger.

10

But nothing brings more joy than to live well
in serene high sanctuaries fortified
by wise men's learning—where you can look down
on other men, see them wandering round
in all directions, roaming here and there,
looking for a path in life, competing
in their natural gifts, striving for honours,
seeking with all their effort night and day
to rise to the top, to win great power.

[10]

O wretched minds of men, O blinded hearts!

20

In what living darkness, what great dangers,
you spend your lives, however long they last!
Do you not notice nature barking out
her one demand, that pain be kept away,
divorced from body, so that, free from care,
free from fear, she may derive enjoyment
in her mind from a sense of pleasure?

Hence, we see that for our body's nature
only a few things are truly needed—
the ones which do away with any pain.¹

[20]

30

Now, although there are also many things
which can more agreeably at times
provide us many pleasures, for her part
nature does not seek them—if houses lack
golden statues of young lads with right hands
holding flaming torches out, so that light

NATURE DOES NOT SEEK
MATERIAL PLEASURE

¹Here and in the lines following Lucretius refers to the Epicurean teaching that the best life is one lived free of pain. The most important pleasures are those of the mind when it has no worries. This principle is different from the common misconception that Epicureanism always involves living wholly for active physical or material pleasures.

Book Two

may be provided for nocturnal feasts,
if the home does not glitter with silver
or gleam with gold, or if harps do not make
gilded panels on the ceiling echo, 40
nonetheless, when, in their own company,
men lie beside a river on soft grass, [30]
under the branches of a towering tree,
and, with no great effort, enjoy themselves,
they restore their bodies, especially
when the weather smiles and annual seasons
scatter flowers across the greening turf.
If you are tossing on embroidered sheets
dyed deep purple, hot fevers will not leave
your body faster than if you are forced 50
to lie on common bedding. That is why,
since riches, high rank, and ruling glory
are of no advantage to our bodies,
it therefore follows that we must assume
they also bring no profit to our minds,
unless perhaps, when you see your legions [40]
marching keenly onto the Campus fields,
as if going off to war, with many men
held in reserve and strongly reinforced
with cavalry, and you draw up your troops 60
armed and ready, all equally inspired
with a common will, or when you observe
your ships swarming out, spreading far and wide,
then your religion, shocked by these events,
runs from your mind dismayed, and timid fears
of death leave—your heart is clear and carefree.¹
But if we see this is sheer foolishness,
a mockery, that, in fact, those worries,
the fears that follow men, are not afraid
of noisy weapons or of brutal spears— 70
they boldly live with kings and those who rule [50]
in our affairs and have no reverence
for glittering gold or glorious splendours
of purple garments—then why do you doubt
that all power to help us with these things
belongs to reason? That is especially true, 80
since our whole life is struggling in the dark.
For just as children in the dead of night
tremble and are afraid of everything,
so we, too, in the daylight, sometimes fear
things which should no more frighten us than those
which scare children in the night, those terrors

¹The manuscript has minor corruptions in two lines here. I have adopted the suggestions of Munro. The “Campus” into which the legions are marching is the Campus Martius (Field of Mars) outside Rome, where armies often practised maneuvers or put on displays. The point here is that sometimes military displays fill men with such enthusiasm they forget their normal fears. Lines 62 and 63 in the English (the reference to watching ships) are sometimes omitted or inserted elsewhere.

Book Two

they believe will happen. Therefore, this fear,
this darkness in the mind, must be dispelled,
not by the sun's rays or shafts of daylight, [60]
but by the face of nature and by reason.

Come now, I will explain how, through motion, MATTER IN MOTION
creative matter in material stuff
produces various things and, once produced,
breaks them down, then how a force compels them 90
to act this way, and what motive power
has been given to them, so they can travel
across huge empty space. So remember
to set your mind on what I have to say.
For clearly matter in its compact form
does not stick together, since we observe
every object getting smaller—we see,
over a long expanse of time, all things,
as it were, melting—old age removes them
from our sight. However, the total sum 100 [70]
we see remains unchanged, and this is why:
when particles leave, they diminish things
they are moving from and increase the size
of what they are moving to. They force one
to decay but, by contrast, they compel
the other one to grow. But nonetheless,
they do not stay there. So in this manner,
the grand sum of things always is maintained.
By mutual exchange among themselves
mortal men live on: one race increases, 110
another is reduced. In no time at all,
generations of living creatures change
and, like racers, hand off the torch of life.¹

If you think primary elements of things [80]
can cease moving and, in a state of rest,
produce new motions in material stuff,
you are meandering a long, long way
from proper reasoning. Those particles,
first elements of things, since they travel
through empty space, must all be moved along 120
by their own weight or perhaps by impact
with other particles. And when they meet
in numerous collisions at high speed,
what happens is they quickly bounce apart
in various directions. That is not strange,
for they are very hard, with solid weight
and nothing from behind obstructing them.
So that you may more readily discern

¹Smith notes that this image comes from a contest in Athens in which riders on horses carried a torch in a relay race.

Book Two

that all corporeal matter is pushed
here and there, recall there is no bottom
to the whole universe, nor any point
where primary particles stand still, for space
is without limit, without boundaries,
and, in its immensity, stretches out
in all directions everywhere. This point
I have discussed at length—it has been proved
by flawless reasoning.

 This being the case,
it is clear that elementary particles
throughout deep empty space receive no rest.
Instead, always driven by different motions,
some, after colliding, bounce very far,
others rebound a short way from the blow.¹
All those pushed to closer, denser unions
spring back short distances and get caught up
in their own united combinations.
These form powerful basic roots for rock,
brute stuff of iron, and other things like them,
not very numerous, which wander off
through enormous empty space. All the rest
fly far apart and rebound long distances,
with large gaps between them. These particles
provide us glorious sunlight and thin air.
And through the huge void many more of them,
thrown from matter in combination, move on,
or, if absorbed, are still quite unable
to link their movements. As I perceive it,
an illustrative image of this matter
is always moving right before our eyes.
For look carefully whenever sunlight
pours its piercing rays into dark places
of the house: in light from those same rays
you will see many tiny particles
in empty space mixed up in many ways,
as if waging war in endless battles,
group by group, not conceding any pause,
constantly stirred up by their collisions
and their moving apart. From this image
you can infer how primary elements
of stuff are constantly being tossed around
in huge empty space. That shows how small things
can illustrate large concepts and provide
traces by which they can be understood.
So it is all the more appropriate
for you to turn your mind to those bodies

¹Lucretius is here talking of distances within objects made up of different first particles: some substances formed by collisions have particles more closely packed than others.

Book Two

one observes moving in great disorder
in the sun's rays, because such confusion
shows there is also motion in matter
going on underneath, hidden and unseen.
For you will see many particles there,
struck by invisible blows, change their path, 180 [130]
as they are pushed, forced to reverse themselves,
sometimes in one way, sometimes another,
in all directions everywhere. No doubt,
this roaming motion in all particles
comes from primordial elements of things,
for by themselves these primary elements
are moved, and then from that motion bodies
in small compounds, those which are, as it were,
closest to the force of primary matter,
are set in motion by the impulses 190
of blind collisions with those particles,
and then they themselves stir compound bodies
of slightly larger size. And thus, motion
rises from basic particles and goes,
little by little, up to our senses,
so that those things we can see in sunlight [140]
are shifted, too, although we do not see
the impacts that bring about their motion.

And now, Memmius, from what follows here
you may briefly learn what speeds are given 200
to material bodies. When Dawn first spreads
new light upon the earth and various birds
fly in pathless woods through delicate air,
filling whole regions with their liquid song,
we see how the sun, suddenly rising
at such a moment, is in the habit,
as it pours forth, of clothing everything
in sunlight—that is clearly manifest
to all. However, that clear light and heat [150]
that the sun sends out do not travel through
an empty space. That is why they are forced 210
to move more slowly, while they, so to speak,
cut through waves of air. Particles of heat
do not move one by one but are combined,
joined together in a mass, and therefore
slow each other down and at the same time
are hindered by external matter, and thus
they are forced to move at a slower rate.
But all primary stuff is simple solids,
and when these move through vacant empty space, 220
no outside object slows them down, and so,
with all their parts a unit, they are carried,
moving forcefully, to the single place [160]

Book Two

towards which they began.¹ It is quite clear
they have to travel at the highest speed,
carried at much faster rates than sunlight,
rushing through much greater areas of space
in the same period of time it takes
bright sunlight to fill up the heavenly sky.
[.]² 230
. . . nor do [gods] follow each primordial element
to see the reasons everything takes place.
But some men oppose these views, ignoring
[that particles of matter on their own
keep on moving—time does not wear them down.]³
They claim that without power of the gods
nature could not, in ways which match so well
the needs of man, change seasons of the year, [170]
produce the crops, and other things as well,
which sacred pleasure urges mortal men 240
to undertake, while she herself, life's guide,
leads on and coaxes them to reproduce,
through acts of Venus, their generations,
lest the human race die out.⁴ When they think
gods produced each thing for human beings,
they seem, in all respects, to have fallen
a long, long way from proper reasoning.
For even if I were quite ignorant
about primordial elements of things,
I would, on the basis of the sky itself 250
and many other reasons, dare to claim
and to assert the nature of the world
was not, in any way, designed for us [180]
by the power of gods, for as it stands,
it has enormous flaws. But these issues,
Memmius, we will clarify for you
later on—at this point I will explain
what there is still left to say on motion.⁵

¹Compound matter moves more slowly through air because the particles within it are moving and obstructing each other and also because external particles of air are hindering it. Primordial elements lack that inner motion and, when they move through vacant space, any external obstacles; hence, the latter move more quickly.

²A number of lines are lost here. Bailey makes the plausible suggestion that they probably dealt with other reasons for the rapid speed of elementary particles and with what Lucretius earlier promises to explain, how the motion of primordial particles makes objects smaller. The two lines after the gap are the conclusion of an incomplete sentence. Munro offers the suggestion that in the lost passage, there is a reference to the gods not being disposed to follow the movements of every atom, an idea which makes good sense of the incomplete sentence after the omission.

³I follow Bailey and others by inserting here a line in the Latin.

⁴Venus, the Roman equivalent of the Greek Aphrodite, is the goddess of sexual desire. Fowler points out that the structure of the sentence invites us to see natural desire in charge of the goddess, rather than the other way around.

⁵Lucretius deals with the issue of the imperfections of the earth in Book 5 (lines 156 ff. in the Latin).

Book Two

In my argument, this is now the place,
I think, to confirm for you that nothing
made of corporeal stuff is able,
through its own force, to be carried upward,
or to rise—in case those fire particles
give you a false idea, for they are born
and grow in upward motion. Moreover,
shining crops and trees also grow upward,
though all their weight, however much they have,
is carried downward. And when fires jump up
towards the roof and rushing flames consume
beams and rafters in the home, you must not think
they do this on their own, without some force
driving them. It is the same sort of thing
blood does when emitted from our bodies—
it arches high up, with violent spurts,
spattering gore. Have you not also seen
the force with which liquid water spits up
planks and timbers? For the more we force them,
with many of us pushing them straight down—
and that is difficult, it takes great force—
the more eagerly water throws them out
and sends them back, so they leap up, rising
more than half their length. But we do not doubt,
I think, that these objects, given the weight
inherent in them, are all carried down
through empty space. And therefore flames as well
must be able, under pressure, to rise
up through the breezy air, although their weight,
all which they possess, strives to lead them down.
Do you not perceive how, in the heavens,
night fires fly up, drawing long fiery trails
wherever nature gives them for their motion?
Surely you see stars and constellations
falling towards earth? And the sun, as well,
from way up in the sky, spreads out his heat
in all directions and sows fields with light.
Thus, the sun's heat tends down towards the earth.
And you see lightning flashing across rain—
fires burst from clouds and rush, now here, now there,
and all around us fiery forces crash to earth.

In these matters there is also something
we are eager for you to understand:
when particles are borne by their own weight
on a downward path straight through empty space,
at undetermined times and random places,
they swerve a little—not much, just enough
so you can say they have changed direction.
Unless they had this habit of swerving,
all of them would fall through deep empty space

260

DIRECTION OF MOTION
IN ELEMENTARY PARTICLES

[190]

270

280

[200]

290

[210]

300

ELEMENTARY PARTICLES SWERVE

[220]

Book Two

like drops of rain—among first elements
no impacts or collisions would be made,
so nature never would have made a thing.¹ 310

For if anyone happens to believe
that heavier bodies, since they are carried
straight down though empty space more rapidly,
could hit the lighter ones from up above
and in this way generate collisions,
which could then create productive motions,
he is moving backwards and is far removed
from truthful reasoning. For all objects
which sink through water, even through thin air, 320 [230]
must, depending on their weight, move faster
in this fall, since the material substance
in water and the nature of thin air
can hardly hold back each thing equally:
heavier bodies will overpower them,
and they will move aside more rapidly.
But, by contrast, an empty space cannot
hold back a single thing at any time
or any place, since it keeps giving way,
as its own nature forces it to do. 330

That is why all bodies set in motion,
even though their weights may be unequal,
must be carried through unresisting void
at the same rate. And so the heavier ones
can never fall down from above and hit
the lighter ones and, on their own, create
those collisions which make motions vary,
and through which nature carries on her work.² [240]

Thus, to repeat myself, these bodies must
change course a little—but nothing greater 340
than the minimum, so we do not seem
to be imagining oblique movements,
and truth should prove this picture incorrect.
For we know it is manifestly clear
that heavy bodies, in and of themselves,
when they fall down from above, cannot
move obliquely, as you can plainly see.
But that there is nothing that swerves at all

¹This chance alteration in the direct linear movement downward (the swerve of the elementary particles) has, as one learns a few lines later, enormously important consequences, since it frees nature and human beings from rigid determinism and accounts for freedom of will. However, Lucretius, as Fowler points out, uses the existence of free will to demonstrate the validity of the idea of the swerve in the basic particles, rather than the other way around. Serres maintains that this chance swerve (which has often been viewed with suspicion or scorn) is the heart of Epicurean science and the birth of modern physics. The concept of the random swerve means that for Lucretius there are three basic causes of motion in matter: the downward motion caused by weight, the swerve, and collisions.

²This notion that objects fall in a void at the same rate is an interesting anticipation of one of the most famous stories of early modern physics, Galileo's experiment from the top of the tower of Pisa (c. 1590).

Book Two

from the straight direction of its descent— [250]
 what man is capable of seeing that? 350

Then, too, if all movement is always linked, FREEDOM FROM DETERMINISM
 new motions always rising from old ones
 in a set order, and if primary stuff
 does not, by swerving from its downward path,
 begin specific movements which can break
 the laws of fate, so there does not follow
 an endless sequence of cause after cause,
 where does this freedom of the will arise
 in all living creatures throughout the earth?
 Where, I ask, does it come from, that free will 360
 we rip from fate and thanks to which we go
 wherever the will leads each one of us?
 We change our motions in a similar way,
 not at predetermined times and places,
 but as our minds propose. There is no doubt [260]
 that in these matters a man's own free will
 provides the start, for from his will motions
 are conducted through the limbs. Moreover,
 surely you see how, in the quick moment
 when gates open, a horse's eager strength 370
 still cannot, in that instant, charge ahead
 in the way even its own mind demands?
 For through its whole body the full supply
 of matter must be contacted, so that,
 energized in every limb, it can strive
 to follow inclinations of its mind.
 Thus, you may see how the start of movement
 created from the heart emerges first [270]
 from free will in the mind and after that
 is spread through all the body and its limbs.¹ 380
 And this is not the same as when we move
 under the impact of a blow given
 by the forceful strength or great coercion
 of someone else, for then, quite obviously,
 all the material in our whole body
 is shoved forward against our will and moves,
 until our will controls it in our limbs.
 So do you now see that, though outside forces
 push many men, often compelling them,
 to move unwillingly and be carried off 390
 headfirst, still there is something in our heart
 able to struggle against that motion,
 resist it, something whose judgment sometimes [280]

¹The image is taken from horse racing, where just before the start of the race the animal is behind a gate. Fowler makes the point that the horse cannot move, even though his mind wants to and his body is fully ready to, until the will initiates motion. Hence, there is a distinction between the will and the mind, and motion originates in the elementary particles of the former. For the Epicureans the spirit (*animus*), where free will (*voluntas*) originates, is in the chest.

Book Two

compels our store of physical matter
to turn to one side through body and limbs
and, when pushed forward, to be held in check
and settle down in place again. And thus,
we must concede that with material seeds
things are like this, too—besides their own weight
and collisions, there is another cause 400
of motion, and from that originates
this power innate in us, since we know
nothing can be created out of nothing.
For weight reveals that all things are not caused
by impact, as if from some external force.
But that the mind, in everything it does, [290]
itself has no necessity within
and is not forced to suffer and endure,
as if it were completely overwhelmed—
what creates this is that tiny swerving 410
of the primordial elements of things
at no set time or predetermined place.

Nor were supplies of matter ever pressed
more compactly or, by contrast, spread out
at greater intervals. Material stuff
does not increase, nor does any perish. MATERIAL MOTIONS HAVE NOT
CHANGED
And, therefore, those primordial elements
in the past moved around in the same way
as they do now, and for all time to come
will be transported in a similar way. 420
Whatever has been habitually produced [300]
will be produced with the same conditions—
it will exist, grow, and gain strength—each thing
to the extent that natural law allows.
No force can change the total sum of things,
for there is no place any form of matter
can flee outside the universe or from which
some new force can arise and move inside it,
altering the entire nature of things,
transforming how they move.

In these matters, 430
there is nothing amazing in the fact
that, though all primary elements of things
are in motion, the total sum still seems
to be at rest, except whatever moves
with its own body.¹ For the whole nature
of primary stuff lies far below our senses.
That is why, since you cannot now perceive
these things themselves, they must also conceal

[310]

¹Although the primary elements are always in motion, the object of which they are composed looks at rest, unless its whole body is in motion.

Book Two

their motions—above all, since what we can see
nonetheless still often hides its movements 440
when set far from us in a distant spot.
For woolly sheep grazing in fine pastures
often move slowly on the hill to spots
where tempting grasses sprinkled with fresh dew
call each of them, and lambs, their bellies full,
play games and leap about delightedly. [320]
From far away all this appears to us
somewhat hazy—a dazzling patch of white,
as it were, resting on green hills. And then,
when great legions charge and fill all places 450
in the field, stirring images of war,
a brilliant glitter rises to the sky,
the land sparkles on every side with bronze,
while beneath the power of soldiers' feet
a sound arises from below, the hills,
once noises hit, echo the shouting back
to stars in heaven, while those on horses
wheel around and then, without a warning,
gallop across the middle of the fields,
shaking them with the fury of their charge. 460 [330]
Yet from a certain place high in the hills
they seem a bright patch standing on the plain.

Come now, learn next about the particles
from which all things begin—what they are like,
how they differ greatly in their structure,
how they have shapes of many different kinds.
It is not that only a few of them
Have similar shapes, but that, in general,
they do not all resemble one another.
And no wonder, for the supply of them 470
is so enormous, that, as I have shown,
there is no limit to them, no grand sum.
Thus, clearly they are not identical, [340]
completely alike, so that they all have
a similar size and shape. Moreover,
the human race, mute schools of swimming fish,
fat cattle, savage beasts, and various birds
which flock together in joyous places
by waters of river banks, springs, and lakes
and fly soaring through forest wilderness— 480
go on and select any one of these,
a single group, whichever one you wish,
you will still discover that among themselves
they have different shapes. That is the only way
young offspring can recognize their mothers,
and mothers know their offspring. And we see [350]
they can—they do recognize each other,
in just the same way human beings do.

Book Two

For often in front of a god's temple,
some richly decorated shrine, a calf, 490
slaughtered by incense-burning altars, falls,
hot rivers of blood spurting from its heart,
but its mother wanders through green pasture
in the woods, without her child, and searches
for tracks of cloven hoof prints in the ground,
her eyes exploring every single place—
if only she could somewhere catch a glimpse
of her lost young one, and then, standing still,
she fills the leafy woods with sounds of grief.
She keeps on going back, time after time, 500
to her enclosure, transfixed with longing [360]
for her new-born calf. Tender willow shoots,
grasses fresh with dew, streams gliding past,
filled with water up to the riverbanks—
not one of these can divert her spirit,
ease her sudden apprehension. The sight
of other young calves in joyful pastures
cannot distract her mind, relieve her care,
so great is her need for the child she knows
and recognizes as her own. Then, too, 510
tender young goats with tremulous voices
know their horned mothers, and young butting lambs
know flocks of bleating sheep—that's why they run
almost always to their own milky teat,
as nature bids. Finally, take some crops, [370]
any type you wish, and you will observe
that, though the grains are one variety,
among themselves they are not all the same—
there still will be some differences in form.
And we perceive the same with types of shells 520
embroidering the bosom of the earth
in places where the sea with gentle waves
strikes curving shores of thirsty sand. And so,
in the same way, to make the point once more,
since primordial elements of matter
are set by nature and not made by hand
to fit a single form, some must fly around
with shapes which do not match the others. [380]

Our minds find it quite easy to explain,
using this sort of reasoning, why fire 530
from lightning penetrates much more than flames
from our torches here on earth. You can say
heaven's lightning fire, being more subtle,
is made of smaller shapes, and that therefore
it makes its way through openings which our fire
cannot penetrate, since it comes from wood
and is made by torches. And furthermore,
light passes right through lanterns made from horn,

DIFFERENT SHAPES PRODUCE
DIFFERENT CHARACTERISTICS

Book Two

but rain drops are repelled. Why would that be,
unless particles of light were smaller 540
than those in nourishing liquid water? [390]
We see wine will travel very quickly
through a sieve, but sluggish oil, by contrast,
moves slowly, because, as is obvious,
either it has larger particles, or else
they are hooked and more closely intertwined.
And thus it happens that these particles
cannot, as single units, so quickly
be separated from each another
and flow through single holes in anything. 550
Add to this that liquid milk and honey
held in our mouths feel pleasant to the tongue,
but, in contrast, wormwood's bitter nature [400]
and acrid centaury with their foul taste
make our mouths grimace.¹ So it is easy
for you to recognize that substances
which can affect our senses pleasantly
are created from smooth, round particles,
and, on the other hand, all substances
we find tart and bitter are held together 560
by hooked elements, combined more closely,
and thus routinely tear the passageways
into our senses and, as they move in,
break through the body.

And finally,
all things agreeable to the senses
and those unpleasant when we contact them
are made of different shapes and are opposed
to one another—just in case you think,
as perhaps you do, that the harsh noises [410]
of screeching saws consist of particles 570
as smooth as those in melodious music
which performers make by awakening sounds
on strings, shaping them with their deft fingers,
or believe that primordial elements
with the same shape enter human nostrils
when nauseating corpses burn as when
the stage has been freshly strewn with saffron
from Cilicia, and near by altars breathe
Panchaeon incense, or take for granted
that lovely colours which can feed our eyes 580
consist of the same seeds of things as those
which prick our sight and force us to shed tears [420]
or appear abhorrent and disgusting—

¹Wormwood and centaury are species of bitter tasting herbs commonly used as natural medical remedies.

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the sight of something foul.¹ For every shape
which gratifies your senses all the time
must not be made of primordial matter
without some smoothness. On the other hand,
whatever we find rough and irritating
has not been created from material
which lacks coarse elements. There are, as well, 590
particles which are not considered smooth—
and justly so—but which have no bent points
and are not completely hooked. Instead, they have
small corners projecting out a little,
so that they can titillate our senses,
rather than injure them. This type includes
wine lees and the taste of elecampane.² [430]
And then warm fire and cold frost, both with teeth,
penetrate our body's senses differently—
the way each feels is evidence of that. 600
For by the sacred powers of the gods,
touch, yes touch, is physical sensation,
either when something from outside pushes
its way in, or when something created
in the body hurts us or brings delight
when it comes out, as in those fruitful acts
of making love, or when the seeds collide,
get disturbed inside the body itself,
and then, in their mutual agitation,
confuse our senses. This you may witness 610
if you should happen to hit any part [440]
of your own body with your hand. And thus,
primary elements which are capable
of producing various sensations
must have very different shapes.³

Moreover,
substances we find hard and dense must be
more closely interlocked and keep themselves
together tightly packed, as if their parts
were branches. Among this sort of matter,
adamantine rocks come first in the front ranks— 620
they have the habit of resisting blows—
tough flint stone as well as hard, strong iron,

¹Watson notes that, “theatres were sprinkled with saffron mixed with wine, as Pliny relates.” Cilicia is a coastal region of Asia Minor, part of modern Turkey. Panchaea was an imaginary Arabian island, famous, among other things, for incense-bearing sand.

²Elecampane (also called horse heal and elfwort) is a herb with a slightly bitter taste, once used in medicine and in food recipes as a condiment.

³Touch is, as we learn in more detail later (particularly in Book 4), the primary sense, since all the others (sight, hearing, taste) depend upon particles touching the appropriate sense organ.

Book Two

and squealing brass bolts that resist their locks.¹ [450]
Those substances which make liquid matter
and fluids must consist of more rounded,
smoother parts. For poppy seeds, like water,
are poured out easily—several round grains
do not hold each other back and, when spilled,
also roll away downhill. Finally,
all substances which you see diffusing 630
very quickly—like vapour, smoke, and flames—
must, if they do not totally consist
of round, smooth particles, still not be checked
by complex ones, so they can pierce bodies, [460]
penetrate rocks, and yet not stick together.

Thus, you can easily see that all things
we notice biting into our senses
are not made up of tangled elements
but of pointed ones.² And that you observe
something bitter which is also liquid, 640
like sea water, is not the least bit strange.
For since it is a fluid, it consists
of smooth, round particles, but intermixed
with smooth particles are rough ones, as well,
which bring us pain. But still these elements,
even if hooked, must not cling together—
though they are rough, as you must understand,
they consist of spheres, so they can roll on
and yet at the same time hurt our senses. [470]
So that you may more readily believe 650
that rough primordial elements are mixed
with smooth ones and that Neptune's body
consists of such a bitter mix, there is
a way of separating them and then
observing them apart.³ For sea water,
tastes sweet when filtered many times through earth.
It flows into a trench and then grows soft,
for in the surface layers of the ground
it leaves behind harsh particles of brine—
being rough, they cling more readily to earth.⁴ 660

Since I have proved that point, I will go on
to another point whose truth stems from it: SHAPES OF BASIC PARTICLES

¹Adamantine is a mythical rock of legendary hardness. As an adjective the word adamantine refers to something very hard and bright (like diamond). The image of squealing brass refers to a metal hinges or bolts on a door or gate.

²To penetrate the body's sense organs, the particles must be small (i.e., not intertwined in larger and more complex combinations)—otherwise they would be blocked—and yet they must have "points" (i.e., not be totally smooth) in order to register harshly on our senses.

³Neptune is god of the sea; hence, his body is sea water.

⁴Bailey conjectures that after this line a section is missing, one in which Lucretius argues that the basic particles were limited in size.

Book Two

though primary elements of things vary,
there is a set limit to the number [480]
of their shapes. For if this were not the case,
some seeds would, as a result, have to have
bodies of infinite size. With the same seed,
in the one small size of any particle,
shapes cannot vary much among themselves.
For suppose primary particles consist 670
of three minuscule parts, or, if you wish,
add a few more, then once you have arranged
all these parts within a single body,
placing each one on top and underneath,
shifting them to left and right, obviously
you will have tried out all the different ways
in which each arrangement may demonstrate
a form for the shape of that whole body. [490]
As for the rest, if you should wish perhaps
to change those shapes, you will then have to add 680
other parts. And from that it will follow,
for similar reasons, that if, by chance,
you wish to keep on varying the shapes,
the structure will need other elements.
Therefore, an increase in the body size
will follow the creation of new forms.
And thus you cannot claim those seeds possess
an infinite diversity of shapes,
or else you force the size of some of them
to be immense, a claim which earlier 690
I have already shown cannot be proved.¹

If not, by now you would have cast aside
barbarian clothing and shining purple
from Meliboea, with colours steeped [500]
in shell-fish dyes obtained from Thessaly,
[and those displayed] by golden peacock broods
bathed in smiling loveliness—all replaced
by the new colour of things.² You would spurn
the odour of myrrh, the taste of honey.
The song of swans, the artful melodies 700
of Phoebus' strings, for similar reasons,
would have been overcome and sound no more.³
For something finer would have been produced,
surpassing all the rest. But then again,

¹This proof, one assumes, was part of the gap in the manuscript earlier (see the footnote immediately above); however, Lucretius may be referring back to what he says in Book I (lines 599-634 in the Latin text).

²I follow Munro's suggestion that some words are lost in the Latin here. The insertion is in square brackets. Meliboea was a town in Thessaly, in north-east Greece. Purple dye comes from certain shellfish. Lucretius is here insisting that if basic particles could have an infinity of shapes, there would be no end to the marvellous new objects which would make those things we now consider beautiful inferior by comparison.

³Phoebus is another name for the Greek god Apollo. He was associated with playing the lyre.

Book Two

each object could decline to something worse,
in the same way we said they could improve.
For, if things regressed, there could also be [510]
something more disgusting than the others,
fouler to our nostrils, ears, eyes, and taste.
Since this is not so and a fixed limit 710
assigned to matter keeps extremes in check
in both directions, one must acknowledge
the amount of variation in shapes
of material stuff is limited, as well.
Lastly, from fires to freezing winter frost
the distance has been fixed. By the same means,
it has again been measured in reverse.
All heat and cold and intermediate warmth
fall in between the two and, by degrees,
fill in the total. Thus, they all are made 720
and differ within determined limits,
since two points designate the two extremes, [520]
at one end fire, at the other rigid frost,
and these are hostile to material things.¹

Since I have proved that point, I will go on
to something else whose truth derives from it:
the number of first elements of things
with shapes like one another is endless.
Since differences in form are limited,
those which are the same must be infinite, 730
or else the amount of material stuff
has limits—an assertion I have shown
is not the case, by proving in my verse
that corporeal substances maintain
the total sum of things eternally, [530]
with a constant series of collisions
on every side. For although you notice
certain animals are less numerous
and see nature is less fertile in them,
yet in other places, in some region 740
of a far-off land, there may be a lot
of just that kind to make up their numbers.²
We see that in classes of quadrupeds,
above all with snake-handed elephants,
whose many thousands keep India fenced in
with an ivory wall, so there is no way
one can move into its interior—

¹The point of the example of temperatures, as Watson notes, is to demonstrate that in nature things (like the shapes of particles or degrees of heat and cold) can have much variation but that there are fixed limits beyond which they cannot go. These limits are “hostile” to matter because at the extremes they help dissolve it.

²Bailey here makes reference to the doctrine of Epicurus that things are equally distributed in the universe (there is an equal number of things of the same sort), so that what is rare in one area must exist in larger numbers elsewhere—if not in this world, then somewhere else.

Book Two

that shows how numerous those wild beasts are.
Yet we see very few examples of them.¹

[540]

But so I may allow this point, as well, 750
let there be, if you like, one single thing
living in its natural body all alone,
with nothing like it in any region
of the entire world—but nevertheless,
if there were not an infinite number
of materials from which it could be
conceived and born, it would be impossible
for it to be produced and, beyond that,
for it to feed itself and grow. In fact,
if in addition I assume this point, 760
that particles from which one single thing
is born are being tossed around through space
in a finite number, where would they meet
and join together? Where would they come from?
What would force them there? How would that happen,
in such a huge sea, such a strange tumult
of materials? I think those particles [550]
have no way of forming combinations—
just like those times when many large shipwrecks
have taken place, and benches, empty holds, 770
yard arms, prows, masts, and swimming oars are tossed
in mighty seas, so one can see stern fittings
floating on all coastal shore lands, giving
mortal men a warning: they should resolve
to shun the faithless sea, with its deceit,
violence, and treachery, and never more
have faith in its devious seductions,
when the calm sea smiles.² With this example, [560]
you can rest assured, if you ever claim
that certain elementary particles 780
have a finite number, then the movement
of various materials must scatter them,
tossing them around for all eternity,
so they can never be forced together
and meet in combination, or remain
combined, or grow by adding matter on.
But clear and obvious experience
shows us that both activities occur:
objects can be produced, and, once produced,
can grow. Therefore, with any group you like 790
the primordial elements of its stuff,

¹The phrase “snake-handed” is a reference to the elephant’s trunk.

²The point of this example seems to be that if the elementary particles were finite in number they would be tossed around the universe like the parts of wrecked ships in the sea and, just as it would be impossible for a ship to be assembled from the flotsam and jetsam by the movement of the water, so it would be impossible for any objects to be formed from the random movements of a limited amount of disconnected matter in space.

Book Two

with which every substance is provided,
are clearly infinite.

For this reason,
destructive motions cannot prevail for ever
and bury things in an eternal tomb,
nor, in turn, can motion in materials,
which generate and make things grow,
preserve created things perpetually.
Thus, an equal battle is being waged,
and has been from time immemorial,
among the basic particles. Sometimes
forceful vitality of things wins out,
now in one place, now another, and then,
in turn, is overcome. The wailing cries
young children raise when they first look upon
the shores of light mix in with funeral songs.
No night has followed day or dawn the night,
which has not heard, mingling with those weak howls
from infants, groans accompanying death
and gloomy funerals.

GROWTH AND DISSOLUTION

[570]

800

In these matters,
it is also good to have one thing sealed
and firmly stored in your mind's memory—
none of those things whose nature we can see
before our eyes is made up of one type
of primary stuff, nor is there anything
which is not formed by mixing different seeds.
And whatever contains within itself
in a greater amount many powers
and properties, in that way demonstrates
there is in it the greatest quantity
of different types of primordial matter
of various shapes. Firstly, within itself
earth has those primary particles from which
cool springs well up and constantly renew
enormous seas. It has materials
from which fires arise. For in many spots
earth's soil is on fire underneath and burns,
while violent Etna rages on with flames
from down below.¹ But earth also contains
elements which enable her to raise
delightful orchard trees and polished fruits
for races of mankind and to provide
rivers, foliage, and joyful pastures
for races of wild beasts roaming the hills.
Thus, earth is the only one who is called
“the gods' great mother,” “mother of wild beasts,”

810 [580]

ALL THINGS HAVE DIFFERENT
ELEMENTS

820

[590]

830

¹Etna is an active volcano in Sicily.

Book Two

and “maternal parent of our bodies.”

The old and learned poets of the Greeks [600]

sang that she, [carried on high and] seated

in a chariot, drives on a pair of lions, 840

thus teaching that great earth hangs suspended

in airy space and earth cannot be placed

on earth.¹ They added wild creatures to show

that any offspring, no matter how ferocious,

should be mollified, subdued by favours

from its parents. And the top of her head

they circled with a crown depicting walls,

since she sustains those cities fortified

in select locations. And now, furnished

with this sign, Sacred Mother’s image is borne 850

far and wide across the earth, inspiring awe.

Various nations, following ancient rites [610]

of worship, call her “Mother of Ida”

and produce for her throngs of Phrygians

as her companions, since, from those regions,

they claim, crops first began to be produced

throughout the world. And they assign to her

the Galli, eunuch priests, because they wish

to signify that those who violate

the Mother’s sanctity and have been found 860

ungrateful to their parents must be thought

unsuitable to bring living children

into regions of the light.² In their hands,

sounds of tight-stretched drums and hollow cymbals

boom all around, horns ring out raucous threats,

and with their Phrygian rhythms hollow flutes [620]

stir up the soul. In front of them they hold

their weapons, signs of violent fury,

to frighten wicked hearts and thankless minds

among the crowd by making them afraid 870

of what the goddess’ powers could achieve.³

Thus, as soon as the goddess is brought in

to mighty cities and, without a word,

offers mortal men her silent blessing,

they strew all the roadways along her route

¹Lucretius is referring here to the goddess Cybele, the great mother goddess of Asia Minor. A cult dedicated to her began in Rome in 210 BC, and the Senate adopted Cybele as an official state goddess in 203 BC. Cybele is often confused or identified with Rhea, in Greek mythology the mother of Zeus, perhaps because both are associated with a Mount Ida (one in Asia Minor, just outside Troy, and one in Crete). The chariot freely moving through the air, as a symbol of the earth, suggests that the earth is not supported by some other solid mass. Phrygia is an area in Asia Minor. Some editors conjecture that one or two lines have been lost right after line 600 in the Latin. The part in square brackets is an insertion prompted by a suggestion by Munro.

²The Galli were voluntary eunuchs who worshipped Cybele. Roman citizens were prohibited from becoming Galli (until the first century AD). According to one account, the name derives from the river Gallus in Phrygia, a stream whose waters, it was said, drove anyone who drank them so insane that he would castrate himself on the spot.

³Smith notes that the weapons mentioned here are the knives with which these men castrated themselves.

Book Two

with brass and silver coins, enriching her
 with massive contributions. They snow her
 with showers of roses, covering Mother
 and her companion throng. Here an armed band,
 men who are called, according to the Greeks, 880
 Phrygian Curetes—for among themselves,
 they now and then play games with weapons— [630]
 dance in rhythmic motion, and dripping blood,
 they shake the terrifying helmet plumes
 as they nod their heads. These men represent
 Curetes from Dictaea, who, they say,
 in earlier days in Crete concealed the cries
 of infant Jupiter, when armed young boys
 in a swift-moving dance around the child
 struck bronze on bronze in rhythm, so Saturn 890
 would not catch him and devour him, giving
 his mother's heart an everlasting wound.¹
 That is why Great Mother is accompanied
 by men with weapons, or they mean to show [640]
 what the great goddess has proclaimed—that men
 must resolve to defend their fatherland
 with arms and courage and prepare themselves
 to be a guard and tribute to their parents.

Now, though well set down and superbly told,
 this is still a long way from true reasoning. NATURE OF THE GODS
 900
 For the whole nature of gods, in itself,
 must for all time enjoy the utmost peace—
 far removed and long cut off from us
 and our affairs, and free from any pain,
 free from dangers, strong in its own power, [650]
 and needing nothing from us, such nature
 will not give in to those good things we do
 nor will it be moved by our resentment.²
 If a man decides to call the sea Neptune,
 grain crops Ceres, and wishes to misuse 910
 the name of Bacchus rather than call out
 the name appropriate to the liquid,
 let us concede he might as well declare
 the earthy sphere the mother of the gods,
 provided, for the sake of truth itself,
 he refrain from tarnishing his own mind

¹Accounts of the Curetes typically mix together the tales of Rhea, mother of Jupiter (the Greek Zeus), with those of Cybele (the Great Mother from Asia Minor). In Greek mythology Rhea concealed the infant Zeus in Crete, hiding him from his father, Cronos (the Roman Saturn), who, in order to protect his power, ate his children; the Curetes were Rhea's attendants, whose loud cries and music helped to stifle the wailing of the baby god, so that his father would not know where he was.

²These lines (901 to 908 in the English) occur earlier in the poem as well (in 1.54 ff). They are much more appropriate here.

Book Two

with repulsive doctrines of religion.¹

[660]

But earth is always without sensation,
and since it holds the primary elements
of many substances, it brings them out
into the sunlight in all sorts of ways.

920

Hence, woolly flocks, warrior breeds of horses,
and horned herds of cattle will often graze
on grasses from a single field, beneath
the same roof of the sky, and quench their thirst
drinking water from a single river,
yet they live on looking quite different.
They keep their parents' nature, imitate
their habits, each according to its kind.
That shows how many different materials
there are in any sort of grass and stream.
What's more, any single living creature
has bone, blood, veins, heat, water, flesh, sinew—
and these things are, in turn, very different,
all created from primordial matter
of dissimilar shapes. Further, all those things
which are set on fire and burned have stored up
in their bodies, if nothing else, at least
that material stuff which enables them
to hurl up flames, send out light, shoot off sparks,
and scatter embers far and wide. And so,
if with similar reasoning you go through
all other substances, you will find out
that in their bodies they have hidden seeds
of many things and contain various shapes.

930

[670]

940

Then, too, you notice there are many things
which have been endowed with taste and colour
as well as smell—especially most gifts
[you burn as special offerings to the gods.]²
Hence, these things must consist of different shapes,
for burning odours penetrate our frame
where colours cannot go, and colours, too,
find their own way, as does taste, to contact
our senses—that is how you can infer
their primary particles have different shapes.
Thus, elements with dissimilar forms
join into a single sphere, and matter
is composed of seeds in compound mixtures.

[680]

950

¹Lucretius, of course, uses a god's name like this from time to time himself, particularly Venus, and later (in Book 5) he frequently treats the earth as the Great Mother, who creates and sustains all things on earth.

²A line seems to be missing here. I have followed (with some variation) Bailey's suggestion about the missing material.

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And besides, everywhere in my own verse 960
you see many letters shared by many words,
though you must admit that words and verses [690]
in themselves consist of different letters,
sometimes of these ones, sometimes of others.
It's not that a few common elements
run through all the words or that, of all words,
no two have exactly the same letters,
but that, in general, all words do not match
each other. And therefore with other things
it is the same—though many of them have 970
several primary elements in common,
they can still consist of combined totals
different from each other, so one could say,
with justice, that the human species, fruits,
and joyful orchard trees are each made up
of different elements.

But you must not think [700]
that everything can form combinations
in every way, for then you would observe
amazing monsters produced everywhere—
things which look half-human, half-animal 980
would come into existence, tall branches
would sometimes grow out from living bodies,
many parts of land animals would join
those from creatures of the sea, and nature
would nourish through all-generating earth
those chimaeras which from their ghastly mouths
spout fire.¹ But it is manifestly clear
that none of these things happens, since we know
each thing created from specific seeds
and a specific parent, as its grows, 990
can maintain its kind. And we can be sure
this must take place by some established law. [710]
For with each entity, from all its food
those particles which suit it, once inside,
break off, pass into its limbs, and, combine,
to make appropriate motions. By contrast,
we notice nature throwing back to earth
foreign particles, and many substances
with elements we cannot see escape
from bodies, forced out by collisions. 1000
These could not be combined with anything
or adapt to inner vital motions
and copy them. However, just in case
you happen to think only living things
are governed by these laws, a certain rule

¹The chimaera in Greek mythology is a fire-breathing monster made of different animals: the head and body of a lion with a snake at the end of its tail and a goat growing out of the middle of its back. In Book 5, Lucretius explains why such compound monsters could not have been created.

Book Two

sets boundaries to all things, for just as [720]
each created thing is, in its whole nature,
quite different, so all of them must consist
of different shapes in their primordial stuff.
It's not that only a few are given 1010
a similar shape but that, in general,
every one is not like all the others.
Moreover, since seeds vary, there must be
differences in their spacing, passages,
connections, in their weights, impacts, motions,
and collisions—things which not only make
bodies of living creatures quite distinct,
but also distinguish land and all the sea
and keep the entire sky distinct from earth.

Come now, listen to what I am saying 1020 [730]
from those things my pleasing work has shown me, COLOURS
in case you happen to believe that things
which your eyes perceive as white and shining
are made of white primordial elements,
or that black particles make objects black,
or think objects tinged with other colours,
any ones you wish, have a tint like that
because the colour of their basic stuff
resembles theirs. For particles of matter
have no colour at all—they are not like 1030
colours of substances or unlike them.
If perhaps it seems to you impossible
for any mind to be projected here,
into these particles, you are wandering
a long way from the road.¹ Since those born blind, [740]
who have never gazed upon the sunlight,
still distinguish substances by touching
and from their earliest years never link them
to any colour, we, too, may recognize
that we can turn our minds to contemplate 1040
the idea of objects without colour
painted over them. Besides, with objects
we touch when we ourselves are in the dark,
unable to see, we do not notice
that they have colour.

Since I am arguing PRIMARY PARTICLES
that this is so, I will now demonstrate LACK COLOUR
[that primary particles lack all colour.]²
For every single colour is transformed

¹Bailey notes that this phrase about mental projection refers to Epicurus' doctrine that the mind, although its particles are normally stirred by other particles from sensation, can spontaneously "project itself upon" images and form new conceptions. Lucretius is here challenging the notion that we cannot form a mental image of colourless particles because we have no experience of seeing something without colour.

²A line is apparently lost in the Latin here. The translated text provides the general sense of the lost text.

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to any other. But first elements [750]
should not have any way of doing this, 1050
because something must remain unaltered,
or all things will be utterly reduced
to nothing, for whatever has been changed
then moves beyond its own proper limits,
which is instant death for what it was before.
And therefore with seeds of things, be careful
not to sprinkle them with colours, or else
you will see all things totally reduced
to nothing.

Besides, if no natural colour
has been given to primary particles 1060
and if they are endowed with various shapes
from which they then create and modify
all types of colour—since it is crucial [760]
what all seeds combine with, what arrangements
they are placed in, and what mutual motions
they receive and give—you can show at once,
without the slightest trouble, the reason
those objects which, a short moment ago,
were coloured black can, in an instant, turn
a dazzling marble white—just as the sea, 1070
when immense winds whip up its calm waters,
changes to white waves of shining marble.
For you could say that what we often see
as something black, once its material
has been mixed up and the arrangement
of its primordial elements transformed,
with certain matter added and removed, [770]
immediately is made into something
which we see as brilliant white. However,
if the unruffled waters of the sea 1080
were made of sky-blue seeds, there is no way
they could turn white, because no matter how
you shake up matter which is coloured blue,
it could never change that colour into white.
However, if the seeds which make the sea
one pure shining white are soaked in colours
of various different kinds, in the same way
one often makes the form of just one square
from various different shapes, it then follows
that, just as we see there are different shapes 1090 [780]
inside the square, so we should perceive
in the untroubled waters of the sea,
or in any other pure white shining thing,
various colours, all completely different
from each other. Moreover, with the square
the unlike shapes do not block or hinder
the whole outline from being square, but in things,

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the different colours do get in the way:
they prevent the object from displaying,
in its entirety, one single lustre. 1100

In that case, too, the reason which prompts us
and leads us sometimes to assign colours
to those first elements of things, is gone,
since white substances are not created [790]
from white elements, nor those we call black
from black ones—instead they are created
from things of various colours.¹ And, in fact,
it is far more likely that white objects
will be born and rise up from elements
that contain no colour than from black ones, 1110
or from any other colour you wish,
which opposes white and fights against it.

Moreover, since colours cannot exist
where there is no light and primary bodies
do not come in the light, you may infer
they are not wrapped up in any colour.²
What quality of colour could there be
in blinding darkness? And, in fact, colour
is transformed by light itself, depending
how it reflects direct or slanted light 1120 [800]
which strikes it, like the way a dove's plumage
appears in sunlight, with those feathers placed
behind its neck and those around its throat.
For sometimes they become a bright gold red,
like bronze, and sometimes, from a certain view,
they seem what looks like a combination
of green emeralds and dark blue. Peacock tails,
when fully bathed with light, change their colours
in a similar way, as they move around.
Because these colours are brought out by light 1130
striking a certain way, you may conclude
it must be impossible for us to think
they could arise without it. Moreover,
since the pupil of the eye receives a blow [810]
of a certain kind on its inner part
when it is said to sense the colour white
and then impacts of other different kinds
when it sees black and all the rest, and since
when you touch objects, it does not matter

¹Lucretius' point in this long discussion is that colour is not a property of the primary particles. Any assumption that it is leads to certain contradictions with sense experience or reason or both. Colour thus results from changes in the combinations of primordial elements, not from inherent properties of colour in the particles themselves. The claim that the particles may be many different colours contradicts our sense experience and, besides, the latter theory effectively denies the notion that black things are black because they are made up entirely of black particles.

²The reasoning here is rather odd, as Watson notes, since primary elements exist on the surface of things as well as in the interior and therefore "come in the light."

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what colours they may happen to possess 1140
 but rather the types of shapes they have,
 you may understand that first elements
 do not need colour—with their various shapes
 they produce different varieties of touch.
 Moreover, since no fixed natural colour
 has been given to particular shapes
 and since primordial elements combined
 in all configurations can exist
 in any colour you wish, why are things
 created from them, for the same reason, 1150
 not suffused with every sort of colour
 in all their types? Then it would be fitting
 that flying crows, as well, often displayed
 the colour white from their white feathers
 and that black seed made swans the colour black,
 or any colour you wish, one or many.¹

[820]

Besides, the more any object is cut up
 into small parts, the more you can observe
 its colour vanish—little by little
 it disappears, for this is what takes place 1160
 when some purple fabric is torn apart
 in tiny pieces: once it is shredded
 thread by thread, the purple and scarlet shades,
 which are by far the most brilliant colours,
 are totally destroyed. You can conclude
 from this that small parts discard all colour
 before they are reduced to seeds of things.
 Lastly, since you admit not all bodies
 send out a sound or smell, it then follows
 you would not, for that reason, attribute 1170
 sounds and smells to every object. And so,
 since we cannot see all things with our eyes,
 we may conclude that certain things exist
 which lack colour, just as certain objects
 have no odour and never make a sound,
 and yet a keen mind is no less able
 to understand these objects than to note
 materials which lack other qualities.

[830]

[840]

But just in case you happen to believe
 the only thing that primary elements 1180
 do not possess is colour, they also
 are completely devoid of warmth—they have
 no cold or scalding heat—and are carried
 empty of sound and destitute of taste.

1180
 PRIMARY ELEMENTS
 HAVE NO SMELL

¹Lucretius is insisting that what matters is the shape of the primary material, not any given colour. The shapes themselves are not coloured. Here he is refuting the idea that shapes of primary elements come with many colours. If that were the case, he says, then the crow particles, which have a certain shape, should sometimes be a colour other than black, just as swans should change their colour.

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And from their bodies they do not emit
any odour of their own. It is just like
when you start to make enticing perfume
from marjoram, myrrh, and flower of nard,
which exhales nectar to our sense of smell.
First, you must look for some oily substance 1190
whose nature has no smell—to the extent [850]
you can and are allowed to do so—something
which diffuses no smell to our nostrils,
so, as much as possible, it cannot,
with its own strong odour, corrupt those scents
boiled in and compounded with its substance,
polluting them with its own pungent smell.¹
Likewise, primary elements of matter
must not, when things are created from them,
add their own sound or odour—for they can 1200
send out nothing from themselves, nor can they,
for the same reason, bring any taste at all,
nor any cold or heat, warm or scalding.²
[To emit such things, substances must be
composed of particles in combination
and, like perfumes, hold in them vacant space]
and other things made up in such a way
that they are mortal—soft and pliant stuff,
brittle from decay, hollowed out and thin— [860]
all substances one must keep separate 1210
from primary matter, if we wish to set
an eternal foundation under things,
on which their entire preservation rests,
so you do not see everything reduced
entirely to nothing.

Now, we must admit

that all those things we see as having sense
are nonetheless in every instance made
from basic elements that lack sensation.
Clear evidence does not refute this claim,
and those things we openly acknowledge 1220
do not deny it. Instead, on their own,
they lead us by the hand, compelling us
to accept what I just said, that matter
endowed with life comes from material stuff [870]
which is insensible. For you may see
living worms born out of disgusting dung,

¹Myrrh is a resin from various trees and used in certain forms of incense and scents; marjoram is a Mediterranean herb from the same family as oregano; nard is a mountain plant, used in aromatic ointments.

²A number of lines appear to be lost here. Giussani, according to Bailey, suggests that in the missing lines Lucretius is arguing that only matter which contains vacant space (void) can emit things like smell and heat and that he then offers a list of such matter. In light of this suggestion I have inserted a short bridge passage (between square brackets) to make the transition to the point where the text recommences in mid-sentence.

Book Two

when earth, soaked by unseasonable rains,
acquires a rotten smell. And furthermore,
all things change themselves in the same manner.
Rivers are transformed to foliage on trees, 1230
and joyful fields into herds of cattle.
The cattle alter their material stuff
into our bodies, and from our own flesh
wild beasts and birds with power on the wing
often increase their size. Thus, nature converts
all foods to living bodies and from this
produces every sense in living things. [880]
Her method does not differ very much
from how she makes dry logs give rise to flames
and turns them all to fire. So now, therefore, 1240
surely you see it matters a great deal
how all the primary elements of things
are set in an arrangement and what things
they are connected to in those motions
they receive and give?

Besides, what is it
which so strikes your very spirit, worries you,
and forces you to state in various ways
you do not think that something having sense
is born from things that are insensible?
No doubt it is that stones and wood and earth, 1250
no matter how one mixes them together,
still cannot give rise to vital senses. [890]
And therefore you will have to keep in mind,
in dealing with these things, I do not claim
that sensations and things possessing sense
are readily produced from all materials,
without exception, from which things are made,
but that it matters a great deal, first of all,
how small the bodies are which do create
sentient things, what shape they have been given, 1260
then what they are in motion, arrangement,
and position. These factors we do not see
in wood and lumps of earth. And yet these things,
when they are, as it were, made putrescent
by showers of rain, then give birth to worms:
their corporeal matter, once shaken
out of its old structure by something new, [900]
combines in such a way it must produce
living creatures.

Then, too, those who believe
things with sensation [only] can be formed 1270
from substances with sense, and these, in turn,
tend to get sensation from other stuff
[with sense, turn particles producing feeling

Book Two

to something mortal] when they make them soft,
for all sense is joined to flesh, sinews, veins,
materials that form a mortal body,
which, when we look at them, are always soft.¹
But let us grant, for now, these particles
can last forever. Then they must, no doubt,
have sensation either the way parts do 1280
or be considered like whole living things.
But by necessity it must be true [910]
that parts cannot have feelings in themselves,
for all sensation in the limbs depends
on something else—a hand cut off from us
has, on its own, no ability at all
to feel things, nor has any body part.
Thus, it follows that they must resemble
complete living beings, so they are able
to share vital sensation in each part. 1290
If this were so, primordial elements
would have to perceive, in the same manner,
the same things we feel. But then how can they
be called primary elements of matter
and avoid the path to death—they are alive,
and living things are one and the same as those
which perish?² But let us assume they can.
They will make nothing when they meet and join [920]
but huge crowds of living things, in the same way
human beings, cattle, and wild creatures, 1300
as you know, cannot give birth to something new
by breeding with themselves. But if it happens
that they give up their sense and then acquire
a different one, what use was it giving them
what then is taken away?³ Furthermore,
to refer to what we said earlier,
since we see that animal eggs are changed
to living chicks and that, when the foul stench
of rotting seizes earth after too much rain,
it swarm with worms, you may well understand 1310
that sentient objects can be created
from elements that lack all sensation. [930]
But if someone, by chance, were to point out

¹A line is apparently missing after line 903 in the Latin. I have adapted Bailey's suggestion for the missing material, and I have added the word "only" to line 1272 in the English text to clarify the sense of the passage.

²Lucretius is continuing to refute the notion that elementary particles have sensation. If they do, then they must be either like parts which register sense (e.g., a sore toe) or like the entire living creature which feels the soreness. But parts, he argues, have no feeling without reference to a total living creature (a severed toe would not, in itself, register feelings of pain). So if they have sensation, they must be complete living creatures. And if they are alive, then they must die.

³If the primary elements of things have sensation, then they must be alive. In that case, they could, like all living creatures, produce nothing but living beings. But if, in creating things, they lose their sensation, why did they have it in the first place?

Book Two

that sensation could at least come from things
deprived of sense by some transformation,
or through, as it were, some form of birthing
which brings out sensation, it will be enough
to make plain and prove to him that no birth
happens unless some previous act of union
has occurred, nor does any matter change
without some combination. First of all,
before the nature of the living thing
is itself formed, no body can possess
sensation, because, as is quite obvious,
its scattered materials are held in air,
rivers, lands, and objects earth produces,
and these have not united and combined
in such a way among themselves that they
meet in that vital motion thanks to which
all-perceiving senses are set alight
and serve to guard each thing that is alive.

1320

[940]

1330

Moreover, any blow which is more intense
than nature can endure immediately
knocks any creature down and quickly numbs
all sensations in the body and the mind.
For positions of the basic elements
are disturbed, and deep within the body
vital motion is checked, until all matter
badly shaken by the shock within the limbs,
releases those bonds of the living soul
from the body and then expels the soul,
scattering it outside through every opening.
For what else do we think inflicted blows
can do, except shake everything apart
and dissolve it? This also can happen—
when the impact of the blow is less severe,
often the vital motions which remain
have a habit of winning through, prevailing,
calming the immense disruption brought on
by the blow, leading all things back again
to their own proper paths, and, so to speak,
dispelling death's movements in the body,
as those now gain control, and rekindling
those sensations almost lost. For how else
could bodies have their minds restored and move
from the very door of death back to life,
rather than keep moving on and pass away
to where their race already almost ends?

EFFECT OF BLOWS ON
SENSATION

1340 [950]

1350

[960]

Besides, since pain comes when material stuff,
shocked by some force through living flesh and limbs,
is disturbed in its location deep inside,
and a relaxing pleasure is produced

PAIN AND PLEASURE

1360

Book Two

when it moves back in place, you may conclude
that primary matter cannot be attacked
by any pain or gather any pleasure
from itself, because it is not composed
of any elementary particles
by whose new motions it might suffer pain [970]
or get some delight from genial pleasure.
Thus, such matter must lack all sensation. 1370

Then, too, if in order for all living things
to be able to register sensations,
we must now attribute sense of feeling
to their first elements, what of those seeds
out of which the race of humans has grown
in its own special way? Well, that seems clear:
they are shaken up with trembling laughter
and cackle aloud and sprinkle face and cheeks
with dewy tears and are very clever
at saying many things about mixtures 1380
of matter, then seeking out what might be
their first beginnings. Because they are made [980]
to resemble complete mortal men, they, too,
must themselves consist of other elements,
and these, in turn, of others, so that you
will never dare to offer a conclusion.
In fact, I will keep this up—whatever
you may say speaks and laughs and understands
must be made up of other particles
which do the same. But if we recognize 1390
this reasoning is insanely stupid,
because someone can laugh without being made
of laughing elements, can understand
and give reasons in educated words,
and yet not be made up of particles
which are eloquent and clever, then why
cannot every sentient thing we notice
be a compound mixture of seeds which lack [990]
any sense at all?¹

And then each of us
arises from celestial seed—there is 1400
this common father for us all, from whom,
once our nourishing Mother Earth receives
wet, watery drops and then grows pregnant,
she gives birth to shining crops, joyful trees,
and the human race. She bears every tribe
of savage beast and offers food with which
they all feed their bodies, lead pleasant lives,

¹If primary elements have to display the emotional characteristics of the creatures they make up, then we reach an absurd conclusion. Lucretius' logical technique here is similar to his treatment of Anaxagoras in Book 1.

Book Two

and bear their offspring—that is the reason
she has justly acquired the name Mother.
What has previously arisen from the earth 1410
also sinks back into earth. What was sent [1000]
from regions of the air is carried back
and taken in by spaces in the sky.
And death does not destroy materials
in such a way it kills what makes them up—
instead it breaks down their compound unions,
and then it joins one thing to another
and sees to it that all substance alters form,
changes colours, and acquires sensation,
and in an instant gives them back again, 1420
so you may know how it really matters
what primary elements of things combine with,
what kind of arrangements they are placed in,
what mutual motions they receive and give,
and do not assume that what we observe
floating on the surface of materials,
sometimes being born and quickly dying,
could be inextricably connected [1010]
to primary particles, which do not die.¹
Indeed, even in my verse it matters 1430
what every letter is combined with
and in what arrangement it is placed,
for the same letters signify the sky,
sea, land, rivers, sun, and these same letters
indicate crops, trees, and living creatures.
If they are not all alike, the greatest part,
by far, remains the same, and, nonetheless,
their position gives them different meanings.
So with things themselves, in a similar way—
when their spacings, pathways, bondings, weights, 1440 [1020]
collisions, meetings, arrangements, motions,
shape, placement are adjusted, then matter
must also be transformed.

But now set your mind, THE CELESTIAL UNIVERSE
I pray, to true reason, for a new issue
is struggling eagerly to reach your ears
and a new face of things to show itself.
But there is nothing which is so simple
that it is not harder to believe at first,
and, in the same manner, nothing so great,
so marvellous, that all men's amazement 1450
does not gradually lessen. First of all,
the clear bright colour of the sky and all [1030]
it holds in it—stars roaming here and there,

¹The constant death and renewal of objects on earth should not lead us to assume that primary particles are also always being destroyed and reborn.

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the moon, sun's brilliant, illuminating light—
if all these now, for the very first time,
were there for mortal men, quickly thrown down
without a warning, what could one declare
was more wonderful or, before this happened,
what would nations have ventured to believe
less than that? In my view, nothing at all—
this sight would have been so astonishing.
But think how no one now, tired from looking
at it so much, considers it worthwhile
to gaze up at bright spaces in the sky.
And therefore if the very novelty
in an argument gives you cause to fear,
then stop ejecting reason from your mind.
Instead, you must weigh it more judiciously,
and if it seems to you legitimate,
give it your hand, or else, if it seems false,
prepare to fight against it. Given that
the totality of space beyond the walls
of our own world is infinite, my mind
seeks to understand what exists out there,
far away, where the spirit always yearns
to look ahead, those places into which
fly off the free projections of our mind.

1460

[1040]

1470

To start with, we know that in every part,
in all directions and on either side,
above and below and throughout all space,
there is no limit, as I have explained,
and facts themselves announce it on their own—
the nature of deep space is very clear.
Since infinite space lies empty on all sides
and seeds in countless numbers fly around
through the deep universe in various ways,
driven by eternal motion, we must not,
in any way, now think it probable
that only this one sphere of earth and sky
has been created, that beyond us here
all those many particles of matter
do nothing at all, especially since earth
was made by nature. Seeds of things themselves,
jostling freely here and there in various ways
and forced to random, confused collisions,
produced nothing—then finally those ones
suddenly united which could become,
every time, the beginnings of great things,
land, sea, sky, the race of living beings.
And so, to repeat myself, you must grant
that there are other aggregates of matter
similar to this in other places,
which aether clutches in its keen embrace.

DEEP SPACE

1480

[1050]

PLACES SIMILAR TO EARTH
EXIST IN SPACE

1490

[1060]

1500

Book Two

Further, when large quantities of matter
are on hand and there is sufficient space,
with no causal factor standing in the way,
we may be sure that things must be produced
and their full development completed.
Now, if supplies of seed are so enormous
that all the years of living animals
could not count the total, and if nature
and the same force remain which could collect
the seeds of matter into every place
in the same way they are thrown together here,
one must grant there are other earthly spheres
in other regions, with different races
of human beings and classes of wild beasts.

LIFE IN OTHER PLACES

[1070]

1510

Add to this that in the whole universe
no single thing exists all on its own—
nothing is born unique and flourishes
as the single specimen of its kind.
No, it is always a member of some race,
and those of the same kind are numerous.
If, to begin with, you direct your mind
to living creatures, you will discover
this is true for living varieties
of savage animals which roam the hills,
true for human offspring, and it is true
for mute herds of scaly fish and all bodies
of things that fly. Thus, one must acknowledge,
that, in the same way, sky, land, sun, moon, sea,
and all the other objects that exist
are not unique—instead their quantity
is beyond all counting. Since for these things
the deep-set boundary stone of life awaits,
they are as much a body that was born,
as every class of substance here on earth
overflowing with things of its own kind.

OBJECTS ON EARTH ARE
NOT UNIQUE

1520

[1080]

1530

If you grasp these points well and hold to them,
you will see at once that nature is free,
liberated from her proud possessors,
doing all things on her own initiative,
without divinities playing any part.
For by the sacred hearts of gods, who spend
a calm eternity, a serene life,
in tranquil peace, who can administer
the limitless universe? Who can hold
in his controlling hand the mighty reins
of the abyss? Who can turn all heavens
at the same time and keep all fertile lands
warm with celestial fires, or be present

[1090]

1540

GODS PLAY NO ROLE IN
NATURE

1550

Book Two

in all places all the time, so as to make
darkness with clouds and rattle tranquil skies [1100]
with thunder, then throw down bolts of lightning,
which often shatter his own sanctuaries,
and move back to the desert, in his rage,
to use that weapon, which so often spares
the wicked and kills off the innocent,
those who do not deserve to be destroyed?

Since the moment earth was first created, 1560
that day sea, land, and rising sun were born,
many particles have been added on
from areas outside. All around them,
seeds which the immense universe has joined
by hurling them about have been attached.
Because of that, sea and lands could increase,
the mansion of the sky could gain more space [1110]
and raise its high roof far above the land,
and air could flow there. For from everywhere,
all bodies are distributed by impacts 1570
to places fit for each of them and move
to their own kind—moisture goes to moisture,
earth grows larger from particles of earth,
fire is produced from elements of fire,
and aether from particles of aether,
until nature, who produces matter
and brings it to completion, leads all things
to the limit of their growth. This takes place
when what goes to the inner veins of life
does not exceed what flows off and departs.¹ 1580
Here, the age of growth must halt for all things. [1120]
Here nature, by her own force, checks increase.
For all those things you see enjoying growth,
getting larger, and climbing by degrees
to full maturity, take into themselves
more matter than they send out from the body,
as long as nourishment goes easily
to every vein and they are not spread out
so wide they throw off many particles
and make what they are casting off greater 1590
than what their age of life requires as food.
For there is no doubt we must acknowledge
that many elements do flow away
and withdraw from things, but then more bodies
must attach themselves, until the moment
those things attain their greatest peak of growth. [1130]
From then on old age gradually breaks down
their full-grown power and strength, which waste away
as they decline. In fact, with anything,

¹Smith notes that ancient medicine believed the veins carried blood and the arteries carried air.

Book Two

the larger and wider its substance grows, 1600
once its growth has stopped, the more particles
it sends out from its body, releasing them
in all directions everywhere. Its food
is not easily discharged to every vein
and is not sufficient to allow matter
to be made in sufficient quantities
to make up for the large flow it gives off.
For nourishment must repair all objects
and restore them, food must provide support,
food must sustain each thing. All for nothing. 1610
For veins do not provide what is needed,
and nature does not give what they require.
And so by rights they die, when what flows out
has made all matter scarce and they succumb [1140]
to outside blows, for food eventually
fails extreme old age, while external things
never cease from pounding any substance,
wearing out its body, overpowering it
with harmful blows.

And thus, in the same way,
the great world's walls will be attacked, as well, 1620
from every side, will fall into decay
and crumbling ruins. Even nowadays [1150]
the age of earth is broken and worn out.
Earth once produced all species, giving birth
to huge bodies of wild animals, and now
has trouble making any living beings,
even small ones. For in my opinion,
it was no golden chain from up above
which let living things come down from heaven
onto the fields, nor did the sea or waves 1630
which strike against the rocks create them. No—
that same earth gave birth to living things and now
nourishes them from her own materials.¹
Then, on her own initiative, earth herself
for mortal creatures first made shining crops
and joyful vineyards, she herself produced
for mortal beings sweet fruits and happy fields,
which these days scarcely grow, for all the help
our hard work provides. We wear out cattle [1160]
and our farmers' strength; we grind down iron 1640
by ploughing fields which scarcely offer us
what we require—and thus the land, reluctant
to produce its fruits, makes us work all the more.
So now, the ancient ploughman shakes his head
and sighs, again and again, that hard work

¹In Book 8 of Homer's *Iliad*, Zeus talks of attaching a golden chain to the world. The passage was interpreted in some quarters as a way of explaining the creation of the earth and of life on it.

Book Two

of his hands has been wasted and compares
his present days with those from ages past,
often praising the good luck his father had.
The man who plants a shrivelled, worn-out vine
for the same reason sadly blames the times, 1650
how things are going, and makes heaven tired,
muttering how older races, full of piety, [1170]
led easy lives, although they had less land,
for what each one received in earlier days
was a far smaller piece of ground. That man
does not understand that gradually all things
waste away and, weary from advanced old age
after so much time, move on to the grave.¹

¹The gloomy image of an earth getting very old contrasts with other parts of the poem (especially in Book 5) where Lucretius indicates that, in his view, the earth and the world are comparatively young.

BOOK THREE

[Tribute to Epicurus; fear of the afterworld upsets human life; concern about death leads to unjust actions; mind is part of the body; doctrine of the mind as harmony is false; mind has feelings; heat and vital air in the mind; mind and soul; soul responds to feelings in the mind; mind and soul are physical; material composing mind is minute, round, and small; soul occupies very little space and has hardly any weight; soul contains air; a fourth element in the soul is the "soul of the soul"; unity of the four element of the soul; heat, cold wind, and passive air in the mind; different combinations of these elements; body and soul not separate from body; mind and soul essential for life; actions of soul, mind, and body in sensation; disagreement with Democritus; placement of particles of soul; proofs of mortality of body and soul; death not something to be concerned about; the mythical stories of punishments in Hades are foolish; all great men from the past have died; importance of understanding the source of one's fears; all human life has a limit.]

O you who were the first man capable
of raising such illuminating light
out of such deep darkness and making clear
the truest things in life, I follow you,
great glory of the race of Greeks, and now,
in those deep tracks you made I firmly place
my footsteps, not from any strong desire
to be your rival, but because with love
I yearn to emulate you.¹ For why should
the swallow struggle against the swan?
Or in a race what could young goats achieve,
on their tottering limbs, which might compare
with mighty powers of a horse? You are
our father, discoverer of truth. For us,
you supply in full a father's teaching,
and from your writings, you illustrious man,
as bees in flowery woodland pastures sip
from every plant, in the same way we feed
on all your golden words—yes, all of gold,
always most worthy of eternal life.
For once that philosophy which arose
in your godlike mind has begun to speak
about the nature of things, then terrors
in the mind disperse, world's walls fall open,
I see what is going on in all the void,
the majesty and calm habitations
of the gods reveal themselves in places
where no winds disturb, no clouds bring showers,
no white snow falls, congealed with bitter frost,
to harm them, the always cloudless aether
vaults above, and they smile, as far and wide
the light spreads out. Then, too, nature provides
plentiful supplies of all things—their peace
is not disturbed by anything at any time.
The regions of Acheron, by contrast,
are nowhere to be seen, and earth presents
no barrier to a full view of events

INVOCATION TO
EPICURUS

10

[10]

20

[20]

30

¹ The invocation is addressed to Epicurus.

Book Three

going on throughout the void lying underfoot.¹
Godlike pleasure and awe take hold of me
up there with these things, to think that nature,
through your genius, is laid out so clearly,
so openly exposed on every side.

40

[30]

Now, since I have shown for every substance
what its primordial particles are like,
how they differ in their various shapes,
as they fly spontaneously, driven
by eternal motion, and how all things
can be produced from them, following this,
it seems that in my verses I must now
clarify the nature of mind and soul
and drive away that fear of Acheron
headfirst—it utterly disturbs the life
of human beings at its foundation,
filling all action with death's black darkness,
leaving no pleasure clean and free of stains.²

50

FEAR OF DEATH AND HADES

[40]

For though men often claim that sicknesses
and a shameful life are more to be feared
than death and Tartarus and that they know
the nature of the soul is blood or wind,
if, by chance, their inclination tells them
that is the case, and say they have no need
of any part of our philosophy,
from this you may be sure all their remarks
are tossed out more to earn them praises,
and not because they take them for the truth.³
For these same men, driven from their country
and exiled far away from human sight,
polluted by some filthy crime, afflicted
with every kind of hardship, still live on,
and no matter where they may end up
in their wretched state, they nevertheless
make sacrifices to the dead—they kill
black cattle and send offerings to gods
who rule the dead and, in their distress, turn
their minds more keenly to religion.
And that is why it is more revealing
to see a man in doubt and peril, to learn
who he may be in hostile situations,
for only then are truthful words squeezed out
from the bottom of his heart—his facade

60

[50]

70

80

¹ Acheron is one of the rivers of the underworld, where, in Greek and Roman religious belief, souls go after death. The implication here is that Epicurus's philosophy gives Lucretius a godlike freedom and tranquilly to survey the world without a glimpse of what religion claims is the traditional abode of the dead.

² The terms mind (*animus*) and soul (*anima*), as we shall see in this section, were not always clearly distinguished in antiquity and were often used interchangeably.

³ Tartarus is the lowest point in the underworld. Some ancient philosophers held that the blood was the main location of consciousness (e.g., Empedocles); others that it was the breath (e.g., Anaximenes).

Book Three

is torn away, what he truly is remains.
Furthermore, avarice and blind desire
for honours, which drive miserable men [60]
to go beyond the limit of what's right,
and, as servants or accomplices, sometimes
to work day and night as hard as possible
to reach the height of power—these feelings,
these living wounds, are fed not least of all
by their fear of death. For shameful contempt
and biting poverty to most men seem 90
far removed from a sweet and stable life—
they are, as it were, simply a delay
before the gates of death. And when people,
driven by false terrors, desire to flee
far away and set them at a distance,
they heap up treasure with civil bloodshed [70]
and, in their greed, double their own riches,
piling slaughter upon slaughter, cruelly
rejoicing in a brother's mournful death,
hating and fearing the banquet tables 100
of their relatives.¹ For the same reasons,
and often moved by the same fear, these men
are eaten up with envy that someone
powerful, someone looked on with respect,
passes by, right before their very eyes,
with fame and honour. Then they complain
they are wallowing in dirt and darkness.
Some squander their lives, ruining themselves
for the sake of statues and a famous name.
And through their fear of death, hatred of life 110 [80]
and of seeing the sunlight often seizes
men so forcibly that, with anguished hearts,
they kill themselves, forgetting that this fear
is the origin of their trouble, this fear
[encourages men to all kinds of crime],²
corrupts their honour, breaks bonds of friendship,
and, in brief, urges them to cast aside
their sense of duty. For men have often
betrayed their country, their loving parents,
by seeking to avoid the realms of Acheron.
And just as children shake and are afraid 120
of all things in blinding darkness, so men
sometimes fear things in the daylight—but these
should no more terrify us than those things
which make young children tremble in the dark,
imagining what might happen. Therefore, [90]
we must dispel this terror in the mind,
this darkness, not with rays of sunlight

¹ Watson notes that this is a reference to their fear of being poisoned for their money.

² I follow Munro in inserting a line into the Latin here.

Book Three

or with glittering arrows of the day,
but with reason and the face of nature.

First, I say that mind, which we often call
the understanding and in which is placed
the guiding and directing power of life,
is no less part of man than hands, feet, and eyes
are parts of one whole living animal.

[And yet many philosophers have thought]
mental sensation is not located
in a specific place, but is instead
a certain vital habit of the body,
what Greeks call harmony, which causes us
to live with a capacity for sense,
although the mind has no determined place.¹

Just as people often say a body
possesses excellent health but this health
is not a part inside the healthy man,
so these people locate a sense of mind
in no specific spot. In saying this,
they seem to me to be wandering off,
straying a long way from the road. For often
our body is ill—we see that clearly—
yet we feel pleasure in some other part
hidden within. Often the reverse takes place,
as well, when, by contrast, a man whose mind
is sad feel pleasure in his whole body.

In the same way, if a man's foot pains him,
perhaps at the same time his head may feel
no pain at all. Moreover, when our limbs
surrender to soft sleep and our body,
relaxed and heavy, lies there without sense,
at that very time there is something else
inside us still, which is, in various ways,
stirred up and which receives within itself
all motions of joy and vain cares of heart.

And now, so you also can understand
that soul is in the limbs and that body
is not in the habit of sensing things
by harmony, firstly, it so happens
that, when a large portion of our body
has been removed, frequently in our limbs
life still remains. But, on the other hand,
when a few particles of heat have left
and some air has been forced out from the mouth,
that same life instantly abandons veins
and leaves the bones. From this you can infer
that particles do not all have equal roles,

130

THE MIND AS PART OF
THE BODY

[100]

140

DOCTRINE OF THE SOUL
AS HARMONY

150

[110]

160

[120]

170

¹At least one line is missing in the manuscript at the start of this sentence. The general sense of the missing text, however, seems clear.

Book Three

they do not equally maintain our health,
and that these seeds of wind and warming heat
have more to do with life staying in our limbs.
Thus, in the body itself there is heat
and vital wind, which depart our bodies
as we die. And therefore, since we have found
the nature of the mind and of the soul
is like a part of man, you must give up
that term harmony, which was handed down
to musicians from lofty Helicon,
or they themselves dragged it from somewhere else
and then reassigned it to this object
which at that time lacked its own proper name.¹
Whatever the case, let them keep the term,
and listen to the rest of what I say.

180 [130]

Now, I claim mind and soul are held united
and together form a single nature,
but the main one, which, as it were, has power
in the entire body, is our judgment,
which we call the mind or understanding,
fixed in place in the mid-part of the chest.
For here throb fear and terror. Soothing joys
move round this region, too. And therefore here
are mind and understanding. Of the soul,
all other parts, dispersed through the whole body,
obey and are moved in accordance with
the will and inclination of the mind.
Only the mind by itself has knowledge
for itself and rejoices in itself,
when no single thing is agitating
either soul or body: just as those times
attacks of pain make our head or eye hurt,
yet we do not ache in our whole body,
so mind sometimes is troubled on its own
or feels strong pleasure, when soul's other parts
throughout the limbs or body are not stirred
by any new sensation. But when mind
is shaken by some more violent fear,
we see the whole soul act in sympathy
throughout the limbs—we lose colour and sweat
in all our body, our tongue is broken,
our voice vanishes, our eyes grow darker,
our ears ring out, and limbs give way beneath.
Then, too, we often see how men collapse
from terror in their minds, so that from this
anyone can easily see that soul
is closely joined to mind: when force from mind

190

MIND OR JUDGMENT IS IN THE
CHEST

[140]

200

[150]

210

220

¹Helicon is a mountain in Boeotia, near the Gulf of Corinth; its springs were considered, in the popular imagination of the ancient Greeks, the source of poetic inspiration.

Book Three

affects the soul, soul then strikes the body
and makes it move. [160]

This same reasoning shows
the nature of the soul and of the mind
is physical. When we see this nature
moving limbs, rousing bodies out of sleep,
changing expressions, turning and guiding
the entire person, and we understand
that not one of these effects can happen
without touch and, furthermore, that touch
cannot occur without material stuff, 230
surely we must concede that soul and mind
have a nature which is made of matter?
Moreover, you observe our mind suffering
with our body, having common feelings
with the body. If a spear's brutal force
drives it in deep, exposing bones and sinews,
and does not take one's life, still what follows
is a fainting spell, a sluggish tendency
to sink down to the ground, and on the ground 240
a giddiness of mind occurs, and sometimes,
as it were, an uncertain wish to rise.
Therefore, the nature of the mind must be
material, since it is afflicted
by a blow and by material weapons.

SOUL AND MIND ARE PHYSICAL

[170]

240

And now, I will move on in this discourse
to offer you an argument about
what kind of matter the mind consists of
and how it is made up. And first, I say
that the mind is extremely fine, composed
of very tiny particles. If you wish
to pay attention to what follows here,
you should be able to appreciate
that this is so. We see nothing happens
faster than those things which mind imagines
taking place and which it itself begins.
Thus, mind rouses itself more rapidly
than any other substance whose nature
we perceive in front of us. But since it works
so quickly, it must be composed of seeds
which are extremely round and very small,
so that, when a slight impulse acts on them,
they can be set in motion. For water
under very slight contact is moved
and ripples back and forth, since it is made
of small, round particles. But, by contrast,
honey has a firmer nature—its fluid
is more sluggish, its movements more delayed,
its whole supply of particles adheres

NATURE OF MIND

250 [180]

260

[190]

Book Three

together more, because, quite obviously, 270
it consists of elements which are not
as smooth, as fine and round. And, as you know,
a tiny breath of air can force tall piles
of poppy seeds to scatter from the top,
but, by contrast, even the south-east wind
cannot do the same to a pile of rocks.
And so particles will move more freely [200]
the more they are extremely small and smooth.
But on the other hand, all elements
which prove to be heavier and more rough 280
will be that much more difficult to move.
Thus, since we have found the nature of mind
more mobile than the rest, it must consist
of very small, smooth, rounded elements.

And, my good friend, once you understand this,
you will find it helpful with many things
and think it good to know. The following fact
points out as well the nature of the soul,
how thin its texture is, how small a space
it might be kept in, if it could be compressed: 290 [210]
as soon as the serene repose of death
has seized a man and what makes up his soul
and mind has left him, from the way he looks
or from what he weighs, you cannot perceive
that any portion has been taken away
from his whole body. Death preserves it all,
except for vital sense and warming heat.
Thus, the entire soul must consist of seeds
which are very small, interconnected
through veins, flesh, and sinew, since, by the time 300
it has completely left the entire body,
the external outline of the limbs remains
intact, and there is not the slightest loss [220]
of weight—like those times when the smell of wine
has vanished, or the sweet scent of ointment
disappears in air, or the flavour leaves
from any matter. The substance itself
still does not appear smaller to our eyes,
nothing seems to be taken from its weight,
clearly because in the whole body of things 310
taste and scent are made by many tiny seeds.
Therefore, to state the issue once again,
you may know that naturally mind and soul
are made up of extremely minute seeds,
because when they depart they take no weight [230]
away with them.

.
Still, we must not believe
this nature is an uncompounded mix,

Book Three

for a certain delicate wind leaves men
when they are dying—it's combined with heat,
and heat draws air with it: there is no heat
without some air mixed in combination. 320
Since the nature of heat is rarefied,
then many primary particles of air
must move around in it.

Thus, to this point
we have found that the nature of the soul
has three parts, but these three things together
are not enough to create sensation,
since facts do not accept that any of these
could produce those motions that generate
our senses [and thoughts moving through our minds.]¹ 330 [240]
Thus, to these three substances we must add
a certain fourth nature, as well, something
that has no name at all. But there is nothing
more agile or more tenuous than it,
or made of smaller, smoother elements.
This matter first sends out through the body
those motions which activate sensations.
For since it is composed of tiny shapes,
it is the first substance stirred, and from it
heat as well as the hidden force of wind 340
acquire motion, and from that air, as well.
After that everything is mobilized—
blood is roused, and then all flesh feels it, too,
and bones and marrow get it last of all, [250]
whether pleasure or a burning torment
of the opposite kind. And pain cannot
easily penetrate as far as this,
nor any bitter evil move within,
without all matter being shaken up,
so much so that there is no room for life, 350
and through all the openings of the body
the parts of soul disperse. But generally,
a limit is set to motions, as it were,
on the surface of the body—that is why
we stay strong enough to maintain our lives.

Now, though I am keen to give an argument
showing how these parts are mixed together
how, once arranged, they act effectively,
the poverty in my native language [260]
holds me back against my will. However, 360
I will touch upon the subject briefly,
as best I can. These primary substances,

¹Part of line 240 in the Latin is corrupt. The translation in square brackets provides the general sense of the missing words. The three elements introduced so far are wind, air, and warmth.

Book Three

through motion of primordial elements,
move among themselves, so no single one
can be cut out, nor can its power become
set off from the rest by any space. They are,
as it were, many forces of one body.
Just as in the flesh of any creature
anywhere at all there is an odour,
a certain heat, and taste, yet from all these
a single corporeal mass is formed,
so heat, air, and hidden power of wind
create in combination one nature,
together with that active force which sends
out from itself to those three parts the start
of movements from which arise those motions
which first bring sensation to the tissues.
This fourth nature lies completely hidden,
far inside—in our whole body nothing
is deeper down than this. And furthermore,
it is the very soul of all the soul.
Just as in our limbs and our whole body
the mind's force and the soul's power exist
in a hidden mixture, since they are made
from a few small particles, so, you see,
this force without a name, which is composed
of minute elements, lies there hidden.
Beyond that, it is itself, so to speak,
the very soul of the entire soul—it rules
throughout the body. In a similar way,
wind, air, and heat all combined together
throughout the limbs must act effectively—
one being more subservient to the others
or more prominent, but in such a way
that all of them seem to create one thing,
so heat and wind, without the other parts,
or the power of air all by itself,
could not separate from other portions
and abolish and dissolve sensation.
Another thing—there is in mind that heat
which it takes on when it boils up in rage
and fire flashes more fiercely in the eyes.
There is much cold wind, too, fear's companion,
which starts a trembling in the limbs and stirs
the body. There is in mind also a state
where that air is passive—it comes about
when heart is undisturbed and face serene.
But there is more heat in those living things
whose fiery hearts and passionate minds are quick
to boil in fury. The prime example
in this group is the fierce power of lions,
who frequently, when they give out a cry,
break their hearts with roaring and cannot hold

370

[270]

380

[280]

390

400

[290]

410

Book Three

inside the chest the torrent of their rage.
But the cold mind of deer contains more wind
and is more quick to rouse throughout its flesh [300]
the chilling breeze which in the limbs creates
the start of quivering motion. In oxen,
their nature subsists more on peaceful air—
anger's smoking torch is never applied 420
to rouse it to excess, suffusing it
with shades of blinding cloud, nor is it dull,
impaled on freezing spikes of fear. It sits
midway between deer and savage lions.
The race of men is just like that, as well.
Though education does make some of them
equally refined, it still leaves in place
nature's first vestiges in each man's mind.
And we should not think that evil habits [310]
can be plucked out by the roots, for one man 430
will rush more readily to bitter rage,
a second one will be somewhat faster
to succumb to fear, a third take some things
more calmly than is right. And differences
among various natures of human beings
and in the habits which arise from them
must exist in many other matters.

I cannot now explain hidden causes
of these differences nor come up with names
for so many shapes of primary elements 440
which create this diversity in things.
But in these matters I do see one thing
I can affirm—the remaining traces [320]
of those natures which reasoning cannot
remove from us are so slight, that nothing
stops us living a life worthy of the gods.
This nature, then, is held in our whole body
and is itself the body's guardian
and its source of health. For body and soul
mutually cling to one another 450
and have roots in common, and, we notice,
cannot be torn apart without destruction.
Just as it is difficult to cut out
the odour from pieces of frankincense
without also wiping out its nature,
so it is not easy to pull the substance
of mind and soul from the entire body [330]
without dissolving all things. They arise,
at their first origin, from elements
so closely intertwined among themselves, 460
possessing a life they share together,
and it does not seem that body or soul
can have power to sense things on their own,

Book Three

without the other's force, but that sensations,
after being kindled by common motions
of the two of them acting on each other,
catch fire throughout the tissue.¹ Moreover,
body is never formed nor does it grow
on its own, and we do not observe it
lasting after death. It is not like water, 470
whose moisture often radiates the heat
which has been given to it and is not, [340]
for that reason, shaken apart itself,
but stays intact. No, our bodily frames,
I say, once left abandoned by the soul,
cannot tolerate the separation,
and, after the soul has been wrenched away,
they perish utterly and rot. From the start,
when life begins, mutual interactions
of body and soul acquire those movements 480
which give vital force, even when lying
in the mother's womb inside her body,
so their separation cannot take place
without disease and death. Since, as you see,
what keeps them living is their combination,
their matter must also be united.
As for the rest, if anyone denies [350]
that body has capacity for sense
and thinks that soul, mixed in all the body
sustains this movement we call sensation, 490
he is resisting true and obvious facts.
Who will ever explain what body feels,
unless it is something which facts themselves
have obviously revealed and taught us?
You may say that once soul has been scattered,
all body lacks sensation. That is true,
for it loses what, during its lifetime,
did not belong to it, and what is more,
before soul has been driven out from life
a body loses many things.²

Moreover, 500
to assert that eyes cannot see a thing,
but that mind looks through them, like open doors, [360]

¹ There's a slight problem with Lucretius' vocabulary here. Having set up the division between mind (*animus*) and soul (*anima*), with the former in the chest and the latter dispersed throughout the body, Lucretius now, in discussing the relationship of these two elements with the body, uses the word *animus* to refer to the combination of mind and soul. I have used the word soul for this meaning of *animus*, so that here (and elsewhere) soul refers to the combination mind and soul, two elements which, in his earlier discussion, were kept separate.

² The sense of these four lines is awkward and disputed (some editors have rejected them). The point seems to be that the soul and body are both required for sensation. When death scatters the soul from the body, sensation ends in the body, but the body loses other things before the soul leaves (as Munro observes), like strength, vigour, health, and so on.

Book Three

is difficult, since our sense in the eyes
contradicts this claim, for that sensation
draws us forcibly to a sense of sight
in pupils of our eyes themselves. In fact,
often we cannot look at brilliant things
because their brightness impedes our eyesight.
This does not occur with doors. When we look
through open doors, they suffer no distress. 510
Moreover, if our eyes were just like doors,
mind, it appears, should perceive things better
with the eyes, our very doorposts, ripped out
and removed.

In considering these things, [370]
you cannot accept at all the theory DEMOCRITUS
in the revered views of great Democritus
that individual primary particles
of body and of soul are put in place,
alternating one after the other,
and shape our limbs, holding them together.¹ 520
For since the basic particles of soul
are much tinier than those making up
our tissues and our body, their number
is also smaller and thinly scattered
throughout our frame. What you can claim is this: LOCATION OF SOUL
the primary particles of soul are spaced
in intervals at least as far apart
as the size of the smallest substances
which, when thrown against a body, can first
start motions of sensation in that body.² 530 [380]
For sometimes we do not feel any dust
clinging to the body or sense that chalk
has been shaken on our limbs and settled.
Nor do we feel a mist at night, or sense
a spider's slender web get in our way
when we get tangled in it as we move,
or notice its wrinkled web has fallen
on our head, or feathers from birds, or seeds
flying from plants, which have so little weight,
they usually have trouble falling down. 540
We do not feel the tracks of all creatures

¹ Democritus (c. 460 BC-c.370 BC), a Greek philosopher, is credited as the first to propose a detailed atomic theory. Democritus claimed that atoms of body and soul were equal in number and united in pairs throughout the human body.

² Physical sensation, which always arises from material contact, starts, as Lucretius has explained earlier, in something which energizes a particle of soul, which is scattered through the body. But, as he goes on to argue, parts of our body can be touched without any sensation arising. Thus, not every part of the body contains soul, and the soul particles must have intervals between them no greater than the size of the smallest substances which, when they contact the body, create sensation. Substances smaller than that may contact the body without affecting soul, since they may not hit a soul particle or rouse the body's other particles.

Book Three

that creep along our body, or notice
each and every footstep along our skin
taken by gnats and other bugs. In fact, [390]
so many things in us must be dislodged
before the basic elements of soul,
mixed throughout the framework of our bodies,
sense that primary particles have been hit,
move across the gaps they have between them,
then, in sequence, collide, come together, 550
and bounce back once more.¹

And mind does more MIND IS MORE VITAL
to maintain bands of life and govern life THAN SOUL
than does the power of soul.² For without
mind and understanding, no part of soul
can stay, even for the briefest moment,
inside the body—it quickly follows them, [400]
as their comrade, and scatters in the air,
leaving cold limbs to icy death. And yet
anyone whose mind and understanding
remain behind continues on with life, 560
although his body has been maimed, with limbs
cut off all round. If on every side his soul
has been removed and has left his limbs,
the trunk still lives and breathes celestial air
which gives him life. A large part of his soul
is gone, but not the whole of it—he still
holds on and clings to life. It's like the eye—
if there are wounds around it but the pupil
stays intact, the living power of sight
remains, but only if you do not hurt 570 [410]
the entire eyeball, leaving the pupil
alone and cutting round it. But slicing it
cannot be done without also destroying
the eye as well.³ And if that tiny part
in the middle of the eye is punctured
light leaves instantly, and darkness follows,
though the bright orb is otherwise unhurt.
That shows how closely soul and mind are linked,
bound together in a lasting union.

Come, so you can learn that delicate souls 580

¹ This passage is a summary statement of Lucretius' notion of how physical sensation occurs. A sufficient number of the primary particles making up our bodies must be stirred to rouse the scarcer particles of soul, so that the latter can begin to move across the intervals separating them and collide, thus transporting the sensation through the body. If the number of primary particles roused by initial contact is insufficient, then the particles of soul will not be activated, and no sensation will register (e.g., with a spider's web).

² Lucretius returns here to the distinction between the mind (*animus* or *mens*), located in the chest, and the soul (*anima*), scattered throughout the body.

³ The exact meaning of this sentence is debated. Lucretius seems to be saying either that cutting around the entire eyeball destroys the sight or that cutting the pupil will destroy the sight.

Book Three

and minds in living things are born and die,
I will now proceed to set down verses
worthy of your life, ones I have long sought
and then produced in work which brought me joy. [420]
You see to it that you link soul and mind
under one name, and when, for example,
I go on to speak of soul, establishing
that it is mortal, understand I speak
about the mind, as well, since their substance
is made up of one mutual combination.¹ 590

Now, first of all, I have revealed that soul
is thin and consists of minute particles
created from primordial elements
much tinier than clear liquid water
or mist or smoke—it far surpasses these
in its mobility, and it is moved
more easily when struck by slighter blows,
since it is set in motion by images [430]
of smoke and mist, those times when, for instance,
we are lulled to sleep and look at altars 600
exhaling steam and sending smoke on high,
for there is no doubt that these things send out
images to us.² Now, since, when jars crack,
you see water flow in all directions
and liquid seeping out, since mist and smoke
disperse in air, therefore you must believe
that soul, too, is diffused and perishes
at a much faster rate and is dissolved
into primary elements more quickly,
once it has been removed from someone's limbs 610
and has departed. Indeed, since body,
which is, as it were, the soul's container, [440]
cannot keep the soul intact, once something
weakens and thins it out by having blood
removed from veins, how then can you believe
that any air can keep the soul inside,
because air is thinner than our bodies
and [thus less able to contain it]?³
Then, too, we sense mind comes into being

¹As he states here, Lucretius is now going back to ignoring his earlier distinction between mind and soul. So from this point on the word soul in this section of the translated text refers to both mind (in the chest) and soul (distributed throughout the body). At this point, Lucretius moves on to what is (for him) obviously a central part of his entire book—the various proofs (seventeen in all) that the soul is mortal. The immortality of the soul is, clearly enough, one of the central claims of the many religious doctrines which Lucretius is determined to eradicate.

²Images come from objects, contact the body, and affect the soul in such a way as to produce dreams. Lucretius deals with this issue of images later in Book 4. His point here is that the basic particles of soul are so slight and sensitive that they are moved, not merely by mist and smoke, but even by images of mist and smoke (which must be even more tenuous than those substances themselves).

³The words within square brackets are prompted by a suggestion from Bailey.

Book Three

together with body, matures with it, 620
and, like body, ages. Just as children
totter on with weak and tender bodies,
so judgment in the mind accompanying them
is frail. After that, when they grow older,
into a strong, robust maturity,
their understanding is enlarged, as well, [450]
their strength of mind is more comprehensive.
Later, when their bodies have been shattered
by the potent force of time and their frame,
its powers exhausted, has broken down, 630
then natural abilities are crippled—
tongue prattles, mind totters, and every part
fades away at the same time and fails.
Thus, it is appropriate that all matter
of the soul should also be dissolved,
like smoke, in upper breezes of the air,
since we see it is produced with body,
grows with it, and, as I have shown, with age
they both fail and fall apart together.
Then add to this the fact that we observe 640
that, just as the body itself is prone
to frightful illnesses and severe pain, [460]
so mind has bitter worries, grief, and fear.
Thus, it makes sense that mind experiences
death, too. Besides, when our body is ill,
the mind often roams around aimlessly,
for it raves on and utters senseless things,
and sometimes, in a heavy lethargy,
is carried to a deep eternal sleep,
its eyes and head nodding, as it sinks down 650
to where it hears no voices and has lost
power to recognize the look of those
who stand around, recalling it to life,
wetting their faces and their cheeks with tears.
Thus, since morbid sicknesses reach the soul,
you must concede that it, too, is dissolved, [470]
for both disease and pain are harbingers
of death, as we have learned from countless men
who perished in the past.¹ And why is it,
when the shrewd force of wine gets in a man 660
and its spreading heat moves throughout his veins,
there follows a heaviness in the limbs—
as he reels to and fro, his feet trip up,
his tongue becomes thick, his mind grows tipsy,
his eyes swim, and shouts, sighs, and fights arise, [480]
and all the other actions which result
from this behaviour—why does this happen,

¹ Following other editors, I have omitted two lines here (474-475 in the Latin). One of them recurs at line 510 of the Latin below.

Book Three

unless the overpowering force of wine
has the habit of disordering the mind
inside the body itself? But those things 670
which can be overthrown and blocked reveal
that, if a somewhat stronger cause pushed in,
they would then perish, robbed of future life.
Moreover, when the force of a disease
overcomes someone, often he falls down
unexpectedly right there in front of us,
as if hit by lightning—he foams at the mouth,
moans, trembles in his limbs, acts foolishly,
jerks his muscles, twists, has trouble breathing, [490]
exhausts his body twitching back and forth, 680
clearly because the force of the disease
spreading throughout his frame affects his soul
and shakes it, so that it foams—just as waves,
beneath the winds' strong fury, boil over
on the briny sea. He is forced to groan,
for his limbs are wracked with pain, above all
because vocal particles, grouped together,
are expelled, carried from his mouth, and move,
as it were, on their customary road.
Madness sets in, for force of mind and soul 690
is broken, and, as I have shown, ripped up, [500]
torn apart, and split by that same poison.
Later, when what brought on the sickness leaves
and bitter fluid in the ailing body
has retreated to its hiding places,
then, as if staggering, he first gets up,
gradually returns to all his senses,
and regains his soul. Thus, when mind and soul
are, even inside the body, disturbed
by such serious illnesses and suffer, 700
pulled apart in such miserable ways,
why do you believe that without body,
in open air among the blustering winds,
they could continue living?

And since we see [510]
mind is cured, just like a suffering body,
and observe it can be changed with healing,
this also reveals that mind is mortal.
For anyone who comes along and starts
to transform the mind or seeks to alter
some other substance, whatever it may be, 710
must either add parts or change their order
or take away at least some small portion
of the whole. But anything immortal
does not allow its parts to be transferred
or the least part to be added or removed.
For whenever something changes and moves

Book Three

beyond its limits, that is instant death [520]
for what it was before. Therefore, the mind,
whether it is sick or changed by healing,
gives evidence of its mortality, 720
as I have shown. That shows how much real facts
are seen to contradict false reasoning,
to cut off an escape for anyone
hostile to truth, and with a two-edged proof
to overthrow his falsehood.

And furthermore,
frequently we see that someone dying
gradually loses vital sensation
limb by limb: first, on his feet toes and nails
turn black, then feet and legs expire, and then,
through the rest of him, we see, step by step, 730 [530]
the tracks of icy death. Since this substance
of the soul is divided up in parts
and does not emerge all at once intact,
we must think of it as something mortal.
But if you perhaps believe that the soul,
all on its own, could throughout the body
pull itself back inside, contract its parts
into one place, and in this way withdraw
sensation from every limb, then that place
where such a large amount of soul collects 740
should seem to have more feeling. Such a place
does not exist. Thus, it is obvious,
as we said before, that soul is torn apart,
dispersed outside, and therefore perishes.
What's more, even if we agreed to grant [540]
a falsehood and conceded that the soul
could be collected inside the bodies
of people who, when they die, leave the light
one part at a time, you must still admit
that soul is mortal—it makes no difference 750
if it dies dispersed in air or is pulled
into one place from all its parts and then
becomes inert, once sensation has left
all parts of the whole man and everywhere
less and less life remains.

And since mind
is one part of a man and remains fixed
in a specific place, like ears and eyes
and all other senses which guide our lives,
and, just as hand and eye or nose cannot, [550]
once detached from us, sense things or exist, 760
but are soon melted by decay, so mind
cannot live on its own without body
and without the man himself, who appears,

SOUL CANNOT FUNCTION
WITHOUT BODY

Book Three

so to speak, the container of the soul,
or whatever else you might imagine
more closely linked with it, since it adheres
to body in such a close connection.
And vital power of body and mind,
when combined, are strong and delight in life,
for the nature of mind, without body, 770
cannot alone and by itself produce [560]
vital movements, and yet, deprived of soul,
body cannot last and use its senses.
We know that, just as eyes torn from their roots,
cut off from the whole body, cannot see
a single thing, so soul and mind are seen
to have no power on their own, because,
quite clearly, when mixed up with veins and flesh,
through bone and sinew, they are both contained
by all the body—their basic elements 780
are not free to leap around, spacing themselves
at large intervals. Hence, confined like this,
they are stirred in motions for sensation, [570]
movements which after death they cannot make,
once they have been thrown outside the body
into the air, because they are not held
in the same way. If the soul is able
to keep itself together in the air
and to contain in itself those motions
which it carried out before in sinews 790
and in the body itself, then the air
will be a living entity. That is why,
to repeat myself, when the whole covering
of body has collapsed and vital breath
has been expelled outside, you must agree
sensations in the mind and soul dissolve,
since for body and soul, the cause of death
is linked inseparably.

Moreover,
when body cannot bear a separation [580]
from the soul without smelling disgusting 800
and turning rotten, why do you then doubt
that soul's power, rising from deep within,
has moved away and been dispersed, like smoke,
and thus the body, changed enormously
by putrefaction, falls in ruins, because
its foundations have completely shifted
from their location and soul has flown out
through limbs, through all the winding passages
inside the body, and departed through the pores?
So you can ascertain in many ways 810
that soul's substance, divided into parts,
has withdrawn and left the body and that

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it has been torn up into parts itself
inside the body, before it slipped away, [590]
gliding out into the airy breeze.

And furthermore, while soul is still turning
within limits set by life, for some reason,
when it is disturbed, frequently it seems
to move and to be seeking deliverance
from the entire body—the face appears 820
to grow listless, as at the time of death,
and on the bloodless body all the limbs
fall limp. That's what happens when people say,
“The mind is damaged” or “His heart has gone”—

when there is great concern and everyone
strives to keep grasping the last thread of life.
For then the mind and all force in the soul [600]
are broken apart, and they both collapse,
together with the body, too, so that
a slightly stronger cause can then dissolve them. 830

Why, may I ask, do you doubt the frail soul,
driven outside body, robbed of shelter,
in the open air, not only could not last
forever, but could not sustain itself
for any length of time, however short?
For no one who is dying seems to feel
soul leaving his whole body all at once—
first rising to his neck, then to his throat—
no, he feels it fail in a certain place, [610]
a fixed location, just as he discerns 840
other senses being dissolved, each one
in its own spot. But if mind were immortal,
it would not, in dying, complain so much
that it was being dissolved, but rather
that it was going outside, abandoning
its covering, like a snake.

Then, too, why
are mental judgment and understanding
never produced in head or feet or hands,
but cling to a single place, a position
fixed for all men, unless for everything 850
a certain location has been assigned
where it is born and where, once created,
each thing can then survive and stay alive?
[Thus, our body must follow the same law],
with such a varied structure in its limbs
that their arrangement could never be upset.¹ [620]
That shows how much one thing always follows
something else. And it is not customary
for fire to be born in streams of water

¹ The addition in square brackets is a suggestion by Munro.

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or cold to be conceived in flames.

Besides, 860
if soul's nature is immortal and able
to feel sensations outside our body,
I think we must assume it is endowed
with all the senses. We cannot envisage
for ourselves in any other way those souls
roaming the lower world of Acheron.
That is why writers from past generations
and painters, too, have represented souls [630]
possessing senses in this way. But eyes,
nostrils, even hands, are not capable, 870
without body, of existing for the soul,
nor are tongues or ears, not all on their own.
Therefore, souls cannot sense things or exist
all by themselves.

And since we do perceive
vital sense in our whole body and see
it all as a living thing, if some force
with a rapid blow across the middle
suddenly sliced through, so as to cut it
into two separate parts, undoubtedly
the soul's force will be cut in half, as well, 880
divided at the same time as the body.
But what is split up and then separates [640]
into any parts clearly demonstrates
that its nature cannot be eternal.
People talk about chariots armed with scythes
growing hot in a promiscuous slaughter
and often slicing limbs so suddenly
that what is severed from the frame falls down
and is seen to quiver on the ground, although,
given the swiftness of the wound, in the man 890
his mind and spirit cannot feel the pain.¹
Since, at the time, his mind is focused on
the fury of the fight, it keeps on going
with the remnants of the body, seeking
battle and slaughter, often unaware
that the left arm with the shield is missing,
sliced away by wheels and ravenous scythes [650]
among the horses, while another man,
as he climbs up and keeps charging forward,
does not know his right arm has fallen off, 900

¹The scythes extended straight out from the hub of the chariot wheel and cut down soldiers when it drove through their ranks. Smith notes that neither the Greeks nor Romans used such chariots, but that they were a feature of eastern armies.

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and yet another man, with his leg gone,
attempts to rise, while nearby on the ground
his dying foot wiggles its toes, and a head,
severed from the warm and living torso,
maintains down on the ground a living look
with its eyes open, until it gives out
all the soul that still remains. Moreover,
if, when faced by a snake with flicking tongue,
menacing tail, and extended body,
you decide to take an axe and chop up 910
its tail and body into numerous pieces,
you will see all the separate sliced-off bits [660]
writhing from the recent wound and sprinkling
earth with blood and the front part, mouth open,
seeking its own tail, so that, struck with pain
from the agonizing wound, it can soothe it
with its teeth.¹ Shall we say that complete souls
exist in all those smaller parts? If so,
by this reasoning, it will then follow
that in its body one living creature 920
had many souls. [But since this is absurd,]
the soul which lived as a combined unit
with its body has been divided up.
Thus, you must think of them both as mortal,
for each of them has been broken apart
in the same way into many pieces.²

Moreover, if the nature of the soul [670]
is immortal and is placed in bodies
when we are born, why are we unable
to remember those periods of our lives 930
from earlier times? Why do we not retain
any traces of past events? For if
the power of mind has been changed so much
that all remembering of things gone by
has passed away, then, in my view, this change
is not far removed from death. And therefore,
you must admit that what it was before
has been destroyed and that what now exists
has been created now.

And furthermore,
if, as a rule, the living power of soul 940 [680]
is set in place at the moment of birth,
when we move across the threshold into life
once all our body is already formed,
it would not be appropriate that it
seems to grow together with the body

SOULS ARE NOT INSERTED
FROM OUTSIDE

¹The text in the first part of this sentence is uncertain and disputed. Some words may be missing.

²Following Munro, I have added the phrase in square bracket to clarify the logic of the sentence.

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and the limbs, in the very blood—instead,
it would be natural for it to live
by itself, as if in some enclosure.¹
But obvious facts reveal the opposite.
For soul is so mixed in with veins and flesh, 950
with bones and sinews, that even our teeth
share in sensation, too, as is revealed [690]
by toothache, cold-water shock, or hard stones
hidden in food when we bite down on them.²
Thus, to repeat myself, we must not think
souls have no beginning and do not face
the law of death. For we cannot believe
our souls could be so closely interlinked
with our bodies, if they were inserted
from outside. Since souls are so closely joined, 960
it does not seem they could come out intact
and without damage extricate themselves
from every sinew, bone, and joint. But if,
perhaps, you think that soul, once inserted
from outside into us, has the habit
of seeping through limbs, and then uniting
with the body, there is a greater chance
that it will die, since what spreads out dissolves, [700]
and therefore perishes, for it is passed
through all the passages inside the body. 970
Just as food dies off, when distributed
in all the limbs and portions of the body,
producing from itself another substance,
so soul and mind, no matter how intact
they are when entering a new-made body,
are still dissolved as they are moved around,
while, as it were, through every opening are sent
into our limbs particles which produce
this nature of mind now ruling in our body,
which was born out of what was then destroyed, 980
while being distributed throughout the limbs. [710]
And thus, we see the nature of the soul
does not lack a moment when it is born,
nor is it exempt from death.

Moreover, SOULS DO NOT SELECT
are particles of soul left in a body BODIES TO ENTER
which is dead, or not? If they do remain
and are still inside, we cannot justly call
the soul immortal, since when it went away
it lost particles and was diminished.

¹I have followed Munro in omitting line 585 of the Latin, which seems an unnecessary interruption in the idea.

²Following some other editors, I have moved lines 690 to 694 in the Latin (“For soul . . . down on them”) up to this point (lines 686 to 690 in the Latin).

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But if, when carried off, the soul escaped 990
while limbs were still complete, so that it left
no parts of itself inside the body,
then, once the innards rot, how do corpses
bring forth worms? How do such large quantities [720]
of living creatures lacking bones and blood
swarm through bloated limbs? If, by any chance,
you think souls are inserted in these worms
from outside, each one able to go in
its own body and do not consider
why souls should gather in many thousands 1000
where one soul has departed, it still seems
we should investigate and determine
whether all those souls really do chase down,
all on their own, the seeds of tiny worms
and build themselves a place in which to live,
or whether they are, as it were, inserted
into bodies already fully made.
But one cannot give a reason why souls
would work so hard making themselves bodies, [730]
for when they lack bodies, they flit about 1010
without being disturbed by cold, illness,
and hunger, because body is more prone
to suffer from these pains, and mind acquires
many ills through contact with the body.
But still, suppose it is really useful
for these souls to manufacture bodies
which they may enter, there still seems to be
no way that they could do it. And thus souls
do not make limbs and bodies for themselves.¹
However, they cannot be inserted 1020
into bodies which are made already,
for they will not be able to exist
in those delicate connections or make,
by mutual contact, shared sensations. [740]

Then, too, why does raging fury appear
in grim broods of lions? Why are foxes sly?
Why is running away passed down to deer
from fathers, so their father's timidity
makes their limbs move quickly? As for the rest,
other things like this, why are all produced 1030
at the earliest moments of existence
in limbs and temperament, if not because
a force of mind set by its own seed and race
also grows along with the whole body?
But if soul is immortal and, as a rule,

¹The point here seems to be (perhaps) that souls would not be able to shape matter into bodies since they would not have the physical equipment to do that (e.g., fingers and hands), just as they could not (according to an argument Lucretius has already made), enjoy sensation on their own, because they would lack organs of sense.

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changes bodies, then living animals
would have changeable dispositions—dogs
made from Hyrcanian seed would often flee [750]
a charging stag with horns, up in the air
a hawk would tremble and fly off in fear 1040
when doves came near, people would lose their minds,
and savage tribes of beasts grow rational.¹
For it is faulty reasoning to claim,
as some men do, that an immortal soul,
when it switches bodies is transformed,
since what is changed dissolves, and therefore dies,
for its parts are moved, their arrangement shifts.
Therefore, they must also be capable
of being dissolved through every limb, so that
in the end they all die with the body. 1050

If some men claim that souls of human beings [760]
always enter into human bodies,
I will still ask why, after being wise,
a soul can then become so idiotic,
why no child is clever, why no mare's foal
is as well trained as bold strength in a horse.²
They will, no doubt, seek refuge by saying
that in fragile bodies minds are fragile.
But if that is the case, you must admit
soul is mortal, since it has been altered 1060
so greatly in the body and has lost
its earlier vitality and sense.
In what way will the power of the mind
be able to grow strong along with body
and reach the longed-for prime of life, unless [770]
it is body's partner from the very start?
Or why would soul desire to go away
once the limbs grow old? Is the soul afraid
to stay enclosed in a decaying body
in case its domicile, now undermined 1070
by the long interval of years, might fall
and bury it? But there are no dangers
for a thing which is immortal.

Besides,
for souls to be standing there when wild beasts
are born or have sex appears ridiculous—
immortal souls in countless numbers waiting
for mortal limbs and in hot contention
among themselves which one will be the first [780]

¹Hyrcania, a remote region south of the Caspian Sea (which the Greeks called the Hyrcanian Sea) was famous for its fierce wild animals. The doctrine that the immortal soul could after death live on in a different creature (palingenesis) is most commonly associated with the Pythagoreans.

²Line 763 in the Latin has been omitted. It is the same as line 746 (line 1034 in the English text) above and is commonly removed.

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to be inserted well before the rest,
unless perhaps a treaty has been forged 1080
among the souls—whichever one flies up
and gets there first will be the first one in—
so that there is no fight of any kind,
no mutual test of strength.

Furthermore,
a tree cannot live in aether, or clouds
deep underwater, or fish in farmlands,
or blood exist in wood, liquid in stones.
There is a fixed arrangement where each thing
belongs and grows. Thus, the nature of mind 1090
cannot arise without body, or live
on its own, apart from blood and sinew.
If—and this is far more likely to occur—
the power of mind itself were able [790]
to live in the head, or heel, or shoulder,
or could be born in any part you wish,
it would still be accustomed to remain
in the same man, in the same container.
However, since we see in our bodies
where the mind and soul can exist and grow
in their own place, so we must all the more 1100
deny they can be born and continue
totally outside the body. Therefore,
when body dies, you must admit that soul,
pulled apart inside the entire body,
also perishes. In fact, as you can see, [800]
to join the mortal with the immortal,
to think that they can work in harmony
and be acted on by one another
is foolish. For what can one imagine
more paradoxical, more inconsistent, 1110
a greater inherent contradiction,
than that something mortal should be combined
with something immortal and eternal
and, united with it, should then endure
raging storms? Besides, what lasts forever
must either, being made of solid stuff,
fend off attacks and not let anything
penetrate inside it which could loosen
close-packed inner parts, like material stuff
whose nature I have previously shown, 1120 [810]
or it must be able to continue
through all ages, because it is exempt
from blows, just like the void, which stays intact
and does not suffer the slightest damage
from collisions, or else because there is
insufficient room around it in which,
so to speak, its material could disperse

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and be dissolved, in the same way the sum
of all things is eternal—there is no space
beyond it where its matter could escape, 1130
nor are there any substances able
to strike and fracture it with a strong blow.¹

But if perhaps soul is thought immortal
more because it is kept well fortified
from things fatal to life or else because [820]
objects which threaten its security
do not appear at all, or those which come
for some reason move away, driven back
before we can perceive what harm they do,
[facts clearly show that this cannot be true, 1140
for many harmful things affect the soul.]²
Besides falling sick when body is ill,
something often happens to vex the soul
about what will happen in the future,
to keep it anxious and disturbed, worn out
with worries, and, when past evil actions
are long over, the guilt brings on remorse.
Then, too, mind has its own form of madness
and can become oblivious to things,
besides those times when it keeps sinking down 1150
beneath black waves of lethargy.

Death, therefore,
is nothing to us, does not concern us [830]
in the least, since the nature of the mind DEATH IS OF NO CONCERN
we consider mortal. Just as in the past
we felt no pain when Carthaginian troops,
massing for battle, advanced from every side,
when all things, disturbed by war's fearful noise,
shook with dread under high heavenly skies,
in doubt on which of the two sides would fall
power to rule all men on sea and land, 1160
so, when we cease to be, when soul and body,
whose union makes us one single being,
part company, it is clear nothing at all [840]
can happen to us or rouse our feelings,
not even if earth is mixed in with sea
and sea with sky—for then we won't exist.³
And even if the nature of our mind

¹This final point, about the totality of the universe remaining eternally complete, Lucretius has argued earlier. Some editors (Munro included) omit the passage (lines 806 to 818 in the Latin).

²At least one line is missing in the text at this point. The words in square brackets provide an English text which completes the sense of the sentence.

³The Carthaginians, inhabitants of North Africa, fought three major wars with Rome (the First, Second, and Third Punic Wars, from 264 BC to 146 BC). The final defeat and demolition of Carthage was the most significant and celebrated military event in the history of the Roman Republic. The point of the reference is that if we are not alive, then nothing, no matter how serious, affects us.

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and power in the soul have sensations
after they are split off from our body,
that still means nothing to us, who consist 1170
of a united combination, joined
by an arrangement and in a marriage
of body and soul. And if time gathered
our material stuff after we have died
and brought it back again as it is placed
right now and if light of life were given
back to us—even if these things were done— [850]
it would not matter to us, when memory
of what we once were had been disrupted.
Even now we are not at all affected 1180
by who we were before, in earlier times—
worries about that do not alarm us.
For when you look back on all past ages,
on that immeasurable length of time,
and think how various the movements are
in material stuff, then it is easy
to accept the fact that those same particles
of which we now consist have before this
often been set in the same arrangement
as they are now. Yet we are unable 1190
to recover that in our mind's memory,
since a pause in life has been interposed, [860]
and all movements have wandered aimlessly
far from sensation. For if by chance a man
is to live in misery and sorrow,
then at the time he also must exist
in person, so trouble can afflict him.
Since dying prevents this and ends existence
for the man who could be swamped by troubles,
we can know that there is nothing to fear 1200
in death and someone who does not exist
cannot be sad—it makes no difference at all
whether he was even born at any point,
once immortal death has taken away
his mortal life.

Thus, if you see a man [870]
concerned about himself, that after death
he will either rot away, once his body
is buried in the ground, or be destroyed
by flames or wild creatures' jaws, you will know
his words do not ring true and in his heart 1210
there is some hidden torment, even though
he himself may say he does not believe
he will have any feelings once he's dead.
For, in my view, he is not following
what he claims is his belief or its reasons—
he does not withdraw from life, removing

FEAR OF WHAT HAPPENS AFTER
DEATH

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himself completely, but, in ignorance,
assumes that something of himself lives on.
For any living person who proposes
to himself what will take place in future, 1220
that wild beasts and birds will mutilate him
once he is dead, is pitying himself. [880]
He has not separated himself from death,
nor pulled away from the cast-off body
far enough. He imagines it is him,
and standing there, he mixes in the corpse
his own feeling. Thus, he resents the fact
he was created mortal and does not see
that when his death really comes there will be
no second self which, still alive, can mourn 1230
to him of his own death, standing in grief
that he lies there being mangled or burned up.
For if, when one has died, it is painful
to be chewed up by wild beasts' jaws and teeth,
I do not see how it is not painful
to be laid out in searing flames and burn, [890]
or be immersed in honey and then choked,
or grow stiff with cold, as one is lying
on top of a flat, frozen rock, or crushed
and buried by the weight of earth above.¹ 1240

“Now, your joyful home and excellent wife
will no more welcome you, your sweet children
will not come running up to snatch kisses
and touch your heart with secret joy. No more
will you be able to live prosperously
and protect your own. You unhappy man,
sadly one hostile day has taken from you
all the numerous privileges of life.”

So people state, but in saying these things,
they do not add this,

“And there now remains 1250 [900]
left over in you no yearning for these things.”

If they perceived this clearly in their minds
and followed it in what they said, they would
relieve themselves in their own minds of fear
and great anxiety.

“Indeed, just as now
you are asleep in death, so will you be
for all time to come, free of all suffering

¹Lucretius is here mentioning various treatments of the corpse in burial. Honey was sometimes used for embalming.

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and pain, but close by we lamented you
inconsolably, as you burned to ashes
on the dreadful funeral pyre, and no day
will rid our hearts of everlasting grief.” 1260

Therefore, we should ask the man who says this
what is so harsh: if death is a return [910]
to repose and sleep, how could anyone
pine away in constant lamentation?
And often men even behave like this
when they lie down to eat, hold up their cups,
put garlands on their faces, and cry out
from the heart,

“This pleasure is but fleeting
for us, we insignificant men—soon 1270
it will be over and then afterwards
will never be recalled.”¹

As if in death
this would be their principal misfortune,
that thirst would burn them in their misery
and parch them dry, or that they would be seized
by longing for something else. For no man
has the least thought about himself or life
when mind and body are both at rest in sleep. [920]
For all we care, such sleep may last forever—
no desire about ourselves affects us, 1280
and yet at that time throughout our body
none of those primary elements wander
far from motions which create sensation,
since a man, when roused suddenly from sleep,
can gather himself together. Therefore,
we should think of death as much less to us,
if something can be less than what we see
is nothing. For when we die, there follows
a greater scattering of dispersed matter,
and no man is woken up and rises 1290
once overcome by that cold halt to life. [930]

Furthermore, what if the nature of things
suddenly spoke and personally rebuked
any one of us in the following words:

“Why is your distress so great, you mortal,
that you indulge in sorrowful laments
to such excess? Why do you moan and weep
at death? For if the life you had before,

¹Kelsey points out that the sentiment here is like the slogan “Eat, drink, and be merry, for tomorrow we die,” associated with Epicureanism. Lucretius, who upholds a sterner and older tradition has little sympathy for this view.

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which is now over, was pleasing to you,
and all its good things have not leaked away, 1300
as if stored in containers full of holes,
and disappeared without delighting you,
why do you not take your leave like a guest
well satisfied with life, you foolish man,
and with your mind at ease accept a rest
which will not be disturbed? But if all things
which you enjoyed have been frittered away [940]
and come to nothing and life offends you,
why seek to add on more which, once again,
may all be squandered foolishly and leave 1310
without providing pleasure? Instead of that,
why do you not end your life and troubles?
For if I can discover or invent
nothing more to please you, then everything
always is the same. And if your body
is not yet shrivelled up with years, your limbs
not yet worn out and torpid, still all things
will stay the same, even if you keep going
and outlast all living races, or even more,
if you should never die.”

What do we reply, 1320 [950]
except that nature makes a valid charge—
what she alleges in her speech is true?
But if an older man, more advanced in years,
in his misery should complain of it,
wailing about death beyond all reason,
would nature with more justice not call out
and in a sharp voice chastise him:

“You wretch,
end those tears at once, and stop complaining.
After going through all rewards of life,
you are ailing, but since you always want 1330
what is not there and spurn what is at hand,
an incomplete and disagreeable life
has slipped from you, and, before you can leave
richly content and satisfied with things, [960]
unbeknownst to you, death is standing there,
beside your head. But now you should give up
all those things inappropriate to your age—
come now, and, as you must, surrender them
with grace and a calm mind. You have no choice.”

She would be right, in my view, to say this— 1340
right to rebuke and criticize the man.
For old things, driven out by what is new,
always yield, and one must renew one thing
with something else. So no one is sent down

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into the abyss and black Tartarus.
Material is needed for the growth
of later generations—yet all of them,
once their life is over, will follow you.
Men have died before and will die again,
just like you. Thus, one thing will never cease
being born from something else. Life is given
to no man as a permanent possession—
instead all men receive it as a loan.

1350 [970]

Look back once more at how past centuries
of infinite time prior to our birth
have meant nothing to us. This, therefore,
nature offers to us as a mirror
of time to come, once we are dead and gone.
What appears so horrifying about it?
Does anything seem gloomy? Is it not
more free of misery than any sleep?
There is no doubt that all those things they say
are deep in Acheron are in our lives.
And wretched Tantalus is not afraid
of the huge rock suspended in the air
above him, rigid with futile terror,
as the story says.¹ It is more the case
that in life our vain terror of the gods
oppresses mortal men, who fear the blow
which chance may bring to each of them. And birds
do not eat their way into Tityos,
as he lies there in Acheron—in fact,
they could not uncover things to scavenge
in his huge chest for an eternity.²
No matter how vast his sprawling body,
which, with its spread-eagled limbs, might cover
not just nine acres, but the whole extent
of our earth's sphere—nevertheless, he still
will not be capable of suffering pain
forever, always offering nourishment
from his own flesh. But for us Tityos
is here, a man who lies down sick with love,
whom vultures rip and anxious cares consume
or worries slice up with some other passion.
And Sisyphus is in our life, as well,
right before our eyes, a man who chooses
to solicit people for the *fascēs*
and savage axes and always comes back

MYTHS OF THE AFTERLIFE ARE
FICTIONS

1360

[980]

1370

[990]

1380

¹Lucretius now surveys some of the major legendary sinners who were punished in Hades, especially those mentioned in Homer's *Odyssey* (Book 11), both to debunk the legends and to remind his readers that hellish punishments comparable to these legends occur in life for those who do not have their desires and fears under control. Tantalus was eternally tormented with thirst and hunger and threatened by a rock whenever he reached for food.

²Tityos was a huge monster punished in Hades by having vultures eat his liver.

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defeated and depressed.¹ Seeking power,
which is unfulfilling and never granted, 1390
and always toiling in pursuit of it—
this is straining to push uphill a stone [1000]
which, with gathering speed, still comes rolling down
once more from the summit and keeps on going
to the level surface of the plain. And then
to give constant nourishment to a mind
which shows no gratitude, to cram it full
with fine things, yet never satisfy it—
an offering which the seasons of the year
provide for us when they come round again, 1400
bringing their fruits and various delights,
while we still feel we never get enough
of life's pleasures—this, in my opinion,
is the story they tell of those young girls,
in the flower of life, who pour water
into leaky jars, yet there is no way [1010]
they can fill them up.² But then Cerberus,
the Furies, lack of light, [are idle tales,
as are Ixion's wheel and black] Tartarus
vomiting horrific fire from his jaws— 1410
these things are not to be found anywhere
and, in fact, cannot exist.³ But in life
there is a fear of punishment for crimes
one has committed—major penalties
for major crimes—atonement for misdeeds:
prison, the dreadful toss down from the rock,
and floggings, executions, the rack, pitch,
red-hot metal, as well as brands of fire.⁴
And though these may be absent, yet the mind,
conscious of its deeds and apprehensive, 1420
prods and torments itself with goads and whips,
and does not see meanwhile how its distress [1020]
could end, what final limits there might be
to punishment, and is instead afraid

¹Sisyphus is another character punished in Homer's vision of Hades. He has to push a huge rock uphill, but every time he is almost at the top the rock rolls back down again. The "*fasces* and savage axes" are the symbols of political authority in Rome (the *fasces* is a bundle of round sticks bound together to symbolize the unity of the state; the axes symbolize the power of the state). The adjective "savage" indicates Lucretius's sense of the harsh demands of seeking and holding political office in republican Rome.

²This is a reference to the famous daughters of Danaus, who killed their husbands on their wedding night. Their task of filling leaky jars is a symbol of their useless, wasted lives and, beyond that, of the lives of those who are never satisfied with the good things of life.

³The words in square brackets are Munro's suggestion (more or less) for missing material. Ixion was the first human being to murder another and later was punished for trying to have sex with Hera, Zeus wife. Zeus had him bound to a spinning wheel of fire.

⁴Cerberus, in Greek and Roman mythology, is the famous dog with many heads which guards the gates of the underworld. The Furies are the dreaded goddess of blood revenge, whose special task is to avenge family murders. The "toss down from the rock" is the Roman punishment for traitors, who were thrown from the Tarpeian Rock, a cliff in Rome. Some editors suggest there are a few lines missing after line 1010 in the Latin (line 1410 in the English text above).

Book Three

these same penalties may grow more serious
once one is dead. And here the life of fools
becomes an Acheron at last.

Then, too,
you could from time to time say to yourself,

“Even splendid Ancus with those eyes of his
went from the light of life, a finer man, 1430
in many ways, than you, you worthless rogue.¹
Since him, many other kings and rulers
have perished, men who ruled mighty nations.
Even that man who once built a roadway
over the great sea, providing a path
for legions to cross the deep, teaching them [1030]
to go on foot above the salty gulf,
with prancing horses showing his contempt
for the ocean’s roar, that man lost the light
and from a dying body poured out his soul.² 1440
The son of Scipio, war’s thunderbolt,
who terrorized the Carthaginians,
gave his bones to earth, just as if he were
the lowest household slave. Then add to these
those who made discoveries in learning
and the graceful arts, then add companions
of sisters from Mount Helicon, with whom
Homer, holding unique authority,
rests in the same sleep as all the others.³
Then, too, after mature old age advised 1450
Democritus that observant powers [1040]
in his mind were failing, with his own hand
he personally offered death his head
and went to meet him.⁴ Even Epicurus,
when he had travelled through his light of life,
also died, a man whose genius surpassed
the human race, eclipsing everyone,
just as the sun, when rising in the sky,
extinguishes the stars. So will you still
hesitate and resent going to your death? 1460
You, whose life, while you still live and see,
is almost death, you, who squanders away

¹Ancus (Ancus Marcius) was, according to tradition, the fourth king of Rome, (642 to 617 BC); he was called “Ancus the Good.” The line about his eyes leaving the light is taken from a poem by the celebrated Latin poet Ennius, to whom Lucretius pays tribute in Book I.

²This is a reference to the Persian emperor Xerxes, who invaded Greece by land in 480 BC. His expedition involved building a bridge across the Hellespont so that his enormous army could cross out of Asia Minor.

³Scipio (Publius Cornelius Scipio Africanus, 236 to 183 BC) was the victorious Roman general in the second Punic War. He defeated the Carthaginian general Hannibal at the battle of Zama in 202 BC. The sisters from Helicon are the Muses, divine patronesses of the arts.

⁴Democritus (c. 460 BC to c. 370 BC), Greek philosopher, founded the school of materialistic atomism. Whether he committed suicide or not is unclear.

Book Three

most of your years in sleeping and then snores
when you are wide awake, who does not stop
seeing idle dreams and has a mind distressed
by empty terrors—you cannot find out [1050]
what it is that often makes you anxious,
when many troubles press from every side,
and, in your misery, you wander round,
like a drunkard, with an unsteady mind, 1470
floundering in uncertainty.”

And thus,
with men who clearly feel there is something
weighing on their minds which is so oppressive
it wears them out, if they could also grasp
the causes which have brought this feeling on
and where it originates, that huge mass
of evil, as it were, living in the chest,
they would not carry on their lives the way
we generally see them now, each one
not knowing what he wants, always seeking 1480
to change places, as if by doing that
he could set aside his burden. Often
a man bored with staying at home will leave [1060]
his huge residence for some other place,
then suddenly return, since going away
does nothing to improve the way he feels.
He rushes to his villa, urging on
his galloping horses, as if desperate
to bring help to a house on fire, but then,
once he sets foot on the building’s threshold, 1490
he quickly yawns and falls in a deep sleep,
seeking oblivion, or even rushes off
demanding to get back to the city.
In this way, each man flees himself—and yet,
as is commonly the case, we observe
he cannot flee the self, he clings to it
against his will, and he dislikes himself,
since he is sick and does not know the cause [1070]
of his disease. If he saw that clearly,
he would leave aside all other matters 1500
and would seek, first of all, to comprehend
the nature of things, for what is at stake
is his condition, not for just one hour,
but for eternity, the state in which
every generation of mortal men
must continue, whatever is still left
after they have died.

And finally,
what evil longing for life is so strong
that it forces us with such compulsion

Book Three

to remain confused, in doubt and danger? 1510
A certain limit has been fixed to life
for mortals. We cannot avoid our death,
but must move on to meet it. Moreover,
we keep spinning around, always staying [1080]
with the same things, and, as we go on living,
we forge no new pleasure. But while we lack
what we desire, that seems to matter more
than all the rest, and, when we obtain that,
we crave something else. That same thirst for life
always keeps us with our mouths wide open. 1520
We are in doubt about what fortune time
may bring to us in future, or what chance
has ready for us, or what our end will be.
By prolonging life, we do not shorten
the time we spend when dead, and we cannot
remove a thing which might enable us
to stay dead perhaps a shorter length of time.
Thus, you may live on and on and bury [1090]
as many generations as you will,
that eternal death will still be waiting, 1530
nonetheless—nor will he who ended life
with this day's light lack all existence
for a shorter period of time than he
who perished many months or years ago.

BOOK FOUR

[Invocation to his own poetry; images of things exist, sent out from objects with a form just like the object; material of the image very small; images can shatter or be reflected; images move extremely quickly; sounds, smells, and taste are also particles sent out from things; images enable us to see how far away things are; images in a mirror; seeing things from light and darkness; shadows; senses do not deceive us; optical illusions; error of scepticism; how senses work; different sounds; penetration of sound and vision and smell; different tastes; different animals require different food; variety in odours; images affecting the mind; senses not made to serve living; explanation of physical motion; what happens in sleeping; nature of dreams; origin of human sexuality; nature of sexual activity; pleasures and problems of sex; transmission of hereditary features; causes of infertility; familiarity can lead to love.]

I am wandering through trackless regions
of the Pierides, where no man's foot
has ever gone before.¹ It gives me joy
to approach those fountains never tasted
by anyone and to drink from them.

THE PURPOSE OF THIS POETRY

I love to pick fresh flowers and obtain
a splendid garland for my head in places
from where Muses have never crowned the brows
of any man before. First, because I teach
important things and seek to free the mind
from constricting fetters of religion.

10

And then because the verses I compose
about dark matters are so luminous,
investing all things with poetic grace.
And that, too, does not seem unreasonable.

[10]

For just as healers, when they try to give
young children foul-tasting wormwood, first spread
sweet golden liquid honey round the cup,
so at this age the unsuspecting child,
with honey on his lips, may be deceived
and in the meantime swallow down the drink
of bitter gall—he may have been misled,
but he is not hurt—with such deception
he may be restored instead, grow stronger.

20

In the same way now, since this reasoning
seems generally too bitter for those men
who have not tried it and the common crowd
shrinks back in fear, I wanted to explain
what I have to say to you in verses,
sweet-spoken Pierian song, as if I were
sprinkling it with poetry's sweet honey,
if, with such a method, I could perhaps
get your attention on my verse, until
you see the entire nature of things
and recognize how useful that can be.

[20]

30

But since I have explained those particles
from which all substances originate,

¹The opening twenty-five lines in the Latin are an almost exact repetition of the lines in Book 1 (1.925 ff in the Latin). The Pierides is another name for the Muses, derived from the place near Mount Olympus where they were alleged to have been born.

Book Four

what they are like and how, all on their own,
they move around, in various different shapes,
driven on by everlasting motion, 40
and how all things can be produced from them,
since I have shown what our mind's nature is,
the substances of which it is composed,
as it grows and thrives along with body,
and then how, when separated from it,
mind goes back to its primary elements,
now I will begin to set out for you
something extremely pertinent to this.

There are what we call images of things
stripped off the surface layers of substances,
like membranes—these fly to and fro in air.¹
These same images, when they contact us,
make our minds fearful while we are awake
and in sleep, as well, when we often see
strange shapes and images of dead people
deprived of light. Frequently they rouse us
from our sleep, as we lie there slumbering,
and terrify us. We must not assume,
by some mistake, that souls from Acheron
have got away, or that their shadows flit 60
here among the living, or that some part
could still remain from us once we are dead,
when our body and the substance of our mind
have been destroyed together and reduced
to their own various primary particles. [50]
So, then, I say thin shapes and likenesses
of objects are sent out by those objects
from their top surfaces. These we can call,
as it were, membranes or bark, for each one
possesses an appearance and a form 70
that resemble whatever the object is
from which we say it was shed and wanders.
This we may understand from what follows,
no matter how obtuse our minds may be.

IMAGES OF THINGS
50 [40]

60

[50]

SURFACES OF OBJECTS SEND OUT
MEMBRANES

70

First, many things we see all around us
send out particles, sometimes thinly scattered,
as when wood produces smoke and fires heat,
and sometimes more compact and more condensed,
as cicadas now and then in summer
discard their smooth outer layer, young calves, 80
after they are born, shake off the membrane

¹ Lucretius' theory of perception relies upon this concept of images (in his Latin text the word is *simulacra*). These images are material stuff (i.e., made up of the same elements that make up the objects of the world). They are not, in any sense, illusions or insubstantial pictures. There is in the Latin text some confusion in lines 30-39, with repetitions and some lines clearly in the wrong place. Hence, there is no line number [30] to the right of the text above.

Book Four

from the outer surface of their bodies,
and, in the same way, the slippery snake [60]
strips off its outer skin among the thorns,
for frequently we see bramble bushes
full of fluttering hides from those animals.
Since this takes place, things also must emit
from their surface layer a thin image,
for why those substances should fall away
from things and leave rather than thin membranes 90
no one is able to enlighten us,
above all since on their outer surface
objects have many minute particles
which can be thrown off in the same order
in which they were arranged and thus preserve
the outline of their form. They can do this
much more quickly, for there are few of them [70]
and, being placed on the outer surface,
they are less hemmed in.¹ For we truly see
many things detach and cast off matter, 100
not only, as we previously mentioned,
from deep inside, but frequently as well
from their surfaces, including colour.
And this commonly occurs with awnings—
yellow and red and dark blue coverings—
which, when stretched out across large theatres
and spread everywhere on poles and timbers,
flutter and flap around, for their tint affects
the audience below them on the benches,
the whole appearance of the scenery 110
and men and women below, forcing them [80]
to quiver in their colours.² And the more
they are enclosed all round by theatre walls,
the more all these things inside, when daylight
catches them, are filled with colour and smile.
Since from its outermost layer the cloth
sends out these tones, all other substances
must also send out subtle likenesses—
in both examples something is cast off
from the outer surface. It then follows 120
there are certain outlines of shapes, endowed
with subtle textures, which fly all around,
but which cannot be perceived on their own
as individual objects. Moreover,

¹ As Lucretius has explained earlier, all particles in an object are in constant motion and therefore can, under some circumstances, leave the object or be detached from it by impact. Those on the surface are obviously much more likely to do this than particles on the inside, which are more tightly enclosed by other particles.

² Part of this sentence is apparently illegible in the Latin. I have translated it as “men and women underneath” to retain the sense of the sentence. In Rome popular theatres were temporary structures made from poles, beams, and awnings. The light from the sky shining through the coloured awnings changes the colours in the audience below.

Book Four

all odour, smoke, heat, and other things [90]
 like these flow off objects and get dispersed,
 since, while they are rising from deep within
 and moving out through twisting passages,
 they are torn up—the path they move along
 lacks direct openings where they could try 130
 to make their way out in a single mass.
 But, by contrast, when the slender membrane
 of colour is cast off from the surface,
 there is nothing which can mutilate it,
 since its location on the very top
 leaves it ready to fall off. Finally,
 whenever images appear to us
 in mirrors, water, all bright surfaces,
 they must consist of images sent out,
 because on the exterior they possess 140
 an appearance resembling the objects. [100]
 Therefore, there are slim shapes and likenesses
 similar to objects—although no one
 can see them individually, they still
 are thrown back in constant, successive waves,
 while being reflected from flat surfaces
 of mirrors and then give the image back.¹
 It seems there is no other way that shapes
 can be preserved, so that for everything
 reflected forms are very accurate. 150

Come now and learn how thin the substance is [110]
 which makes up an image. And first of all, IMAGES ARE THIN
 since primary elements are far below
 what we can sense and so much tinier
 than those things which our eyesight first begins
 to be incapable of noticing,
 you must grasp in a few words how minute
 the particles are of all elements
 from which all things begin, so that I now
 may confirm this point, as well. To start with, 160
 some living creatures are so very small
 one cannot, by any means at all, see
 a third of them. How must we imagine
 the nature of their internal organs?
 What of the round ball of their hearts or eyes?
 What about their limbs? Or parts of their frame?
 How minute are they? And then, what about [120]
 all the primary particles which must form
 their souls and the material of their minds?
 Surely you perceive how small and slender 170
 they must be? Moreover, all those objects

¹ Following other translators, I have omitted lines 102 and 103 in the Latin. They are identical to lines 65-66 of the Latin (lines 89-91 in the English).

Book Four

whose bodies give off a powerful smell—
nasty wormwood, pungent abrotanum,
bitter centaury, and panacea—
if you happen [to press] any of these
gently with two [fingers, the smell will stay
for some time, although you will not see
anything at all.¹ Thus, you may realize
how minute the primary particles are
which create the smell and then] understand 180
more readily that many images
of objects float around in many ways
without any force and without being seen.

But in case you may perhaps imagine
those images of things which roam about
are, in fact, only those which are detached [130]
from things, there are also images produced
spontaneously—they generate themselves SPONTANEOUSLY PRODUCED
in this vault of heaven we call the air. IMAGES
They are formed in many ways and carried 190
in the air. Being fluid, they do not stop
changing their appearance, converting it
to all varieties of outlined shapes,
just like the clouds we see from time to time
which have no trouble gathering way up high,
spoiling the calm face of the firmament,
and which, as they move on, caress the air.
Often giants' faces seem to fly past
and spread shadows far and wide, and sometimes
huge mountains and boulders ripped out from them 200
appear to move above our head and pass
before the sun—then some huge wild beast seems
to drag out and lead on other storm clouds. [140]

Now, [I will explain] how quick and easy
the process is by which these images
are made, how they constantly flow from things,
slip off, and leave.² For some of the surface
always streams from things—matter is cast off.
And when this discarded material
meets certain substances, it passes through— 210
glass is the best example—but when it strikes

¹ Wormwood is a wild plant used for making medicines, tea, and wine; abrotanum (Southernwood) is a wild plant used as an antiseptic; centaury (named after the centaur Chiron) is a wild herb used in medicines; panacea is a fabulous plant reputed to cure all diseases. There appears to be a gap in the manuscript after line 126 in the Latin. In order to complete the sense, I have used (and reworked slightly) the substitute passage supplied by Bailey (who states that the gap may amount to about 50 lines). That insertion is in square brackets. Copley suggests that the missing passage included more proofs of how invisible particles affect the senses, part of Lucretius' argument about the minute size of the particles which make up the images.

² There is evidently a gap in the manuscript of at least one line in the middle of this sentence (at line 144 in the Latin). I have added the phrase in square brackets to complete the sense.

Book Four

rough rocks or wooden things, it shatters there
immediately, so it cannot provide
a single image. However, when objects [150]
which are bright and dense are placed in its way—
the finest illustration is a mirror—
neither of these alternatives occurs,
for the image cannot travel through it,
as it can with glass, nor is it shattered,
since the smooth surface carefully preserves 220
the image safely. That's why images
happen to flow back from these surfaces
to us. And any time you set something,
however quickly, against a mirror,
its image will appear, so you may grasp
that thin shapes of things, with fragile textures,
always stream out from an object's surface.

Therefore, many images are produced
in a short space of time, and one may say,
with justice, that their origin is swift. IMAGES ARE PRODUCED
SWIFTLY
230 [160]
Just as the sun must send out numerous rays
in a brief moment, so that all places
may always be full of light, so from objects
many images of things must be carried,
in many ways, out to all locations
everywhere, in an instant, given that,
no matter where we direct the mirror
towards the surfaces of some objects,
the mirror will reflect those objects back
with the same shape and colour. Moreover, 240
when the weather in the sky has just been
extremely clear, it can very quickly turn
into such a nasty storm, you could think
all darkness had everywhere left Acheron
and filled up the mighty vaults of heaven. [170]
That's how much the outlines of black terror
rise up in the ghastly night of storm clouds
and hang high above us. And yet how small
a part of these their image is no one
could explain or put in words.¹

Come, now, 250
how quickly images are carried off
and what mobility they are given,
as they swim through air, so that they travel
huge distances in a brief length of time
to whatever place each one is aiming for
from the specific impulse it receives—

¹ The point of this rather awkward example is presumably to stress that very grand events, like the clouding of the entire sky, can happen very quickly. Hence, the development of an image of the event, which must be inexpressibly smaller than the event itself, can also be very rapid.

Book Four

all this I will set down: the lines I write
will not be many, but they will sound sweet, [180]
just as a swan's brief song is preferable
to the scream of cranes scattering through clouds 260
high in the southern air. First, we can see
that light things made of tiny particles
are very often fast. This group includes
the sun's light and heat, for they are composed
of minute primary elements which are,
so to speak, knocked out and have no trouble
moving through the intervening gap of air,
driven by a blow from those which follow,
for light is immediately replaced with light,
and brightness is goaded on by brightness, 270 [190]
as if in strict succession.¹ And therefore,
images must, in a similar way,
be capable of rushing in an instant
across spaces we cannot imagine,
firstly, because there is a minute cause
some distance behind, which pushes them on
and propels them forward, then secondly,
because they are carried on so swiftly
thanks to their light weight, and finally,
because they are sent out with a texture 280
so fine that they can easily pass through
any substances you like and, as it were,
break their way through the intervening air.²

Then, too, if tiny particles of things
which are dispatched outside from deep within, [200]
like the sun's light and heat, are seen to spread
across the entire extent of heaven
in one brief instant—they fly over sea
and land and flood the sky—what then happens
with those particles which now stand ready 290
on the surface, when they are ejected
and nothing hinders them from being discharged?
Do you not see they must move more quickly
and go further, racing through many times
the extent of space in the same length of time
the sunlight takes to fill the sky?³ This, too,

¹ Lucretius' understanding of sunlight, which he explains in more detail later, is an interesting concept of pulses or waves sent out in a continuous series, so that the particles are always being pushed by those behind them.

² The "minute cause" which propels the image from behind is the initial blow which detaches the image from the surface of the object, a force which comes from the always moving particles inside the object. Lucretius has already discussed in Book 2 how very small particles can move extremely quickly through air, because they are not impeded as much by internal movements of their parts (as compared with larger and more complex compounds).

³ Particles which move from the inside of an object to the surface before being expelled (like the particles of heat and light from the sun) have to, as it were, fight their way to the surface of the object and therefore lose some of their motion before they leave. Particles on the surface do not have to do this; they "stand

Book Four

seems a true and excellent example
of how swift the motion is which carries [210]
images of things along: as soon as
a bright water surface is first set out 300
in the open air under starry skies,
the world's calm and radiant constellations
respond at once, appearing in the water.
Do you not now see in how short a time
the image falls from regions of the sky
to places here on earth? For this reason,
to repeat myself, you must concede the fact
that bodies are sent out which strike our eyes,
then stimulate our vision, and [these move
all the time with astonishing rapidity].¹ 310

And smells constantly flow from certain things,
just as cold from rivers, heat from the sun,
and spray from sea waves, which consumes the walls [220]
around the shoreline. And different noises
keep flying through the air incessantly.
Then, too, when we are strolling near the sea,
often a salty tasting moisture comes
into our mouths; by contrast, when we watch
wormwood being diluted in a mixture,
something bitter makes contact with our mouths. 320
That shows how much all that material
is carried away from every object,
dispersed in all directions everywhere.
And in this flow there is no slowing down,
no respite, for we feel it all the time—
we can always see and smell all objects
and hear their sounds.

In addition to this, [230]
because we know a shape we feel by hand
in the darkness is the same one we see
in clear and brilliant light, then touch and sight 330
must be kindled by similar causes.
If, then, we now handle a square object
in the dark and it stimulates our sense,
then in daylight what square thing can contact
our sense of sight other than its image?
Thus, it is clear that the cause of seeing
is in images and that without them
we would not be able to see a thing.

Now, these images of things I talk of

ready" to leave. Hence, Lucretius argues, their speed will be greater. Since these are the particles which make up the images, then images will move faster than sunlight.

¹ The addition in square brackets is prompted by a comment from Munro about some words missing at this point in the manuscript.

Book Four

are carried everywhere—they are cast off
and dispersed on every side. However,
since we can see only with our eyesight,
it therefore happens that no matter where
we turn our sight, all objects on that side
strike it with their shape and colour. What's more,
the image enables us to see how far
each thing is away from us and makes sure
we can distinguish that. For the image,
when it is sent, immediately disturbs
and pushes forward whatever air stands
between it and the eyes, and all this air
thus glides through our eyeballs and, as it were,
brushes the pupils and so keeps moving.
And thus it comes about that we perceive
how far distant each object is: the more air
pushed before the image and the longer
its breeze moves past our eyes, the further off
each thing is seen to be. You can be sure
these motions are produced by some process
which is extremely fast, so that we see
what something is and, in the same instant,
how far away it is.

340 [240]

DISTANCE OF OBJECTS

350

[250]

360

In these matters,
we should not think it at all wonderful
that, while those images which strike the eye
one by one cannot be perceived, we do see
things themselves, for when the wind, too, strikes us
with successive gusts and when bitter cold
flows over us, normally we do not sense
each separate particle of wind and cold,
but rather all of them collectively,
and we then feel just as if our body
were being subjected to some injury,
as if some object were striking at us
and making us aware that it is there,
outside of us. And then, when our fingers
strike a stone, we make contact with the rock
on its extreme outside and the colour
on the surface, but we do not perceive
the colour with our touch, but rather feel
the very hardness deep inside the stone.

[260]

370

380

Come now, and learn why we see an image
beyond the mirror, because the truth is
the image seems displaced deep within it.
It is like those things we really do observe
outside, when a door gives us a clear view
through it and from within the house lets us
look at many things outside. For this view

MIRROR IMAGES

[270]

Book Four

is produced by two twin waves of air, as well.
In this case, we first sense the wave of air
on our side of the door posts, then follow 390
panels of the doors themselves, left and right,
then the outside light brushes through our eyes
and the second wave of air, and those things
we really see outside. In the same way,
when the image of the mirror first moves
out towards us, while it is still coming [280]
to our eyeballs, it strikes and pushes on
all the air located between itself
and our eyes, and does so in such a way
that we are able to feel all this air 400
before we sense the mirror.¹ However,
when we also see the mirror itself,
the image which is carried out from us
reaches the mirror instantaneously
and, once reflected, comes back to our eyes—
pushing and rolling on in front of it
another wave of air—and it does this
so that we sense the air before we see
the image. That's why it seems so distant
from the mirror. Hence—to repeat myself— 410
it is not right to be at all surprised
[that how we sense things happens in this way
both for the things we truly see outside
and also] for those which give back an image [290]
from the level surface of a mirror,
since in both cases the effect occurs
by the two waves of air.²

Now, in mirrors

those parts of our limbs which are on the right
are so arranged we see them on the left,
because when the image comes up and strikes 420
against the flat surface of the mirror,
it is not reflected without being changed—
instead it bounces back in a straight line,
just as with a plaster mask if someone
pressed it against a pillar or a beam
when it was not yet dry and, at that moment,
it still retained its proper shape in front,
and the mould then turned itself inside out,
that will cause what was the right eye before
to be now on the left and, in the same way, 430 [300]

¹ This first image we get is of the mirror itself. That pushes a wave of air against our eyeballs. Our image is reflected from the mirror, pushing on a second wave of air.

² A line appears to have been lost here. I adopt Bailey's suggestion for the missing Latin. In this explanation I have at times inserted the phrase "waves of" in front of the word "air" in order to make clearer sense of the explanation. Lucretius simply uses the word "air."

Book Four

the left eye will now become the right.¹

It so happens as well that an image
may be passed on from mirror to mirror,
so that five and even six images
are commonly produced. For when objects
are hidden back in an interior room,
no matter how remote and deep within
and how tortuous the path, one can still,
using several mirrors, lead them all out
through twisting passageways and then observe 440
that they are in the house. That shows how well
the image is passed on from one mirror
to another, and when what is on the left
is sent on, it then changes to the right,
and from there it then changes back again,
shifting to the same place it was before. [310]

In addition, all mirrors with bent sides,
which have a curving shape like our own torso,
for that very reason send back to us
an image with our right side on the right, 450
either because the image is transferred
from one part of the mirror to another
and then, after being reflected twice,
flies back to us, or because the image,
as it gets to the mirror, is reversed—
the curving shape of the surface leads it
to spin about towards us.²

Moreover,
you should know our images move forward
step by step, setting their feet as we do,
mimicking our actions. If you walk away 460 [320]
from any section of the mirror
then at that instant images cannot
be reflected back from there, for nature
requires all objects to be carried back
and to rebound from things in such a way
they are sent back at an equal angle.³

¹ When we look in a mirror, our right eye is on the left side of the face which looks back at us.

² Lucretius is here talking of a mirror with a laterally concave surface facing us, one which therefore curves outwards away from us, “like our torso.” Such a mirror will produce an image in which the parts are on the correct side of the face (looking outward from the mirror), an effect opposite to the orientation on a flat mirror but the same as a double reflection from two flat mirrors.

³ That is, the same angle at which they struck the mirror. This requirement is now a general law in physics: a light ray striking a mirror so that it makes an angle with the line perpendicular to the surface must be reflected from the surface at the same angle to the perpendicular, (i.e., the angle of incidence is equal to the angle of reflection). This translation, however, has been disputed, since Lucretius does not use the word for “angle” (*angulus*) but a word meaning “turning” or “shifting” (*flexus*). Watson, for example, claims that “Lucretius had no thought of equal angles.” This objection, so far as I can tell, has not persuaded many

Book Four

Moreover, the eyes avoid bright objects
 and refuse to look at them. The sun, too,
 is blinding, if you strive to keep your gaze
 directly on it, since its force is great 470
 and its images are carried from high up
 through clear air—they strike the eye, disrupting
 its connections. Then, too, any object
 dazzlingly bright frequently burns our eyes
 because it contains many seeds of fire, [330]
 which move into the eye and make it hurt.
 What's more, all things those with jaundice look at
 become ghastly yellow, for many seeds
 of yellow flow from their bodies to meet
 the images of things, and many seeds 480
 are also mixed inside their eyes, and these,
 thanks to their contagion, paint everything
 with their own pallor.

Now, from darkness

we see things in the light, since, once black air
 of darkness, which is closer, enters first
 and takes possession of our open eyes,
 bright, clear air immediately follows [340]
 and, as it were, cleanses them, scattering
 black shadows of that former air, because
 that bright air is many times more agile, 490
 many times smaller and more powerful.
 As soon as it fills pathways of the eyes
 with light and opens those which the dark air
 earlier had blocked, images of things
 located in the light arrive at once
 and stimulate our eyes, so that we see.
 But, by contrast, we cannot do the same
 looking from the light into the darkness,
 because the air which comes to us later
 from the darkness is more dense—it fills up 500
 all the openings in the eyes, obstructing
 its passageways, so that no images [350]
 of any objects can strike or stir them.

When we look at a city's square towers
 from a long way out, it often happens
 that they look round, because every angle,
 when seen from a distance, appears blunted,
 or rather is not even seen at all,
 and its impact dies away: the impulse
 does not glide through our eyes, for its image, 510

OBJECTS SEEN FROM A
DISTANCE

modern translators. Munro thinks Lucretius is probably referring to this law and points out that it was well known to Greek and Roman mathematicians.

Book Four

while carried through large quantities of air,
is forced, by frequent impacts with that air,
to flatten out. Hence, when every angle [360]
escapes our senses simultaneously,
that causes us to see these stone towers
as if they had been rounded on a lathe.
But they are not like things that, seen up close,
are truly round. However, they do seem
somewhat the same—their outline, so to speak.
Similarly our shadow seems to us 520
to move in sunshine: it follows our steps
SHADOWS
and imitates our gestures, if, in fact,
you believe that air deprived of light
can walk ahead, copying how men walk
and bear themselves, for what we usually call
shadows cannot be anything but air
which has no light, and it is obvious [370]
that in particular places the ground
is successively deprived of sunlight,
wherever we, in wandering around, 530
obstruct it, and similarly the part
we moved from is filled in with light again.
That, then, is the reason it so happens
that what was the shadow of our body
always seems to stay the same and follow
directly across from us. For new rays
of light pour out all the time—the first ones
die away, like spun wool pulled into fire.
In this way the ground is easily robbed
of light and then easily filled again 540
and washes away its own black shadows.

However, in this we do not admit
that the eyes are in any way deceived. OUR EYES DO NOT
For their purpose is to see all places DECEIVE
where there is light and shade, but whether [380]
it is the same light or not, or whether
it is the same shadow which was here
that now wanders over there, or whether
what takes place is rather what I mentioned
a short while ago above, these matters 550
the reasoning of the mind, all on its own,
must sort out. The eyes cannot understand
the nature of things. And therefore, do not
falsely attribute to the eyes this failing
in the mind.¹ When we travel aboard ship,

¹ This is an important caveat. Lucretius has repeatedly emphasized, as a core component of his materialist theory, that sense experience is the only criterion we have for checking our theories about the natural world. Therefore, he needs to reassure us that the senses themselves do not deceive us; our interpretation of our sense experience, however, can be wrong. The list of illusions he now provides is meant to underscore this warning.

Book Four

it is carried forward, although it seems
to be standing still, while another boat
which remains tied up is, so we believe,
moving past us. When we drive our ship on [390]
and fly under full sail, then hills and fields
560 appear to run off to the stern. All stars
in the celestial vault seem fixed in place,
quite motionless, yet every one of them
is always moving, since they rise, and then,
when their bright bodies have crossed the heavens,
they return back to their distant settings.
So, too, the sun and moon in the same way
seem to remain in place, but facts themselves
indicate that they are carried forward.
And from far away mountains jutting up 570
in the middle of the sea where there is
between them a large, free strait for shipping
standing open, nevertheless still seem
a single island, a union of the two.
It also happens that when young children
have stopped twirling themselves in circles,
rooms seem to spin and pillars run around, [400]
so much so they can hardly now believe
the whole roof is not threatening to fall
580 right down on them. Moreover, when nature
starts to lift on high the rays of the sun,
ruddy with twinkling fires, raising them high
above the mountains, those peaks over which
it seems to you the sun is standing then
so close, with his blazing fire touching them,
are hardly far away from us—a distance
of two thousand arrow flights, and often
scarcely five hundred javelin throws—and yet
between those mountains and the sun there lie [410]
immense expanses of the sea, stretched out
590 beneath vast regions of the heavenly sky,
many thousands of lands are there, as well,
inhabited by various human types
and races of wild animals. And then,
a pool of water with a depth no greater
than one finger width, which has collected
on a paved road among the stones, gives us
a view down underneath the earth as great
as the high mouth of heaven opens up
above the earth, so that you seem to see
600 clouds and heaven and celestial bodies
hidden underground in an amazing sky.
Then, when we are on a spirited horse [420]
stuck fast in the middle of some river
and we look down at the rushing waters
of the stream, some force appears to carry

Book Four

the horse's body, which is not moving,
sideways to the current, to be driving it
rapidly upstream. And no matter where
we turn our eyes, all objects seem to us
to be carried and to flow in the same way.¹
And though dimensions of a colonnade
are the same throughout, and it is standing
supported from one end to the other
by equal columns, yet when we look down
at its entire length from the top portion,
it gradually shrinks down to the tip
of a tapering cone, joining roof and floor
and all things on the right and on the left,
until it brings everything together
at the apex of the cone and disappears.
Then, it happens for sailors out at sea
that the sun seems to rise out of the waves
and sink down into the waves, burying
its light, because, given their location,
they see nothing except sky and water,
and so you must not casually suppose
their senses have completely gone astray.
But to those who know nothing of the sea,
ships in port, as they work against the waves,
appear handicapped by broken fittings,
for every section of those oars lifted
above the salt foam of the sea is straight,
and the rudder but above the waterline
is also straight, everything submerged
below the water appears all fractured—
turned around, twisted and sloping upwards,
bent back, almost floating to the surface
of the sea. And when winds carry thin clouds
across the sky at night, then brilliant stars
seem to glide in the opposite direction
against the clouds, moving high above them
on a path very different from the one
they really travel. And it so happens
that if, by chance, we position our hand
underneath one eye and then press it down,
by some kind of sensation everything
we observe seems to be duplicated
as we look—two lights blossoming with flames
in lanterns, twin pieces of furniture
doubled all through the house, and people
with duplicate faces, double bodies.
And then, when sleep has overcome our limbs
with sweet repose and our whole body lies

610

[430]

620

630

[440]

640

[450]

650

¹ This illusion created here by moving water has been called the “waterfall effect.” After looking at something moving in one direction, a person who then fixes on a stationary object will think it is moving in a direction opposite to the original motion.

Book Four

completely quiet, yet at that moment
to ourselves we appear to be awake,
to move our limbs, and we believe we see,
even in blinding darkness of the night,
the sun and light of day, and from the space
in which we are enclosed, we seem to change 660
to sky, sea, rivers, mountains, and to move
on foot across the fields, to hear noises, [460]
although the solemn quiet of the night
remains intact everywhere around us,
and to utter words, though we do not speak.

We witness many other things like this,
to our astonishment, and all of them
seek, as it were, to violate our faith
in sense perception, but do not succeed.
Most of them deceive thanks to opinions 670
of the mind which we bring to bear on them,
so that we think we have perceived some things
which our senses have not seen. For nothing
is more difficult than to distinguish
what we clearly see from what is doubtful,
things which the mind, by acting on its own,
immediately adds on.

And furthermore, SCEPTICISM AND SENSE
if anyone thinks that nothing is known, EXPERIENCE
he also does not know this can be known, [470]
since he claims he does not know anything. 680
So I will decline to debate this issue
with a man who is standing upside down,
his head located where his feet should be.¹
But if I, too, agreed he does know this,
I would direct this one question at him:
since he has seen no truth in things before,
where did he find out what it means to know
or, then again, what not to know might mean?
What condition has created knowledge
of truth and falsity? What circumstance 690
demonstrates that what is doubtful differs
from what is certain? You will discover
the idea of truth is first created
from our senses, that sense experience
cannot be disproved. We would have to find
something more trustworthy that, on its own,
could overpower falsehood with the truth.
What, then, must we hold as more credible [480]
than our senses? Will reason which arises

¹ Lucretius is here addressing the scepticism which denies that genuine knowledge is possible, a tradition well established in classical philosophy.

Book Four

from false sense experience be strong enough
to speak against the senses, when reason
emerges entirely from sensations?¹
If those are not true, then all reasoning
is false, as well. Or will our ears be able
to refute our eyes? Or touch rebut our ears?
Or, then again, will our mouth's sense of taste
contradict this touch? Or will our nostrils
show touch is false, or our eyes disprove it?
In my view, things are not like that. Each sense
has its own separate power, its own force.
Thus, we must perceive what is soft or cold
or hot in one way and various colours
of objects in another, along with
all those things we must include with colour.²
In the same manner, our mouth's sense of taste
has its own separate force; smells are produced
in their own way, and sounds are separate, too.
And thus it must be the case that senses
cannot disprove each other. Moreover,
they will not be able, all on their own,
to refute themselves, since we must always
place equal trust in them.³ Hence, anything
which they have, at any moment, perceived,
is true. If reasoning is unable
to analyze the causes why those things
which, when we are close beside them, are square,
and round when we observe them far away,
still it is better to use faulty reasons
and make mistakes in explaining causes
for both shapes, than in any way to let
slip from our hands what we have clearly seen,
to undermine the grounds for our belief,
and to rip up the entire foundation
on which life and our well-being depend.⁴
For not only would all reasoning fall down,
life would itself collapse at once, as well,
if you did not choose to trust the senses
and to stay away from perilous cliffs
and other things like that one should avoid,
and to go after very different things.
Thus, you should realize that all those words
drawn up in fine array against the senses
are a hollow army. And finally,

700
WHAT IS MORE TRUSTWORTHY
THAN SENSE?

710 [490]

720

[500]

730

[510]

740

¹ Reason, for Lucretius, arises from sense experience and is not prior to it. Hence, if sense experience is inherently deceptive, how can we rely upon reasoning?

² These things would include other visual attributes, like shape.

³ Since the senses are all equally reliable they cannot refute each other. We cannot use one sense to confirm the truth or falsity of another.

⁴ Lucretius here and elsewhere in the poem repeatedly stresses that particular sense experience of nature is much more important than any theories designed to explain why events happen the way they do.

Book Four

as with a building, if some measuring rod
is inaccurate at first, if the square
is false and deviates from the right line,
and if the level anywhere is off
the slightest bit, all the structure must be
warped and faulty—irregular, sloping,
leaning to the front or back, the whole thing
out of alignment, so that some portions
appear to want to fall, or some do fall,
all betrayed by the first wrong measurements.
Therefore, in your reasoning about things
whatever comes from false sense experience,
must, in the same way, be false and crooked.

750

[520]

Now, what remains is an explanation
how other senses each perceives its object,
an argument by no means hard to make.
First of all, every voice and sound is heard
when it has come into the ears and struck
that sense with its own material substance.
For you have to concede that voice and sound
are physical matter, too, since they can
impinge upon the senses. Moreover,
the voice often scrapes against the pharynx
and, as it emerges, its loud sound makes
the windpipe rougher for this reason:
when primordial elements of voices,
rise up in a larger throng together
through a narrow passageway and begin
to move outside, then, with their channels crammed,
the entrance obviously is scraped, as well.
Hence, there is no doubt that words and voices
consist of primary particles and thus
can cause us pain. Nor are you unaware
how much is taken, in the same process,
from the body, from men's very sinews
and strength, by continued public speaking,
lasting from rising splendours of the dawn
to shadows of black night, especially
if it comes pouring forth in a loud shout.
And so the voice must consist of matter,
since the man who speaks a great deal loses
part of his bodily stuff. What is more,
roughness in the voice is created from
roughness in its primordial elements,
and smoothness is similarly produced
from smoothness in the voice's particles.
Primary matter does not penetrate
the ears in the same form when the trumpet
booms out its heavy muffled tone, stirring
and sending back raucous barbarian sounds,

SOUND, VOICES, ECHOES

760

[530]

770

780

[540]

790

Book Four

as when from rushing waters of Helicon
swans raise clear tones of sorrowful lament.¹
And therefore, when we force up these voices
from deep inside our bodies and send them [550]
straight out from our mouths, then our nimble tongues,
skilled at making words, articulate them,
and the shape the lips take on, for its part, 800
forms them. Thus, when there is no great distance
between where every voice originates
and where it reaches us, the words themselves
must also be heard clearly, distinguished
sound by sound, for sounds maintain their pattern
and keep their form. But if between the two
the intervening distance is too great,
words moving through great quantities of air
must be shaken up, and a voice flying
through the breezes must become distorted. 810
And thus, it comes about that you can hear [560]
the sound and not understand the meaning
of the words—that's how confused and scrambled
the voice is when it reaches you. Then, too,
a single word sent from a herald's mouth
often excites the ears of everyone
in an assembled crowd. Therefore, one voice
can quickly spread out into many voices,
since it splits itself into each man's ear,
stamping on its words a clear sound and shape. 820
But portions of those voices which do not fall
into the ears themselves are carried past
and perish, vainly scattered in the air.
Some voices strike firm places, are sent back, [570]
and return the sound, at times playing tricks
with a word which echoes. When you grasp this,
you can then provide an explanation
to yourself and others about the way
rocks in solitary places echo back
the same forms of words in proper order. 830
When we are searching for lost companions
wandering among the shadowy mountains
and we call out to our scattered comrades
with a noisy yell, I have observed places
returning six or even seven shouts,
when you sent out just one—that demonstrates
how hills themselves bounced words back to the hills
and kept repeating words which had been trained
to come back once again. And those people
who dwell around such places imagine 840
nymphs live there and goat-footed satyrs, too. [580]

¹ There are some problems with the Latin in lines 546-548, and translations of these lines tend to be very different. Helicon is a hill in Boeotia associated with Apollo and the Muses.

Book Four

They claim also there are fauns whose noises
and sporting play, which wander through the night,
shatter the tranquil silence—most of them
affirm the truth of this—and there are sounds
of chords, and sweet melodious notes ring out
from flutes, whose stops musicians' fingers press,
and far and wide the tribe of country folk
listen, while Pan, shaking the pine garland
on his half-savage head, often races 850
over open reeds and from his curving mouth
pipes never cease to pour forth woodland song.¹
They speak of other miracles like this, [590]
other portents, perhaps in case men think
they live in remote and lonely places,
regions abandoned even by the gods.
So when they talk to people they throw in
amazing things. Or some other reason
guides them, as it does the whole human race
in its excessive greed for ears which listen.² 860
As for the rest, it need not surprise us
how voices come and stimulate our ears
in places where our eyes cannot see things
standing in plain view. Often, too, we notice
a conversation going on through closed doors.
There is nothing strange in this, for the voice
can pass intact through winding passageways
in things, but images refuse to do so, [600]
since they are broken up, unless they pass
through direct openings, like those in glass, 870
which every image can fly through. Moreover,
voices are divided up in all directions,
since some of them are produced from others,
where one voice comes forth once and then splits up
into many voices, just as a spark of fire
has a frequent habit of spreading itself
in many separate fires.³ Thus, places
kept concealed from view are full of voices—
things reverberate on every side and move
with sound, but all images keep going, 880
moving in a straight line, once they are sent out.⁴
That's why no one can see beyond a wall, [610]

¹ Pan is, in Greek mythology, the god of shepherds, flocks, and woods. He has the legs, horns, and hindquarters of a goat and is associated with, among other things, playing music on shepherd's pipes made out of hollow reeds.

² Perhaps they make up stories because, like all human beings, they are desperate to have people listen to what they have to say.

³ This splitting of a single voice into many is another reference to the fact that one voice can enter many ears at once and to the echo phenomenon which Lucretius has just discussed.

⁴ Because visual images have to move in a direct line, they cannot wriggle through twisting passages within the material of the wall; whereas, sounds can get through these passages. Hence, we can hear sounds from inside the room, but we cannot see anything through the wall.

Book Four

but can hear voices on the other side.
 But still, while going through a building's walls,
 the voice itself is also weakened and comes
 distorted to our ears—instead of words
 we seem to hear the sound.

As for the palate and the tongue, by which we distinguish taste, these do not require much further effort or a longer explanation. First of all, we perceive taste in the mouth: we press it out by chewing food, just as, for example, someone begins to press and dry by hand a saturated sponge. What we press out is then all distributed through openings within the palate and through winding paths inside the porous tongue. In this manner, when the particles of flowing liquid are smooth, their touch is pleasant, and contact brings delight to all the open places, around the tongue moistened with saliva. But, by contrast, the more the particles become completely rough, the more they prick and lacerate the sense, as they emerge. Then, too, pleasure from taste is limited to the palate. In fact, once the juices pass down through the throat, there is no pleasure while all of them are being distributed into the limbs. Nor does it matter at all what meal feeds the body, provided only that you are able to digest it, break down what you consume and then distribute it into all the limbs, while holding steady the levels of moisture in the stomach.	TASTE 890 [620] 900 [630] 910
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Now I will set down an explanation, so we may appreciate the reasons why different animals have different foods, why what is nasty and bitter to some can still seem delectable to others. Here the various differences are so great that what some animals consider food is for others toxic poison. There is, for instance, a snake which dies on contact with human spit—it commits suicide by eating its own body.¹ And hellebore, which is severely venomous to us,	DIFFERENT ANIMALS HAVE DIFFERENT FOODS 920 [640]
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¹ This observation, Munro notes, was also later made by Pliny, *Natural History*, Book VII.

Book Four

makes goats and quails put on more fat.¹

And now,
so you can understand how this takes place,
first, it is appropriate to remember
what we discussed before: in substances 930
there are primordial elements combined
in many different ways. And furthermore,
just as all living things which take in food
have outer differences and are limited
by the contours of their exterior limbs,
each according to its kind, so they also
consist of particles of different shapes.
Moreover, since these seeds are not the same,
in every limb spaces and passageways, [650]
which we call openings, must be different, 940
as well as in the mouth and palate, too.
Some openings must be smaller, some larger,
in some beings they must be triangular,
in others square, with several round ones,
some with many angles in many shapes.
For according to what is demanded
by the relationship of shapes and motions,
so the forms of openings must be different,
and the passageways must therefore vary
as does the texture which encloses them. 950
Because of this, when matter which is sweet
to some is bitter to others, for those
who find it sweet, the smoothest particles
must enter the pathways of the palate [660]
with a pleasing touch; on the other hand,
with those who find the same stuff sour inside,
the particles going in their passageways
are clearly rough and hooked.²

Given these details,
it is now easy to analyze each case.
For when a fever develops in someone 960
from an excess of bile—or something else
causes the force of a disease to rise—
then his entire body is soon disturbed,
and so the primary particles all change
arrangements. And therefore it comes about
that substances which pleased his sense before
do not please it now and that some others
fit better and can make their way inside

¹ Hellebore is the name for a species of plant frequently used as a medicine in ancient times, in spite of the fact certain types are poisonous. According to some historical accounts, Alexander the Great died from taking hellebore as a medicine.

² This difference, one assumes, must come about because of the size and structure of the passageways, which determine which particles can enter the palates of the two individuals.

Book Four

and produce disagreeable sensations. [670]
For both elements mingle in the taste 970
of honey, as I have already shown
and demonstrated to you many times.

Come now, I will consider how odours ODOURS
contact the nostrils. Firstly, there must be
many substances from which various streams
of scent flow off and fly away. We must grant
that smells are sent out, move off, and scatter
all around. But some are better suited
to certain living beings than others,
given their different shapes. And therefore bees 980
are led through the air from long distances
by the smell of honey, and carrion birds [680]
by corpses. A powerful sense of smell
sent out in advance leads on hunting dogs
wherever a wild creature's cloven hoof
has left his track, and from a distant place
the white goose, who rescued the citadel
of Romulus's sons, senses the smell
of human beings.¹ In this way, different smells
lead different creatures, each to its own food, 990
and make them recoil from harmful poison.
This process protects races of wild beasts.
This very odour, then, which stirs the nostrils,
can, in some instances, be given off
for greater distances than in others,
but still, none of them can be transported
as far as sound or voice, not to mention [690]
those things which strike the pupil in our eye
and stir our sense of vision. For odour
wanders about, moves slowly, and, spreading 1000
easily through airy breezes, soon dies
little by little, because, first of all,
since it comes from deep within an object
an effort is required to send it out,
for we know that odours flow off and leave
from well inside an object, since all things
seem to have a stronger smell when fractured,
pulverized, or broken down in fire.² Then, too,
you can see that odour is created
from larger particles than vocal sounds 1010
because it does not penetrate stone walls,
which sounds and voices usually pass through. [700]

¹ "Romulus' sons" are the Roman people. According to an old legend, recorded by the historian Livy, the geese in the temple of Juno saved Rome from the Gauls, around 390 BC, by cackling when they were disturbed by the invaders.

² Lucretius has already argued that primary particles that have to come from deep inside an object before being emitted lose some of their velocity in the struggle to get to the surface of the object and hence move more slowly through the air once they are emitted.

Book Four

For this reason you will also notice
it is not so easy to investigate
the location of something from its smell,
because in moving slowly through the air
the impact cools—what carries a report
about the object does not rush in heat
towards the senses. That is the reason dogs
are often wrong and have to search for tracks. 1020

But this does not occur only with smells
and assorted tastes, for colours and shapes
of things, in a similar way, are not all
well fitted to the sense in everything,
so that some of them, in certain creatures,
are harsher on the sight than other ones.¹
For instance, fierce lions cannot stand up to [710]
and gaze upon a rooster, whose flapping wings
drive out the night and who, with his shrill voice,
habitually calls up the dawn, for lions 1030
immediately think of scampering off.²
This is not strange, for in a rooster's body
are certain particles which, once sent out
to lions' eyes, bore into the pupils
and cause sharp pain, so that even wild beasts,
though fierce, cannot bear to stand against them,
although these seeds cannot in any way
cause damage to our eyes, either because
they do not penetrate or else because,
once they do get in, they are provided 1040
a free outlet from the eye and therefore [720]
cannot injure any portion of it
by remaining there.

Come now and find out
what substances affect the mind, and learn,
in a few words, where those objects come from
which move into our mind. First, I say this:
many images of things wander round
in all sorts of ways in all directions
everywhere. These delicate images
easily join together in the air 1050
if they should meet, like cobwebs or gold leaf,
for these images possess a texture
much thinner than the ones which strike our eyes
and stir our vision, since they penetrate [730]
porous openings in the body, provoke
the delicate substance of the mind inside,

¹ As Bailey and others point out, this verse paragraph seems out of place. Its logical position in the argument would seem to be one verse paragraph earlier.

² Munro observes that a number of classical writers refer to this curious behaviour of the lion: Pliny (in *Natural History*, Book VIII), Aelian, Plutarch, and others.

Book Four

and rouse the senses. Hence we see centaurs,
Scylla's limbs, dog faces of Cerberus,
and images of people who have died,
whose bones the earth contains.¹ For images 1060
of every kind are carried everywhere—
some of them are spontaneously produced
in air itself, some always fly off things
of various kinds, and some are produced
by shapes created out of both of these.
For, of course, the image of a centaur
is not produced from any living thing,
since the nature of such an animal
has never lived. But when, by chance, images [740]
of a horse and man have come together, 1070
they easily cohere immediately,
as we said before, because their nature
is subtle and their texture delicate.
All other images like this are made
in the same way. They are carried quickly,
because they are extremely light, something
I demonstrated earlier, and thus
with a single impact one thin image
of any of them quickly stirs our mind,
because the mind itself is sensitive 1080
and set in motion with astounding speed.

That these things happen as I have described
is easily seen from the following point:
since what we view with our minds resembles [750]
what we see with our eyes, they must be made
in the same way. So now that I have shown
I see lions, for instance, through images
which always stir my vision, we can know
how the mind is moved in a similar way—
it sees a lion and all other things 1090
by means of images, no more or less
than do our eyes, except that it perceives
more tenuous images. When sleep flows
through our limbs, understanding in the mind
is wide awake for no other reason
than that the same images stir our minds
as when we are not sleeping, so much so,

¹ Centaurs are fabulous creatures with the head and torso of a man and the body of a horse; Scylla is a monster with six heads who lives in the rocks at the strait between Sicily and Italy; Cerberus is a dog with several heads (usually three) who guards the entrance to Hades. Lucretius seems to be claiming that since images like these are not derived from real objects, our “sense” of them comes from combinations of very delicate, tenuous particles which enter our bodies and affect our minds, so that we “perceive” them. This process also (as Lucretius mentions in Book 5) appears to be the way in which we come to have a visual sense of the gods (i.e., thanks to material images which we cannot see with our eyes, but which enter our body and affect our minds)—with this important difference, the gods do exist; whereas, the images of these compound, fabulous creatures are formed in the air from various images combining, not by being stripped away from living animals.

Book Four

that we seem clearly to observe a man [760]
who has left this life and now been taken
by death and earth. And nature forces this 1100
to happen, since all our body senses
are obstructed in our limbs and resting—
they cannot argue against what is false
with genuine evidence. Moreover,
in sleep the memory is inactive
and indolent—it does not disagree
and say that the man our mind now believes
it sees alive was seized by death and fate
a long time past.

As for other matters,
it is not strange that images can move 1110
and wave their arms and other limbs around
in rhythm, for in sleep it does happen [770]
that an image is seen to act like this,
since after the first one has died away
and another in a different posture
has later been produced, the first image
seems to have changed the way it holds itself.
We must, no doubt, assume a quick process
brings this about—the motion is so fast,
the supply of things so large, and so great, 1120
in any single moment of perception,
the profusion of minute particles
from which they can be readily supplied.

In these matters there are several questions
to be asked, and we need to clarify
many things, if we want a plain account.
First of all, we ask why, when we desire
to think of anything at all, the mind
thinks of that very thing immediately. MIND AND PERCEPTION
Do images watch our will? If we want [780]
to think of sea, land, or sky, do images 1130
arise in us as soon as we desire?
Assemblies of men, parades, banquets, fights—
does nature make and hold all things ready
for a word from us, especially when
all minds in the same place and region
are thinking of completely different things?
And then what about when we are sleeping
and we perceive images coming forward
in rhythmic motion, moving graceful limbs, 1140
and with rapid, alternating gestures [790]
stretching their supple arms, and for our eyes
repeating foot motions made in harmony?
Do images really have artistic skill
and with this education wander round,

Book Four

so that in the night they can go dancing?
Or will it be closer to the truth to say
that in the one moment we perceive it—
that is, the time it takes to say one word—
lie hidden many moments, which reason 1150
ascertains are there, and thus it happens
that at any instant there are images
present and prepared in all locations,
so great is the supply and speed of things?
And thus, when the first image dies away [800]
and another is created later
in a different pose, what was there before
seems to have changed its posture. Moreover,
since images are tenuous, the mind
cannot see them distinctly, other than 1160
the ones it makes an effort to perceive,
and thus, except for these, they all perish,
apart from those for which the mind itself
has been organized by its own efforts.
The mind, then, makes itself ready, hoping
things will take place so that it can perceive
what follows on from each particular thing.¹
So that is what takes place. And furthermore,
have you not seen how eyes, when they begin
to look at some delicate object, strain 1170
and prepare themselves, and how, without that,
it would be quite impossible for us [810]
to see things clearly?² Even with objects
openly in view, you can still notice
that if you do not turn your mind to them,
then it is as if things were not near you
all the time, but remote and far away.
Therefore, why is it so strange if the mind
overlooks all other things, apart from
those where it has focused its attention? 1180
Then, too, from tiny signs we draw conclusions
which are very sweeping and lead ourselves
to snares of self-deception. Sometimes, too,
it happens that an image is supplied
which is not of the same kind as the first—
what was a woman previously appears
to have been altered by our own powers, [820]
so that a man seems present, or faces
and ages follow one after another.

¹ This awkward sentence is proposing that in a short but perceptible space of time (e.g., the time it takes to utter one word), there are many smaller moments intelligible to reason, and in these very short times images can change, so as to suggest continuous motion. The passage also seems to be suggesting that the mind to some extent shapes what it sees in accordance with what it hopes to see. As Copley notes, the explanation is “not too lucid.”

² Line 808 in the Latin has been omitted. It is the same as line 804 in the Latin (lines 1162-4 in the English).

Book Four

But then sleep and oblivion guarantee
we do not find this strange. 1190

In these matters
you must desire with all your eagerness
to shun this mistake and with keen foresight
to avoid this blunder: do not assume
that bright light was created in the eyes
so we might be capable of vision,
or that the top parts of our thighs and shins
above our feet can bend, so we could take
long strides, or, yet again, that our forearms
are joined to our strong upper arms and hands 1200
and provided on both sides to help us, [830]
so we could do what we would need to live.
All other ideas like this which men declare,
on the basis of preposterous reasoning,
transform effects to causes, since nothing
in the body was made with a purpose,
so that we could use it. No. What was born
created its own use. There was no seeing
before light in the eyes was born, no words
to speak before the tongue was made. Instead, 1210
the tongue originated long before
any spoken words, ears were created
a long time before any sound was heard.
In short, all the limbs, in my opinion, [840]
existed well before they had a use.
And therefore, they could not have developed
in order to be used.¹ But, by contrast,
to join in fighting battles with one's hands,
to tear limbs apart and stain the body
with streams of blood existed long before 1220
bright spears flew. Nature forced men to avoid
being hurt before the left arm ever learned
the skill of holding a protective shield.
And we know for certain that setting down
our tired body to rest is far older
than soft bed cushions, and quenching one's thirst
was born before the cup. Therefore, these things, [850]
which were devised to serve the needs of life,
we can well imagine being invented
in order to be used. Nevertheless, 1230
those other things are separate from them:
they were first born themselves, and afterwards

¹ Lucretius is here emphatically rejecting the notion that there is a purposeful design in the creation of the body. We were not given eyes in order to see. We happen to be able to see because we have eyes. The present uses of various organs developed after the organs were created. This, of course, is in line with modern biological thinking, which claims that new organic structures are produced fortuitously and have a better chance of being passed on if they serve a useful purpose in survival or reproduction or both. They were not created with the purpose of assisting survival.

Book Four

gave us some ideas about their uses.
First in this group, we see limbs and senses.
That is why, to repeat myself once more,
it is impossible for you to think
they were produced for their utility,
because they had a function.

Similarly,
it is not strange that the very nature
of body in all living beings seeks food.
For I have shown that many elements
flow off from things in many ways and leave,
but most must go from living animals,
since particles are disturbed by motion,
in sweating many are squeezed and carried
from deep inside, and many are exhaled
through the mouth when exhausted creatures pant.
In these ways, then, body is diminished,
its entire nature undermined, a state
which brings on pain. That is why the body
takes in food—to sustain limbs, to renew
strength once food moves inside, and to allay
in limbs and veins the gaping wish to eat.
Liquid also moves down to every part
requiring fluid—the moisture scatters
the many piled up particles of heat,
which produce a burning in our stomach,
moving in and extinguishing them, like fire,
so arid heat is no longer capable
of burning up our frame. In this way, therefore,
panting thirst is washed out of the body
and our hungry longing is satisfied.

HUNGER AND THIRST

1240

[860]

1250

[870]

1260

Now I will explain how it comes about
that we can propel our footsteps forward
when we wish, how we have been provided
the means to move our limbs in various ways,
and what it is that habitually shifts
this heavy weight of our body forward.
Listen to what I have to say. I claim
that, first of all, images of moving
fall into our mind and keep pushing it,
as we said before. From that arises will,
for no one starts to do anything at all
before his mind decides what it desires,
something the mind determines in advance,
so that there is an image of that thing.
And therefore, when the mind has thus been roused
so that it wants to move, to stride forward,
it strikes the power of soul immediately
in the whole body, spread through limbs and frame.

HOW HUMAN BEINGS MOVE

[880]

1270

1280

Book Four

This is easily done, since soul and mind
 are held in combination.¹ After that,
 soul goes on to strike the body. And so, [890]
 little by little, the whole mass is pushed
 and moves ahead. Moreover, the body
 then becomes more porous, as well, and air
 comes through the open spaces—as, in fact,
 it must do, given how it is always
 so quick to move—and large amounts of air
 penetrate the passageways and scatter 1290
 to all minute portions of the body.
 Thus, in this way body is made to move
 by two separate causes, just like a ship
 with sails and wind.²

We should not be surprised

in these matters, however, that particles
 so tiny can swing around a body
 of such size and redirect our whole mass. [900]
 For although the wind is, in fact, composed
 of delicate and subtle substances,
 it drives and pushes forward a huge ship, 1300
 which takes great effort, and a single hand
 guides the ship, no matter how rapidly
 it may be moving, and turns one rudder
 in whatever direction it desires.
 With wheels and pulleys a machine can move
 many very heavy things, lifting them
 with little effort.

The ways that sleep floods rest

throughout the limbs and lets cares of the mind
 escape the chest I will now clarify
 in my verses—these will not be numerous 1310
 but will instead sound sweet, just as brief songs [910]
 from swans are better than the screech of cranes
 spreading through southern clouds, high in the sky.
 Give me your subtle ear and eager mind,
 so you do not deny that what I say
 is possible and leave me, with your heart
 rejecting my true words, when you yourself
 are in the wrong and cannot understand.
 First, sleep occurs when power in the soul
 is spread out through the limbs and part of it 1320
 has left the body, after being sent out,

¹ Lucretius is here reverting to his earlier distinction (in Book 3) between the mind in the chest and the soul distributed throughout the body.

² There are problems with the text here, which may account for the poor analogy to a ship. Gassendi (according to Munro) suggests “with oars and wind” (*remis ventoque*) because these are, in fact, two separate ways of moving a ship forward, whereas wind and sails are only one way. It is, however, still not entirely clear how the inrush of air would help propel the body forward.

Book Four

and another part is pushed further in
and has withdrawn deep inside the body,
since at that very point the limbs unwind
and grow relaxed. For there can be no doubt [920]
that we have this capacity for sense
thanks to the soul. When sleep obstructs our sense,
we must assume our soul has been disturbed
and sent outside. But not the entire soul,
for then the body would lie there immersed 1330
in the eternal iciness of death.
Since no part of soul would remain concealed
within the limbs, the way fire lies concealed
under piles of ash, how could sensation
be suddenly rekindled in the limbs,
like flames that rise up from a hidden fire?¹
However, I will explain how this new state
is produced in matter, and how the soul
can be disturbed, the body grow relaxed. [930]
Take care I am not scattering my words 1340
into the winds. First of all, the body,
given its close contact with the airy breeze,
must be beaten on its outer surface
and struck by frequent impacts with the air.
That is the reason almost everything
is covered with hide, shell, hard skin, or bark.
The air also beats against that region
inside the body, when during breathing,
it is drawn in and then blown out. And thus,
since the body is lashed in these two ways 1350
and the blows enter through tiny openings [940]
in our bodies to reach the basic parts
and primordial elements, what takes place
is, so to speak, a gradual dissolution
in our limbs. The alignments of the soul
and primary particles are shaken up.
After that, part of the soul is drawn away,
part retreats inside and conceals itself,
and part is also ripped up into pieces
throughout the body and cannot maintain 1360
its mutual combinations or go through
the motions it reciprocates, for nature
interferes with passages and movements.
Hence, once impulses are changed, sensation
moves away, deep inside. And since there is
nothing which, as it were, props up all the limbs, [950]
the body becomes weak, and every part
grows slack—arms and eyelids droop, and knees
give way, letting their energies relax,

¹ Smith points out that Lucretius makes no mention of how the soul regains that part of itself which goes outside the body during sleep or makes up for the loss.

Book Four

often while someone is still reclining. 1370
Then, sleep follows after meals, because food,
while being distributed to all the veins,
has the same effect as air. And that sleep
which you take when you are full or weary
is the heaviest by far, for at those times
most of the particles are disordered,
crushed by great exertion. In the same way,
part of the soul is driven deeper down,
a larger part of it is thrust outside, [960]
and in itself it grows more divided, 1380
more torn apart within.

And for the most part,
whatever actions each man carries out
and clings to, or whatever activities
we have spent much time on previously
where the mind has been more keenly active,
in general, we seem, when we are sleeping,
to go over matters which are much the same—
lawyers seem to plead causes, challenge laws,
generals seem to fight, march into battle, 1390
sailors to wage collective war with winds,
and I constantly to pursue this work
and seek out the nature of things, and then,
once that is discovered, to set it down
in my own native tongue. And thus, in sleep, [970]
all other arts and studies mostly seem
to control and mock our minds. And if
men ever pay unwavering attention
for several days without interruption
to public shows, we generally see
that even when they cease to grasp these things 1400
with their senses, in their minds still remain
open pathways through which can penetrate
the same images of things, and therefore,
for many days they see those same objects
pass before their eyes, so that they appear,
even while awake, to see the dancers [980]
moving graceful limbs; their ears seem to hear
the cithara's speaking strings, its liquid song.¹
They appear to see the same crowd gathered
and, at the same time, shining splendidly, 1410
the various decorations of the scene,
so great is the influence of effort
and preferences and those occupations
men do habitually.

And not just men,

SLEEPING ANIMALS

¹ The cithara is a stringed instrument, somewhat like a small harp or a lyre, used by professional musicians.

Book Four

but indeed all animals, for you will see
brawny horses stretch out their limbs in sleep,
and yet they continually sweat and pant,
as though exerting all their energy
to win the prize or [striving to race ahead],
as though the gates had opened.¹ Hunting dogs, 1420
while gently resting, often twitch their legs [990]
unexpectedly and suddenly send out
their baying call—their nostrils sniff the air
repeatedly, as if they had just found
and were pursuing some wild creature's tracks.
And often, after they are woken up,
they chase imaginary images
of deer, as if they were seeing them turn
to run away, until the deception
is shattered and they recover themselves. 1430
And the fawning breeds of young puppy dogs
used to staying at home start to rouse themselves
and lift their bodies from the ground, just as if
they were seeing strange shapes and faces.²
The more ferocious any breed may be,
the more it must display its rage in sleep.
And various birds fly off and suddenly,
during the night, disturb sacred thickets,
if, in their tranquil sleep, they notice hawks
on the wing, chasing and offering battle. 1440 [1010]

Then, too, human minds which, with great effort,
achieve important things often, in sleep
carry on performing the same actions—
kings launch attacks, are captured, join battle,
raise a shout, as if, that very moment,
their throats were being slit. Many fight hard,
groan aloud in pain, and with their huge cries
completely fill all the space around them,
as if they were being chewed by leopards
or savage lions. In sleep, many men 1450
talk of serious things and have often made
confessions about something they have done.
Many meet death. Many are terrified, [1020]
as if their whole body were being hurled
from high mountains down to the earth below,
and have trouble, as though their minds were gone,
recovering from sleep, as they tremble
from the agitation in their bodies.
In the same way, a thirsty man sits down
beside a river or a pleasant spring 1460

¹ I have adopted (more or less) the suggestion of Munro for a textual difficulty here. The image is from the start of a race in which each horse is behind a gate.

² Lines 1000 to 1003 in the Latin have been omitted. They are identical to lines 992-995 (lines 1419-1425 in the English). Hence, there is no line [1000] above.

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and almost drains the whole stream down his throat.
 Often, clean, decent people, bound by sleep,
 if they think they are beside a toilet
 or a chamber pot, lift up their clothing,
 and their whole body pours out filtered liquid,
 saturating the splendid magnificence
 of coverlets from Babylon. And then,
 for those in whose vital raging waters
 for the first time semen begins to flow,
 when maturity of age creates it
 throughout their limbs, external images
 from anybody gather, bringing reports
 of a superb face and lovely colouring.
 These stimulate and rouse swollen places
 with lots of seed, so that, as if doing
 the whole act, often it comes bursting out,
 in great waves of semen, and stains the clothes.
 That seed which we just spoke about above
 is stirred in us when adult maturity
 for the first time makes our limbs much more robust.
 Now, some things are roused and stimulated
 by one thing, and different things by others.
 Human force alone draws human sperm from man.¹
 Once it is forced out from those locations
 where it sits, it moves off, shifting away
 from all places in the body through limbs
 and frame. It gathers in appropriate spots
 in the tissues and instantly excites
 the body's sexual parts themselves, and these
 once roused to action, swell up with semen,
 creating the desire to eject the seed
 in the place ill-fated lust strains to reach,
 and the body searches out the object
 which stabbed the mind with love.² For normally,
 all men collapse towards a wound, the blood
 spurts out towards that place where we received
 the blow, and if our enemy is close by
 the crimson liquid spatters him. Therefore,
 when a man is hit by bolts from Venus—
 whether a boy with girlish limbs strikes him
 or some woman exudes sensual passion
 from her whole body—he then moves towards
 the place from which he has received the blow
 and is keen to copulate, to discharge
 from his body the liquid gathered there,
 inside the body, for passion, though mute,
 still speaks of pleasures yet to come.

HUMAN SEMEN

[1030]

1470

1480

[1040]

1490

[1050]

1500

¹ These three lines are somewhat elliptical. The point seems to be that in men it is only other people (the implication is both male and female) who stimulate the physical reactions of sex which draw sexual seed distributed through the body to the genitals.

² Line 1047 in the Latin has been omitted. It is the same as line 1034 (lines 1472-3 in the English).

Book Four

This pleasure
we call Venus. From it Love gets his name.
And from it, too, has dripped into our heart
the first drops of that seductive allure
of Venus and then chilling anxiety
has later followed. For if the one you love
is absent, those images are still present,
and that sweet name still hovers at your ears.
However, you must flee such images,
scare away what nourishes your passion,
turn your mind to something else, and discharge
your collected fluid into bodies
anywhere—you must not hang onto it,
once you have changed to loving only one,
and thus reserving trouble for yourself
and certain pain.¹ For the festering sore
comes alive and settles in with feeding.
Day by day delirium increases,
hardship weighs you down, unless you confuse
those wounds you sustained at first with new blows
and heal them while still fresh, by wandering
with a Venus who wanders everywhere,
or can shift your mind to other matters.
A man who avoids love is not without
delights of Venus, but rather chooses
those whose benefits bring no punishment.
For there is no doubt that for healthy men
sexual pleasure is purer than for those
sick with love. In fact, in the very moment
of possession lovers' passion fluctuates,
it wavers, strays here and there, undecided
where eyes and hands should first reap their delight.
What lovers desire, they crush hard, causing
physical pain, frequently sinking teeth
in little lips, pressing mouths together,
because their pleasure is not pure—there are
hidden goads driving them to inflict pain
even on the thing, whatever it is,
which first aroused those seeds of frenzy.
But with a light hand Venus mitigates
these penalties of passion, by mixing in
seductive joys which curb their biting teeth.
For there is hope in this—that at the source
of passion fires can be put out, as well,
by the same body.² But nature protests

¹ Promiscuous sex with anyone satisfies the physical desires, while avoiding the emotional complications of romantic love. Hence, for the Epicurean, who is seeking mental tranquillity above all else, the former is to be preferred.

² Brown points out the implied metaphor here of controlling a passionate horse and underlines the distinction between frenzied, painful passion (which inflicts pain) and the gentler sexual pleasures

Book Four

that what happens is completely different.
This is the one thing where the more we have,
the more ill-fated lust burns in our hearts. [1090]
For food and drink are taken in our limbs,
since these can settle in certain places
and one can quite easily satisfy
desire for bread and wine. But from humans
the face and lovely colouring transfer
nothing to the body to be enjoyed 1560
except frail images, and frequently
these woeful hopes are snatched off by the wind.
Just as a thirsty man, when he's asleep,
desires a drink and receives no liquid
that could quench the burning in his body,
but keeps seeking images of water,
struggling in vain, still thirsty, as he drinks
in the middle of a boiling river— [1100]
that's how, in matters of love, Venus mocks
lovers with images, and they cannot 1570
satisfy their bodies by gazing at them
face to face, nor can their hands, which wander
randomly all over the whole body,
scrape anything away from tender limbs.¹
And when at last their bodies intertwine
and they take pleasure in their bloom of youth,
while flesh is now sensing more delights to come
and Venus has prepared herself to sow
the ploughed field in the female, the lovers
fixate on the body greedily, their mouths 1580
linking their spit, and breathing heavily,
with teeth pressing against each other's lips.
But there's no point. For they cannot scrape off [1110]
anything from there or penetrate inside
and with their entire body move into
the other body, for sometimes they seem
to want that and struggle to achieve it.
That is how passionately they stay there
locked in Venus's embrace, while their limbs,
loosened by the power of pleasure, melt. 1590
At last, when desire, pent-up in the penis,
is released, there is, for a brief period,
a short let up in their violent passion.
Then the same madness returns, that frenzy

associated with Venus. Hence, sex is a combination of pleasure and pain. The obvious point to this passage about human sexuality is not that sex is bad (its pleasures are to be welcomed), but rather that it can be dangerous and inherently unsatisfying, especially for someone who places a very high value on living without mental anxiety.

¹ Images cannot satisfy the demand of physical passion for pleasure, nor can looking at the body of one's lover in the flesh. Unlike food, these actions do not transfer anything material into the body which satisfies the craving. The emphasis on sexual desire as driven by a craving for physical possession or assimilation is remarkable.

Book Four

comes back once more, when they strive to attain
what they themselves desire, and they cannot
discover what technique may overcome
what's wrong—that shows how much they waste away,
in great uncertainty, from hidden wounds. [1120]
And besides all this, they exhaust their strength, 1600
worn down by their exertions and then add
that they spend their life at the beck and call
of someone else. They neglect their duties,
and their tottering reputation sickens.
Meanwhile, their possessions slip away,
converted into scents from Babylon,
while lovely slippers from Sicyon laugh
on the lady's feet, and, you may be sure,
enormous emeralds, all sparkling green,
are set in gold, and her purple garment 1610
is constantly being ripped and roughly used,
as it soaks up the sweat of sexual passion.¹
The father's well-earned wealth is then transformed
to ribbons and scarves and sometimes is changed [1130]
to robes and goods from Chios and Elis.
Banquets are prepared with gorgeous carpets
and fine food, games, repeated drinking bouts,
perfumes, wreaths, and garlands. All for nothing.
For in the midst of this fountain of delights,
a certain choking bitterness wells up, 1620
even among the flowers, when mind itself,
sensing guilt, feels the strong bite of remorse
for living such a slothful life, wasted
in debauchery, or when she throws out
a word and leaves the sense ambiguous
and, fixed in a passionate heart, it grows
like fire, or when he thinks she casts her eyes
and glances at another man too much,
or sees a trace of mockery in her face. [1140]
And these problems are those one finds in love 1630
which is lasting and fully prosperous.
But when love is desperate and destitute,
with your eyes shut you can grasp the troubles—
they are innumerable. So it is better
to be cautious in advance, as I have shown,
and to be careful you are not seduced.
For to avoid being drawn into love's nets
is not as hard as to escape the mesh
and break through those mighty knots of Venus,
once you have been ensnared. But nonetheless, 1640
although you get entangled and caught up,
you can still evade the danger, unless [1150]
you stand in your own way and overlook,

¹ The purple colour is a sign of extravagance, since the dye was very expensive.

Book Four

right at the start, all the imperfections
of mind and body in the one you want,
the woman you are chasing, because men,
for the most part, proceed from blind desire
and give women delightful attributes
which are not really theirs. And so we see
those who are in many ways misshapen 1650
and repulsive are dearly loved and thrive
in utmost favour. And some people laugh
at others and urge them, since they are trapped
in foul sexual passion, to placate Venus,
and yet often those people, the poor fools,
do not think of their own tribulations,
which are excessive. A dark woman is [1160]
“honey coloured,” a filthy one who stinks
is “unpretentious,” one who has gray eyes
is “small Athena,” a sinewy one 1660
who looks like wooden sticks is “a gazelle,”
a squat, dwarfish girl “one of the Graces,”
“all genuine charm,” a large and lumpy one
“impressively imposing,” “dignified.”
If she has a stammer and cannot talk
she “has a lisp,” if mute, she is “modest,”
if a fiery, hateful gossip, she becomes
“a flaming torch.” If she is so skinny
she can hardly stay alive, she becomes
“a slender darling,” if about to die 1670
from coughing fits, then she is “delicate.”
A fat bosomy one is “Ceres herself
after giving birth to Iacchus,”
a snub-nosed girl “a female Silenus
or a satyr woman.” One with thick lips
becomes “a living kiss.”¹

It would be tedious
to try listing all other things like this. [1170]
But let her face even be as lovely
as you wish, and let the power of Venus
radiate from every limb, nonetheless 1680
there are surely other women, as well,
surely we lived without this one before;
surely she carries out all the same things
ugly women do—and we know she does.
The woman drenches her miserable self
with disgusting odours. Her slaves run off
some distance and laugh at her in secret.
But the tearful lover who is shut out

¹ This is obviously a list of poetical clichés and is a satire on conventional love poetry as much as on certain male attitudes in courtship. The Graces, in Greek mythology, are three divine goddesses of charm and gracefulness. Ceres is a Roman goddess of farming and cereal crops. Iacchus is a common name for the Greek god Dionysus or Bacchus, the god of wine. Silenus is a companion of Dionysus.

Book Four

buries the threshold with frequent flowers
and garlands, and with scent of marjoram 1690
anoints her haughty doorposts, plants kisses
on the doors, the miserable fool, and yet
if once he were let in and just one whiff [1180]
hit him as he entered, he would seek out
decent reasons to be gone.¹ The sad song
drawn from deep within and reflected on
so long would disappear, and then and there
he would curse his foolishness. He would see
he had bestowed on her more than is right
to give any human being. Our Venuses 1700
are not unaware of this, and so they use
their utmost efforts all the more to hide
all that goes on behind the scenes of life
from those they wish to keep bound up in love.
All in vain. For in your mind you can drag
everything into the light, search all smiles,
and if her mind is good and free from spite, [1190]
then, for your part, let her go, and pardon
those features which make her a human being.
And when a woman heaves a sigh of love, 1710
she is not always faking. While embracing,
she joins her lover's body to her own
and holds it. As they suck lips, she keeps his moist
with kisses. Often she acts from the heart,
and, seeking mutual delight, stirs him
to complete love's race. For there is no way
that in birds, cattle, horses, savage beasts,
and sheep, females could crouch under the males,
if their natures did not bring them in heat,
burn to overflowing, respond with joy, 1720
as the penis mounts them. Do you not see [1200]
how those whom mutual pleasure often links
are also tortured in the chains they share—
how often dogs at crossroads really strive
with all their eager strength to separate,
to go their different ways, while all the time
they are stuck together in the strong chains
of sexual lust? This they would never do,
unless they experienced those shared joys
which can throw them into a delusion 1730
and hold them bound. So, to repeat myself,
I say pleasure comes to men and women.
And when, during the mingling of the seed,
the female happens to overcome male force
with sudden power and has seized control, [1210]

¹ Brown notes that there has been much scholarly discussion about the emphasis here on the woman's smell: suggestions have included perfume, body odour, flatulence, menstruation, vaginal fumigation, and medical treatments for hysteria. Whatever the precise reference, Lucretius' main point here is that all women, no matter how beautiful or ugly in public, in the privacy of their own rooms smell disgusting.

Book Four

then children are born from the mother's seed,
looking like the mother, just as children
from the father's seed look like the father.¹
But those you see who look like both of them,
with mixed parental features side by side, 1740
grow from father's body and mother's blood,
when sexual seeds, once roused through the limbs
by the pricks of Venus, flow together,
unite in harmonious, mutual passion,
and neither one of them is dominant,
and neither one submissive.² Sometimes, too,
children can be created who look like
their grandparents and frequently bring back
the features of their grandparents' parents,
because many primordial elements 1750 [1220]
mixed in many ways are often hidden
in the bodies of their parents, and these,
from the first beginnings of the family,
fathers pass on to fathers, and from them
Venus, by drawing different lots, creates
their shapes and brings back facial expressions,
vocal sounds, and hair of their ancestors.
And the race of females may well spring up
from the father's seed, and men may be born
shaped by their mother's body, since, in fact, 1760
these are no more made by one parent's seed
than are our faces and our trunk and limbs.³
For birth always consists of double seeds,
and whatever is born which resembles [1230]
one of the two parents more possesses
a more than equal portion of that parent.
And whether the offspring is from the male
or has its origin in the female—
that is a feature you can distinguish.
And the powers of gods do not withhold 1770
from any man the planting of his seed,
so that sweet children may never call him
father and he may live out all his days
in a barren marriage. But usually
men believe they do, and in their sadness,
spray altars with streams of blood and cover
high places with their gifts, hoping they may,
with prodigious quantities of their seed,
impregnate wives. In vain they wear away

FERTILITY AND INFERTILITY
IN HUMAN BEINGS

¹ This sudden seizing of power refers to the female seed overpowering the male seed when they mix, not to the woman overpowering the man during sex.

² The origin of hereditary traits was much discussed in ancient times, with various debates about the different roles of male and female sexual "seed" and about the precise location of the hereditary material (in the blood or sexual fluid).

³ (57) I follow Munro in moving line 1227-8 in the Latin to 1225-6.

Book Four

the majesty of gods and sacred lots.¹ 1780
For some men who are sterile have semen [1240]
which is too thick; in others, by contrast,
it is thin, more watery than it should be.
Thin seed cannot firmly fix itself in place—
it leaves immediately, sinks back, withdraws,
its attempt aborted. And then again,
seed which is too thick because it spurts out
in a denser form than is appropriate
either does not get discharged with a thrust
that goes far enough, or is less able 1790
to work its way into the right places,
or, having penetrated these, mixes
poorly with the female seed. For we see
many differences in those sexual acts
which work out well—some men can impregnate
some women more easily than others,
while other women more readily take on
the load from different men and grow heavy. [1250]
And many women have been infertile
in several previous marriages and yet 1800
afterwards have discovered men from whom
they could bear children and enrich themselves
with tender offspring. And for those men, too,
whose wives at home, though fertile, had often
been unable to give birth previously,
an appropriate partner has been found,
so they could fortify their older years
with children. That shows how crucial it is
that seeds suitable for reproduction
are mixed, that thick seeds bond with liquid ones, 1810
and liquid seeds with thick. And on this point,
the food we eat, by which life is maintained, [1260]
is truly relevant. Some substances
condense the seed inside the limbs; others,
in turn, make it thinner and destroy it.

And the ways in which the charming pleasure
is carried on also really matter.
For people generally believe that wives
conceive more easily if they have sex
like wild animals, following the style
of quadrupeds, for that way, with chests down
and sex organs raised, appropriate parts
can take in seed. And wives do not require
the slightest sensual motions. A woman
stops herself conceiving and resists it,
if for pleasure's sake she herself draws back

¹ (58) Sacred lots (Brown notes) were pieces of wood on which were written prophetic utterances. The divination proceeded by lottery.

Book Four

from her husband's penis with her buttocks [1270]
and then, with her whole body limp, begins
to move in rhythm, for she throws the furrow
from the pathway and the straight alignment 1830
of the ploughshare, altering the impact
of the seeds away from the right places.¹
Prostitutes will often move this way,
for their own reasons, to stop conceiving
too many times, lying around inactive
in pregnancy, while simultaneously
to make sex for men itself more pleasing.
Our wives, it seems clear, have no need of this.²
And sometimes, by no action of the gods
or arrows from Venus, it does happen 1840
that some mediocre little female
with a less favourable shape is loved,
for a woman, thanks to the way she acts [1280]
and to her accommodating manner
and well-tended body can now and then
make you become easily accustomed
to spending life with her. As for the rest,
familiarity gives rise to love,
for whatever is struck repeatedly
by any blow, however slight, will at last, 1850
with a long lapse of time, be overcome
and concede. Do you not observe also
how, after a long period of time,
falling drops of water bore holes in rocks?

¹ The anatomical details of this procedure have prompted a certain amount of comment, some people seeing here a reference to anal intercourse, with the wife pulling the man's penis with her buttocks. That, however, seems unlikely, given the context of the discussion (how to avoid conception during regular heterosexual intercourse). It seems more a matter of the woman's pulling herself back somewhat from the man's penis (by moving her buttocks) and then swaying around so as to alter the angle of entry.

² This curious link between a woman's active participation in sexual motion during copulation and her infertility may, as Brown suggests, be linked to the notion that it was considered improper for a decent wife to get too carried away during sex, in spite of the fact that it makes the act more pleasurable.

BOOK FIVE

[Tribute to Epicurus; comparison with deeds of Hercules; intention to account for the formation of the world and life on earth; future destruction of earth and sky; mind's place in body; no divine places of the gods in the world; tenuous nature of gods; futility of thinking humans can benefit gods; doubts about divine creation of things; defects in the creation of the world; world created from mortal substances; world is relatively young; war between different parts of the world; first materials separate out, creating different regions; reasons why stars move; earth merges with air underneath; size of sun and moon; causes of sun's heat; annual and daily motion of sun and moon; changes in light from sun and moon; causes of solar and lunar eclipses; first plant growth on earth; creation of animal life from earth; earth produced monsters; animals which cannot cope die out; no composite animals produced; first humans lived off wild nature; acquisition of huts, fire, customs; development of language; growth of towns, division of land; murder of kings, creation of laws through mutual agreements; origin of religion; uselessness of worship; discovery of metals; use of animals in battle; development of clothing and agriculture; origin of music; changes in diet; development of sailing, poetry, writing, other arts.]

Who has the power in his mighty heart
to frame a poem worthy of these things
we have found out and of the majesty
of what we are discussing? Who has words
strong enough, so he can fashion praises
which could match the quality of the man
who bequeathed such things to us, these prizes
imagined and searched out in his own heart?
In my view, no one born with mortal flesh
will have that power. For if we must speak
as the known majesty of things demands,
that man was divine, noble Memmius,
a god, who first set down that plan for life
we now call wisdom, who, thanks to his skill,
took life from such great turmoil and darkness
and set it in such peace, in such clear light.¹
For compare the divine discoveries
of others from long ages past with his.
Now, Ceres, so they say, taught mortal men
about grain crops, and Bacchus liquid juice
grown on the vine.² But life without these things
could still go on, as certain races live
even today, according to reports.
But men could not have lived successfully
without pure hearts, and that is why we claim
this man is more justly thought a god—from him
life's tender consolations now extend
even to mighty races and assuage
the minds of men. But if you think the deeds
of Hercules are more remarkable,
you will be carried even further off
from proper reasoning. For what damage
would that mighty gaping Nemean lion
and that terrifying Arcadian boar

TRIBUTE TO EPICURUS

10

[10]

20

[20]

30

¹ Lucretius is here paying tribute, once more, to Epicurus. In this tribute we are reminded again that the great value of Epicurus's teaching for Lucretius is not only the knowledge it reveals of the world, but, more importantly, the ethical implications of that knowledge: it enables us to live properly.

² Lucretius uses the name Liber rather than Bacchus, a traditional Italian god associated with farming. Later Liber was identified with the Greek god of wine, Bacchus.

Book Five

do to us now?¹ What of the Cretan bull
or that Lyrnaean pestilence, the hydra,
guarded by her wall of venomous snakes?
What could they do to us? And the power
of those three chests on the triple body
of Geryon? [How could those birds] who live 40
in [foul] Stympthalian [swamps] have injured us
so much, or steeds of Thracian Diomedes, [30]
with nostrils snorting flames beside the coasts
of Bistonia and Ismara?² And the snake
protecting the glistening golden apples
of the Hesperides—that fierce creature
with a lethal gaze, who coils his vast shape
around the tree trunk? In the end, what harm
could he have done by the Atlantic shore
and its harsh seas, which none of us comes near 50
and no barbarian will dare approach?
And all the other monsters of this kind—
who, if they were not overcome, were killed—
what damage could they finally inflict,
if they were still alive? In my opinion,
none at all. As it is, the earth is full, [40]
even nowadays, of savage creatures,
crammed with alarming terror in the woods,
immense mountains, and deep forests, but we,
for the most part, have the power to shun 60
such places. However, unless our hearts
are purified, what battles and dangers
must then insinuate themselves in us,
against our will! What bitter cares then rend
men torn apart by passion! What other fears
do just the same? What of arrogance, filth,
depravity? What ruin they produce!
What of luxuriousness and indolence?
And so the man who has overpowered
all these and driven them out of his mind— 70
not by weapons but by words—should this man [50]
not be rightly found worthy of inclusion
among the gods, especially because
it was his custom to say many things,
in an elegant and inspired manner,
concerning the immortal gods themselves,
and in his teachings to elucidate

¹ Hercules is the major human hero of Greek mythology and (as Bailey points out) an important figure for the Stoics, whose ideas Lucretius repeatedly attacks. As a punishment for killing his wife in a fit of madness, Hercules was given twelve tasks: killing the Nemean lion, slaughtering the nine-headed Lernaean hydra, capturing the golden hind of Artemis, capturing the Erymanthian boar, cleaning the Augean stables, killing the Stympthalian birds, capturing the Cretan bull, stealing the horses of Diomedes, getting the girdle of the queen of the Amazons, stealing the cattle of the monster Geryon (who had three torsos, hence he was “triple-bodied”), getting the apples of the Hesperides, and capturing Cerberus (the dog guarding the gates of Hades).

² The text of the Latin is commonly rearranged here to make the list more coherent. Munro conjectures a line has been lost before line 30 of the Latin. The suggested additions are in square brackets in the English above.

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the entire nature of things?

While treading
in his steps, I pursue his reasoning,
and in the things I say I teach the law 80
by which all things are produced and by which
they must continue—they have no power
to break mighty statutes of the ages.
In this group, first of all, it has been shown
that the mind's nature, from the very start, [60]
is a substance which was born—it cannot
stay intact for long periods of time—
but images, in sleep, habitually
deceive the mind, when we appear to see
a man whose life has left him. And so now, 90
in what remains, my train of argument
has now brought me to this point, where I must
set down an explanation how the world
is a mortal substance and was born,
how a collection of materials
established earth, heaven, sea, stars, sun,
and the moon's globe, then what living creatures
sprang from earth, as well as those never born [70]
at any time, how the human race began
to employ among themselves various words 100
by giving names to things, and ways in which
that fear of gods slid into human hearts,
which preserves sacred places on earth's sphere—
shrines, lakes, groves, altars, images of gods.
Moreover, I will explain the power
by which pilot nature steers the sun's course
and the wandering of the moon, just in case
we may perhaps believe they circle round
their eternal pathway between heaven
and earth of their own free will, graciously 110
increasing growth of crops and living things, [80]
or think they orbit there thanks to some plan
devised by gods. If those who rightly teach
that gods pursue a carefree life still wonder,
from time to time, about how all these things
can work the way they do, especially those
we see in heavenly regions overhead,
they are carried back to old religions
once again and adopt stern overlords,
who are, they claim, in their unhappy state, 120
omnipotent—they are still ignorant
of what can and cannot be and, in short,
of why each thing has limited power [90]
and deep-set boundary stones.

As for the rest—

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so we avoid delaying you any more
 with promises—you must first consider
 seas and lands and sky, their threefold nature,
 three bodies, Memmius, three different forms,
 three such excellently created things—
 these in one day will be given over
 to destruction, the huge mass and fabric
 of the world, standing for so many years,
 will fall in ruins. My mind is quite aware
 of the new and astonishing effect
 this point has upon the understanding—
 the future destruction of earth and sky!—
 and how difficult it is for me to prove
 by what I say. But that is what occurs
 when you convey something to people's ears
 they did not know before, yet you cannot
 set it in open view before their eyes
 or place it in their hands, those ways in which
 the paved road of belief leads most directly
 to the heart and open places in the mind.
 But still I will speak out. It may well be
 that facts themselves will validate my words,
 and you will observe earthquakes breaking out,
 all things, in one brief moment, badly crushed.
 But may helmsman Fortune steer these troubles
 far away from us, and may reasoning,
 rather than brute fact, lead us to believe
 that all things can be overcome and fall
 with a horrifying, resounding crash.

FUTURE DESTRUCTION OF THE
 EARTH

130

[100]

140

150

But in these matters, before I begin
 to pour forth about fate with more sanctity
 and with far more coherent reasoning
 than the Pythian priestess, who prophesies
 from Phoebus' tripod and his laurel tree,
 I will, in my learned discourse, explain
 many consolations to you, in case,
 curbed by religion, you perhaps suppose
 that lands and sun and sky, sea, stars, and moon
 must last eternally, for their substance
 is divine, and thus you believe it right
 that, like the Giants, all those should suffer
 some punishment for their abhorrent crime
 who with their own reasoning undermine
 the ramparts of the world and wish to quench
 the splendid sun in heaven by branding
 immortal things with mortal words.¹ But these,

[110]

160

[120]

170

¹ The Pythian priestess is the prophetess of Phoebus Apollo, at his shrine in Delphi. The Giants, in Greek mythology, were monstrous children of Earth, who fought against the Olympian gods; the latter prevailed with the help of Hercules, and the Giants were all destroyed or imprisoned. This section (starting in line 110 of the Latin) is a digression from the announced intention to explain the material formation of the earth, a subject to which Lucretius returns at line 235 of the Latin.

Book Five

in fact, are quite separate things, far distant
from godlike majesty, so unworthy
of being reckoned among the gods,
that they could, by contrast, be looked upon
as providing evidence of objects
without vital motion and sensation.

For obviously we cannot just assume
that the nature and judgment of the mind
could exist in any body at all,
just as a tree cannot live in aether,
clouds in salty seas, fish cannot survive
in farmland, blood cannot exist in wood,
and sap in stones. Each thing has a set place,
a predetermined spot, where it belongs
and grows. And therefore, the nature of mind
cannot be born by itself without body,
cannot exist far from blood and sinews.

For if the very powers of the mind—
and this is far more likely—could exist
in head, or shoulders, or below the heels,
born in any part you wish—in the end,
it might grow accustomed to remaining
in the same man or vessel. However,
since it is determined where soul and mind
can grow even in our bodies—and we see
that this is fixed—then we must all the more
deny that mind could totally survive
outside the body and the form of things
which are alive, in rotting lumps of earth
or in the fire of the sun, or water,
or in soaring regions of the aether.
Thus, these things do not exist possessing
divine sense, since they are incapable
of being brought to life with vital feelings.¹

MIND AND SOUL CANNOT EXIST
EVERYWHERE

180

[130]

190

[140]

200

In the same way, it is impossible
you could believe this point—that there exist
sacred dwelling places for deities
in any regions of the world. For in gods
nature is tenuous and far removed
from our sensations—hardly perceptible
to the understanding of human minds.
It eludes what our hands can feel or strike,
and so it must not contact anything
which we can handle. For nothing can touch
which may not be touched itself. And therefore,
their homes must also be unlike our homes—

MATERIAL MAKING UP THE GODS

210

[150]

¹ The point here seems to be that since the mind cannot live just anywhere in the body but has its own designated place, then there is all the more reason to believe that it cannot survive outside the body in things which are always inanimate (earth, water, sun, fire). Lucretius is arguing against the notion that nature is somehow filled with divine attributes or sensation.

Book Five

tenuous, just as their bodies are. All this I will set out in a long discussion for you later on.¹ Moreover, to state gods wished, for the sake of human beings, to make the glorious nature of the world, and for that reason we should praise their work as something worthy of our commendation, thinking it immortal and eternal, and that it is at any time profane to use any force to shake from its seat what the ancient reasoning of the gods has set for races of human beings for all eternity, or to attack, using arguments, and overthrow it from top to bottom—to invent and add up all sorts of other things like this, Memmius, is ridiculous. For what benefits could our gratitude give blessed beings who live forever, so that they would try to accomplish anything on our behalf? Or when they were previously resting, what novelty could have attracted them to desire so long afterwards to change their earlier life? It seems clear that someone whom old things irritate should find delight in new things, but in the case of someone to whom nothing sorrowful has happened in times past, when he led a pleasant life, what could have set alight in such a one a passion for new things? Am I to think gods' lives lay immersed in grief and darkness until the origin of created things first dawned? And if we never had been made, what evil would that be for us? It's true that someone born must wish to stay alive as long as enticing pleasure holds him, but for someone who has never tasted the love of life, who has not been counted among the living, what harm would there be if he was not created? Furthermore, how was there first implanted in the gods some example of giving birth to things and that conception of human creatures, so that they knew what they desired to do and saw it in their minds? How did the gods ever learn the force of primary elements and what they could make with alterations in their mutual arrangements, unless nature herself presented the idea	GODS DID NOT MAKE THIS WORLD 220 [160] 230 240 250 [180] 260	[170] [170]
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¹ Lucretius never does deliver on this promise. He returns briefly to the gods later in this book (lines 1642 to 1646), but never clarifies precisely the nature of their material substance.

Book Five

of creating things?¹ There are so many
primary particles of things forced by blows
in many ways for endless lengths of time
pushed and driven along by their own weight—
these have grown accustomed to being carried, 270
to combining in every sort of way, [190]
and to trying out all possibilities
for producing things in mutual unions.
Thus, there is nothing strange about the fact,
if they have also fallen in those patterns
and have arrived at the type of movements
by which this grand totality of things
is now sustained and constantly renewed.
Even if I did not already know
what primary particles are, nonetheless, 280
from the very workings of the heavens
I would venture to point out and to insist
from many other facts there is no way
the nature of things has been made for us
by the work of gods, for it possesses
such enormous flaws. First, of all the space
which the huge expanse of heaven covers [200]
part is taken up by greedy mountains
and forests of wild beasts, deserted pools
and rocks have taken over, as has the sea, 290
which keeps the coasts of different areas
so far apart. Besides, almost two thirds
is stolen from mortal men by scorching heat
and falling frost which never goes away.
As for what is left for farming, nature
with her own force would even cover that
with shrubs, if the strength of human beings,
to make life possible, did not fight back
against it, once men had grown accustomed
to groan over strong hoes and carve up earth 300
by leaning on the plough. If we did not turn
productive lumps of earth with our plough shares [210]
and cultivate earth's soil and make things grow,
they could not spring up in the flowing air,
not on their own. And even then, sometimes
when things now achieved with laborious work
come into leaf and all of them are blooming
through the land, either the sun in heaven
shrivels them with excessive heat, or else
sudden rains and chilling frosts destroy them, 310
and with a violent storm the blasting winds

¹ Lucretius here seems to be assuming that gods are incapable of imagining or coming up with anything entirely new. And, unlike some later thinkers influenced by this poem, Lucretius does not link the gods with the rules by which nature proceeds, making them the creators of a world which operates on material principles which they have established (one common way of linking a scientific understanding of the universe with religious faith), nor, as he goes on to say, does he see in the way nature works any evidence of a divine design.

Book Five

inflict great damage. Then, why does nature
nourish and foster horrible species
of wild beasts hostile to the human race
on land and sea? Why do annual seasons [220]
bring sicknesses? Why does death stalk around
before his time? And there's the child, as well—
like a sailor tossed up from cruel waves,
he lies there naked on the ground, speechless,
needing every help to go on living, 320
once nature brings him through his mother's pain
out of her womb to regions of the light,
and he fills the space with tearful wailing,
as is fitting for someone who is waiting
to live through so many distressful things.
But different flocks, herds, and wild animals
grow and have no use for baby rattles,
they do not require some fostering nurse
to utter gentle broken words to them, [230]
nor do they seek different clothing to suit 330
the seasons of the sky, nor do they need
weapons or lofty walls to guard their own,
since earth herself brings forth abundantly
all things for all of them, and nature, too,
that skilful artisan.

Now, to resume. DEPLETION AND RENEWAL
OF MATTER
Since the body of the earth and water
and pleasant breaths of air and searing heat,
from which we see this sum of things is made,
are all made up of matter which was born
and which will die, we must accept the fact 340
that the whole nature of the world consists
of similar substances.¹ For obviously
with things whose parts and members we perceive [240]
are produced from a body which was born
and from mortal natures, these very things
without exception we see as mortal
and, at the same time, as being born, as well.
Thus, since I see the chief parts and portions
of the world are consumed and then reborn,
I may be certain that, in the same way, 350
there has also been for heaven and earth
a certain moment when they first began
and there will be a moment when they die.
And in case you think that in this matter
I stole that point for my own purposes,
because I have assumed that earth and fire

¹ Lucretius here returns to the argument he originally announced about the formation of the world, ending the digression which begins on line 110 of the Latin. The opening phrase he uses "first of all" (*principio*) has no connection with the verses immediately preceding this new section; hence, that phrase has been changed in the English text above to "Now, to resume."

Book Five

are mortal and have not shown any doubt
that air and water die and have stated
that these same things are born and grow again, [250]
first of all, some parts of the earth, when baked 360
by constant sunshine and trampled over
by the force of many feet, give off haze
and flying clouds of dust, which gusting winds
disperse all through the air. And furthermore,
rain removes part of the soil in flooding,
and rivers graze upon and chew away
their banks. Then, too, whatever nourishes
something else is, in its turn, replenished,
since we do understand, without a doubt,
that earth, universal mother of things, 370
is at the same time their common graveyard.
Thus, you see that earth is eaten away,
then once more increases and grows larger. [260]
And furthermore, it needs no words to show
that seas, streams, and springs are always filling
with new moisture and that waters well up
all the time. Great downward flows of water
from every region make that clear enough.
But surface liquid is taken away
continually—and so it comes about 380
that, in the end, there is no excess water,
in part because strong breezes, as they blow
across the seas, diminish them and rays
of the aetherial sun draw moisture off,
in part because it is distributed
below the ground in every land. The salt
is filtered out, the liquid stuff runs back,
gathers at the head of every river, [270]
and then, in a fresh current, flows again
over the land, along the river beds 390
which, once hollowed out, have carried waters
on their liquid march downstream.

Now I will speak
of air, which every single hour changes
in its entire body in countless ways.
Whatever flows from things is all carried
all the time into the enormous sea of air,
and if it did not, in its turn, give back
material to things and restore them
as they flow off, all things would already
have been eroded and turned into air. 400
Thus, it never stops being made from things
and going back to things, since, as we know,
all substances flow off incessantly. [280]
In a similar way, that plentiful source
of pure light, the aetherial sun, constantly

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inundates the sky with fresh-born brilliance
and instantly supplies the place of light
with new light, for every flash of brightness
which comes before, no matter where it falls,
is lost. You may learn this from what follows. 410
As soon as clouds first start to move across
below the sun and, as it were, to break
the rays of sunlight, all their lower part
immediately perishes, and earth
is cast in shadows everywhere those clouds
are carried. Therefore, you can understand [290]
how things continually need fresh brightness
and all the previously projected light
disappears. There is no way that we can see
things in sunlight, unless the fountain head 420
of light itself constantly supplies it.
Besides, you also see night lights on earth,
like hanging lamps and resinous torches
bright with fluttering fires, similarly strive,
in great darkness, assisted by their flames,
to supply new light, keen to keep their blaze
still flickering, so eager that the light
is not broken off and absent anywhere—
that's how fast its destruction is concealed [300]
by rapid birth of flames from every fire. 430
Thus, we must accept that sun, moon, and stars
in the same way give off light from sources
which rise up and are steadily renewed
and always lose all their earlier flames,
just in case you should happen to believe
that these continue on without being damaged.¹
And then do you not see that even rocks
are overpowered by time, high towers
fall in ruins, stones crumble, images
and shrines of gods decay and fall apart, 440
and that the power of gods cannot extend
limits set by fate or struggle against [310]
laws of nature? Besides, do we not see
ruined monuments of men [still asking,
on their behalf, if you believe these men
ever could grow old] and granite rocks torn
from soaring mountain slopes come crashing down,
unable to stand up against and bear
the overwhelming force of finite time?
For surely they would not be torn away 450
and fall so suddenly, if they withstood
from time immemorial all blows of age

¹ This rather awkwardly expressed example is part of Lucretius's argument to show that the world is constantly changing, with material always shifting around and being used up. There is nothing permanent or lasting about light, since it requires the constant use of new matter.

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and never cracked.¹

And then look at the sky,
which overhead and all around contains
all earth in its embrace. If it gives birth,
as some maintain, to all things from itself [320]
and takes them back once they have been destroyed,
then, in its totality, it was born
and possesses a body which will die,
for whatever increases and sustains 460
other substances out of itself must
be diminished and, when it takes things back,
must be replenished.

Then, too, if there were EARTH IS NOT VERY OLD
no moment of birth for earth and heaven
and they had always been here forever,
why, apart from the tearing down of Troy
and the Theban War, have other poets
not sung of other happenings as well?
Why have so many of men's achievements
so often disappeared? And why are they 470
not celebrated anywhere, embossed
on monuments of everlasting fame?
Well, in my opinion, the truth is this—
the entire universe is not that old, [330]
the nature of the world is new, as well;
it did not begin all that long ago.
And that is why certain arts, even now,
are being refined, even now still growing.
In recent years many innovations
have been made in ships, just a few years past 480
musicians gave birth to tuneful harmonies,
and only lately has this reasoning,
this nature of matter been discovered,
and I, the very first able to turn it
into my native tongue, have only now
been found. But if you happen to believe
that all things that existed earlier
were the same as these, but generations
of human beings died in scorching heat,
or that, by some great world-shattering act, 490 [340]
cities have collapsed, or that constant rains
made rapacious rivers move across the earth,
overwhelming towns, then so much the more
you must yield, conceding that earth and sky
will collapse as well. For when such great ills
and such major dangers were battering things,

¹ A corrupt line (line 312 of the Latin) has been emended by Munro, who points out that Lucretius is being sarcastic here. The monuments are asking the observer if he thinks it is possible for the memory of these men to disappear, and yet the monuments themselves are in ruins and will soon be gone.

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at that time they would have gone to ruin,
with massive devastation far and wide,
if a more disastrous cause had fallen
on them. And we can see that nothing else 500
shows that we are mortals more than this point—
we all get sick from the same diseases
as those whom nature has removed from life. [350]
Then, too, all objects which last forever
must either possess a solid body,
repel blows, and not let any substance
penetrate inside them that could loosen
the close-packed parts within—like those bodies
of things whose nature we discussed before—
or they must be able to carry on 510
for all time because they are not exposed
to blows, like empty space, which stays untouched
and does not suffer the slightest impact,
or because there is insufficient room
around them into which material could,
so to speak, move out and then be dissolved— [360]
just as the grand totality of all things
is eternal, for there exists no place
outside it where substances may split off,
and there are no objects which could hit them 520
and pulverize them with a mighty blow.
But I have shown the nature of the world
is not solid matter, since empty space
is intermixed in things, and it is not
like vacant space. In fact, there is no lack
of bodies which could perhaps assemble
out of infinite space and overwhelm
this sum of things with a violent whirlwind
or bring in some other dangerous hazard.
And there is no lack of natural places 530 [370]
or room in the abyss of space in which
the bulwarks of the world could be dispersed.
Or else things could be attacked and perish
from whatever other violence you wish.
And thus death's door is not kept shut against
heaven, or sun, or earth, or deep waters
of the sea, but stands ajar, facing them
with massive gaping jaws. And that is why
you must grant these same things were also born.
For objects which have a mortal body 540
could not have defied the powerful force
of boundless age for such an infinite time
up to the present day.

And furthermore,
since the most important portions of the world [380]
fight so much among themselves, incited

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to unsanctioned internecine warfare,
surely you see some limit could be set
to their lasting enmity—for instance,
when the sun and all its heat have drunk up
all water and prevailed?¹ They are striving
to achieve this, but have not yet won out
in their attempt—rivers supply so much
and threaten to do more, to flood all things
from the deep gulf of the sea. All in vain.
For winds, as they blow across the waters,
reduce them, as does the aetherial sun,
whose rays unweave their fabric—sun and wind
are confident they can dry everything
before the waters can achieve the goal
of their endeavours. Both sides manifest
such great enmity in their equal fight,
as they battle each other to decide
this mighty issue. Still, once fire prevailed,
and once, so they say, water ruled the fields.
For fire triumphed, consuming and burning
many things, when the rapacious power
of the horses of the sun charged off course,
carrying Phaeton through the entire sky
and past every land.² But roused to fierce rage,
the omnipotent Father quickly hurled
high-spirited Phaeton from his horses
down to the ground with a bolt of thunder.
Sun met him as he fell and took from him
the world's enduring light, then pacified
the scattered horses, and, as they trembled,
put them in harness. He led them from there
on their proper path and restored all things.
That, at least, is what old Greek poets sang,
although it is extremely far removed
from proper reasoning. Fire can prevail
when it can gather more materials
out of infinite space, and then its force
grows smaller, overpowered in some way,
or its materials are consumed, burnt up
by scorching air.³ In the same way, water,
so people say, once gathered and began
to win the battle when it overwhelmed

BATTLE BETWEEN WATER
AND FIRE

550

[390]

560

570

[400]

580

[410]

¹ The war between the different part of the earth is “unsanctioned,” as Smith observes, because the combatants are all part of the same world (i.e., their strife is like a civil war).

² Phaeton, in Greek mythology, was the son of Helios, god of the sun (Lucretius uses the name of the old Roman god of the sun, Sol). Phaeton tried driving the sun's chariot and horses on their usual route across the sky but lost control. As a result, the sun came too close to the earth, burning it and creating deserts. In order to save the earth, Zeus had to destroy Phaeton with a thunderbolt.

³ Lucretius seems to be conceding that there may have been a devastating fire, of the sort the Phaeton myth describes, but the only true explanation is a physical one: fire needs material fuel in order to burn and, once that fuel is used up, the fire goes out, unless it has been put out in some other way first. He goes on to make a similar concession with the well-known myth of the great flood.

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many human cities. Then, once the force
which had collected from limitless space
was somehow turned aside and ebbed away,
the rains stopped, the rivers' force diminished.

590

However, I will now set down in order
the ways in which assembled materials
laid foundations for the earth and heaven,
the ocean depths, and paths of sun and moon.

FIRST DEVELOPMENTS IN THE
FORMATION OF THE EARTH

For clearly primary elements of things
did not organize themselves, with each one
in position according to some plan
or some perceptive mind. And obviously

[420]

they did not enter into an agreement
about the motions each of them should have.

600

But the numerous first particles of things
have been driven by blows of many kinds
from time immemorial, moved forward
by their own weight, and have grown accustomed
to being carried, to form combinations
in every sort of way, and to attempt
everything they could possibly create
by mutually uniting, and therefore

it comes about that, by being spread around
for such a long time and by trying out
every sort of movement and arrangement,

610

at last those particles come together
which, suddenly combined, often become
the beginnings of great things—earth and sea,
heaven, and the race of living beings.

[430]

At this point, then, the sun's high soaring disk
with its abundant light could not be seen,
nor could the stars of this enormous world,
or sea, or sky, or even earth and air.

620

There was nothing to observe similar
to what we have now, but only some sort
of a new storm and shapeless mass arising
from primary elements of every kind,
whose disorder was a battle being waged,
which disturbed their internal passageways,
connections, weights, impacts, collisions,
and motions, since, given their different forms
and various shapes, they could not all remain
joined the way they were or meet together
and set mutually harmonious movements.

[440]

630

Then parts began to separate, and things
joined up with other things just like themselves,
dividing up the world, partitioning
its component parts and sectioning off
the major portions. That is, they set earth
apart from lofty heaven and the sea

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off by itself, so its waters could spread
in their own separate place. In the same way,
they placed the aether's fires in their own spot, 640
uncontaminated all by themselves.

For at first all the substances of earth,
being heavy and closely linked, gathered [450]
in the middle, and all of them took up
positions lower down. The more they mixed
and interlocked, the more they forced away
material stuff which would produce the sea,
stars, sun, moon, and the walls of the huge world.
All these are made from smooth, round elements,
much smaller than the particles of earth. 650
So in parts of earth the fiery aether
first burst out through porous openings, rose up,
and, being light, carried away with it
many fires, in a way not so different
from what we often see when golden sunlight [460]
first blushes on turf glittering with dew
in early morning and when pools of water
and always-flowing rivers exude mist—
just as we sometimes perceive earth itself
give off steam. When all these materials 660
gather overhead, bodies of clouds form
high up, weaving their web beneath the heavens.
Thus, in this way the light, diffuse aether
its body now cohering, was then stretched
all around and curved in all directions,
spreading far and wide in every region
on all sides, and in this process embraced
all other things in its voracious grip. [470]

Then there followed the first developments
of sun and moon, whose spheres move through the air 670
between earth and aether. They were not drawn in
by earth or lofty aether—for they lacked
sufficient weight to sink down and settle
and were not light enough to float along
through the highest regions. Still, they are set
between the two in such a way they turn
their lively bodies and exist as parts
of the whole world, just as in our bodies
certain limbs may remain in place at rest,
while there are other ones which move about. 680

With these substances removed, earth at once [480]
sank down to where the sea's vast blue surface
now stretches, and a flood of brine immersed
the trenches. And then every day, the more
encircling aether's currents and sun's rays,
with their repeated blows on every side
DEVELOPMENTS IN
EARTH AND SEA

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along earth's outer edges, compressed it
into a dense mass, so with this pounding
earth became closely packed and collected
around its centre, the more salty sweat 690
squeezed from its body, as it trickled out,
enlarged the ocean and fields of water,
all the more those many particles of heat
and air escaped by flying away, making
the high glittering spaces of the heavens, [490]
far away from earth, more dense. Fields sank down,
the height of soaring mountains grew, for rocks
could not subside, nor could all parts move down
to the same level equally.

And thus, THE AETHER
the heavy, solid body of the earth 700
was produced, and all the world's heavy sludge,
as it were, slid down to the lowest point
and settled on the bottom, just like dregs.
Then sea, then air, then fiery aether itself
were all left pure and unmixed substances,
some lighter than the others. The aether,
purest and lightest of all, floats above [500]
the airy breezes, and its clear matter
does not join with gusting currents of air.
It lets all matter underneath be stirred 710
by tempestuous whirlwinds, allowing them
to be upset by shifting storms. It bears
its own fires itself as it glides ahead
in its unvarying forward motion.
That aether can keep flowing evenly
with one steady effort, the Black Sea proves,
for it moves with an unchanging current,
and, as it flows on, constantly maintains
an uninterrupted single motion.¹

Let us sing now the causes of motion 720
of the stars.² First of all, if the great sphere
of heaven rotates, then we must conclude [510]
an air presses its axis at both poles
and confines it from outside, closing it
at either end, and then another air
flows above, moving in the same direction
in which the stars of the enduring world
turn as they go through their twinkling motion, MOTION OF THE STARS

¹ This is a reference to the steady flow of water towards the Hellespont, something reported on later by Pliny the Elder, and picked up from there (in Holland's English translation) by Shakespeare: "Like to the Pontic sea,/ Whose icy current and compulsive course/ Ne'er feels retiring ebb, but keeps due on/ To the Propontic and the Hellespont. . . ." (*Othello*, 3.3).

² This passage on the movement of the stars, as many editors have observed, seems out of place, since it interrupts the description of how the world developed.

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or another current of air below
flowing in the opposite direction 730
makes the sphere rotate, just as we perceive
streams turning waterwheels and scoops.¹ Then, too,
all heaven could also remain in place,
while bright constellations are borne ahead,
either because swift currents of aether
are enclosed and, as they seek an exit, [520]
move around and thus make their fires revolve
everywhere in open spaces of the night
throughout the sky, or a current of air
from somewhere else, from some external place, 740
makes the fires turn, or they can creep along
on their own, to whatever place their food
summons each of them, as they move around,
inviting them to feed their fiery mass
in every region of the heavenly sky.²
As for which of these causes is at work
in this world, it is hard to say for sure,
but what could happen and what does happen
throughout the universe in various worlds
formed in various ways—that is what I teach, 750
and I proceed to set down several causes,
which could account for motions of the stars [530]
throughout the universe. Of these, however,
there must also be one which in this world
is the cause which generates the motion
of the constellations. But to declare
which one of them does this is not the task
of any man proceeding step by step.³

Now, in order for earth to stay at rest
in the world's central part, it is fitting
that its weight should gradually get smaller
and decrease underneath, that it should have,
from the very start of its existence,
another nature down below, interlinked
and closely joined with those airy regions

EARTH IS AT REST

760

¹ Lucretius considers here different possible explanations for why the stars move. The first idea is that the world (i.e., our part of the cosmos), which is spherical, moves like a giant water-wheel, with a fixed axis held in place by the pressure of air, which is then turned by another current of air from either above or below. The lower current will be in a direction opposite to the movement of the upper portion of the circle (as in a waterwheel).

² As Lucretius has remarked more than once, since the stars are fires, they require a constant supply of fuel.

³ Lucretius here acknowledges an important principle for him. He has already stated that whatever the senses confirm is true and whatever the senses contradict is false. However, theories which seek to explain natural phenomena are all equally true unless they are denied by sense experience. Even though in this world there may be only one cause, in a different world the same natural event might happen for a different reason. Hence, his task is not to determine one single explanation in cases where different accounts all agree equally well with sense experience, as in the discussion of the four possible causes for the motion of the stars. This point helps to underscore the priority Lucretius gives to sense experience rather than to a single theoretical explanation of that experience. After this short discussion of the motion of the stars, Lucretius returns to the formation of the earth.

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of the world in which it is placed and lives.¹
Thus, it is not a burden and does not
weigh down the air, just as in every man [540]
his own limbs do not weigh him down, his head
is no burden to his neck, and, in fact, 770
we do not sense that all our body weight
rests on our feet. But any loads imposed
on us from outside are painful to us,
though often they are a great deal smaller.
That shows how crucial it is what each thing
is capable of doing. Hence, earth is not
a foreign object suddenly brought in
or thrown from somewhere else into strange air,
but was conceived together with that air
at the world's original creation 780
and is a fixed part of it, just as we see
our limbs are part of us. Moreover, earth,
when suddenly shaken by loud thunder, [550]
with its motion makes all things above it
shudder, and this it could not do at all,
if it were not linked to airy regions
of heaven and the world. For these places,
given their common roots, merge together,
combine, and form into one entity
from the very start of their existence. 790
Do you not also see that no matter
how much our body weighs, force in our soul,
which is very tenuous, supports it,
because soul is so closely joined to it
and with it forms a single unity?
Finally, what can lift the body up
with an agile leap except force of mind, [560]
which controls the limbs? Do you now perceive
how much influence a delicate substance
can have, when joined to a heavy body, 800
the way the air is interlinked with earth
and force of mind with us?

And sun's disk and fire
cannot be much larger or smaller
than they seem to our senses. For with fires,
from whatever distance they are able
to throw off their light and breathe out their heat
onto our limbs, they lose nothing material
from their flames in the intervening space,
and the appearance of the fire does not

SIZE OF THE SUN
AND MOON

¹ As Munro notes, Lucretius does not here mention the overall shape of earth, but these remarks suggest that he thinks of it as a having flat surface above and below. Its material gradually gets less dense under the top surface, so that on the bottom the material merges or becomes one with the air below (Bailey uses the image of a spring mattress to describe the idea). This phenomenon keeps earth in place because it forms an almost organic entity with the material below, as the word "lives" and the following analogy to the human body suggest (although elsewhere Lucretius is insistent that the earth is not a living creature).

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get any smaller.¹ Thus, since the sun's heat
 and the light it pours out reach our senses
 and caress the regions on which they fall,
 the shape of the sun and its size, as well,
 when we look at them from earth, must be seen
 in their true dimensions, so you cannot
 change it in any way to enlarge it
 or make it smaller. Whether the moon, too,
 as it is moved forward, shines on places
 with light from some other spot or throws off
 a specific light from its own body,
 whatever the case, it is borne along
 with a shape not one bit larger than the one
 we recognize when we look up at it
 with our own eyes. For everything we see
 from far away through a great deal of air
 seems blurry in appearance before its size
 gets smaller. Therefore, since the moon presents
 a bright face and a clearly outlined shape
 from here on earth, we must see it high up
 just as it is formed by its outer edge
 and exactly the same size. Finally,
 since with every fire we observe on earth,
 as long as its bright light is clear to see
 and we feel its heat, we see that its size
 sometimes changes very little either way,
 depending how far distant it may be,
 we may know that all those aetherial fires
 we can observe from here on earth could be
 an extremely minute fraction smaller
 or larger to a slight and small degree.
 Also there is nothing astonishing
 the way the sun, which is itself quite small,
 can send out such great quantities of light
 that it completely inundates all lands,
 seas, and heaven, and washes everything
 in its warm heat. For it may be the case
 that from the sun a single fountainhead
 opens up for the entire world and flows
 in large quantities, shooting out its light,
 because elements of heat gather here
 from every side of the whole world—this mass
 of particles streams forth in such a way
 that in this place the heat comes flowing out
 from just one single source.² Do you not see

810 [570]

820

[580]

830

[590]

840

850

[600]

¹ Bailey (along with many others) notes the curiosity of these statements about how the size of fires does not apparently change with distance and the inference that the sun and moon must be more or less the same size as they appear to be when we look at them from the earth. Copley refers here to "the great central weakness of Epicureanism, its total lack of mathematics. . . ." Lucretius is, of course, relying upon his basic claim that the senses do not deceive us; hence, the celestial fires must be more or less the same size as they appear to us because they are so clear and distinct. Still, the logic does seem strange.

² Line 596 in the Latin has been omitted. It is the same as line 584.

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how a small spring of water also spreads
far and wide over meadows and sometimes
floods the fields? Then, too, it may also be
that heat from the sun's fires, although not great,
with their warm fiery blaze fills up the air,
if the air which happens to be present 860
is combustible and sufficient, so that,
when struck by tiny particles of heat,
it can catch fire, just as we sometimes see
all parts of a field of crops or stubble
caught in a huge fire from a single spark.
Perhaps also the sun shining on high
with his rosy torch has there around him [610]
large amounts of invisible hot fire,
which does not display any light at all,
so that it brings only heat and strengthens 870
the impact of the rays considerably.¹

There is no plain, direct explanation MOTION OF THE SUN AND MOON
which clarifies how the sun moves forward
from his position in the summertime
to Capricorn, his winter turning point,
and how, coming back from there, changes course
to the solstitial point in Cancer, or how
the moon is seen within a single month
to traverse that distance, which takes the sun
a period of one whole year to cross. 880
No plain reason, I say, has been given [620]
for these phenomena.² For first of all,
it appears that what could happen is what
the revered thinking of great Democritus
proposed—the closer each constellation
is to earth, the less it can be carried
by heaven's whirling winds, for lower down
their swift, keen force gets less and disappears,
and so the sun is gradually left behind
among the constellations at the back, 890
because it is at a much lower height
than those fiery signs.³ The moon even more—
the lower her path, the further she is
from heaven, and the closer to the earth, [630]
the more incapable she is of keeping
her course level with the constellations.
Also, since moon is lower than the sun

¹ The third possibility is the notion of invisible heat. The area around the sun might produce heat without our being able to see any flames; hence, the fact that the sun presents the appearance of a small burning disk is less important. Munro calls attention to modern scientific parallels to this passage.

² As Bailey points out, Lucretius is here attempting to account for two motions of the sun, its annual circuit in which it moves through the constellations, and its movement up and down in its daily orbit around the earth. His explanation, Bailey observes, is somewhat confused because he offers his reasons for these two different phenomena as alternatives, rather than as two explanations for two different features of the sun's movement.

³ The constellations (sometimes called signs) are, of course, the signs of the zodiac.

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and the whirling wind which bears her onward
is less energetic, the more all signs
catch up all around and overtake her. 900
Thus, it happens that we observe the moon
coming back to every constellation
faster than the sun, since those signs move up
more swiftly to her. It could happen, too,
that from those regions of the world which cross
the pathway of the sun two air currents
may alternately stream, each one of them
at a specific time, so one of them
could push the sun away from constellations
of the summer down to the turning point 910
of the winter solstice and freezing cold, [640]
and one may thrust him back from cold darkness,
all the way to the heat-bearing regions
and the fiery constellations. Likewise,
we must assume that alternating airs
from opposite regions could shift the moon
and those stars which move in massive circles
for thousands of normal years.¹ Surely you see
how contrary winds also blow the clouds,
moving the upper and the lower ones 920
in different directions? Why should the stars
be less capable of being carried
through their huge orbits in the aether
by currents pushing them in different ways?
Further, night shrouds earth in murky darkness, [650]
either when sun, after his long passage,
comes to the most distant parts of heaven
and, in his exhaustion, blows out his fires,
shaken in their journey and made weaker
by large amounts of air, or else because 930
the same forces which carried the sun's orb
above the earth compel it to change course
and move below the earth.

In the same way,
the goddess of the morning, Matuta,
at a certain moment sends rosy Dawn
through aetherial regions and spreads her light,
either because that same sun, returning
back under the earth, keen to set the sky
blazing with his rays, seizes it too soon,
or because at a particular time 940 [660]
fires collect and many heat particles

¹ To describe the time of these orbits, Lucretius uses the term "great years," which, as mentioned before, is a time equivalent to many hundreds or thousands of solar (or sidereal) years (the exact number was disputed). Cicero (in *De natura deorum*, 45 BC) defined the Great Year as follows: "On the diverse motions of the planets the mathematicians have based what they call the Great Year, 'which is completed when the sun, moon and five planets having all finished their courses have returned to the same positions relative to one another. The length of this period is hotly debated, but it must necessarily be a fixed and definite time.'"

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by habit flow together, and these cause
new sunlight constantly to be produced.
And therefore, from Mount Ida's lofty peaks,
so people say, one can see scattered fires,
as sun's light rises, and then they gather,
as it were, in one ball and make a sphere.
Nor in these matters should it surprise us
that at a predetermined time these seeds
of fire can stream together and renew 950
the brilliance of the sun. For we observe
with all things many events occurring
at set times—trees blossom at a set time, [670]
and at a set time they shed their flowers.
At a time no less firmly fixed, our age
instructs our teeth to fall and the young lad
to acquire the soft hair of puberty
and let a tender beard grow equally
down both his cheeks. Moreover, lightning, snow,
rain, clouds, winds—these occur at times of year 960
which we can surely more or less predict.
Since from the first origin of causes
that has been the case, and things have happened
this way from the beginning of the world.
They also come back now in a fixed order,
one following the other.

Similarly,
days may grow longer and nights get shorter, [680]
and daylight diminish, while nights increase,
either because the same sun, as it runs
in different circuits above and below 970
the earth divides the aetherial regions
and splits the sphere into unequal parts,
and when he takes from one of the two parts
he adds to the other the same amount,
as he moves around, until he reaches
that constellation in the heavenly sky
where the yearly node makes the shades of night
equal the light of day.¹ For as sun moves
into the middle of the blasts of wind [690]
from north and south, heaven keeps his two goals— 980
[points where sun rises and then later sets]—
at equal distances, given the placement
of the whole orbit of the constellations,
which the sun, as he glides around, completes
in one full year, lighting the earth and sky
with his slanting rays, as is clearly shown
by the plans of those who have noted down
all those places in the sky which are marked

¹ The "yearly node" is the equinox, which occurs twice a year when the path of the sun's annual movement crosses the earth's equatorial plane.

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by the sequence of the constellations.¹
 Or then again, since the air is denser 990
 in certain regions and below the earth
 sun's tremulous rays of fire are therefore
 held back and cannot easily break through
 and move toward the place where dawn appears,
 in winter time long nights keep dragging on,
 until the bright signal of day arrives, [700]
 or because in alternating seasons
 of the year, fires which make the sun arise
 in a certain region of the heavens
 have a habit of streaming together 1000
 more quickly or more slowly, and therefore,
 it happens that men seem to speak the truth
 [when they claim a new sun is born each day.]²
 And the moon could shine because she is struck
 by rays from the sun and day by day turns
 that light more towards our sight, as she moves
 further from sun's sphere, until she is placed
 across from him, has shone her bright, full light,
 and, as she rises high, has seen him set.
 Then, in the same way, she must, as it were, 1010 [710]
 move back and gradually hide her light,
 the more she now glides close to blazing sun
 from a different region through the circle
 of the constellations, as those men claim
 who imagine the moon is a like a ball
 and stays on her path underneath the sun.
 There is also a way moon could revolve
 with her own light and show various phases
 of illumination. For there may be
 another body which is borne forward, 1020
 glides with her, and in all sorts of ways blocks
 and obscures her. This cannot be observed,
 because it moves on without light. Or else
 the moon might spin round, perhaps something like [720]
 a spherical orb flooded with bright light
 on half its surface, and, by revolving,
 the sphere manifests its various phases,
 until, turned to our watching, open eyes,
 it reveals that part which is all burning,
 and then gradually turns back, withdrawing 1030

¹ The first part of this sentence is confusing and its meaning has been disputed. Is Lucretius talking about the annual orbit of the sun through the cosmos or about its daily rotation around the earth? In the first case, the "two goals" would be the solstices, and the sentence would mean (as Munro points out) that when the sun is midway between the solstices it is midway between the solstices; in the latter, the "two goals" would be the rising and setting of the sun, and the sentence would mean that when the sun is midway between the solstices day and night are equal in length. I have followed Munro's suggestions, and, in order to clarify the passage somewhat, have added the line "points where sun rises and then later sets" in square brackets.

² The words in square brackets are Bailey's suggestion for a line which appears to be missing. The final explanation for why some days are shorter or longer than others assumes that the sun is remade each day. The other two assume that the sun passes below the earth during the night. Once again, Lucretius offers a selection of theories but does not adjudicate among them, since they all satisfy our sense experience.

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that portion of its sphere which gives us light,
as Babylonian doctrines of Chaldeans
attempt to prove, when they contest those claims
astronomers have made and deny them,
as if what both of them are fighting for
could not be equally right, or there were
some reason why you might desire to take [730]
one explanation rather than the other.

And then again why could not a new moon
always be produced every single day, 1040
with a preset sequence in her phases
and fixed shapes, so that each created moon
disappears each day, and then in its place
another is formed? This is hard to prove
by reasoning or demonstrate in words,
but [you see] so many things created
in a certain sequence.¹ Spring and Venus
walk along, with Venus's winged herald
marching on in front, and Mother Flora,
right beside the footsteps of Zephyrus, 1050
strews the whole road in front of them, spreading [740]
the finest colours and scents.² Next in line,
come scorching Summer and her companion,
dusty Ceres, with the yearly breezes
of the northern winds. Then Autumn follows,
and inspired Bacchus walks along there, too.³
Then come other storming winds and tempests—
loud roaring Voltumnus as well as South Wind,
whose power is lightning.⁴ And finally
the solstice brings on snow and fetches back 1060
numbing cold. Winter follows with the frost
that makes teeth chatter. It is not so strange,
therefore, if moon is born at a fixed time
and at a fixed time is once more destroyed,
for many things occur at preset moments. [750]

You must assume for similar reasons ECLIPSES
that eclipses of sun and moon, as well,
can be brought about from several causes.
Why should the moon be able to close off
earth from the sun's light, placing her high head 1070
in front of him in line with earth, hurling
her dark sphere before his burning rays,

¹ The words in square brackets are commonly added to the text to make better sense of the sentence. And I have changed the conjunction from since to but, in order to clarify the logic of the argument.

² As many editors point out, this passage seems to be a description of an illustration or a pantomime of some sort. Zephyrus is the west wind, normally the gentlest and most welcome of the winds; Flora is the Roman goddess of flowers.

³ Lucretius uses the Latin *Eubius Euan*, a phrase denoting Bacchus, god of wine and the grape harvest. Ceres is the goddess of grain crops.

⁴ Voltumnus is a river god, but the name is often conflated or confused with Vultumnus, one of the wind gods.

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yet at the same time we should not believe
 some other body which always moves on
 without being lit up could not do the same?
 Why could the sun, at a certain moment,
 not grow sluggish and lose his fires and then,
 as he moves through the air, renew his light,
 after he has passed beyond those places [760]
 which act against his flames and cause his fire 1080
 to be put out and die? And why should earth,
 in turn, be able to deprive the moon
 of light and, in addition, block the sun
 above her, while moon, in her monthly course,
 glides through the hard-edged shadows of the cone,
 and at the same time some other body
 not be able to move beneath the moon
 or slide above sun's sphere, to intercept
 sun's rays and the light he sheds?¹ Moreover,
 if the moon really shines with her own light, 1090
 why, as she is passing through those places
 hostile to her light, could she not grow dim
 in a particular region of the world?² [770]

As for what remains, since I have explained
 how all things can occur in the blue sky
 of this great world, so we could understand
 what forces and causes might bring about
 different courses of the sun and journeys
 of the moon, and how, with their light blocked out,
 they could be eclipsed and drape in darkness 1100
 the unsuspecting earth, when, so to speak,
 they wink and then open their eyes once more
 and look on every place with clear, bright light,
 I will now return to when the world was young,
 to the tender fields of earth, to those things YOUTH OF THE EARTH
 they chose, with their new creative power, [780]
 to raise first into regions of the light,
 entrusting them to the uncertain winds.

First of all, earth gave out types of grasses
 and splendid greenery around the hills 1110
 and over all the plains—the flowering fields
 shone a brilliant green. After that, in trees
 of various kinds great longing was unleashed
 to race up through the breezy air and grow
 unbridled. Just as feathers, hair, and bristles
 are first produced on limbs of quadrupeds
 and bodies of strong-winged birds, so new earth

¹ Bailey points out that Lucretius' ability to understand eclipses is severely hampered by his insistence that the sun and the moon are the same size as we observe them in the sky (i.e., much smaller than earth). In such an arrangement the "cone" of the shadow cast by the earth on the moon could not be formed.

² Line 771 of the Latin has been omitted. It is the same as line 764 of the Latin.

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then began by raising shrubs and bushes [790]
and after that created many tribes
of mortal animals, which were produced 1120
in numerous forms in every sort of way.
For living beings cannot have fallen
from the sky, and terrestrial creatures
cannot have come out of salt-water pools.
It then follows that earth has rightly earned
the name Mother, since all created things
exist from earth.¹ And many animals,
even now, are born from earth, taking shape
thanks to rain and warming heat of sunshine.
So it is less surprising if back then 1130
more creatures were born and they were larger
and matured when the earth and air were young. [800]
Firstly, the race of animals with wings,
the different birds, would move out from their eggs,
which hatched in springtime, just as nowadays
in summer cicadas leave their smooth shells
on their own, seeking life and sustenance.
At that time, you should know, earth first produced
tribes of mortal beings.² For in meadows
heat and moisture were plentifully supplied, 1140
and thus when any area appeared
which was appropriate, there wombs would grow
with roots attaching them to earth, and when,
in the fullness of time, the infants' warmth,
fleeing moisture and searching out the air, [810]
would open these, nature would turn the pores
within the earth to these spots, forcing them
to pour from their open veins a liquid
just like milk, the way every woman now,
when she has given birth, has much sweet milk, 1150
since all the current of her nourishment
is directed to her breasts. For the young,
the earth provided food, heat a garment,
and grass a place to rest, richly supplied
with plentiful soft down. But in its youth
the earth produced neither cruel freezing,
nor too much heat, nor very violent winds.
For everything acquires power and grows [820]
at the same time and to the same degree.³
Thus, I repeat, earth has justly received 1160

¹ In classical times the idea that the first human life was born in the earth was widespread; as Blundell puts it, "No other basic hypothesis, so far as we know was ever put forward in scientific philosophy" (quoted by Campbell).

² The cicada emerges from the ground in the summer heat, climbs up a plant stalk, and sheds its thin skin. This was taken by some as evidence of earth generating living things spontaneously. The birds' eggs mentioned, one assumes, were first produced by earth. There is some ambiguity about whether Lucretius sees a creation sequence, with human beings coming after birds or whether he sees the creation of animal life all occurring at the same time. Campbell insists that the emphasis is on simultaneous creation of animal and human species.

³ This sentence seems to mean that because the earth was young, therefore the various natural forces (wind, cold, and so on) were also young and weak.

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and keeps the name of Mother: she herself
 produced the animal and human races,
 pouring forth, almost at a preset time,
 all animals which run wild everywhere
 among huge mountains and, along with them,
 air-borne birds of various shapes.¹ But then,
 since she must reach some end of giving birth,
 she stopped, just like a woman exhausted
 by the passing years. For time does transform
 the nature of the entire world—all things
 must shift from one condition to another,
 and nothing keeps on going the way it is.
 All things move from where they are, and nature
 alters everything, forcing it to change
 to something else. For one thing rots away
 and, feeble with age, grows limp, and later,
 from a scorned condition another thing
 bursts forth and grows. And therefore, in this way
 age changes the nature of all the world,
 one state on earth is followed by another—
 so that what could bear life then now cannot,
 and what could not bear life before now can.²

CHANGES IN NATURE

1170

[830]

1180

At that time earth also strove to bring forth
 numerous monsters, produced with bizarre looks
 and limbs—hermaphrodites, intermediate
 types between the sexes, yet neither male
 nor female, remote from each, some creatures
 without feet or, then again, lacking hands.
 Some even had no mouth and turned out dumb,
 others, without eyes, were blind, still others
 were hampered by the way their limbs adhered
 to their whole body: they lacked the power
 to do anything, move anywhere, shun trouble,
 or obtain the things their needs demanded.
 Other monsters like this and prodigies
 kept being produced, but it was futile,
 for nature put a stop to their increase.
 They strove to bloom in full maturity
 but were unable to—they could find no food
 or unite in sexual reproduction.
 For we know many factors must combine
 so things can breed and propagate their race:
 first comes nourishment, and then sexual seed

MONSTERS

[840]

1190

1200

[850]

¹ The “preset time” refers to the youth of the world. The organic metaphor at work here in the description of the origin of living things, of the youth of the earth, and of earth as a birth mother is somewhat at odds with the notion of random, mechanical collisions and combinations as the events which create all things. Campbell notes that Lucretius appears to have a dual vision of earth in its early days: on the one hand, a procreative, soft, and caring mother and, on the other hand, a hard and cruel stage for the survival of the fittest.

² As Campbell points out, the sense here is that the earth once could produce all sorts of living beings which it cannot produce any more, and things which could not produce life at first (i.e., the newly emerging animals) now, through sexual reproduction, can.

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throughout the body must have ways to flow,
once limbs relax, and then, for the female
and the male to be able to have sex,
both must have organs which enable them
to share their mutual joy between them.¹

Back then many races of animals
must have died off—they could not procreate
and sustain their breed. For with all beings
you see breathing vital air, either craft,
or courage, or speed has kept them alive,
protecting their race from the beginning.

And there are many which commend themselves
to us by their usefulness and remain
entrusted to our care. Firstly, courage
has protected the fierce race of lions
and ferocious breeds, cunning saves foxes,
and swiftness to escape preserves the deer.

But light-sleeping dogs with trustworthy hearts
inside their chests, along with every race
produced from the seed of beasts of burden,
and woolly flocks, as well as breeds with horns—
all these beasts, Memmius, have been entrusted
to care of human beings. These creatures,
eager to run from savage animals,
sought peace and generous quantities of food,
which they get without working on their own
to find it, food we give as a reward
for their utility. But those whom nature
has not assigned these qualities, the ones
who cannot survive by themselves or give us
useful benefits, so that we would allow
their kind to feed and survive in safety
under our protection, these quite clearly,
all handicapped by their own lethal chains,
fell prey and spoil to others, until nature
led those races on to their extinction.

But there were no centaurs.² And animals
with a double nature, a dual body
assembled from limbs of different beings,
so that the powers in this and that part
could be sufficiently alike—such creatures
could not exist at any time. This fact
one can understand from what follows here,
no matter how obtuse one's mind may be.

SURVIVAL OF ANIMALS

1210

[860]

1220

1230 [870]

1240

COMPOUND ANIMALS

[880]

¹ Here (and in what follows) is an interesting anticipation of the rudiments of natural selection: nature produces a wide variety of types, and those which cannot support themselves or reproduce die out. The fittest survive because they have a physical advantage of some kind. Campbell notes, however, that we must not be too quick to see here an anticipation of Darwin's theories, in part because Lucretius has no sense of evolution and of the development of new species out of old ones. The production of these varieties took place only in the youth of the world.

² As mentioned previously, a centaur is a creature with the head and torso of a man and the body of a horse.

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First, a horse near three years old is full grown,
in its prime. A child is obviously not,
for often at that age, while in his sleep, 1250
he still seeks out his mother's milky teat.
Later, when a horse's vigorous power
and its strong limbs get weak in its old age
and, as vitality departs, grow frail,
at that time for the young man finally
the bloom of youth begins and coats his cheeks
with a soft down. So you cannot accept
centaurs could be created or exist, [890]
put together by chance from human beings
and load-bearing progeny of horses, 1260
or Scyllas, with bodies half sea creatures
enclosed by ravenous dogs, and all other
monsters of this sort, those whose limbs we see
do not match each other, for they do not
mature or acquire full bodily strength
or lose that to old age at the same time,
they do not burn with the same sexual fire,
do not share a single common habit,
and the same things are not pleasurable
throughout their bodies.¹ For you may notice 1270
bearded goats often grow fat on hemlock,
which is bitter poison to human beings.² [900]
Besides, since flame has a habit of singeing
and burning tawny bodies of lions,
as well as every kind of flesh and blood
living on the earth, how could it happen
that the chimaera, one single body
in three parts—with a lion in the front,
a snake at the rear, and in the middle,
as her name suggests, a goat—could spew out 1280
with her mouth fierce flame from her own body?³

And therefore, anyone who still believes
that when the earth was new and sky was young,
such animals as these could have been made
and rests his case upon mere novelty,
an empty term, may, using this reason,
let his mouth prattle on of many things— [910]
he may say that back then rivers of gold
flowed everywhere across the lands, and trees
used to bring forth jewelry for blossoms, 1290
or man was born with limbs of such great strength

¹ Scylla is a composite monster with six heads and dogs attached to the body living in the rocks in the straits between Italy and Sicily.

² The reference here is to satyrs, composite creatures which, in Roman times, were pictured as part human being and part goat.

³ The Chimaera is a legendary fire-breathing monster made up of three different animals. Chimaera is the Greek word for she goat.

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he could plant his footsteps across deep seas
and with his hands turn all heaven round him.
For in the era when earth first produced
living creatures, though there were in the ground
many seeds of things, that is still no proof
that compound beasts could have been created
and limbs of different animals combined.
For types of grasses, crops, and fertile trees [920]
which, even nowadays, grow up from earth 1300
in rich abundance still cannot be formed
into compound mutual creations,
but each arises in its own manner,
and by a predetermined natural law
all keep their characteristic features.
But that human race was much sturdier
EARLY HUMAN BEINGS
in the fields, as was natural for a group
the hard earth made. It was built up inside
from larger and more solid bones, attached
to powerful sinews through the tissues. 1310
They were not easily hurt by heat or cold
or new food or any bodily harm. [930]
And then through many circuits of the sun
rolling across the sky, they went through lives
of wandering, the way that wild beasts do.
There was no hardy farmer to manage
the curving plough, no one who understood
how to cultivate the fields with iron, or set
young plant seedlings in the earth, or cut off
old branches from high trees with pruning knives. 1320
What sun and rains provided, what earth made
all on its own—these gifts were sufficient
to satisfy their hearts. And their bodies
these men would replenish, for the most part,
among acorn-bearing oaks. At that time,
the earth produced wild strawberries, as well,
in huge quantities, larger than the ones
you now see in winter, as they ripen [940]
to a rich red colour. Then, too, the world,
in its blossoming youth, also gave them 1330
many coarse foods, enough to gratify
mortals in a wretched state. But rivers
and springs would call to them to quench their thirst,
the way that water now cascading down
large mountains clearly calls from far and wide
the thirsty races of wild animals.
And then, as they roamed around, they would stay
in the nymphs' familiar forest spaces,
where they knew that flowing brooks of water
washed slippery rocks with a generous stream, 1340 [950]
trickled on wet stones, and from up above
dripped down on verdant moss, and here and there

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burst out and flowed across the level plain.
Back then they did not know how to use fire
or to cover their own bodies with pelts
from wild animal hides. Instead they lived
in forest groves and mountain caves and woods,
sheltering their filthy limbs in bushes,
forced to avoid the scourging winds and rain.
They could not look toward the common good 1350
and did not know how to make for themselves
any laws or customs. A man would take
whatever prize fortune might throw his way, [960]
with each one trained to look out for himself
and get by on his own. And in the woods,
Venus would join bodies in sexual acts,
for each woman was either overwhelmed
by mutual lust, or by the violent force
and reckless passion of the man, or else
by some reward—acorns, or strawberries, 1360
or fine pears. And trusting in the power
of their hands and feet, which was amazing,
they went after wild beasts in the forest
by throwing rocks and with large, heavy clubs.
They brought down many, but there were a few
they avoided in their hiding places.
When night overtook them, they would settle
their naked, savage limbs down on the ground, [970]
like feral pigs, wrapping leaves and branches
all around them. Nor did they moan a lot, 1370
demanding daylight and the sun, wandering
the fields in terror through the shades of night.
Instead they stayed quiet, buried in sleep,
until sun with his rosy torch brought light
into the sky. For since, from childhood on,
they were used to seeing light and darkness
always being produced at alternate times,
it could not happen they would ever wonder
or feel apprehensive that the sunlight [980]
might be permanently withdrawn and then 1380
darkness would possess the land forever.
But what did give them more cause to worry
was that tribes of wild creatures frequently
made quiet rest unsafe for wretched men.
Driven from their shelter, they would run off
from their rocky home when a foaming boar
or mighty lion came too close—trembling
in the dead of night they gave up their beds
of piled up leaves to their ferocious guests.
Back then mortal beings would not have left 1390
sweet light of failing life in greater numbers
than they do now. True, any one of them [990]
was more likely to be seized and offer

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wild beasts a living meal, chewed by their teeth,
and would have filled the woods, groves, and mountains
with his screams, as he watched his living flesh
buried in a living tomb. And those men
who, with mangled bodies, had saved themselves
by running away would hold shaking hands
over ghastly wounds and later call out 1400
in horrifying cries for death, until
savage writhing pain took away their lives,
for they did not know how to help themselves
and were ignorant of what their wounds required.
But many thousands of men were not led
under army banners to their slaughter
in a single day, and stormy waters [1000]
of the ocean did not hurl ships and men
against the rocks. The sea would often rise
and rage in random, vain futility, 1410
then lightly set aside its empty threats.
The seductive charms of calm sea waters
could not lure any man to his destruction
with their deceptive, smiling waves, for then
the reckless art of seamanship remained
as yet unknown. Then, too, a lack of food
would deliver their weakened limbs to death—
and now, by contrast, an excess of things
destroys. Back then, men, in their ignorance,
would often pour out poison for themselves— 1420
and now more skilful men give it to others. [1010]

Then, once they had acquired huts, hides, and fire
and woman linked up with man and moved
into one [home and] learned [marriage customs],
and they saw themselves creating offspring,
at that point the human race first began
to soften.¹ Fire meant their freezing limbs
could no longer tolerate the cold so well
under heaven's roof, sexual habits made
their strength diminish, and children soon 1430
shattered the stern character of parents
with their endearing charms. And then neighbours
began to join in mutual agreements,
seeking not to harm each other or be harmed, [1020]
and they entrusted children and the race
of women to the care of all, pointing out
with vocal sounds, gestures, and broken words
that it was right for all to have pity
on the weak. And though they could not create
universal harmony, nonetheless, 1440
large numbers would faithfully keep their word,

¹ A line is evidently missing after line 1012 of the Latin. The words in square brackets provide the general sense.

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or else the human race would, even then,
have been entirely killed off, and breeding
could not have kept up their generations
to this very day.

But nature drove men	ORIGIN OF LANGUAGE
to use their tongues to send out various sounds, and convenience then brought in names for things, in much the same way we see a failure	[1030]
to use their tongues for speech pushes children to gestures, when it makes them indicate with their fingers objects in front of them.	1450
For all animals sense how they can use their own abilities. Before horns emerge and sprout on a calf's forehead, it uses them to butt when angry and charges furiously. Panther cubs and lion whelps use their claws, feet, and teeth to fight, when their teeth and claws are still hardly formed. Then with birds we see that every species trusts its wings and seeks fluttering assistance from its feathers.	1460 [1040]
Thus, to suppose that in the past one man allocated names to things and that is how men first learned words is sheer absurdity. ¹ For why was this one man able to mark all things with words and with his tongue to make various noises, and we are to believe that at the same time other men could not do the same? Moreover, if the others were not also using words among themselves, how did the notion of their usefulness	1470
plant itself in him? Where did it come from— the power which was given first to him to know and in his mind to visualize what he wished to do? Furthermore, one man	[1050]
would not have been able to compel the rest, prevailing over them with force, so that they were willing to learn his names for things. It is not all that easy to persuade men who cannot listen and to instruct them what they need to do. They would not bear it	1480
or in any way let the sounds of words they had not heard before keep battering their ears quite uselessly. And finally, in this matter what is so amazing if the human race, which had vigorous tongues and voices, should note things with different sounds in accordance with their different feelings,	

¹ The origin of language was a matter of considerable dispute among classical philosophers. Some of them maintained that one person was responsible for giving names to things (e.g., Pythagoras), in the same way the Bible assigns credit for that to Adam. Lucretius is arguing for a much more natural development of language.

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when mute herd creatures and even races
 of wild animals are in the habit
 of sending out distinctly different sounds 1490 [1060]
 when they feel fear or pain and when their joy
 increases? Indeed, one can find this out
 from well-known facts. When in Molossian dogs,
 their large, loose lips pull back, expose hard teeth,
 and start to growl with anger, then their rage
 menaces with a very different sound
 from when they merely bark and with their noise
 fill every space around them.¹ Then again,
 when they gently try to lick their puppies
 with their tongues or play games by tossing them 1500
 with their paws and then, with their mouths open,
 go after them, pretending, as their teeth
 gently close, that they are swallowing them,
 they fondle those pups with a yelping sound
 of a kind far different from what they howl [1070]
 when left in a building all by themselves
 or when, with their bodies cringing, they creep
 whimpering from blows. Furthermore, does not
 a horse's neigh also appear different
 when a young stallion in the prime of youth, 1510
 urged on by the prick of winged passion,
 rages among mares and, nostrils flared, snorts
 his call to arms and when, at other times,
 he may neigh while all his limbs are trembling?²
 And finally, the race of beasts with wings,
 the different birds—sea eagles, hawks, and gulls—
 which in the sea's salt water waves seek out [1080]
 their food and livelihood, at other times
 give very different cries than when they strive
 for sustenance and fight over their prey. 1520
 And some of them change their raucous squawking
 with the weather, as do the long-lived tribe
 of crows and flocks of ravens, when they cry,
 so men say, for water and rain, sometimes
 summoning winds and breezes. And therefore,
 if different feelings compel animals
 to utter various sounds, though they are dumb,
 how much more reasonable it would be
 that mortal men back then should be able
 to denote different things with different sounds. 1530 [1090]

And just in case, while dealing with these things,
 you are perhaps quietly wondering, ORIGIN OF HUMAN USE
 it was lighting which first carried fire down OF FIRE
 to mortal men on earth—with that all heat

¹ Molossian dogs, well known in ancient times, are now an extinct breed, but they are considered the ancestors of today's large mastiffs.

² The detail about passion having wings is a reference to Cupid (in Latin *Amor*).

Book Five

from flames is generated.¹ For we see
many things ignite and burn up when struck
by fire from heaven, once the bolt transmits
its heat. Then, too, when a tree with branches
is lashed by winds, sways back and forth, presses
and rubs the branches of another tree, 1540
the violent force of rubbing brings out fire,
and while trunk and branches chafe each other,
sometimes the flaming heat of fire ignites. [1100]
Either of these two could have provided
fire to mortal men. And then sun taught them
to cook their food, using the heat of flames
to soften it, because out in the fields
they would see many objects getting soft
once beaten by sun's heat and lashing rays.

Then day after day those men who stood out 1550
for their keen intellect and had strong minds
would, from kindness, increasingly show them
how to exchange their previous livelihood
and former life, for something new. Then kings
began to build towns and found fortresses,
as a defence and refuge for themselves,
and also to divide up and hand out
herds and fields to each man, on the basis [1110]
of his good looks, intelligence, and strength.
For how someone looked was highly valued, 1560
and strength was thought an honour. After that,
wealth was introduced, and gold discovered,
which quickly robbed the strong and beautiful
of their esteem. For people, no matter
how strong they grow or how fine their bodies
are to look at, mostly follow the lead
of those who have more wealth. But if someone
were to guide his life with proper reasoning,
that man would possess great wealth by living
frugally with a tranquil mind, for when 1570
one has few things, there never is a lack.
But what men wanted for themselves was fame [1120]
and power, so their fortune might remain
on a firm foundation and, with that wealth,
they could lead a peaceful life. But in vain,
since, while striving to rise up to the heights
of honour, they made their pathway perilous,
and envy, like a lightning bolt, sometimes
hurls them in disgrace from the very top
down to filthy Tartarus, for envy, 1580
just like lightning, generally sets on fire
the loftiest places, all those which rise

¹ A number of editors observe that this verse paragraph and the next two seem somewhat out of place, since they are not relevant to what comes immediately before or after them.

Book Five

above the others. So it is much better
to stay quiet and obey than to yearn
for regal power and to govern kingdoms. [1130]
Then let men tire themselves out pointlessly
and sweat blood as they fight their way along
ambition's narrow road, since what they know
comes from mouths of others and they search
for things based on what they hear rather than 1590
relying on their own feelings. But doing this
is no more use now than it was before
and will not be in future.

Therefore, kings
were killed, the ancient majesty of thrones
and proud sceptres were cast down and ruined,
the splendid symbol on the monarch's head,
stained with blood beneath the rabble's feet,
mourned the loss of its great reputation,
for what is too much feared in earlier days [1140]
is trampled on with passion. And so things
returned to the utmost dregs of chaos, 1600
when every man sought out ruling power
and dominance for himself. After that,
some taught people to create magistrates
and set up laws, so that they might consent
to follow legal rules. For the human race,
worn out by living in mere violence,
was exhausted by men's hostilities,
and so, on its own, it submitted itself
more readily to rules and binding laws. 1610
For since each man was prepared to punish
in his own cause with greater cruelty
than is now permitted by impartial law,
this fact made men grow sick of living life [1150]
by force. From that the fear of punishment
pollutes the prizes of this life. For harm
and violence entangle everyone,
and, for the most part, they rebound on him
who was their origin. It is not easy
for a man to live a calm, peaceful life, 1620
if his acts contravene the common laws
of peace. For though he may not be noticed
by gods and men, he must still be concerned
whether his secret will remain concealed
forever, since many men frequently
talk in their sleep or grow delirious
from sicknesses and give themselves away,
and, so we are told, publicly reveal
their concealed transgressions and wicked deeds. [1160]

Now, what is it that has spread the influence 1630

Book Five

of the gods through powerful states, filling
cities with altars, and brought it about
that men set up sacred ceremonies,
rituals which today are flourishing
at important times and in great places,
and from which, even now, in mortal men
is placed a dreadful fear which elevates
new temples to the gods in all the earth
and forces men on days of festivals
to gather—to explain all this in words
is not so difficult.¹ For in those days,
races of mortal men already saw,
even while awake, splendid shapes of gods
and, in sleep, these were still more wonderful
for their physical size.² So they gave them
sensation, since they seemed to move their limbs
and utter haughty words appropriate
to their fine appearance and ample strength.
Men gave them eternal life, since their faces
always kept appearing and their figures
stayed the same—and beyond that, above all
because they believed there was no power
which could easily subdue such mighty beings.
And for that reason they assumed these gods
far excelled in happiness, since fear of death
would trouble none of them. At the same time,
while they were sleeping they could see these gods
carrying out many amazing acts,
which for them required no effort at all.
Then, too, they kept observing what went on
in the sky in fixed order—various seasons
of the year returning—and could not see
the causes that made these happen. Therefore,
they found themselves a way out, by linking
all these to the gods, making everything
directed by gods' will. And they created
habitations and spaces for the gods
up in the sky, for they saw night and moon
moving through the heavens—moon, day, and night,
glorious nocturnal constellations,
celestial torches wandering at night,
flying fires, clouds, sun, rain, snow, and wind,
lighting, hail, swift peals and ominous sounds
of menacing thunder.

ORIGIN AND SPREAD OF
RELIGION

1640

[1170]

1650

[1180]

1660

1670 [1190]

¹ Here Lucretius returns to his account of the very early days of human society, a narrative which has been interrupted by the previous three verse paragraphs (on the arrival of fire, the overthrow of kingly rule, and the creation of a legal system).

² Bailey calls attention to the problem of where these images of the divine might originate in a material universe and points out that Lucretius seems to have believed that images of the gods come from a stream of matter passing from them into the minds of human beings. These particles cannot be perceived by the senses but enter the human body and affect the soul. But the evidence, Bailey concedes, makes the issue difficult to resolve.

Book Five

O unhappy race of men,
when they ascribed such actions to the gods
and added to them bitter rage! What sorrows
they then made for themselves, what wounds for us,
what weeping for our children yet to come!
There is no piety in being seen
time and again turning towards a stone
with one's head covered and approaching close
to every altar, and hurling oneself
prostrate on the ground, stretching out one's palms
before gods' shrines, or spreading lots of blood
from four-footed beasts on altars, or piling
sacred pledges onto sacred pledges,
but rather in being able to perceive
all things with one's mind at peace.¹ When we look
at celestial regions of this huge world,
aetherial space fixed above twinkling stars,
and our minds think of paths of sun and moon,
then into hearts oppressed by other ills
fear starts to stir and raise its head, as well,
that perhaps there might exist over us
immensely powerful gods, whose force turns
the sparkling stars in their various motions.
For lack of reasoning attacks the mind
with doubts whether there was an origin,
a beginning of the world, and then, too,
whether there is to be an end—how long
can the world's walls hold up under the strain
of restless motion—or whether, endowed
by the gods with everlasting power,
they can glide through eternal tracts of time
defying the mighty strength of endless age.
Moreover, whose heart does not shrink with fear
of gods, whose limbs do not creep in terror,
when scorched earth shudders from horrific blows
of lightning and rumblings pass through the huge sky?
Do not people and whole nations tremble,
and haughty kings, transfixed by fear of gods,
draw back into their bodies, for fear that,
because of some foul crime or arrogant word,
the dread time of paying full punishment
has come? Moreover, when with utmost force
tempestuous winds at sea sweep the leader
of a fleet across the waves, and with him
strong legions and their elephants, as well,
does he not with vows beg the gods for peace,
pleading timidly in his prayers for winds
to stop and for favouring breezes? In vain—
since often caught up in those stormy winds,

PERNICIOUS EFFECTS OF RELIGION

1680

[1200]

1690

[1210]

1700

[1220]

1710

1720

[1230]

¹ Lucretius is in this sentence describing the various gestures and motions a Roman worshipper goes through in normal worship. The “stone” is a statue of the god.

Book Five

for all his prayers, he is still carried off
to the shoals of death. That reveals how much
some unseen power crushes human things
and seems to trample down and have its fun
with splendid *fascēs* and cruel axes.¹

And then, when all the earth shakes underfoot
and tottering towns fall or their collapse
is threatened and hangs in doubt, no wonder
if races of mortal men hate themselves
and make room for the amazing forces
and immense power of gods here on earth,
so that they have control of everything.

1730

[1240]

Then copper, gold, and iron were discovered,
along with heavy silver and useful lead,
when heat from fires burned up large forests
on massive mountains, from a lightning bolt
sent from the sky, or because men waging war
with each other in the woods brought in fire
among their enemies to create panic,
or because, drawn to the land's fecundity,
men wanted to open up fertile fields
and turn countryside to pasture, or else

DISCOVERY OF METALS

1740

to kill wild beasts and thus enrich themselves
with plunder. For hunting with pits and fires
came before closing off the woods with nets
and chasing beasts with dogs. Whatever the case,
whatever made scorching heat burn up trees

[1250]

with a fearful cracking from their deep roots
and seared the earth with fire, there then flowed out
from boiling veins streams of gold and silver,
as well as copper and lead, which gathered

1750

in hollow places in the ground. Later,
when men saw it solidified, shining
in the ground with a marvellous lustre,
attracted by the smooth, brilliant colour,
they gathered it up. And then they noticed

ORIGIN OF METAL WORK

it had been molded into a figure

[1260]

similar in outline to the hollows

1760

in which each one was located. So then,
it occurred to them that these substances
could be melted down with heat and settle
into the form and shape of anything,
and, in fact, might be molded by hammering
into points and edges as sharp and fine
as one might wish for. Thus they could produce
tools for themselves so they could cut down trees,

¹ The *fascēs* (from the Latin word for a bundle) is a collection of sticks bound together into a cylinder, often with one or more axes included. It was an important symbol of the Roman Republic, indicating the importance of a tight collective unity among the people and the power of the state. In modern times the image has been used as a common symbol for the unity of the state by some countries and political institutions.

Book Five

hew timbers, plane logs smooth, and make holes
 with augers, awls, and drills. First they prepared 1770
 to do this with gold and silver, no less
 than with the fierce strength of sturdy copper.¹ [1270]
 That was no use, since with gold and silver
 their strength kept fracturing and giving out—
 unlike copper, they could not bear hard use.
 So at that time they valued copper more
 and neglected gold—its dull blunt edges
 made it useless. Now copper is ignored,
 and gold has climbed the pinnacle of honour.
 Thus, rolling time changes seasons of things. 1780
 What was once esteemed, later has no worth.
 And then something different follows—it leaves
 its despised place and becomes sought after
 more and more each day, and when found, flowers
 with praise and is held in splendid honour [1280]
 among men.

Now, Memmius, to find out

how the nature of iron was discovered
 is easy—you can do that on your own.
 Ancient weapons were hands, nails, teeth, and stones,
 as well as branches broken off from trees, 1790
 along with flame and fire, once these were known.
 Later, men learned the force of bronze and iron.
 They came to understand how to use bronze
 before they learned of iron, because bronze
 is easier to work and supplies of it
 are larger. Using bronze men worked earth's soil,
 with bronze they launched themselves in storms of war,
 inflicting deep wounds, seizing land and cattle. [1290]
 For everything defenceless and unarmed
 soon surrendered to those who brandished weapons. 1800
 But after that the iron sword gradually
 took over, and the shape of the bronze sickle
 changed to a thing of scorn.² And then with iron,
 men began to plough earth's soil, and contests
 in uncertain wars were rendered equal.³
 Armed men mounted horses' backs, guiding them
 with reins, bravely fighting with their right hands,
 before they undertook the risks of war
 in chariots with two horses. And yoking
 two horses came before men harnessed four 1810 [1300]

BRONZE AND IRON

METAL WEAPONS OF WAR

ANIMALS IN WARFARE

¹ As Copley points out, the Latin word *aer* means both copper and bronze. Bronze is harder than copper and would therefore make good sense here, but bronze is an alloy of copper and tin and does not occur naturally. Hence, the word copper is preferable, since Lucretius is talking about the very early days, when men were working with metallic ores they found in nature.

² As other editors note, this odd reference to a bronze sickle may refer to magical rites.

³ The contests were “rendered equal” because iron weapons became so common they were generally available to all fighting groups.

Book Five

or climbed fully armed into war chariots
equipped with scythes. Then Carthaginians
taught hideous Lucanian bulls—with towers
on their backs and snakes for hands—to suffer
the wounds of battle and create panic
in large groups of fighting martial warriors.¹
Harsh war made one thing after another
to terrify those races of armed men,
and thus increase war's horror day by day.
Men tried to harness bulls to serve in battle 1820
and tried to send out fierce wild boars as well
against their enemy. Some had strong lions
marched out ahead of them, with armed trainers [1310]
and cruel masters who could control them,
keeping them in chains. But that was useless.
For in the confusion of the slaughter
the hot, fierce beasts spread panic in the ranks
of both sides by tossing their fearful manes
around their heads in all directions. Riders
could not calm their horses' hearts, terrified 1830
by roaring lions, or apply their reins
to wheel them round against the enemy.
Female lions hurled their raging bodies,
leaping everywhere, and attacked the face
of those who came against them or seized men
without warning from behind and threw them, [1320]
once in their grip and overcome with wounds,
down on the ground and then ripped into them
with their hooked claws and powerful teeth.
Bulls tossed and stomped their own men underfoot. 1840
With their horns they gored the horses' bellies
and below their flanks, tearing up the ground
in terrifying rage. And wild boars, too,
with their strong tusks, would slaughter their own troops
and in their frenzy spatter blood on spears
broken off in their own muscles, spreading
confused destruction through ranks of soldiers
on horses and on foot. Moving to one side,
horses would avoid the savage onslaughts [1330]
made by tusks, or else would rear up, their feet 1850
pawing air, but that was all quite hopeless,
for you could see them collapse, tendons sliced,
and covering the earth with their heavy fall.
If, before the fight, men had thought those beasts
sufficiently well trained at home, they saw,
once the conflict started, them going berserk
from injuries, screams, flight, fear, confusion,
and they could impose no sense of order

¹ Lucanian bulls are elephants, whose trunks give them "snakes for hands." The Romans used this term because they first saw elephants in Lucania in the wars against Pyrrhus in Italy (in 280 BC). The Carthaginian general Hannibal famously brought elephants with his army over the Alps into Italy from Spain (218-217 BC).

Book Five

on any group of them, for wild creatures,
all the various types, kept on scattering, 1860
the way Lucanian bulls badly hacked with swords
now often scatter, giving their own troops [1340]
many dreadful wounds. Men wished to do this
from their desire, not so much to conquer,
as to give their enemies a reason
to lament before they themselves were killed,
for they lacked confidence in their numbers
and had no weapons. If, in fact, they did this.
But I find it difficult to accept
that before this happened they would not see 1870
and realize in advance how disastrous
it would be for both. You might be able
more plausibly to claim that this was done
out in the universe, in different worlds
created in different ways, rather than
on this one particular sphere of earth.¹

Clothing made from materials tied together [1350]
came before woven garments. Woven clothes
came after iron, for cloth is made with iron—
that is the only way men can turn out 1880
such fine, smooth heddles and spindles, shuttles,
and rattling yard-beams.² Nature forced the males
to work with the wool before the females,
for the male sex far excels in skill and is
much more inventive, until tough farmers
scorned weaving, and then the men were willing
to let the women do that kind of work
and to share equally among themselves
in hard labour, strengthening hands and limbs [1360]
with heavy tasks.

But the creator of things, 1890
nature herself, was the first example
of sowing seed and the start of grafting,
for berries and acorns fell down from trees
and, in due season, produced underneath
a crowd of seedlings. Then from nature, too,
they got the notion of setting young shoots
into branches and planting new saplings
in the ground through all their fields. After that,
they kept trying various ways of tilling
pleasant fields and noticed that with tender care 1900
and gentle cultivation earth would tame
wild fruits. Day by day, men forced the forests
to move further up the mountains, yielding [1370]

¹ Lines 1869 to 1877 in the English have attracted criticism: some editors see them as an interpolation or a marginal comment by someone else and omit them.

² Heddles, spindles, shuttles, and yard beams are parts of the machinery used in weaving with looms.

Book Five

lower parts to farming, so they could have
 meadows, lakes, streams, grain fields, and rich vineyards
 on hills and plains, and dark bands of olives
 could run between, marking the divisions,
 spreading over hillocks, plains, and valleys,
 just as you now see all the land divided
 with various fine things—men make it shine 1910
 by arranging sweet orchard trees in rows,
 and, with fertile shrubs planted all around,
 keep them fenced in.

However, using mouths SINGING AND MUSIC
 to imitate the liquid sounds of birds
 took place well before men could sing in tune, [1380]
 making delightful songs which pleased the ear.
 And winds whistling through hollow parts of reeds
 first taught country people to blow through stalks
 of hemlock hollowed out. From that they learned,
 little by little, the sweet plaintive notes 1920
 which, when players' fingers close off the stops,
 come pouring out. These were heard through forests,
 pathless woods, and thickets, in lonely spots
 of shepherds and places of godlike rest.¹
 Singing soothed their hearts and gave them pleasure, [1390]
 when they had eaten their fill, for at that time
 all things are delightful. As a result,
 they would, as a group, often stretch themselves
 on soft grass beside a stream of water,
 under the branches of a lofty tree, 1930
 and, at no great cost, refresh their bodies,
 above all at those times fine weather smiled
 and seasons of the year painted green grass
 with flowers.² At such times they would enjoy
 jokes, talk, and happy laughter. For back then
 the country muse was young and vigorous.
 Then joyful gaiety encouraged them
 to drape their heads and shoulders with garlands
 of flowers and leaves intertwined and dance, [1400]
 moving ahead with no sense of rhythm, 1940
 shifting their limbs crudely, with heavy feet
 stomping on mother earth. From this arose
 smiles and joyful laughter, for all these things
 were newer then, flourishing, more wonderful.
 And to those who remained awake on guard
 from this came comfort for their loss of sleep—
 letting their voices move through various notes,
 weaving songs, and running their curving lips
 across the pipes. From that, even today,

¹ The exact meaning of this line is uncertain. The two lines immediately after this (1388 and 1389 in the Latin) have been omitted. They appear again at lines 1454 and 1455 of the Latin.

² This passage is almost the same as Book 2, lines 29 to 33 of the Latin.

Book Five

men on watch still keep to these traditions
and have just learned to maintain the rhythm
of the song. But for all that, they derive
no more enjoyment from this sweet delight
than did those forest sons of earth back then.
For if we have not previously known
anything sweeter, then what is present,
here at hand, provides the greatest pleasure
and seems the best, and later, if we find
something better, as a rule it transforms
and kills feelings we had for things before.
And so men began to despise acorns
and abandoned those resting spots covered
with grass and piled with leaves. In the same way,
they shunned clothes made from wild animal hide—
though I suspect that at that time those hides
roused such envy that the man who first began
to wear them was set upon and slaughtered,
and yet because they pulled the hides apart
among themselves and caused so much bloodshed,
they spoiled those skins, so they could not be used.
Then it was hides, and now it is purple
and gold which harass men's lives with worry
and weary them in warfare. And in this,
I think, the greater blame belongs to us.
For cold would torment those earth-born humans,
naked but for wild beasts' hides, but for us
there is nothing harmful about a lack
of purple clothing embellished with gold
and massive symbols, if we still possess
common garments which keep us well protected.
Thus, the human race labours constantly,
without purpose and in vain, consuming
men's lives with empty worries, for clearly
they are ignorant about a limit
to their possessions and about how far
true pleasure can increase. And little by little
this has carried life into deep waters
and stirred up from the very lowest depths
huge seething tides of war.

But sun and moon,
those watchmen moving on with their own light
around the immense revolving spaces
of the world taught human beings that seasons
of the year come back and that what happens
is brought about in a certain order
according to a predetermined plan.
And now they would spend their lives surrounded
by strong fortresses, with land divided,
marked out, and cultivated. The ocean

1950

[1410]

1960

[1420]

1970

1980

[1430]

1990

MODERN CIVILIZATION

[1440]

Book Five

then blossomed with ships flying under sail,
towns had partners and allies, now confirmed 2000
by treaties, poets began to pass down
deeds of men in songs—and just before that
writing was invented.¹ Therefore, our age
cannot look back at what was done before,
unless our reason points out the traces.
Ships and cultivated lands, walls, laws, arms,
roads, clothing, and all other things like this,
all the rewards, all luxuries of life [1450]
without exception—fine polished statues,
poems, paintings—they gradually learned 2010
through practice, along with the experience
of active minds, and advanced step by step.
Thus, little by little time brings in view
each individual thing, and then reason
raises it into regions of the light.
For in the arts things must be clarified
one after another, in due order,
until they reach their highest pinnacle.²

¹ Part of line 1442 of the Latin (line 1999 in the English) is corrupt.

² I have followed Munro's suggested emendation of the Latin in the last sentence.

BOOK SIX

[Tribute to the greatness of Athens and Epicurus; winds and storms; disasters not divine punishment; causes of thunder; lightning faster than thunder; causes and effects of lightning; seasons when lightning occurs more frequently; lightning not divine punishment; origin of presters; formations of clouds; moisture from clouds; causes and effects of earthquakes; reasons for the constant size of the ocean; eruption of volcanoes; odd behaviour of the River Nile; nature of Avernian regions; temperatures in water wells; magnetic powers of lodestone; origin of diseases; the plague in Athens.]

To suffering mortal beings long ago
 Athens, that city with a splendid name,
 first taught ways of producing crops of grain,
 fashioned a new life, and established laws.
 She first offered life's sweet consolations,
 when she gave birth to a man who revealed
 great genius and from whose truthful mouth
 once poured forth all wisdom—his glory,
 even in death, has long been spread abroad,
 through his divine discoveries, and raised
 up to the sky.¹ For when he saw that things
 which mortal men required for survival
 had by now almost all been well supplied,
 that their way of life, as much as possible,
 had a safe foundation, that men possessed
 ample power through wealth, honour, and praise
 and took pride in the fine reputation
 of their children, but that, in spite of this,
 none of them in his own home had a heart
 any less anxious—it disturbed their lives
 by tormenting their minds continuously,
 forcing them to grow enraged, to complain
 about their bitter troubles—he then saw
 that the vessel itself was creating
 the defect and that all things collected
 from outside, however beneficial,
 once they entered, were corrupted inside
 by that fault, partly because he observed
 that the vessel leaked and was full of holes,
 so there was no way it ever could be filled
 and partly because he saw it poisoned
 everything it absorbed within itself,
 as if with a disgusting taste. Therefore,
 he purged men's hearts with words that spoke the truth,
 setting a limit to desires and fears
 and pointing out what was the highest good
 we all are striving for. He showed the road
 by which we can, along a short pathway,
 reach it directly. And what is evil
 in affairs of mortal men everywhere
 he clarified—things which quite naturally
 arise and fly around in various ways,

GREATNESS OF ATHENS
 AND EPICURUS

10

[10]

20

[20]

30

40

[30]

¹ This book opens, as before, with a tribute to Epicurus.

Book Six

whether by accident or by some force,
because that is what nature has arranged—
and the gate through which we should sally forth
to meet each one.¹ And he demonstrated
that in their hearts the human race stirs up
anxious tides of worries, for the most part
with no good reason. For just as children
tremble in blinding darkness and are afraid
of everything, so sometimes in the light
we dread things which are no more to be feared
than those which during the night young people
tremble at, dreaming of what will happen.
Therefore, this terror, this darkness of mind,
must be dispelled, not by rays of sunlight
or bright arrows of the day, but by reason
and the face of nature. In pursuit of that,
I will hasten all the more to finish
what I have been weaving in these verses.

50

[40]

60

Now, since I have shown that the world's regions
are mortal and that the heavens consist
of matter which was born, and for the most part
have discussed all things that happen in it
and which must happen, you should keep listening
to what still remains, since [I have ventured]
this once to climb up in the splendid chariot
[of the Muses and to ascend to heaven,
to explain the true law of winds and storms,
which men, in their folly, ascribe to gods.
People say that gods, when angry, bring on
raging storms and then, when a lull occurs
in the fury] of the winds, that gods' anger
is appeased and everything that was there
has changed back again, now that their anger
has been soothed.² [I will explain] all the rest
which mortals creatures observe taking place
on earth and in the sky, when so often
they are in suspense, their minds full of dread,
things which demean their souls with fear of gods.
These weigh on them and press them to the ground.
Their ignorance of causes forces them
to assign things to the rule of deities
and to concede that gods are in control.³
For if those who have correctly learned that gods

70

FEAR LEADS TO BELIEF IN GODS

80

¹ The metaphor here is a military one: the defenders of the city rush out from behind the walls to attack a threatening enemy.

² At line 48 in the Latin the text is very confusing with some lines evidently missing. I follow Munro's suggested interpolation and translation, given above in square brackets, with some slight changes. I have also followed Munro and Bailey and others in moving lines 48 to 51 in the Latin to a position later on (lines 92 to 95 in the Latin). Hence, there is no line number [50] above.

³ Lines 60 and 61 in the Latin have been omitted here. They appear again at lines 94 and 95 of the Latin. Hence, there is no line [60] above.

Book Six

lead lives free from care still from time to time
wonder how everything can come about,
especially in those events they see
overhead in regions of the aether,
they are carried back to old religion 90
and accept harsh masters, who, they believe,
in their misery, can do everything,
being ignorant of what can and cannot be,
in short, by what law each thing possesses [70]
limited power, a deep-set boundary stone.
And therefore men lose their way even more,
carried away by their blind reasoning.
If you do not spit such things from your mind,
and drive off thoughts unworthy of the gods,
which have no part in their serenity, 100
gods' sacred power, which you have slighted,
will often hurt you—not that one can harm
the supreme majesty of gods so that,
in its anger, it would resolve to seek
harsh punishment, but because you yourself
may well believe that those serene beings
in their calm peace roll out great waves of rage,
and when you approach temples of the gods
your heart will not be calm. You will lack strength
to contemplate with tranquil peace of mind 110
those images borne from divine bodies [80]
into the minds of men as messengers
of their sacred forms.¹ You can imagine
what kind of life would follow after that.

Now, although I have set down many things,
still, in order for the surest reasoning
to hurl such a life far distant from us,
many things remain to be embellished
in polished poetry. We need to grasp
what heaven looks like and the reasons why. 120
We must sing of storms and brilliant lightning,
what they do and what produces each of them,
so you do not section off the heavens [90]
and grow anxious and frantic about where
flying fire came from, or to which part
it has turned itself, or how it passed through
walled areas and, after ruling there,
brought itself back out.² And there is no way
men can see causes for events like this,

¹ This passage is a good indication of Epicurean worship. The gods have no interest in punishing human beings for impiety (for they are unconcerned about human affairs), but human beings who do not understand the nature of the gods hurt themselves because, in their fear of divine punishment, they may become incapable of the only appropriate form of worship, contemplation of the divine images, which, as Lucretius has mentioned before, travel from the gods into the minds of human beings.

² This mention of dividing up the sky refers to the practices of various soothsayers and astrologers, who used these divisions in their interpretations of how storms revealed the wishes of the gods.

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so people believe they are brought about
by power of the gods. And as I race
to the white line which marks my final goal,
point out the pathway lying in front of me,
O Calliope, you ingenious Muse,
you solace for men and delight of gods,
so that, with your assistance, I may win
the crown and with it preeminent fame.¹

First of all, thunder makes the blue sky shake,
because aetherial clouds flying up high
collide when opposing winds are fighting.
For no sound arises from those places
where the sky is clear, but wherever clouds
are more densely packed, from that spot rumbles
more frequently the great roar of thunder.
Then, too, clouds cannot possess a body
as dense as stones and wood or as rarefied
as mists and flying smoke. For then they must
either be brought down by their own dead weight,
like rocks, or else, like smoke, they could not
retain their shape and hold inside themselves
frozen snow and showers of hail. And clouds
also give off sound across the reaches
of the open sky, just as stretched canvas
in large theatres sometimes makes a noise
as it is tossed among the posts and beams,
and at times, when struck by forceful breezes,
it rages wildly and then makes a sound
like crackling paper sheets.² And you can hear
that sort of sound also in the thunder
or when gusting winds beat hanging garments
or flying paper strips and make them rattle
in the air. And it also is the case
that sometimes clouds cannot so much collide
face to face as move past along the side,
scraping their bodies with various motions
slowly on their flanks, and then a dry sound,
which lasts a while, brushes against our ears,
until they move off from that area
in which they are confined.

In this way, too,
all things struck by heavy thunder often
appear to tremble, and the mighty walls
of the spacious world in an instant seem

¹ Calliope is one of the nine Muses. She is most closely associated with heroic poetry, especially with Homer. The position of this address to Calliope varies slightly from one editor to another.

² Lucretius is here referring to sheets of papyrus, the material used in books. These sheets were written on and then rolled up. The papyrus, Smith notes, when being prepared, would be hung up to dry, rather like garments on a clothes line.

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to burst and split apart, when forceful winds
in a gathering storm suddenly twist themselves
inside the clouds and, in that enclosed space,
with their swirling current increasingly
compel the cloud to hollow itself out
in all directions with a thickening crust
around its body and then later on,
when the forceful harsh power of the wind 180
has weakened it, the cloud then splits apart
with a crash, a terrible cracking noise.
This is not surprising, since a small bladder [130]
filled with air often makes a savage noise
if it suddenly explodes.

Moreover,
there is also a way winds may make sounds
when they blow through clouds. For we often see
irregular, branching clouds carried along
in various directions, and we can be sure
it is like those moments when northwest gales 190
blow through dense forest, so that leaves rustle
and branches crack. It can also happen
that sometimes the force of a mighty wind,
as it rushes on, breaks a cloud apart,
attacking from the front and slicing through it,
for what the wind is capable of doing
in the sky is made clear by obvious facts
here on earth, where it is less violent [140]
but still throws down tall trees and rips them out,
deep roots and all. And moving through the clouds 200
there are waves as well, and these, as it were,
in their heavy fall give off crashing sounds,
like the ones created by deep rivers
and by huge seas when their surf breaks onshore.
And sometimes, too, when the fiery power
of lightning cuts from one cloud to another,
if by chance the cloud which receives the fire
contains much moisture, it puts out the flame
at once with a loud noise, just as hot iron
from a burning furnace sometimes hisses 210
when we plunge it quickly in cold water.
Then, too, if the cloud which takes in the flame [150]
is drier, it is set alight at once
and makes a huge noise while it burns, as if
twisting storm winds were pushing flames along
through mountain laurel trees, consuming them
in a massive onslaught—and there is nothing
which makes a more frightful sound when burning
in crackling fire than the Delphic laurel
of Apollo. And then in large high clouds 220
great fractures in the ice and falling hail

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often make a noise, for mountains of clouds
 which are frozen and mixed with hail break up
 when they are pushed together by the wind.
 In the same way there are flashes of light [160]
 when, thanks to their collision, clouds give off
 many seeds of fire, just like when a stone
 strikes stone or iron. For then, as well, a light
 springs out and scatters bright fiery sparks.
 But it so happens that we hear thunder 230
 in our ears after our eyes perceive the flash,
 because things always move towards our ears
 more slowly than things which stir our vision.
 And this you can learn from the following point:
 if from some distance you look at a man
 chopping a large tree with a double axe,
 your eyes will see the blow before its sound
 goes through your ears. So, too, we also see [170]
 lightning flash before we hear the thunder,
 which is given out at the same moment 240
 as the fire and from a similar cause,
 produced from the very same collision.
 And in this way, too, clouds also colour
 places in a fleeting light, and a storm
 flickers with a quivering intensity.
 When by invading and whirling around
 inside a cloud, the wind has made that cloud,
 as I have shown above, hollow and thick,
 its own motion makes it hot, just as you see
 everything gets warmer when its movement 250
 heats it up—even a ball made of lead
 rotating through a lengthy distance melts.
 Thus, once hot wind splits the black cloud apart, [180]
 it scatters particles of fire, as if
 they were all suddenly expelled by force,
 and these produce the pulsing flash of fire.
 The sound then follows. It reaches our ears
 more slowly than those things which make their way
 towards the pupils in our eyes.

These things,

you understand, take place when clouds are thick 260
 and, at the same time, when they are stacked high,
 one above another, a stunning sight.
 Do not deceive yourself because you see
 from down below how widely spaced they are
 rather than how high the pile of clouds extends.
 For you should watch when winds carry these clouds
 with shapes like mountains sideways through the air [190]
 or when you see them massed on mighty peaks,
 heaped on one another and pressing down
 from up above, all firmly fixed in place, 270

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with winds from all directions fast asleep.
Then you can recognize their massive size
and see caverns structured like hanging rocks.
After a storm has gathered and the winds
have filled them and are now enclosed in clouds,
with a loud growling they grow indignant
and threaten like wild creatures in their dens,
at times roaring out from one location
through the clouds, at other times from others,
and, as they seek an outlet, they twist round, 280 [200]
rolling together elements of fire
out of the clouds, and in this way collect
many particles, making flame rotate
in hollow ovens, until they split the clouds
and come bursting out with a brilliant flash.
And for this reason, too, it so happens
that the golden colour of swift liquid fire
flies down to earth, because the clouds themselves
must contain countless particles of flame.
In fact, when there is no moisture in them, 290
their colour, as a rule, is bright and fiery.
For, as one might expect, clouds must absorb
many such particles from the sun's light,
so there is a valid reason they are red [210]
and send out fires. And therefore, when the wind,
as it drives them, pushes them together
and forcefully compacts them in one place,
they squeeze out and emit these particles
that create the flash of flaming colours.
And in a similar way light blazes out 300
when the celestial clouds are thin, as well.
For when winds gently separate the clouds,
as they move and break them up, then those seeds
which make the flash must fall out on their own,
producing light but without horrid fear
or noise or any uproar.

As for the rest, LIGHTNING FIRES
the kind of nature lightning bolts possess
is demonstrated by the blows and marks [220]
their fires burn in things and by the traces
which give off a heavy smell of sulphur. 310
For these are signs of fire, not wind or rain.
Then, too, they also often set on fire
roofs of houses and with their rapid flames
take over even inside the building.
For, as you should know, nature makes these flames,
which produce the most subtle of all fires,
from minute and swift-moving particles,
so nothing at all can stand against them.
A powerful lightning bolt passes through

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walls of houses, as do sounds and voices. 320
It goes through rocks and bronze and, in an instant, [230]
melts brass and gold. It also causes wine
to leak quickly from intact containers—
once its heat arrives, it clearly loosens
all substances around it easily,
thinning the earthy matter of the jar
and moving right into the wine itself.
Its quick motion disperses and dissolves
the elementary particles of wine,
something we see the sun's heat cannot do 330
even in a long period of time,
although its pulsing heat is very strong—
that shows us a lightning bolt possesses
much more speed and power.

Now, how these flashes PRODUCTION OF LIGHTING
are created and acquire such great force STRIKES
that a blow can split fortresses apart, [240]
level houses, tear off planks and timbers,
destroy and scatter human monuments,
annihilate men, wipe out cattle herds
in all directions—the power they have 340
which makes them capable of carrying out
all other things like this I will explain,
and I will not keep you waiting any more
by making promises.

These lightning bolts,
we must assume, are produced from thick clouds
piled up high, for none are ever sent down
from a clear sky or patches of thin cloud.
And there is no doubt that obvious facts
show this to be the case. When storms approach,
clouds form such a dense mass in the whole sky, 350 [250]
that on every side we might well believe
all darkness had abandoned Acheron
and filled up the immense vault of the sky,
so dreadful are the faces of dark horror
hanging high above us, once that foul night
of clouds has gathered and the storm begins
to forge its lightning bolts. And out at sea,
often a black cloud, like a stream of pitch
poured down from the heavens, will also fall
into waves completely filled with darkness 360
some distance off, drawing with it dark storms
weighed down with lightning bolts and hurricanes
and itself so loaded with fires and winds— [260]
more so than all the rest—that even on land
men are afraid and shelter in their homes.
So then we must assume the storm clouds stand

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high above our heads, for they would not shroud
the land in such thick gloom, unless they were
built up in huge numbers on each other
to a great height, extinguishing the sun, 370
and as they move, they could not inundate
the earth with such heavy rain that they make
rivers flood and fields swim underwater,
unless the upper air were filled with clouds
heaped high on one another. And therefore,
all parts are full of winds and fires, so that
they give off thunder claps and lightning strikes [270]
on every side. For I have shown above
that hollow clouds obviously contain
numerous seeds of heat, and they must get 380
many more from the heat of the sun's rays.
Thus, when the same wind which has collected
these clouds by chance in some place or other
has forced out many particles of heat
and, at the same time, has itself mingled
with this fire, an eddy of wind moves in,
twisting around there in the confined space,
and inside the burning furnace sharpens
the lightning bolt. For the wind is heated
in two ways: its own motion makes it hot, 390
as does its contact with the fire. And then,
when the wind's force has grown extremely hot
and the fire's harsh power has entered it,
the lightning is, as it were, fully ripe.
All at once it bursts through the cloud—its fire,
once set in motion, is carried away,
flooding all regions with its pulsing light.
The heavy crash of thunder follows on,
so that it seems to crush open spaces
in the sky, which suddenly has split apart. 400
Then violent shudders run through the earth,
and rumblings race through the heights of heaven,
for at that point the whole storm is shaken,
shuddering and giving off loud noises.
After this commotion comes heavy rain [290]
in huge amounts, so all the upper sky
seems to be turned into rain pelting down
in such a way as to recall the Deluge,
so great is the rainstorm which is discharged
by bursting clouds and windy hurricanes, 410
once thunder flies out from that fiery blow.¹
Sometimes, too, the aroused force of the wind
falls from the outside onto a hot cloud
ready to discharge a flash of lightning.
Once the wind breaks it apart, there shoots out

¹ The Deluge is a reference to the punishment Zeus sent against men for their impiety, the general flood from which Deucalion and his wife Pyrrha escaped.

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immediately that fiery whirl we call
by its ancestral name—the thunderbolt.
The same thing occurs in other places, too,
wherever that force of wind is carried.

There are also times when it so happens
that the power of wind, although sent out
lacking fire, will, after a long distance,
still catch fire in motion. While it proceeds,
in flight it sheds some of its large particles
which cannot keep on moving through the air
the way the others can, and it picks up
other small elements from air itself
and carries them along. These get mixed in
and by their motion create fire, much like
a moving lead ball, which often grows hot,
once it has shed many cold particles
and gathered fire in air. In addition,
it can happen that fire will be kindled
by the very force of the blow itself,
when that power inside the wind which strikes
is sent out cold, without fire, for clearly,
once the wind hits with a forceful impact,
particles of heat can flow together
from the wind itself and at the same time
from the substance which then receives the blow,
just like the times we strike a stone with iron
and fire flies out. Nor do those particles,
those bright fiery sparks, flow off any less
on impact because the iron's force is cold.
So, then, in the same manner an object
must also ignite from a lightning bolt,
if it happens to be combustible
and fit to burn. Nor should we rashly think
that the forceful power in wind can be
fully and completely cold, once discharged
with such strength from high above, but instead,
if it is not already set on fire
earlier in its journey, it still arrives
warm and mixed with heat.

420 [300]
FIRES CAUSED BY
MOTION OF WIND

430

[310]

440

450

[320]

But the lightning bolt
has a high speed and enormous impact.
It almost always charges on its way
in a rapid fall, since its force, once roused,
in every case first gathers itself up,
on its own, inside the clouds, and begins
a massive effort to get out. And then,
when the cloud is unable to restrain
the increased power, its force is expelled
and so escapes at an amazing speed,

VELOCITY OF LIGHTNING

460

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like missiles which are carried off when hurled
 from powerful machines.¹ Beyond all this,
 a lightning bolt consists of particles [330]
 which are small and smooth—it is not easy
 for any object to stand up against
 this kind of substance, for it penetrates
 and makes its way through porous passageways 470
 and thus is not impeded or delayed
 by many obstacles. For this reason,
 it speeds on and falls at a rapid rate.
 Then, too, it is natural that all weights,
 without exception, always fall straight down,
 but when a blow is added, then the speed
 is doubled and that impulse is increased,
 so that the impact of the lightning bolt
 all the more fiercely and swiftly smashes
 whatever gets in its way and hinders it, 480
 as it continues on its journey. Besides,
 since it moves with continuing momentum,
 it must increasingly gain speed, which grows [340]
 as it progresses and makes its huge force
 even greater, strengthening its impact.
 For its speed causes all the particles
 inside the thunderbolt to be carried,
 as it were, towards one place, and forces
 all of them together, as they roll round,
 into that one direction.² And perhaps 490
 the bolt, as it moves, draws from air itself
 certain objects, whose blows increase its speed.
 It goes through some things without harming them
 and with many substances passes through
 leaving them intact, for its molten fire
 slips through open pores. But it breaks apart
 many things when the lightning's particles [350]
 themselves strike an object's basic elements
 where these are held in close combination.
 It melts brass easily and in an instant 500
 makes gold boil, because its power consists
 of smooth and minutely small elements,
 which quickly penetrate and, once inside,
 immediately liquefy connections
 and dissolve all bonds.

The vault of heaven, set with gleaming stars, and the entire earth are violently shaken everywhere, above all in the autumn and the spring,	SEASONS FAVOURABLE TO LIGHTNING
--	------------------------------------

¹ This is a reference to large military catapults.

² As Bailey notes, this passage seems to mean that as the lightning bolt falls the constant motions in all directions of its elementary particles will, because of the duration of the fall and the weight of the particles, increasingly switch to the direction downward, thus increasing the speed of the lightning.

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when the flowers spread themselves in season.
 For in the cold there is a lack of fires, 510 [360]
 and in hot weather winds withdraw, and clouds
 are not so physically dense. And thus,
 when heavenly seasons are between the two,
 then all the various causes of lightning
 come together. For those stormy passages
 during the year themselves mix cold and heat,
 and to produce their lightning bolts the clouds
 need both of these, so matter get disturbed,
 and air, in a great commotion, rages
 and swirls around with fires and winds. And spring 520
 is the time, in part, of the first hot weather
 and, in part, of the last icy freezing.¹
 Thus, at that time unlike things must get mixed [370]
 and fight each other with great turbulence.
 And when the last hot weather rolls along,
 mingled with the initial cold, a time
 which goes by the name of autumn, then, too,
 fierce winters battle against summer heat.
 Thus, we should call these seasons of the year
 times of stormy passage. Nor is it strange 530
 that lightning bolts occur most frequently
 at that time and chaotic storms arise
 up in the sky, since both sides stir themselves
 in dubious battle, one armed with flames,
 one with wind and water mixed together.
 This is how one explores the true nature
 of the fiery lightning bolt and perceives
 the force with which it brings out each effect, [380]
 not by wasting one's time unrolling scrolls
 of Etruscan verses, seeking traces 540
 of some hidden divine will, to find out
 where flying fire came from, which region
 it has gone to from here, how it has pierced
 walled places and, after playing the tyrant
 inside there, has then made its way outside,
 or what harm the blow of a lightning bolt
 from heaven is capable of doing.²
 But if Jupiter and other gods shake
 bright heavenly spaces with dreadful noise
 and hurl down fire to any place at all, 550
 according to what each of them desires,
 why do they not see to it that those men [390]
 who in their recklessness have committed
 abominable acts are struck and stink

¹ Lucretius uses here (and later in line 530 below) the word *fretus*, which, as Munro observes, refers to the strait between two bodies of water and to the turbulent conditions commonly found in such places; hence, the phrase "stormy passages" to describe the seasons of the year favourable to the formation of lightning.

² Etruscans, who lived close to the Romans and influenced them a great deal, were famous for the divinations and prophecies, which they recorded on scrolls.

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of lightning fires from hearts pierced by the bolt,
a bitter precedent for mortal men?
Why instead is the man who is aware
he himself has committed no wrong act
in his innocence entangled and wrapped
in flames, snatched up in fiery hurricanes 560
suddenly sent down from heaven? Besides,
why do they target isolated places
and work so hard for nothing? Or are they
exercising limbs, toning their muscles?
Why do they allow their father's weapon
to be blunted on the earth? Why does he
let that happen and not save the lightning
for his enemies? Why does Jupiter [400]
never hurl down his lightning bolt on earth
or let his thunder peal when skies are clear 570
in all directions? Or as soon as clouds
appear, does he himself go down to them,
so that from there he may guide the impact
his weapons make from close at hand? And why
does he send them into the sea? What charges
does he bring against that liquid mass of waves,
those fields of water? And if he wants us
to beware the stroke of his thunderbolt,
why is he reluctant to arrange things
so we can see it as he hurls it down? 580
But if he wishes to overwhelm us
with his lightning when we are unaware,
why does he thunder from that area,
so we can guard against it? Why does he
first stir up darkness, noises, and rumbling? [410]
And how can you believe he discharges
lightning to many places all at once?
Would you dare to say it never happens
that many strikes occur at the same time?
But that has happened very frequently 590
and must take place—just as rain and showers
fall in many spots, so numerous thunderbolts
are formed at the same time. And finally,
why does he destroy the sacred temples
of the gods and his own splendid dwellings
with hostile lightning and smash to pieces
well fashioned idols of the gods, robbing
his own images of their dignity [420]
with a violent wound? Why for the most part
does he aim at high places, for we see 600
most traces of his fire on mountain tops?

To continue now with this discussion, WATERSPOUTS
from these facts one can quickly understand
those natural things the Greeks called presters,

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which are sent down from the upper regions
and reach the sea.¹ For sometimes it happens
that, as it were, a column from the sky
is sent down and moves right into the sea.
Around it water seethes, roused to fury
by the blasting winds, and any vessels 610
caught up at that time in the turbulence
are shaken and placed in utmost danger. [430]
This occurs when sometimes the force of wind,
once stirred up, cannot burst out from the cloud
it has begun to split apart. Instead,
it pushes the cloud down, so gradually
it looks just like a pillar sent from the sky
down to the sea, as if a thrusting fist
and arm were pushing something from above,
forcing it into the waves. Once the wind 620
has split the cloud, its force bursts out from there
into the sea and agitates the waves
in an amazing way, for the vortex
spins as it descends and carries with it
the viscous body of that cloud. And once
it has pushed the cloud, fully laden, down [440]
to the level of the sea, suddenly
that whole vortex plunges itself fully
in the water, disturbing all the sea
and forcing it into a seething mass, 630
making a tumultuous din. Sometimes, too,
that windy vortex wraps itself in clouds,
gathering particles of cloud from air
and, as it were, imitates a prester
sent down from the sky.² Then, once this vortex
has brought itself to earth and broken up,
with enormous fury it vomits out
hurricanes and storms. But because this wind
is, in general, quite rare and mountains
must hamper it on land, we observe it 640
more often in the wide panorama
of the sea and great stretches of the sky. [450]
And clouds collect when numerous particles
flying high up in this region of the sky
suddenly combine—rougher elements
which are held together by tenuous links
but which still can mutually combine
and keep themselves united. At the start,
these particles cause small clouds to gather,
and then these assemble, merge together 650
and, as they coalesce, increase in size,
and the winds keep carrying them away

¹ A prester, from a Greek word meaning to burn, is a hot whirlwind in a cloud which is pushed down to the sea, where it produces a water spout.

² Watson notes that here Lucretius is referring to a vortex which looks like a prester but which is not hot.

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until at last a savage storm arises.

It also happens that with mountain peaks
the closer they approach the sky, the more [460]
their summits constantly are wreathed in smoke
from murky vapours of yellowish cloud,
because when those clouds begin to gather,
before our eyes can see their tenuous forms,
winds carry them off, driving them to peaks 660
of the highest mountains. And in this place,
when a larger number has collected
and condensed, we can at last perceive them,
and at the same time we see them rise up
from the very summit of the mountain
into the upper air. These facts themselves
and what we observe when we climb high hills
demonstrate that there is plenty of wind
in regions which extend high up above.
Then, too, when clothes hung up along the shore 670 [470]
absorb the moisture which adheres to them,
they show that nature lifts many particles
from the entire ocean. Thus, we perceive
all the more plainly that many of them
could also rise up to augment the clouds
from the salt water in the heaving sea,
for both liquids have a similar nature.

And furthermore, we observe mists and steam
rising from all rivers and from earth, too,
which, after being forced away from there, 680
like a breath, are carried upwards, shrouding
the heavens in darkness and gradually
combining to make clouds up in the sky. [480]
For vapour in the high starry aether
also brings to bear a downward pressure
and by condensing, so to speak, it weaves
a network of clouds underneath the blue.
It happens, too, that from some outside place
there come into this sky those particles
which create clouds, as well as flying storms.¹ 690
For I have shown that their total number
is immeasurable, the full extent
of deep space infinite, and pointed out
how fast bodies fly, how they normally
move unimaginable distances
instantaneously. Thus, it is not strange
if storms and darkness frequently conceal [490]
the sea and land in a short space of time
with such gigantic mountains formed from clouds
hanging overhead, since on every side 700

¹ In this explanation the particles come from outside our world (i.e., from elsewhere in the universe).

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these particles have exits and entrances
through all the passageways in the aether
and, as it were, through the breathing places
of the great universe surrounding them.

Come now, I will show how moisture gathers RAIN
in high clouds and how water is sent down

to earth as rain. First of all, I will prove
that many particles of moisture rise,
along with clouds themselves, from every place
and that both of them, clouds and all water 710 [500]
which the clouds contain, increase together,

just as in us our bodies and our blood
grow at the same rate, and the same is true
for sweat and all the moisture in our limbs.
Also, when clouds are carried by the winds
over the great sea, often they absorb
much water from the sea, like wool fleeces
when they are hung out.¹ In the same manner,
clouds draw water up from every river.

Later, when many water particles 720
have gathered for many reasons and more
have been added on from every quarter,
then swollen clouds seek to discharge water
for two reasons: the power of the wind [510]
forces them together, and the sheer number
of clouds driven into a larger mass

exerts pressure, pushes down from up above,
and makes the rain stream out. Then, too, when winds
thin out the clouds or sun's heat breaks them up
with blows from higher up, they send down rain 730
and drip, just as wax over a hot fire
melts, producing quantities of liquid.

But raging storms of rain occur when clouds
are fiercely pressed by both these forces,
their collective mass and the wind's power.
Rains usually keep pouring down and last
a long time when many water particles [520]
are driven together, when clouds are piled
on one another, when clouds full of water
are borne above them from every region, 740
and when all the steaming earth breathes moisture.

At such times, in the midst of the dark storm,
when the sun's rays have shone right opposite
rain falling from the clouds, then there appears,
standing against the darkness of the clouds,
the colours of the rainbow.

¹ Monserrat and Navarro make the interesting observation that this mention of wool fleeces may be a reference to the practice of hanging them all around a ship and then squeezing them to obtain the fresh water they have absorbed from the sea's evaporation.

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The other things
which are produced and grow all on their own
and all things which, without exception, gather
in the clouds—snow, winds, hail, freezing hoar frosts,
the great force of ice, that mighty power 750
which hardens water and the obstruction [530]
which everywhere holds eager rivers back—
you can easily discover for yourself
and in your mind grasp how all these are made,
the processes by which they are produced,
after you fully know the properties
their basic particles have been assigned.

Pay attention now and learn the reason EARTHQUAKES
there are earthquakes. And first of all assume
the earth below is, like the earth above, 760
full of windy caves everywhere and holds,
within its bosom, many lakes and pools,
cliffs, and broken rocks. And you must suppose
that underneath the surface of the earth
many hidden rivers with strong currents [540]
force waves and submerged rocks to roll around.
For plain facts state that earth should be the same
in every region.¹ And thus with these things
in place and interlinked below the ground,
the earth above shakes when it is disturbed 770
by huge collapses underneath, once time
has turned immense caverns into ruins,
for then the sudden shock makes whole mountains
collapse and tremors spread out far and wide.
That is not surprising, since whole houses
by the street tremble when they are shaken
by wagons, which are not heavy. They shake
just as much if some pebble by the road [550]
disrupts the iron wheel rims on each side.
Then, too, sometimes when a large mass of soil 780
that time has detached from earth tumbles down
into huge extensive pools of water,
the earth is also tossed around and shakes
from the flood of water, just as at times
a container cannot remain steady
unless the liquid it contains has stopped
its unstable motion, shifting to and fro.
Then, too, when the wind which has collected
in cavernous locations underground
blows down from one region and with great force 790
exerts pressure on deep caverns, the earth
tilts in the direction towards which the force [560]

¹ Lucretius is here insisting that the lower half of the earth must be the same as the upper half. This claim is not consistent with his earlier view that the lower part of the earth is composed so that it gradually merges with the aether surrounding the earth and thus keeps the planet suspended in space (see 5.759 ff).

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of rushing wind impels it. Then houses
erected on the surface of the earth,
forced in the same direction, lean over—
and the more each building rises upward
to the sky, the more it tilts—while timbers,
now exposed, are left hanging, suspended there,
ready to drop. And yet men are afraid
of believing that a time of chaos
800
and collapse is waiting for the nature
of this mighty world, even though they see
such a great chunk of earth about to fall.
And yet if the winds do not cease blowing,
no power can hold things back or check them,
as they march ahead to their destruction.
As it is, since these winds now alternate, [570]
easing off and then growing violent,
and, as it were, gather themselves together,
return to the charge, and then, beaten back, 810
withdraw, for this reason the earth threatens
to fall more often than it really does.
For it leans over and shifts back again.
After moving forward, it recovers
its own appropriately balanced state.
And from this cause, therefore, every building
trembles, the top more than in the middle,
the middle more than in the lower parts,
and the bottom to a very small degree.
The same great shaking of the earth also 820
can be caused as follows. When suddenly
the wind, as well as some huge force of air,
gathered outside or buried in the earth,
has hurled itself into hollow places [580]
underground and, to begin with, rages
among huge caverns there, creating havoc
and whirling as it is carried forward,
then later, once its force is fully roused
and energized, it bursts out. As it does,
it splits the earth from deep inside and forms 830
a massive chasm. This is what happened
at Sidon in Syria, and it occurred
at Aegium in the Peloponnese.¹
Such an outrush of air and the earthquake
which ensued overwhelmed these two cities.
Many walled towns have also fallen down
from terrestrial earthquakes. Many cities,
along with their inhabitants, have sunk [590]
to the bottom of the sea. Even if
the air does not break out, nevertheless 840
its very strength and the fierce force of wind

¹ Munro notes that the mention of Aegium is a reference to a famous earthquake which took place in 372 BC.

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are spread, like a quivering ague fit,
 through numerous passageways in the earth
 and thus produce the tremors, just as cold,
 once it penetrates deep inside our limbs,
 shakes them against our will and forces them
 to move and tremble. Thus, men in cities
 are anxious about a double terror:
 they fear the buildings overhead and dread
 the nature of the earth, which, all at once, 850
 may break apart the caverns underground
 and, ripped apart, may open up her jaws
 and in that chaos seek to gorge herself [600]
 on her own ruins. So they may believe
 what they want about how heaven and earth
 will be incorruptible and guaranteed
 eternal safety. Nonetheless, sometimes
 the very power of a present danger
 from some place or other applies this goad
 which makes men fearful that the earth could well 860
 suddenly disappear beneath their feet,
 be carried off to the abyss, and then
 the total sum of things, once overthrown,
 will follow, and the whole world will become
 a chaotic ruin.

To begin with, OCEANS

men find it strange that nature does not make
 the ocean bigger, since so much water
 flows in from all the rivers which reach it [610]
 from every region.¹ Add in wandering rains
 and flying storms, which sprinkle and pour down 870
 on every sea and land. Then add to these
 its own springs. Yet if we compare all these
 to the whole sea, they will increase its bulk
 scarcely by one drop. So it is less strange
 that the great ocean does not grow in size.
 Moreover, with its heat the sun draws off
 large portions of the sea. For we observe
 that with his burning rays the sun dries clothes
 soaked in water. And we well understand
 that there are many seas and these extend 880
 far and wide. And therefore, although the sun
 may at any one location draw up [620]
 from the surface only a small amount
 of moisture, still in such a vast expanse
 it will remove a great deal of water.
 Furthermore, winds sweeping across calm seas
 can also take significant amounts

¹ As a number of commentators note, this passage (lines 608 to 638 in the Latin) seems a very abrupt transition to something unconnected to what precedes it. Bailey suggests that some verses may have been lost which introduced a series of natural paradoxes on the earth.

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of water, for we frequently see roads
 dried out by winds in just a single night
 and soft mud harden into crusts. Besides, 890
 I have shown that clouds also take away
 much water, absorbed from the vast surface
 of the ocean and that they scatter it,
 here and there, in all regions of the world,
 when it rains on earth and winds bring clouds. [630]
 Lastly, since earth is made of porous stuff
 and is in contact with the sea, for earth
 surrounds the ocean shores on every side,
 then water must, just as it moves from land
 into the sea, likewise flow into land 900
 from the briny sea. Salt is filtered out,
 and the liquid material flows back,
 gathering at the head of every river.
 From there it runs back with a fresh current
 over lands through river beds which, once cut,
 take waters on their liquid march downstream.

Now I will explain the reason why fires 910
 sometimes burst out with such tempestuous rage
 from Mount Etna's jaws. For the fiery storm,
 which was no ordinary calamity,
 arose and tyrannized Sicilian fields,
 making near-by people pay attention,
 when they saw all spaces in the heavens
 smoke and sparkle, and in their hearts were full
 of trembling panic at what new changes
 nature was struggling to set in motion.¹
 In such matters your perspective must be
 far and deep. You need to investigate
 over a wide range in all directions,
 so you remember that the sum of things 920
 is beyond all measure and see how small,
 how minutely small, a part of the whole
 one heaven is—not as large a fraction
 as one person is of the entire world.
 If you establish this point properly,
 consider it well, and see it clearly,
 then there will be numerous phenomena
 you will stop wondering about. With us,
 is anyone amazed if a man gets
 a fever in his body which begins 930
 with burning heat, or some illness hurts him
 in his limbs? A foot will suddenly swell up,
 often a sharp pain grabs our teeth or shoots
 right into our eyes. And then that sickness

¹ Mount Etna is an active volcano in Sicily which throughout history up to and including present times has frequently erupted, often with disastrous results. It is not clear whether Lucretius is referring to a particular eruption. There was a major one in 396 BC and another in 122 BC.

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called the sacred fire erupts—it slithers
through the body, and, as it crawls along [660]
inside our limbs, it burns whatever part
it seizes in its grip.¹ For there exist,
not surprisingly, seeds of many things,
and this earth and sky bring us sufficient 940
severe illnesses, and from these can grow
an enormous number of diseases.

Therefore, we must assume all earth and sky
can be supplied out of infinite space
with such objects in sufficient numbers,
and from them earth can suddenly be struck
and shifted and a whirling wind storm sweep
across the sea and land, fires of Etna
can erupt, and heaven burst into flames.

For that happens, too—places in the sky 950 [670]
catch fire. And when particles of water
by chance arrange themselves a certain way,
then more serious rainstorms are produced.²

“But storming fires of Etna,” you may say,
“are too immense.” And that is true. Just as
any river is enormous to someone
who looks at it and who, before that time,
has not seen one greater. So, too, a tree
or man may also appear gigantic.
With all things of every kind the largest 960
that any man has seen he imagines
as prodigious, even though all of them
along with heaven and earth and ocean
are nothing compared to the total sum
of the universal whole.

Now I will show [680]
how that inferno is suddenly roused
and bursts out from those immense furnaces
of Etna. First of all, the whole mountain
is naturally hollow underneath,
supported everywhere on basalt caves. 970
And in all these caves there is wind and air.
For air is transformed into wind once stirred
and set in motion.³ When this wind gets hot
and, as it rages, heats up all the rocks
it makes contact with in its surroundings
and the ground, as well, and draws out from them

¹ The sacred fire has been identified as erysipelas, a severe and very irritating skin infection.

² The point of this rather laboured comment seems to be that, given the infinite number of particles, we should not be astonished that apparently huge natural events (like the eruption of Etna) take place. These seem great to us, but in comparison with infinite space, they are insignificant. Note how Lucretius sees diseases originating from particles which come into our world and onto earth from somewhere in infinite space.

³ Bailey points out that this distinction between wind and air rests on the idea that, for Lucretius, air loses some of its basic particles once it is roused and set in motion and thus is not the same substance.

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a searing fire with swift flames, it rises,
hurling itself high up and thus straight through
the mountain's jaws. And thus it carries heat [690]
long distances, scatters its glowing ash 980
over a huge area, and rolls out
thick, dark, murky smoke, while at the same time
tossing up boulders of amazing weight.
One cannot doubt that these things manifest
the stormy force of air. Then, too, the sea
for the most part diminishes its waves
on that mountain's lower slopes and withdraws
its tide. Caverns extend under the ground
all the way from this sea to the deep mouth
of the mountain. Through these, we must assume, 990
[air enters combined with water, for] facts
compel us [to believe that air comes in
from] the open sea and moves deep inside.
It then blows out, thus pushing up the flames
hurling out rocks, and raising clouds of sand.¹ [700]
For at the summit there are what those men
name craters—features we call jaws and mouths.

There are some things, as well, more than a few,
for which it is not sufficient to set down
one single cause. We must give several, 1000
yet only one of them is the real cause.
Just as if you personally observed
a man's dead body lying some distance off—
it would then be natural to go through
every cause of death, so that you mention
the single cause of that man's death, because
you could not prove that he died from a sword,
or from cold, or from disease, or, perhaps,
from poison, but we know something like that [710]
happened to him. And in many cases, 1010
we can say the same.

The Nile, that river THE RIVER NILE
for all of Egypt, is the only one
on earth which rises in the summertime
and floods the fields. It irrigates Egypt
often in the middle of the season's heat,
perhaps because in summer northern winds,
which at that time of year men give the name
Etesian Winds, confront it at its mouths.²
These blow against the flow and hold it back,
force the waters upstream, fill the channels, 1020
and compel the flowing river to stop.

¹ A line is apparently lost here. I have followed (more or less) Munro's conjecture for the missing material.

² Etesian winds are an annual summer phenomenon in the eastern Mediterranean. They blow steadily from the north-west for much of the summer. The unusual behaviour of the Nile was a subject of great interest in ancient times.

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For there is no doubt that these winds, coming
from the freezing polar constellations, [720]
are carried directly against the stream
flowing from the south, out of those places
which produce great heat. The river rises
in the central region of the daylight,
among human tribes blackened by the sun.
It could also be that when seas are roused
by winds and then push sand into the streams, 1030
great piled up dunes obstruct the river's mouths,
blocking out the waves which move towards them,
which would also make the river's outward flow
less free, and the movement of the water
down the river would be more difficult.
Perhaps it also happens that rains fall
at the Nile's source more during that season,
since at that time northern Etesian winds [730]
blow all the clouds into those areas.
And obviously when the clouds are driven 1040
to the central region of the daylight
and collect there, they are finally pushed
against high mountains in a compact mass
and forcibly compressed. Perhaps the Nile
rises thanks to high Ethiopian hills
far inland, where the sun, whose warming rays
shine everywhere, forces white snow to melt
and flow down to the plains.

Pay attention now, AVERNIAN LAKES AND
and I will show you the kind of nature REGIONS
which all Avernian lakes and areas 1050
possess. First of all, as to the reason
they are called by that name Avernian: [740]
it has been given to them from the fact
that these places are toxic for all birds,
for when they reach these areas and fly
directly over them, the birds forget
to keep rowing with their wings—they slacken
their sails and then, with softly drooping necks,
fall headlong down to earth, if, by some chance,
the nature of the area permits, 1060
or into water, if it so happens
an Avernian lake extends below them.¹
Cumae has a place like that, where mountains
with many hot springs are completely full
of acrid sulphur and give off vapours.
A place like that exists in Athens, too,

¹ The term Avernian is derived from Lake Avernus in Italy, well known for its poisonous vapour, which, so it was believed, killed birds flying over its waters. The name is generally applied to places where birds cannot or will not live. By tradition such regions were closely associated with death and the underworld. The Greek word for "lacking birds" is aornos, and Lucretius seems to hint that this word is related to the name of the lake.

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inside the walls, at the very summit
of the citadel, next to the temple
of Tritonian Pallas, the Nourisher, [750]
where raucous crows on the wing never fly, 1070
not even when the altars smoke with gifts.
That's how much they shun the place—not because
of Pallas's harsh wrath caused by that vigil
Greek poets have sung about, but because
the nature of the place, through its own force,
is enough to bring out this effect.¹ Then, too,
men say in Syria one can see a spot
where even with four-footed animals,
as soon as they first come upon the place,
its force, all by itself, makes them collapse 1080
in a heavy heap, as if, without warning,
they had been killed there as sacrifices
to the gods who rule the dead. All these things,
occur for natural reasons. The causes [760]
which produce them have a clear origin,
just in case men may happen to believe
that the Gate of Orcus is located
in these regions and then we might assume
that gods of the dead perhaps conduct souls
down to shores of Acheron from there, 1090
in the same way men think swift-footed stags,
thanks to their smell, can frequently entice
tribes of wild crawling snakes out of their holes.²
How far this is from valid reasoning
you should learn now, for I will try to state
what really happens.

To begin with, TOXIC SUBSTANCES

I say what I have often said before:
in the earth there are forms of substances [770]
of every kind. Many are good for food
and preserve life, and many can bring on 1100
sicknesses and lead to death more quickly.
And, as we have already pointed out,
in order to maintain life, different things
are better suited to different creatures,
because the natures, interconnections,
and shapes of their primordial particles
are not alike. Many damaging things
pass through the ears, many which are harmful
and damaging to our senses also come

¹ Tritonian Pallas is one of the names given to the Greek goddess Athena. A well known ancient Greek legend claimed that Athena would not allow crows ever to fly above the Acropolis in Athens, as a punishment for bringing her the bad news that the daughters of Cecrops, a mythical king of that city, had failed to obey her instructions. The crow stayed on watch, keeping an eye on the three women (hence the word “vigil”) and informed on them. It is not entirely clear why Athena punished the crow for the disobedience.

² Orcus is the Roman god of the underworld, and the Gate of Orcus is the entrance to the land of the dead. Popular superstition linked this gate to Avernian regions.

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through nostrils, and there are several, too, 1110
 we should refuse to touch, and not a few
 whose sight we should avoid or which possess [780]
 a nauseating taste.

Then you can see

how many things there are whose ill effects
 on human sense are harsh and dangerous,
 toxic and unpleasant. To start with,
 certain trees possess a poisonous shade,
 which is so noxious often they bring on
 headaches in anyone who lies down there,
 reclining underneath them on the grass. 1120
 In the great hills of Helicon, as well,
 there is a tree which, thanks to its flowers,
 which have a nasty smell, has the power
 to kill a man. Clearly these substances
 all spring up out of the earth in this way,
 because the earth holds many particles
 of many things mixed up in many ways
 and sends them out as distinct substances. [790]
 When a night torch has just been extinguished
 and its bitter smell contacts the nostrils, 1130
 it immediately renders unconscious
 a person who, because of some disease,
 keeps falling down and foaming at the mouth.¹
 A woman will collapse and fall asleep
 from the overpowering stench of castor—
 the elegant embroideries will slip
 from her delicate hands—if she smells it
 at the time she has her monthly period.²
 Moreover, in our bodies many things
 relax exhausted limbs and stupefy 1140
 the soul in its location deep within.
 And if you linger too long in hot baths
 and wash yourself when you are rather full, [800]
 how easily and often you can fall
 sitting in the midst of scalding water.
 Also, how readily the heavy force
 and smell of charcoal penetrate the brain,
 if we have not drunk water previously!
 But when it is burning hot and fills up
 the spaces in the house, then the odour 1150
 of that poisonous stuff affects the nerves
 like a deadly blow.³ Surely, too, you see

¹ These are the symptoms of epilepsy.

² Castor (or castoreum) is a liquid taken from small sacs near the anus of the beaver. It has long been used in perfumes and once was a medicinal remedy for various ailments. Pliny the Elder reports that the beaver, when being hunted and aware that the hunter is seeking castor, will chew off its testicles and throw them towards the hunter in order to be left alone.

³ The text is evidently very uncertain here. I have followed Munro's suggestions.

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that sulphur is produced in earth itself
and that bitumen hardens into crusts
with a revolting smell?¹ And furthermore,
when men follow veins of gold and silver,
searching with their picks the hidden regions
deep in the earth, what odours are expelled [810]
underground from mines in Scaptensula?²
What poisonous air comes out of gold mines! 1160
How they change men's faces and complexions!
Have you not seen or heard how those who work
in those places usually die quite soon.
and how the full vital power of life
fails those men whom necessity's strong force
confines to work like that?³ So then clearly
the earth sends all these vapours steaming out
and vents them into clear open spaces
of the sky.

Likewise, Avernian places
must send up vapour which destroys the birds. 1170
It moves up from the earth into the air,
so that it poisons a certain region [820]
of the heavens and, as soon as a bird
on the wing is carried there, it is stopped,
seized by the unseen toxin in the place,
and drops straight down onto the area
the vapour came from. After it falls down,
the same force in that vapour takes away
from all its limbs the vestiges of life.
In fact, the fumes first bring on, as it were, 1180
a certain dizziness. Then, when the bird
falls onto the sources of the poison,
there it must vomit up its life, as well,
because around it is a vast supply
of lethal fumes.

Sometimes it so happens [830]
that this power of Avernian vapours
displaces all the air which is located
between the birds and earth, so that the space
is left almost a void. When flying birds
come directly over such a region, 1190
the power in their wings immediately
ends and is quite useless—on either side
all efforts of their wings have no effect.
And so, when they cannot support themselves
or rely upon their wings, then nature,

¹ Bitumen is a naturally occurring tar-like substance, sometimes called asphalt or heavy crude oil. It contains sulphur.

² Scaptensula is a place in Macedonia famous for its mines. Here the word may be a general name applied to all underground mining.

³ The workers in underground mines were commonly slaves.

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as is clear enough, forces them to sink
under their own weight, downward to the ground.
They fall in what is almost empty space,
and now through all their body's openings
their souls disperse.¹

And furthermore, in wells	1200	[840]
water gets colder in the summertime,		
because the warmth makes earth more rarefied		WATER IN WELLS
and it quickly sends out into the air		
the particles of heat it may contain,		
if it happens to have any of its own,		
and therefore the more earth loses heat,		
the more the moisture hidden underground		
gets colder. Moreover, when all the earth		
is pressed together from the cold, contracts,		
and, as it were, congeals, then obviously,	1210	
as it shrinks, it drives out into the wells		
whatever heat it may itself contain.		
According to reports, there is a spring		
near Ammon's shrine which during the daylight		
is cold and which at night is boiling hot.		
People, amazed at this fountain, believe		[850]
it is quickly heated by fierce sunlight		
below the earth when night has shrouded it		
in fearful darkness. ² But this assertion		
is very far from proper reasoning.	1220	
For if the sun could not warm the water		
on the upper part when it made contact		
with its open surface, even though sunlight		
in air above possesses so much heat,		
how can the sun from underneath the earth,		
which consists of such dense material,		
warm up water, make it intensely hot,		
particularly when his burning rays		[860]
can hardly force heat through walls in houses?		
What, then, is the reason? It is quite clear:	1230	
earth around the fountain is more porous		
than other ground, and particles of heat		
are numerous near that water body.		
And thus, when night, with its dewy shadows,		
covers the earth, immediately the ground		
grows colder deep inside and then contracts.		
As this process takes place, the earth forces		
all heat particles it has within it		
into the fountain, just as if someone		
were squeezing it by hand. This produces	1240	
water which feels hot and its vapour, too.		

¹ At this point it appears that a number of lines have been lost.

² Ammon's shrine is a major religious sanctuary in Libya.

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And later, when rays of the rising sun
have made the ground more loose and rarefied,
as the sun's warming heat grows more intense, [870]
the elementary particles of fire
return once more to where they were before,
and all the water's heat moves to the earth.
That is why the fountain in the daylight
grows cold. Moreover, the liquid matter
in the water is stirred up by those rays 1250
and in the sunlight becomes more porous
from the throbbing heat. And for this reason
it sends out all the particles of heat
it holds inside, just as water often
gives up icy particles it keeps within,
and, by loosening their connections, melts.
There is also a cold spring where coarse flax
held over it is often set on fire,
then at once sends up a flame, and a torch, [880]
kindled in the same manner, casts its light 1260
across the waters, wherever it floats,
pushed forward by the breeze.¹ And this takes place,
we may be sure, because in the water
there are a lot of particles of heat
and, at the bottom, elements of fire
must rise up from the very earth itself
through the entire fountain. At the same time,
these are blown out and move into the air.
However, they are not so numerous
that they can heat the fountain. Moreover, 1270
some force compels these scattered particles
to break out through the water suddenly
and coalesce, once they have moved up.
Near Aradus there is a spring like this [890]
in the sea, where fresh water bubbles up
and pushes aside the salt sea water
all around it.² In many other spots,
the placid surface of the sea offers
thirsty sailors practical assistance,
for in the middle of its salty waves 1280
it vomits up fresh water. And therefore,
in the same way those particles of heat
can burst out through the fountain and disperse.
These elements, once they come together
in the flax or cling onto the body
of the torch, quickly catch fire right away,
because the flax and pine torch also have
many seeds of heat contained inside them.
Do you not perceive as well that a wick [900]

¹ This appears to be a reference to another important religious shrine, the one dedicated to Zeus at Dodona in north-west Greece.

² Aradus is an island off the coast of Asia Minor.

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which has just recently been extinguished, 1290
 if you move it near a night lamp, lights up
 before it can make contact with the flame,
 and that a torch behaves in the same way?
 Besides, many other things catch fire, too,
 at a distance, merely from their contact
 with the heat, before the fire approaches
 and immolates them. We must thus assume
 that this also happens in that fountain.

And now I will proceed to demonstrate MAGNETS
 the natural law by which iron can be drawn 1300
 to that stone the Greeks have called the magnet,
 a name derived from its native country,
 for it originates inside the borders
 of that region where the Magnetes live.¹
 Men are astonished by this stone because [910]
 often it makes a chain of little rings
 suspended from it. In fact, there are times
 one can see five or more of them hanging
 in a line, swaying in the gentle breeze,
 with one attached underneath another, 1310
 suspended there—each ring feels the power
 of the binding attraction of the stone
 through other rings. That shows how much its force
 flows through them all.

With matters of this sort,
 you must clearly establish many things
 before you can provide the principle
 of the thing itself, and you must approach
 by a very long, circuitous road.
 Therefore, I am all the more requesting [920]
 attentive ears and mind. First, from all things— 1320
 no matter what we see—bodies must flow,
 sent out and scattered in a constant stream.
 These strike the eyes and excite our vision.
 From certain objects odours also flow
 continuously, just like cold from rivers,
 heat from the sun, and spray from ocean waves,
 which near the seashore eats away at walls.
 And various noises never stop moving
 through the air. Then, too, when we are walking
 near the sea, a moisture which tastes of salt 1330
 often comes in our mouths, and when we see
 wormwood being diluted in a mixture [930]
 we get a bitter taste. That shows how much

¹ Magnesia is a region of Lydia in Asia Minor. Its inhabitants were called the Magnetes. Watson mentions the story which claims that the name derives from Magnes, the young man who discovered magnetic rocks when he walked over some of them with metal attached to his shoes. The most common naturally occurring magnetic rock is called lodestone, a variety of magnetite.

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certain materials flow from everything,
are carried off, and scattered everywhere.
With this diffusion there is no delay,
no respite, for we can always sense things,
always see and smell them and hear their sounds.
Now I will mention once more how all things
have porous bodies, which I clearly showed 1340
in the first part of my poem, as well.
And though the point is, of course, important
for understanding a great many things,
in the case of this particular matter
which I am going to speak about right now [940]
one must above all establish firmly
that senses do not perceive anything
except matter combined with empty space.
First of all, it so happens that in caves
rocks overhead sweat moisture—they release 1350
water which falls in trickling drops. Likewise,
sweat drips from our entire body, beards grow,
as do hairs on all our limbs and body.
In every vein food is distributed,
which nourishes the body's outer parts
and makes them grow, and that includes our nails.
Similarly, we feel both cold and heat
pass through brass, and we can also sense them
as they make their way through gold and silver,
when we have full cups in our hands. Then, too, 1360 [950]
voices fly through walls of stone in houses,
smells flow through, as do cold and fiery heat,
which has a habit of penetrating
even the power of iron in armour
around the body.¹ And when a tempest
has gathered on earth and in the heavens
and, at the same time, the force of a disease
has also entered, coming from outside,
they both move away, one into the sky,
the other to the earth, and there produce 1370
their natural effects, since there is nothing
which does not possess a porous body.²
To this we should add that all particles
cast off from things are not each provided [960]
with power to stir the same sensations,
nor are they adapted in the same way
for every object. First of all, the sun

¹ The meaning of the Latin is unclear here. There may be, as Bailey points out, a line missing. I have followed Watson's suggestion. The image here is taken from military experience: heat from the fires in war passes through body armour and is felt on the body.

² The sense of the Latin in these lines is not immediately obvious, and different translators have produced widely different readings. The English here is based on Munro's transposition of lines 955 and 956 in the Latin and his overall sense of the passage. The sense seems to be that particles which create storms and others which create diseases both enter from outside and affect us, one in the sky, the other on earth. These are examples of how, given the porous nature of matter, physical substances can move.

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bakes the earth and dries it, but it melts ice
and with its rays compels snow piled up high
on lofty mountains to dissolve. Then, too, 1380
wax turns into liquid if it is placed
in the sun's heat, and in same manner
fire melts bronze and fuses gold, but shrivels
hides and flesh and pulls them all together.
And the liquid stuff of water hardens
iron from fire, but softens hides and flesh
once heat has made them tough. Although there is
no leafy plant which makes more bitter food [970]
for human beings, the wild olive delights
bearded goats as much as if it possessed 1390
flavours of ambrosia dipped in nectar.¹
Also, pigs avoid marjoram and fear
all perfumes, for these are lethal poisons
to bristly swine, although we do perceive
they sometimes give us, so to speak, new life.²
And though to us mud is the foulest muck,
we see that pigs, by contrast, love it so,
that they never have enough of rolling
all around in it.

This one point still remains
which I should speak of before I proceed
to explore matters we are dealing with.
Since the various substances are given
many pores, these openings must be assigned
natures which differ from one another,
with each one possessing its own nature
and passageways, since, as you know, there are
various senses in living animals,
and each of them takes its own material
into itself in its own way—we see
sound comes into us in one way and taste
from flavours by another, and the smells
of vapours by yet another.³ Besides,
we see one thing making its way through stone,
another through wood, another through gold,
and yet other things moving out through glass
and silver. For we notice images
go through the former, and heat the latter,
and through the same passageways certain things
make their way more quickly than do others.
Clearly the nature of the passages
forces this to happen, since it varies
in many ways, as we showed not far above,

¹Ambrosia and nectar are the food and drink of the gods.

² Marjoram is a perennial herb with a strong sweet smell.

³Lines 988 to 989 in the Latin have been omitted. They are repeated at 995 to 996 of the Latin.

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given the different natures and textures
of material things.

And so, once these points
have all been fully settled and set down,
worked out in advance and ready for us,
in what still remains it will be easy, [1000]
using them, to explain the principle
which attracts the power inside iron rings,
and to state openly its entire cause.¹ 1430
First of all, from this stone there must flow off
a great many particles, streams of them,
which by their impacts push aside the air
located between the iron and the stone.
And then, once this space has been vacated
and a large area between the two
has emptied, then the iron particles
at once move forward in a single mass
and fall into that empty space, so that
the ring itself follows and moves that way 1440
with its whole body. For there is nothing
consisting of primordial elements
which contain more intricate connections [1010]
holding it together by its own bonds
than the strong, cold, fearful material
of iron. And therefore what I am claiming
is not so strange: when several particles
move to break out from the iron, they cannot
be carried out into the vacant space,
unless the ring itself moves out with them.² 1450
And that is what it does—it follows on,
until it comes right to the stone itself
and sticks itself to it with hidden bonds.
The same thing occurs in all directions.
Any area where a void is made,
either beside the iron or above it,
neighbouring particles are carried off
immediately into the empty space,
since they are driven onward by impacts [1020]
from somewhere else.³ For they cannot rise up 1460
all on their own into the air. Then, too,
so that this can happen more readily,
these particles are helped along the way
by additional motions and collisions,

¹In the discussion which follows the term iron refers to the material in the rings attracted to or repelled by the magnetic stone. I have added the word “rings” to make that clear here.

²The point here is that the bonds of the iron particles are too strong for individual ones to break free and move away from the ring on their own. So instead they pull the entire ring with them.

³The impacts which drive the iron particles nearest to the empty space out into it would presumably be the particles of iron further away (i.e., on the side away from the magnet), which are constantly moving. And, as Lucretius goes on to mention, the air would also push the particles towards the void.

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because, as soon as air before the ring
is made more rarefied and the region
more void and empty, all air located
behind the ring immediately acts
to push it forward and propel it on,
as if it were blowing on it from behind. 1470
For things are always buffeted by air
surrounding them. But at a time like this,
the air keeps on pushing the iron forwards,
because on one side there is empty space [1030]
which lets the iron enter. And this air
I mention to you subtly penetrates
into the smallest areas of the iron
through their many openings, drives them on,
propelling them ahead, just as the wind
drives a ship and sails. And since substances 1480
have porous bodies and air is placed around
and is in contact with every object,
then all substances must contain some air
inside their physical matter. And thus,
this air, deeply hidden within the iron,
always tossed around in restless motion,
without doubt shakes the ring and from inside
pushes it ahead. And it is quite clear
the ring is carried in the same direction
it has already, once it has started 1490 [1040]
to rush ahead, striving towards the void.
And it also happens that in the iron
material is sometimes pushed away,
back from the stone, for it has a habit
of moving out towards the stone and then,
in turn, going back. I have seen iron rings
from Samothrace even leaping around
and, at the same time, iron filings moving
frantically inside bronze bowls, if one placed
this stone from Magnesia underneath them. 1500
That shows how much the iron seems to yearn
to avoid the stone. Once one places brass
between the two, such a great commotion
is produced because, of course, when the flow
of particles from brass earlier has seized [1050]
and then holds the iron's open passageways,
the stream of elements sent from the stone,
coming later, finds all parts in the iron
completely full—there is no opening
through which it can move, as it could before. 1510
And thus, it is compelled to strike the iron,
to beat against its texture with its waves,
and so to push the iron away from it
and, through the brass, drive off what frequently,

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without the brass, it pulls towards itself.¹
And in these matters do not be surprised
that what streams out from this stone lacks power
to move other things around in the same way.
For they stand still in part through their own weight,
like gold, for instance, and in part because 1520
their material substance is loosely packed,
so that the stream of particles goes through
without impact and they cannot be moved [1060]
in any way. We can observe that wood
is a material of this kind. And thus,
the nature of iron is between the two.
When it absorbs small particles of brass,
then the current from these Magnesian stones
acts to make it move.

And yet these events
are not so foreign to other objects 1530
that I would have much difficulty finding
substances like this which I could mention—
materials adapted to each other
and to nothing else. Firstly, you notice
that only mortar binds stones together.
Wood is joined only with glue made from bulls,
so strongly that veins on wooden timbers [1070]
will frequently split open and then crack
before the binding glue can ease its grip.²
Juices produced from vines dare to mingle 1540
with streams of water, although heavy pitch
and light olive oil refuse to do so.
The only substance purple shellfish dye
can be combined with is wool, so much so
that there is no way it can be removed,
no, not even if you took the trouble
to restore the wool with Neptune's waters,
not if the whole sea wished to wash it clean
with all its moisture. Then, too, is there not
only one substance which joins gold to gold? 1550
Is not tin the only substance which unites

¹ As Munro points out, Lucretius seems to have made an error in his observations and conclusions here, since the actions of a magnet are not affected by placing a non-magnetic substance in between the iron and the magnet. In an email to me, Geoffrey Constable makes the following interesting observations: "On reading this, my instant reaction was puzzlement because bronze is not magnetic – or only very slightly so. However, light dawned when I imagined the construction of the experiment. The magnetised lodestone probably had two poles, North and South, several inches apart. The bronze bowl was possibly of similar size and was located between the two poles. The magnetic 'lines of force' between the poles would cut through the bowl. Replace the bowl with a flat piece of glass and the filings would group into beautiful arcs between the poles. Pop the filings into the bowl and such arcs would be shattered by the bowl's physical obstruction and hence dance about as you describe. The fact that the bowl is made of bronze is unimportant. A bowl of gold, a silver chalice, or a Bakelite Beaker would all cause the same effect. So what Lucretius describes seems to me to be accurate."

² In ancient times bull's hides were an important source of glue.

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brass with brass?¹ How many other cases
might one find like this? What would be the point? [1080]
You do not need such long and winding roads,
not in the least. Nor is it appropriate
for me to devote so much work to this.
No, it is better to be brief—few words
to cover many things: those substances
whose textures mutually correspond,
so that the cavities and material stuff 1560
in one of them match the material stuff
and cavities in the other—these make
the finest unions.² And some things also
can be held in mutual combination,
as if linked together by rings and hooks.
And this seems more likely to be the case
with iron and that stone.

Now, I will explain [1090]
the nature of disease and the reasons
why suddenly the power of illness
can arise, lighting a fire of destruction
for the human race and animal herds. 1570
First, I have shown above that there exist
particles of many things which preserve
our lives. By contrast, there must be many
flying around which bring death and sickness.
And when by chance these happen to gather
and disturb the sky, air becomes diseased.
And all that force of plague and pestilence
arrives, either from outside, moving down
through the heavens like clouds or mists, or else 1580
it often collects itself together [1100]
and rises up out of the very earth,
when, soaked with water from excessive rain
and beaten by the sun, it turns putrid.
Have you not also observed that changes
in air and water affect those people
who travel long distances from their homes
and native lands, because these substances
do not remain unchanged? What do we think
the differences are between the climates 1590
for those in Britain and those in Egypt,
where the world wobbles around its axis?³
How does the climate in Pontus differ

¹ Lucretius uses the Latin phrase for tin, *plumbum album* (white lead), but nowadays white lead is a different substance from tin.

² That is, the best unions are made when the natural irregularities in the two materials fit closely together, like pieces of Lego.

³ Bailey notes that in ancient times people thought the axis of the earth slanted from the upper part in the north down towards Egypt. The verb here (*claudico*), which indicates defective or erratic motion, may possibly be a reference to the axial precession of the earth, the process by which the orientation of the earth's axis rotates (like a wobbling top) and traces out a circular motion in about 26,000 years.

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from the climate in Gades, and so on,
 right up to the races of men baked black
 by the scorching sun?¹ And because we know
 these four climates arising from four winds [1110]
 and four regions of the sky are different,
 so we see that men's colour and appearance
 vary greatly and that disease strikes them 1600
 in different ways, each according to his race.
 There is elephant sickness, which is born
 by the river Nile in middle Egypt
 and nowhere else.² In Attica the feet
 are afflicted with disease, as are the eyes
 in the land of the Achaeans. And thus,
 different areas inflict injuries
 on different parts and limbs, something brought on
 by variations in the air. Therefore,
 when a sky that is, by chance, strange to us 1610
 sets itself in motion, then harmful air
 little by little starts to creep about,
 like mist and cloud disturbing every place [1120]
 where they advance, compelling it to change.
 So it also happens that when that air
 ends up entering our sky, it corrupts it
 and makes it like itself, harmful to us.
 Thus, this new destructive force and sickness
 either quickly falls onto the water
 or even penetrates into the crops 1620
 or into other nourishment for men
 and food for cattle, or else this force stays
 suspended in the very air, so that
 when we breathe we take in air mixed with it
 and, as we do that, must also absorb [1130]
 those sicknesses into our bodies.
 In a similar way a pestilence
 often falls on cattle, too, and sickness
 on dull bleating sheep. Nor does it matter
 whether we go somewhere hostile to us 1630
 and transform the nature of the climate
 which wraps itself around us, or whether
 nature on her own brings us toxic air
 or something else we are not accustomed
 to experience which, when it first arrives,
 can then attack us.

Such a cause of disease, such a poisonous atmosphere, once filled fields in the lands of Cecrops with the dead, emptying roads and draining the city	PLAGUE IN ATHENS [1140]
---	--

¹ Gades is now the city of Cadiz.

² Elephant sickness is elephantiasis, which can cause massive swellings under the skin.

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of its inhabitants.¹ The sickness arose
deep inside the land of Egypt and then,
moving across great portions of the sky
and expanses of the sea, at last reached
the entire population of Pandion,
where it sat, brooding.² Then, group after group
were handed over to disease and death.
First of all, people felt their heads burning
from a raging heat, and both eyes turned red
with a suffused glare. Their throats, black inside,
oozed blood, as well, and the vocal passage,
choked with ulcers, was obstructed, their tongues,
the mind's interpreter, dripped blood, weakened
with disease, hard to move, and rough to touch. [1150]
Once the power of the illness shifted
down through the throat, filled the chest, and gathered
in patients' suffering hearts, at that point
all the bands of life were truly loosened.
The air they breathed out from their mouths gave off
a putrid smell, like the stink emitted
by rotting corpses thrown out unburied. 1660
And all mental powers, all the body,
then quickly weakened, at the very door
of death. This intolerable suffering
always brought with it painful anxiety
and complaints mixed in with cries of anguish.
Frequent dry retching, often day and night, [1160]
forced limbs and sinews to convulse in spasms,
and broke down those who were already tired
and wore them out. And yet you could not see
the outermost surface of the body 1670
on any of them burn with extreme heat.
Instead it produced a tepid feeling
to the touch. At the same time, the body
was completely red, as if burned with sores,
the way it is when sacred fire spreads out
through all the limbs. But people's internal parts
truly were on fire, right down to the bones.
A flame blazed in the stomach, like the fire
inside a furnace. You could not cover [1170]
anyone's limbs with something light and thin— 1680
that offered no relief to anyone—
only wind and cold. Some men plunged their limbs
burning with disease, into freezing streams,
and hurled naked bodies in the water.
Many threw themselves headfirst in deep wells,

¹ The land of Cecrops is Athens and its surrounding territory. This final section of the poem is very closely based on Thucydides' famous description of the plague in Athens at the end of the first year of the Peloponnesian War (430 BC). Munro notes that scholars have come up with a long list of different possibilities for the disease (typhus, bubonic plague, scarlet fever, smallpox, and so on).

² Pandion was a legendary king of Athens.

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with their mouths wide open, seeking water,
for a parching and unquenchable thirst
soaked their bodies and made gigantic gulps
the same as a few drops. With this disease
there was no let up. The bodies lay there, 1690
totally exhausted. The healing arts
muttered in silent dread, for the patients,
on fire with fever, rolled wide open eyes [1180]
over and over, and did not fall asleep.
Then many other signs of death appeared:
minds disturbed by anxiety and fear,
gloomy frowns, a fierce and wild appearance,
ears in pain, as well, and full of noises,
breaths were quick or else deep but rarely drawn,
moist sweat glistening on the neck, saliva 1700
thin and scanty, with a yellowish tint,
and tasting salty, spat out with effort
by coughing it up through rasping gullets.
Sinews in hands did not stop contracting, [1190]
limbs kept trembling, and little by little
cold kept inching its way up from the feet.
And then, in the last moments, the nostrils
were pinched, the tip of the nose sharp and thin,
the eyes hollowed out, the temples shrunken,
the skin icy and hard, the mouth gaping 1710
in a grin, the forehead tense and bulging.
Not long after that, the limbs would lie there
in rigid death. And when the sun shone out
on the eighth day or else when light returned
for the ninth time, they would, for the most part,
yield up their lives. And if any of them,
at that moment, escaped a lethal fate,
then later decline and death still waited
from filthy ulcers and black discharges [1200]
of the bowels, or else, with a head in pain, 1720
a large quantity of corrupted blood
would often pour out of stuffed up nostrils.
Into this the entire strength and substance
of the man would flow. And then, if someone
escaped this violent outflow of foul blood,
the disease still moved into his sinews
and limbs, even to the sexual organs
on his body. Some, excessively afraid
of the gates of death, would keep on living
with these male organs sliced off by a knife, 1730
some still stayed alive without hands or feet, [1210]
and others kept on going without their eyes—
that shows how much a bitter fear of death
had overtaken them. And some were gripped
by loss of memory for everything—
they could not even recognize themselves.

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And although many unburied bodies
lay piled on heaps of corpses on the ground,
still the race of birds and wild animals
would roam some distance off, so as to shun 1740
the nauseous smell or, when one tasted flesh,
it would waste away in a rapid death.
But in those days hardly any birds at all
were to be observed, and the grim species [1220]
of wild creatures did not leave the forests.
Many succumbed to the disease and died.
Above all, faithful dogs in every street
lay prone and, after a struggle, gave up,
for the force of the disease would wrench life
from their bodies. The lonely burials 1750
with no one present proceeded quickly,
like a race. And there was no remedy
which was a certain cure for everyone.¹
What gave one person power to inhale
vital air in through his mouth and stare up
at regions of the sky was lethal to others
and brought on their deaths. But in these events
one thing especially calamitous [1230]
and painful was that once someone found out
he himself was afflicted with the plague, 1760
then, as if he had been condemned to die,
he gave up hope and, his heart full of grief,
lay there gazing at death, surrendering
his soul right on the spot. As it turned out,
there was no pause: people kept being attacked,
corrupted by this rapacious sickness,
one after another, like woolly flocks
and herds of cattle. And this, above all,
piled death on death, for all those who refused
to care for their own sick from fear of death 1770 [1240]
and excessive greed for life were punished
soon afterwards by ruinous neglect
with a harsh and evil death, abandoned
and devoid of help. But those who acted
more responsibly died from the contagion
and from the efforts which a sense of shame
and the soft entreaties of worn-out men,
together with their voices of complaint,
forced them to undertake. As a result,
the best people suffered this kind of death. 1780
Then, too, by now all shepherds and herders,
as well as sturdy farmers who guided
curving ploughs, were falling sick. Their bodies,
thrown in a pile, lay deep inside their huts,
given to death by disease and poverty. [1250]

¹ The transition to this sentence appears abrupt and awkward. Bailey suggests that some lines connecting this sentence with what is immediately before it may be missing.

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At times, you could see the lifeless bodies
of parents on top of lifeless children
and then the reverse, children losing life
lying above their mothers and their fathers.
And to no small degree this disaster
1790 flowed into the city from the country,
carried in by crowds of diseased peasants
who gathered there, affected by this plague,
from every region. They completely filled
all districts and houses, crammed in together.
As a result, death piled them up in heaps—
all the more so in the heat of summer.
Many bodies prostrate with thirst were thrown
into the roadway and lay there stretched out
1800 by water fountains, their breathing blocked off
by the excessively sweet taste of water.
Everywhere in open public places
and in the streets you might see many limbs
hanging down from half-dead bodies, smelling
disgusting, covered in rags, and dying
in their bodies' filth, only skin and bones,
now almost buried in dreadful sores and dirt.
And death had filled all the sacred temples
with lifeless bodies, and all holy shrines
of divine beings were completely full
1810 of corpses everywhere, since these places
the temple keepers had all filled with guests.
And, in fact, by now worship of the gods
and their sanctity did not count for much.
The present suffering overpowered that.
Nor did the funeral customs continue
in the city, rites with which before this
the people had always been accustomed
to be buried. For the whole populace
was confused and in a state of panic,
1820 and each man, in his grief, buried his own
as best he could. And sudden disaster
and need prompted many horrific acts.
For, with mighty cries of sorrow, men placed
their own relatives on funeral pyres
built up for strangers, and applied torches,
often fighting quarrels with much bloodshed
1830 rather than leave the bodies. With corpses
heaped up in different piles people struggled
to bury the multitude of their dead,
and then, exhausted, they went home in tears
and grief. And most of them, in their distress,
would go to bed. At such a dreadful time
no person could be found unaffected

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either by disease, or death, or sorrow.¹

¹ Following the practice of some other editors, I have transferred the last lines here (1728 ff) from their customary position (1247-1251 in the Latin), where they have no clear connection to what immediately precedes them.